Attempts at Hartshorne's *Algebraic Geometry* Exercises

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Chapter 1

Varieties

Chapter 1, Section 1

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2. Let $Y \subseteq \mathbb{A}^3$ be the set $Y = \{(t, t^2, t^3) \mid t \in k\}$. Show that Y is an affine variety of dimension 1. Find generators for the ideal $I(Y)$. Show that $A(Y)$ is isomorphic to a polynomial ring in one variable over k. We say that Y is given by the *parametric representation* $x = t$, $y = t^2$, $z = t^3$.

Proof. We have $I(Y) = (x^2 - y, x^3 - z)$, and Y is irreducible since it is homeomorphic to the affine line $\mathbb{A}^1$ under the map $t \mapsto (t, t^2, t^3)$, thus Y is an affine variety of dimension 1. Consider the map $f : k[x, y, z] \to k[t]$ defined by

$$x \mapsto t, \quad y \mapsto t^2, \quad z \mapsto t^3.$$

It is clearly surjective, and $k[t]$ is an integral domain, which implies $\ker f$ is a prime ideal. Since $\dim k[t] = 1$, the set Y is an affine variety of dimension 1 provided $\ker f = I(Y)$. We obviously have $I(Y) \subseteq \ker f$. By (1.8A), we have

$$\operatorname{height} \ker f + \dim k[t] = \dim k[x, y, z] \implies \operatorname{height} \ker f = 2,$$

and $(x^2 - y)$ is a prime ideal contained in $I(Y)$, thus $I(Y)$ cannot be properly contained in $\ker f$, hence $\ker f = I(Y)$. □

3. Let Y be the algebraic set in $\mathbb{A}^3$ defined by the two polynomials $x^2 - yz$ and $xz - x$. Show that Y is a union of three irreducible components. Describe them and find their prime ideals.

Proof. We claim that

$$Y = Z((x, y)) \cup Z((x, z)) \cup Z((x^2 - y, z - 1)).$$

It is clear the subsets $Z((x, y))$ and $Z((x, z))$ are irreducible. What is less obvious is the irreducibility of $Z((x^2 - y, z - 1))$. Observe that

$$\frac{k[x, y, z]}{(x^2 - y, z - 1)} \cong \frac{k[x, y]}{(x^2 - y)} \cong k[x, x^2] \cong k[x],$$

hence $(x^2 - y, z - 1)$ is a prime ideal, hence $Z((x^2 - y, z - 1))$ is irreducible. If $P = (u, v, w) \in Y$, then $u^2 - vw = 0$ and $uw - u = 0$, so by the second equation either $u = 0$ or $u = 1$. If $u = 0$, either $v = 0$ or $w = 0$, which implies $P \in Z((x, y))$ or $P \in Z((x, z))$. If $u = 1$, then we have $P \in Z((x^2 - y, z - 1))$. The converse direction follows in similar fashion. □

6. Any nonempty open subset of an irreducible topological space is dense and irreducible. If Y is a subset of a topological space X, which is irreducible in its induced topology, then the closure $\overline{Y}$ is also irreducible.

Proof. Let X be an irreducible topological space and let U be a non-empty proper open subset of X. Then $C = X - U$ is a proper closed subset of X, so we have $X = \overline{U} \cup C$, hence $\overline{U} = X$. Conversely, if every open subset of a topological space is dense, and we have $X = C_1 \cup C_2$ for two closed subsets of X with C_1 proper, then $U_1 = X - C_1$ is an open subset contained in C_2, which implies $X = \overline{U_1} \subseteq \overline{C_2} = C_2$, hence $C_2 = X$. It follows immediately that any open subset of an irreducible space is irreducible: if $V \subset U$ are open subsets, then the closure of V as a subspace of U is the intersection of U and its closure as a subspace of X, and V is dense in X, hence it is dense in U. Now, if U is an open subset of $\overline{Y}$ for an irreducible subset Y of any topological space X, then it must meet Y by definition of the closure of a subset, then the closure of U as a subspace of X contains Y since U is dense in Y, hence $\overline{U} \cap \overline{Y} = \overline{Y}$. □

7. (a) Show that the following conditions are equivalent for a topological space X:

(i) X is noetherian;

(ii) every nonempty family of closed subsets has a minimal element;

(iii) X satisfies the ascending chain condition for open subsets;

(vi) every nonempty family of open subsets has a maximal element.

(b) A noetherian topological space is *quasi-compact*.

(c) A subset of a noetherian topological space is noetherian in its induced topology.

(d) A noetherian space which is also Hausdorff must be a finite set with the discrete topology.

Proof.

(a) (i) $\implies$ (ii) If (ii) is false there is a non-empty collection T of closed subsets with no minimal element, and we can construct inductively a non-terminating strictly decreasing sequence in T.

(ii) $\implies$ (iii) Let $U_1 \subset U_2 \subset \cdots$ be an ascending chain of open subsets of X and let $C_i = X - U_i$. Then $\{C_i\}_{i\geq 1}$ is a non-empty collection of closed subsets of X, hence has a minimal element, say C_m. Hence, we have $U_m = U_{m+1} = \cdots$.

(iii) $\implies$ (iv) If (iv) is false, then there is a non-empty collection S of open subsets with no maximal element, and we can construct inductively a non-terminating strictly increasing sequence in S.

(iv) $\implies$ (i) Let $C_1 \supseteq C_2 \supseteq \cdots$ be a descending chain of closed subsets of X and let $U_i = X - C_i$. Then $\{U_i\}_{i\geq 1}$ is a non-empty collection of open subsets of X, hence has a maximal element, say U_m. Hence, we have $C_m = C_{m+1} = \cdots$.

(b) Suppose X is not quasi-compact so that the set Σ of non-quasi-compact closed subsets of X is non-empty. Let Y be a minimal element in Σ. If U is an open subset of Y, then $A = \overline{U}$ and $B = Y - U$ are closed subsets of X contained in Y such that $Y = A \cup B$. Since Y is minimal amongst the set of non-quasi-compact closed subsets in X, A and B must be compact; however, a finite union of compact sets is compact, a contradiction.

(c) Let Y be a subspace of X. Then any open subset of Y is of the form $V = U \cap Y$ for some open subset U of X, so if $V_1 \subseteq V_2 \subseteq \cdots$ is an ascending chain of open sets in Y with $V_i = U_i \cap Y$, then let $U_i' = \bigcup_{j\leq i} U_i$ so that we have the ascending chain $U_1' \subseteq U_2' \subseteq \cdots$ and

$$U_i' \cap Y = \bigcup_{j\leq i} U_i \cap Y = \bigcup_{j\leq i} V_i = V_i.$$

If X is noetherian, then this chain eventually terminates, say at $i = n$, which implies the chain $V_1 \subseteq V_2 \subseteq \cdots$ terminates at $V_n = U_n' \cap Y$. Hence, Y is noetherian.

(d) Every subspace of a noetherian space is compact since every subspace is noetherian, and in a Hausdorff space every compact set is closed, hence every subset is closed, therefore a noetherian Hausdorff space must be discrete. Finiteness follows from quasi-compactness.

□

8. Let Y be an affine variety of dimension r in $\mathbb{A}^n$. Let H be a hypersurface in $\mathbb{A}^n$, and assume that $Y \nsubseteq H$. Then every irreducible component of $Y \cap H$ has dimension $r - 1$.

Proof. Let $Y = Z(\mathfrak{p})$ and $H = Z(f)$ for some prime ideal $\mathfrak{p}$ and irreducible polynomial f in $k[x_1, \ldots, x_n]$. If $Y \nsubseteq H$, then $f \notin \mathfrak{p}$, so the image of f in $A(Y) = k[x_1, \ldots, x_n]/\mathfrak{p}$ is not a zero-divisor. If f is a unit in $A(Y)$, then f does not vanish at any points in Y, which implies Y and H does not intersect, so $Y \cap H$ has no irreducible components (the empty set is defined to be not irreducible). Otherwise, the coordinate ring of an irreducible component of $Y \cap H$ corresponds to a minimal prime ideal $\mathfrak{q}$ in $A(Y)$ which contains f. By Krull's Hauptidealsatz, any such prime ideal must have height 1, and since $A(Y)$ has dimension r, it follows that $A(Y)/\mathfrak{q}$ has dimension $r - 1$ by (1.8A). □

9. Let $\mathfrak{a} \subseteq A = k[x_1, \ldots, x_n]$ be an ideal which can be generated by r elements. Then every irreducible component of $Z(\mathfrak{a})$ has dimension $\geq n - r$.

Proof. We proceed by induction on r. If $r = 1$ and $\mathfrak{a} = (f)$, then the irreducible components of $Z(\mathfrak{a})$ correspond to the irreducible factors of f, which are hypersurfaces in $\mathbb{A}^n$ and therefore has dimension $n-1$. Let $r > 1$ and assume the statement to be true for $r-1$ and suppose $\mathfrak{a} = (f_1, \ldots, f_{r-1}, f_r)$. An irreducible component of $Z(f_1, \ldots, f_{r-1}, f_r)$ corresponds to a minimal prime ideal $\mathfrak{p}$ in $k[x_1, \ldots, x_n]$ containing $(f_1, \ldots, f_{r-1}, f_r)$, and by the inductive hypothesis a minimal prime ideal containing $(f_1, \ldots, f_{r-1})$ have height at most $r-1$, so by Krull's Hauptidealsatz, $\mathfrak{p}$ can have height at most r, hence $A/\mathfrak{p}$ has dimension at least $n-r$, hence an irreducible component of $Z(f_1, \ldots, f_{r-1}, f_r)$ have dimension at least $n-r$. □

10. (a) If Y is any subset of a topological space X, then $\dim Y \leq \dim X$.

(b) If X is a topological space which is covered by a family of open subsets $\{U_i\}$, then $\dim X = \sup \dim U_i$

(c) Give an example of a topological space X and a dense open subset U with $\dim U < \dim X$.

(d) If Y is a closed subset of an irreducible finite dimensional topological space X, and if $\dim Y = \dim X$, then $Y = X$.

(e) Give an example of a noetherian topological space of infinite dimension.

Proof.

(a) If $Z_0 \subset Z_1 \subset \cdots \subset Z_n$ is a chain of distinct irreducible closed subsets of Y, then their closures as subsets of X are also irreducible by Exercise 1.6, that is $\overline{Z_0} \subset \overline{Z_1} \subset \cdots \subset \overline{Z_n}$ is a chain of distinct irreducible closed subsets of X, hence $\dim Y \leq \dim X$. It is indeed distinct since if $x \in Z_{i+1}$ and $x \in \overline{Z_i} = \overline{C_i \cap Y} \subseteq C_i \cap \overline{Y}$ for some closed subset C_i of X, then x must be in $Z_{i+1} \cap C_i \cap \overline{Y} = Z_i$.

(b) By (a) we have $\sup \dim U_i \leq \dim X$. Conversely, if $Z_0 \subset Z_1 \subset \cdots \subset Z_n$ is a chain of distinct irreducible closed subsets of X, then there exists $U \in \{U_i\}$ such that $V_0 = Z_0 \cap U \neq \emptyset$, thus $V_i = Z_i \cap U \neq \emptyset$, so by Exercise 1.6 $V_0 \subset V_1 \subset \cdots \subset V_n$ is a distinct chain of irreducible closed subsets in U, therefore $n \leq \dim U$, hence taking the supremum we have $\dim X \leq \sup \dim U_i$.

(c) Let $X = \{a, b\}$ with topology defined by the open subsets $\{\emptyset, \{a\}, X\}$. Then $U = \{a\}$ is dense since the only closed subset containing U is X, and it has dimension 0, but X has dimension 1.

(d) If Y itself is irreducible, then any chain $Z_0 \subset Z_1 \subset \cdots \subset Z_n$ of irreducible closed subsets of Y can be extended to $Z_0 \subset Z_1 \subset \cdots \subset Z_n \subset Y$ unless $Z_n = Y$, so if $n = \dim Y$ then we must have $Z_n = Y$, and since $\dim Y = \dim X$, Y cannot be a proper subset of X. Otherwise, if Y is not irreducible then Y and therefore Z_n are proper subsets of X, and by the proof of (a) any chain in Y induces a chain of same length in X. Since $\overline{Z}_n$ is a proper subset of X, the chain can be extended to $\overline{Z_0} \subset \overline{Z_1} \subset \cdots \subset \overline{Z_n} \subset X$, contradicting the dimensions of X and Y. Hence, $Y = X$.

(e) Let $X = \mathbb{N}$ with topology defined by the closed sets $C_n = \{1, \ldots, n\}$. If $\{C_{n_\alpha}\}$ is any collection of closed subsets, then the minimal element is C_{n_β} for $n_\beta = \min n_\alpha$, so X is noetherian. Also, every closed set is irreducible, so we have an infinite chain of irreducible closed subsets $C_1 \subset C_2 \subset \cdots$.

□

Chapter 1, Section 2

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1. Prove the "homogenous Nullstellensatz," which says if $\mathfrak{a} \subseteq S$ is a homogenous ideal, and if $f \in S$ is a homogenous polynomial with $\deg f > 0$, such that $f(P) = 0$ for all $P \in Z(\mathfrak{a})$ in $\mathbb{P}^n$, then $f^q \in \mathfrak{a}$ for some $q > 0$.

Proof. We can apply the usual Nullstellensatz by viewing f as a polynomial over $\mathbb{A}^{n+1}$ and $Z(\mathfrak{a})$ as the set $Z(\mathfrak{a})'$ consisting of all $(n+1)$-uples $Q = (a_0, \ldots, a_{n+1}) \in \mathbb{A}^{n+1}$ such that Q is the homogenous coordinate of some point $P \in Z(\mathfrak{a})$. Since f is homogenous, $f(P) = 0$ for $P \in \mathbb{P}^n$ if and only if $f(Q) = 0$ where P has Q as homogenous coordinates. Thus, $f^q \in \mathfrak{a}$ by the Nullstellensatz. □

2. For a homogenous ideal $\mathfrak{a} \subseteq S$, show that the following conditions are equivalent:

(i) $Z(\mathfrak{a}) = \emptyset$;

(ii) $\sqrt{\mathfrak{a}} =$ either S or the ideal $S_+ = \bigoplus_{d>0} S_d$;

(iii) $\mathfrak{a} \supseteq S_d$ for some $d > 0$.

Proof. (i) $\implies$ (ii) Let $S = k[x_0, \ldots, x_n]$. If $Z(\mathfrak{a})$, then all homogenous polynomials in S with degree > 0 are in $\sqrt{\mathfrak{a}}$ by the Nullstellensatz, so we have $S_+ \subseteq \sqrt{\mathfrak{a}}$. If $\sqrt{\mathfrak{a}}$ contains any element in $S - S_+$, then $\sqrt{\mathfrak{a}}$ contains an element in S_0, which is a unit. Hence, $\sqrt{\mathfrak{a}} = S$.

(ii) $\implies$ (iii) Either case $S_+ \subseteq \sqrt{\mathfrak{a}}$, so $x_i^{d_i} \in \mathfrak{a}$ for some $d_i \geq 0$ for each i, which implies $S_d \subseteq \mathfrak{a}$ where $d = \max d_i$.

(iii) $\implies$ (i) Let $d > 0$ be the smallest integer such that $\mathfrak{a} \supseteq S_d$, then we have $\mathfrak{a} \supseteq \bigoplus_{l \geq d} S_l$ so that $x_i^l \in \mathfrak{a}$ for all $0 \leq i \leq n$ and $l \geq d$. If there exists $P = (a_0, \ldots, a_n) \in Z(\mathfrak{a})$, then $a_i^d = a_i^{d+1} = 0$, which implies $a_i = 0$ for all $0 \leq i \leq n$, which is impossible. □

4. (a) There is a 1-1 inclusion-reversing correspondence between algebraic sets in $\mathbb{P}^n$, and homogenous radicals of S not equal S_+, given by $Y \mapsto I(Y)$ and $\mathfrak{a} \mapsto Z(\mathfrak{a})$. *Note:* Since S_+ does not occur in this correspondence, it is sometimes called the *irrelevant* maximal ideal of S.

(b) An algebraic set $Y \subseteq \mathbb{P}^n$ is irreducible if and only if $I(Y)$ is a prime ideal.

(c) Show that $\mathbb{P}^n$ itself is irreducible.

Proof.

(a) Only the last part that states S_+ does not occur in this correspondence is new. Indeed, $I(Y) \supseteq S_+$ for some algebraic set $Y = Z(\mathfrak{a})$, then $Z(\mathfrak{a}) = \emptyset$ b (I, Ex. 2.2), and any constant polynomial $f \in S_0$ vacuously satisfies $f(P) = 0$ for all $P \in Z(\mathfrak{a})$. Hence, $I(Y) = S$.

(b) By the 1-1 correspondence from (a), $I(Y)$ is a homogenous ideal, so it is sufficient to show for any two homogenous elements f, g, that $fg \in I(Y)$ implies $f \in I(Y)$ or $g(Y)$. Indeed, if $fg \in I(Y)$, then $Y \subseteq Z(fg) = Z(f) \cup Z(g)$, thus $Y = (Y \cap Z(f)) \cup (Y \cap Z(g))$ both being closed subsets of Y. Since Y is irreducible, we have either $Y = Y \cap Z(f)$, in which case $Y \subseteq Z(f)$, or $Y \subseteq Z(g)$. Hence, either $f \in I(Y)$ or $g \in I(Y)$.

Conversely, let $\mathfrak{p}$ be a homogenous prime ideal, and suppose that $Z(\mathfrak{p}) = Y_1 \cup Y_2$. Then $\mathfrak{p} = I(Y_1) \cap I(Y_2)$, so either $\mathfrak{p} = I(Y_1)$ or $\mathfrak{p} = I(Y_2)$. Thus, $Z(\mathfrak{p}) = Y_1$ or Y_2, hence it is irreducible.

(c) $I(\mathbb{P}^n) = (0)$, which is a prime ideal.

□

5. (a) $\mathbb{P}^n$ is a noetherian topological space.

(b) Every algebraic set in $\mathbb{P}^n$ can be written uniquely as a finite union of irreducible algebraic sets, no one containing another. These are called its *irreducible components*.

Proof.

(a) $\mathbb{P}^n$ is covered by the open set U_i defined the non-vanshing of the ith homogenous coordinate, each of which is homeomorphic to the affine plane $\mathbb{A}^n$. The affine plane is a noetherian topological space, so it suffices to show a finite union of noetherian topological spaces is also noetherian. An equivalent condition for a space to be noetherian is if every subset is quasi-compact by (A.M. p. 79). If Y is any subset of $\mathbb{P}^n$, then $Y = Y \cap \bigcup U_i$, each of which is quasi-compact in the induced topology of U_i, so Y is a finite union of quasi-compact sets. Hence, Y is quasi-compact.

(b) Follows from (a), (1.5), and (I, Ex. 2.4b).

□

6. If Y is a projective variety with homogenous coordinate ring $S(Y)$, show that $\dim S(Y) = \dim Y + 1$.

Proof. Let $\varphi_i : U_i \to \mathbb{A}^n$ be the homeomoprhism of (2.2), let Y_i be the affine variety $\varphi_i(Y \cap U_i)$, and let $A(Y_i)$ be its affine coordinate ring. If $g(y_1, \ldots, y_n)$ is an element of $A(Y_i)$, then define the map $A(Y_i) \to S(Y)_{x_i}$ as $g(x_0/x_i, \ldots, x_n/x_i)$, or equivalently $g \mapsto \varphi_i^* g = g \circ \varphi_i$; thus we can identify $A(Y_i)$ with the subring of elements of degree 0 of the localized ring $S(Y)_{x_i}$. Then, $S(Y)_{x_i} \simeq A(Y_i)[x_i, x_i^{-1}]$ since every monomial in $k[x_0, \ldots, x_n]$ can be written as

$$x_0^{d_0} \cdots x_i^{d_i} \cdots x_n^{d_n} = \frac{x_0^{d_0} \cdots \widehat{x_i^{d_i}} \cdots x_n^{d_n}}{x_i^{d_0 + \cdots + \widehat{d_i} + \cdots + d_n}} x_i^{d_0 + \cdots + d_n},$$

where $\widehat{\ }$ denotes omission, and the quotient is in the image of $A(Y_i) \to S(Y)_{x_i}$. By (1.8A), the dimension of $S(Y)$ is equal to the transcendence degree of the quotient field of $S(Y)$, which is isomorphic to the quotient field of $S(Y)_{x_i}$. The dimension of $A(Y_i)$ is equal to the dimension to Y_i by (1.7). Therefore, we have

$$\begin{aligned} \dim S(Y) &= \dim S(Y)_{x_i} \\ &= \dim A(Y_i)[x_i, x_i^{-1}] \\ &= \dim A(Y_i) + 1 \\ &= \dim Y_i + 1. \end{aligned}$$

Since the Y_i cover Y, $\dim Y = \sup \dim Y_i$, so $\dim S(Y) = \dim Y + 1$ By (I, Ex. 1.10). □

7. (a) $\dim \mathbb{P}^n = n$.

(b) If $Y \subseteq \mathbb{P}^n$ is quasi-projective variety, then $\dim Y = \dim \overline{Y}$.

Proof.

(a) $\dim \mathbb{P}^n = \dim k[x_0, \ldots, x_n] - 1 = n$.

(b) By Exercise 2.6, we have $\dim Y = \dim Y_i$ if Y_i is non-empty, so by (1.10) $\dim \overline{Y} = \dim \overline{Y}_i = \dim Y_i = \dim Y$.

□

8. A projective variety $Y \subseteq \mathbb{P}^n$ has dimension $n-1$ if and only if it is the zero set of a single irreducible homogenous polynomial f of positive degree. Y is called a *hypersurface* in $\mathbb{P}^n$.

Proof. If Y has dimension $n-1$, then by (1.13) Y is the union of affine varieties Y_i of dimension $n-1$. By the map β in the proof of (2.2) and (1.13), each Y_i is the zero set of an irreducible homogenous polynomial f of positive degree. Hence,

$$Y = \bigcup Y_i = \bigcup_{Y_i \neq \emptyset} Z(f_i) = Z(f_0 \cdots f_n).$$

Conversely, if $Y = Z(f)$ for some homogenous polynomial f of positive degree, then $Y_i = Z(\alpha(f))$ where α is the map defined in the proof of (2.2). Hence, $\dim Y = \dim Y_i = n - 1$ by (1.13) and (Ex. 2.6). □

12. *The d-Uple Embedding.* For given $n, d > 0$, let $M_0, M_1, \ldots, M_N$ be all the monomials of degree d in the $n+1$ variables $x_0, \ldots, x_n$, where $N = \binom{n+d}{n} - 1$. We define a mapping $\rho_d : \mathbb{P}^n \to \mathbb{P}^N$ by sending the point $P = (a_0, \ldots, a_n)$ to the point $\rho_d(P) = (M_0(a), \ldots, M_N(a))$ obtained by substituting the a_i in the monomials M_j. This is called the d-uple *embedding* of $\mathbb{P}^n$ in $\mathbb{P}^N$. For example, if $n = 1$, $d = 2$, then $N = 2$, and the image of Y of the 2-uple embedding of $\mathbb{P}^1$ in $\mathbb{P}^2$ is a conic.

(a) Let $\theta : k[y_0, \ldots, y_N] \to k[x_0, \ldots, x_n]$ be the homomorphism defined by sending y_i to M_i, and let $\mathfrak{a}$ be the kernel of θ. Then $\mathfrak{a}$ is a homogenous prime ideal, and so $Z(\mathfrak{a})$ is a projective variety in $\mathbb{P}^N$.

(b) Show that the image of ρ_d is exactly $Z(\mathfrak{a})$.

(c) Now show that ρ_d is a homeomoprhism of $\mathbb{P}^n$ onto the projective varieety $Z(\mathfrak{a})$.

(d) Show that the twisted cubic curve in $\mathbb{P}^3$ (Ex. 2.9) is equal to the 3-uple embedding of $\mathbb{P}^1$ in $\mathbb{P}^3$, for suitable choice of coordinates.

Proof.

(a) The image of θ is an integral domain, so $\mathfrak{a}$ is a prime ideal, and it is clearly homogenous since each y_i is sent to a polynomial of same degree.

(b) If $Q \in \operatorname{im} \rho_d$, then $Q = \rho_d(P)$ for some $P \in \mathbb{P}^n$, for any $f \in \mathfrak{a}$ we have

$$f(Q) = f(\rho_d(P)) = \theta(f)(P) = 0 \implies Q \in Z(\mathfrak{a}).$$

Before proving the converse direction, consider the case when $n = 1$ and $d = 2$, so that $\rho_d : \mathbb{P}^1 \to \mathbb{P}^2$ is defined as $(b_0, b_1) \mapsto (b_0^2, b_0 b_1, b_1^2)$. Then, the polynomial $y_0 y_2 - y_1^2$ is in $\mathfrak{a}$. If $Q = (a_0, a_1, a_2) \in Z(\mathfrak{a})$, then since k is algebraically closed we have

$$a_0 a_2 - a_1^2 = 0 \implies a_1 = \pm\sqrt{a_0 a_2} \implies \rho_d(\sqrt{a_0}, \sqrt{a_2}) = (a_0, a_1, a_2).$$

Returning to the general case, if $Q = (a_0, \ldots, a_N) \in Z(\mathfrak{a})$, indexing M_i using the stars and bars method, we have $\rho_d(\sqrt[d]{a_0}, \sqrt[d]{a_d}, \ldots, \sqrt[d]{a_N}) = Q$.

(c) ρ_d is clearly bijective, so it will be sufficient to show that the closed sets of $\mathbb{P}^n$ are identified with the closed sets of $Z(\mathfrak{a})$ by ρ_d. Let $Y \subseteq \mathbb{P}^n$ be a closed subset, so $Y = Z(T)$ for some subset $T \subseteq k[x_0, \ldots, x_n]$, then it is easy to see that $\rho_d(Y) = Z(\theta^{-1}(T)) \cap Z(\mathfrak{a})$. Conversely, let W be a closed subset of $Z(\mathfrak{a})$. Let $\overline{W}$ be its closure in $\mathbb{P}^N$. This is an algebraic set, so $\overline{W} = Z(T')$ for some $T' \subseteq k[y_0, \ldots, y_N]$, hence $\rho_d^{-1}(W) = \rho_d^{-1}(\overline{W}) = Z(\theta(T'))$.

(d) Let (S, T) and (X, Y, Z, W) be homogenous coordinates of $\mathbb{P}^1$ and $\mathbb{P}^3$, respectively. Then, the 3-uple embedding of $\mathbb{P}^1$ in $\mathbb{P}^3$ is given by

$$(S, T) \mapsto (S^3, S^2T, ST^2, T^3).$$

Let Y be the twisted cubic curve in $\mathbb{A}^3$ and let $\overline{Y}$ be its projective closure in $\mathbb{P}^3$, then we have

$$\varphi_3^{-1}(Y) = \left\{ \left(\frac{u}{v}, \frac{u^2}{v^2}, \frac{u^3}{v^3}, 1 \right) \middle| u, v \in k,\ v \neq 0 \right\} \implies \overline{Y} = \{(u^3, u^2v, uv^2, v^3) \mid u, v \in k\}.$$

□

14. *Segre Embedding.* Let $\psi : \mathbb{P}^r \times \mathbb{P}^s \to \mathbb{P}^N$ be the map defined by sending the ordered pair $(a_0, \ldots, a_r) \times (b_0, \ldots, b_s)$ to $(\ldots, a_i b_j, \ldots)$ in lexicographic order, where $N = rs + r + s$. Note that ψ is well-defined and injective. It is called the *Segre embedding*. Show that the image of ψ is a *subvariety* of $\mathbb{P}^N$.

Proof. Let the homogenous coordinates of $\mathbb{P}^N$ be $\{z_{ij} \mid i = 0, \ldots, r, j = 0, \ldots, s\}$, and let $\mathfrak{a}$ be the kernel of the homomorphism $\theta : k[\{z_{ij}\}] \to k[x_0, \ldots, x_r, y_0, \ldots, y_s]$ which sends z_{ij} to $x_i y_j$. If $P \in \operatorname{im} \psi$, then $P = \psi(Q, R)$ for some $Q \in \mathbb{P}^r$ and $R \in \mathbb{P}^s$, then for any $f \in \mathfrak{a}$ we have

$$f(P) = f(\psi(Q, R)) = \theta(f)(Q, R) = 0 \implies P \in Z(\mathfrak{a}).$$

Conversely, viewing points of $\mathbb{P}^N$ as $(r+1) \times (s+1)$-matrices, the variety $Z(\mathfrak{a})$ is defined as the vanshing of all 2×2-minors, i.e. $z_{ij} z_{kl} = z_{il} z_{jk}$ for all $0 \leq i, k \leq r$ and $0 \leq j, l \leq s$. This means $\{z_{ij}\} \in Z(\mathfrak{a})$ has rank 1, so it can be expressed as the outer product of two vectors in k^{r+1} and k^{s+1}, which is exactly the mapping defined by ψ. □

15. *Quadric Surface in* $\mathbb{P}^3$. Consider the surface Q (a *surface* is a variety of dimension 2) in $\mathbb{P}^3$ defined by the equation $xy - zw = 0$.

(a) Show that Q is equal to the Segre embedding of $\mathbb{P}^1 \times \mathbb{P}^1$ in $\mathbb{P}^3$, for suitable choice of coordinates.

(b) Show that Q contains two families of lines (a *line* is a linear variety of dimension 1) $\{L_t\}$, $\{M_t\}$, each parametrized by $t \in \mathbb{P}^1$, with the proeprties that if $L_t \neq L_u$, then $L_t \cap L_u = \emptyset$; if $M_t \neq M_u$, $M_t \cap M_u = \emptyset$, and for all t, u, $L_t \cap M_u =$ one point.

(c) Show that Q contains other curves besides these lines, and deduce that the Zariski topology on Q is not homeomoprhic via ψ to the product topology on $\mathbb{P}^1 \times \mathbb{P}^1$ (where each $\mathbb{P}^1$ has its Zariski topology).

Proof.

(a) The kernel of the mapping $k[z_{00}, z_{01}, z_{10}, z_{11}] \to k[x_0, x_1, y_0, y_1]$ as in Exercise 14 is generated by $z_{00}z_{11} - z_{01}z_{10}$. Then $Z(z_{00}z_{11} - z_{01}z_{11})$ is equal to Q for a suitable choice of coordinates.

(b) From here, we assume Q is defined by $xw - yz = 0$ (the author was too lazy to fix his mistake after realizing it at (c)). Consider the Segre embedding $\psi : \mathbb{P}^1 \times \mathbb{P}^1 \to \mathbb{P}^3$ defined by

$$(t_0, t_1) \times (u_0, u_1) \mapsto (t_0u_0, t_0u_1, t_1u_0, t_1u_1).$$

Fixing the first entry as $t = (t_0, t_1) \in \mathbb{P}^1$, we obtain an embedding of $\mathbb{P}^1$ in $\mathbb{P}^3$, which can be identified as the intersection of the zero set of the following linear polynomials

$$t_1x - t_0z = 0, \quad t_1y - t_0w = 0,$$

so we have one family of lines $\{L_t\}$, and we can obtain a second family of lines in the same manner by fixing the second entry of the map ψ, where the line M_u for $u = (u_0, u_1) \in \mathbb{P}^1$ is defined by the intersection of the zero set of the linear polynomials

$$u_1x - u_0y = 0, \quad u_1z - u_0w = 0.$$

Then $L_t \neq L_u$ implies $L_t \cap L_u = \emptyset$ follows from the fact that ψ is an embedding, in particular ψ is injective. To show the intersection of L_t and M_u is a single point for any $t, u \in \mathbb{P}^1$, it suffices to show the linear polynomials defined above intersect at exactly one point. Without loss of generality assume $t_1, u_1 \neq 0$, then setting $\lambda = t_0/t_1$ and $\mu = u_0/u_1$, we have

$$x = \lambda z, \quad y = \lambda w, \quad x = \mu y, \quad z = \mu w \implies L_t \cap M_u = \{(\lambda\mu, \lambda, \mu, 1)\}.$$

(c) Consider the curve K in Q defined by $x = w$. It is clearly not a line since it is the intersection of a nonlinear curve and a linear curve; however it is a closed subset of Q. On the other hand, $\psi^{-1}(K)$ is the diagonal in $\mathbb{P}^1 \times \mathbb{P}^1$, which is certainly not closed since $\mathbb{P}^1$ is not Hausdorff.

□

Chapter 1, Section 3

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2. A morphism whose underlying map on the topological space is a homeomorphism need not be an isomorphism.

(a) For example, let $\varphi : \mathbb{A}^1 \to \mathbb{A}^2$ be defined by $t \mapsto (t^2, t^3)$. Show that φ defines a bijective bicontinuous morphism of $\mathbb{A}^1$ onto the curve $y^2 = x^3$, but that φ is not an isomorphism.

(b) For another example, let the characteristic of the base field k be $p > 0$, and define a map $\varphi : \mathbb{A}^1 \to \mathbb{A}^1$ by $t \mapsto t^p$. Show that φ is bijective and bicontinuous but not an isomorphism. This is called the *Frobenius morphism*.

Proof.

(a) Let Y be the curve $y^2 = x^3$. Then φ clearly maps into Y and is injective since

$$(t^2, t^3) = (u^2, u^3) \implies tu^2 = t^2 u \implies t = u.$$

If $P = (a_1, a_2) \in Y$, then clearly $\varphi(\sqrt[3]{a_2}) = P$ since k is algebraically closed, hence φ is bijective onto Y. To show φ is bicontinuous it is enough to show φ and φ^{-1} of closed sets in $\mathbb{A}^1$ and Y, respectively, is closed. A closed set in either $\mathbb{A}^1$ or Y is a finite set of points, so by the bijectivity of φ, it is also bicontinuous.

To show it is not an isomorphism, it suffices to show the coordinate rings of the affine line $\mathbb{A}^1$ and the curve Y are not isomorphic. Indeed, $A(\mathbb{A}^1) = k[x]$ and $A(Y) = k[x, y]/(y^2 - x^3)$, and while $k[x]$ is factorial, $A(Y)$ is not since $y^2 = x^3$.

(b) It is bijective and bicontinuous for the same reason in (a). If φ is an isomorphism, then it must induce an automorphism $\varphi^* : k[x] \to k[x]$ of the affine coordinate ring $k[x]$ of $\mathbb{A}^1$. However, φ^* is defined by $x \mapsto x^p$, which is not even surjective.

□

3. (a) Let $\varphi : X \to Y$ be a morphism. Then for each $P \in X$, φ induces a homomorphism of local rings $\varphi_P^* : \mathcal{O}_{\varphi(P),Y} \to \mathcal{O}_{P,X}$.

(b) Show that a morphism φ is an isomorphism if and only if φ is a homeomorphism, and that the induced map φ_P^* on local rings is an isomorphism for all $P \in X$.

(c) Show that if $\varphi(X)$ is dense in Y, then the map φ_P^* is *injective* for all $P \in X$.

Proof.

(a) We have already seen that φ induces a homomorphism $\varphi^* : \mathcal{O}(Y) \to \mathcal{O}(X)$. By (2.3), there exists an open affine variety V in Y containing $\varphi(P)$. Then $U = \varphi^{-1}(V)$ contains P, so $\mathcal{O}_{P,X} \simeq \mathcal{O}_{P,U}$ and $\mathcal{O}_{\varphi(P),Y} \simeq \mathcal{O}_{\varphi(P),V}$. By definition f is a regular function defined in some open neighborhood of $\varphi(P)$ in V such that $f(\varphi(P))$ is nonzero if and only if $\varphi^* f$ is nonzero at P. Thus, $\varphi^* f$ is a unit in $\mathcal{O}_{P,U}$, so by the universal property of localization there exists a unique ring homeomorphism $\varphi_P^* : \mathcal{O}_{\varphi(P),Y} \to \mathcal{O}_{P,X}$.

(b) The only if direction is obvious. Conversely, if φ_P^* on local rings is an isomorphism for all $P \in X$, then since isomorphism is a local property (A.M. p. 40), we have an isomorphism $\varphi^* : \mathcal{O}_V \to \mathcal{O}_U$, where V is any affine variety in Y and $U = \varphi^{-1}(V)$. Hence, $\varphi : U \to V$ is an isomorphism.

(c) We want to show for any $P \in X$ that if f, g are elements of $\mathcal{O}_{\varphi(P),Y}$ such that $\varphi_P^* f = \varphi_P^* g$ on some open neighborhood U of P, then $f = g$ in some open neighborhood V of $\varphi(P)$. Let V be an open neighborhood of $\varphi(P)$ with both f and g are defined so that $\varphi_P^* f$ and $\varphi_P^* g$ are defined on $U = \varphi^{-1}(V)$. Then, $\varphi(U) = V \cap \varphi(X)$ is a closed and dense subset of V such that $f = g$, hence $f = g$ on V.

□

4. Show that the d-uple embedding of $\mathbb{P}^n$ (I, Ex. 2.12) is an isomorphism onto its image.

Proof. Let $\rho : \mathbb{P}^n \to Y \subseteq \mathbb{P}^N$ be the d-uple embedding of $\mathbb{P}^n$ with $N = \binom{n+d}{d} - 1$ and image restricted so that ρ is a homeomorphism. By Exercise 2, to show ρ is an isomorphism it suffices to show it is an isomorphism locally. Let $P = (a_0, \ldots, a_n) \in \mathbb{P}^n$ and $\rho(P) = (b_0, \ldots, b_N) \in Y$ and assume $a_0 \neq 0$. By (3.4), we have the following identifications of local rings

$$\mathcal{O}_{P,\mathbb{P}^n} = S(\mathbb{P}^n)_{(\mathfrak{m}_P)}, \quad \mathcal{O}_{\rho(P),Y} = S(Y)_{(\mathfrak{m}_{\rho(P)})},$$

where $\mathfrak{m}_P$ (respectively, $\mathfrak{m}_{\rho(P)}$) is the ideal generated by the set of homogenous $f \in S(\mathbb{P}^n)$ (respectively, $f \in S(Y)$) such that $f(P) = 0$ (respectively, $f(\rho(P)) = 0$). We want to show the induced map $\rho_P^* : \mathcal{O}_{\rho(P),Y} \to \mathcal{O}_{P,\mathbb{P}^n}$ is an isomorphism. By the identification above, the map ρ_P^* can be defined by the canonical projection map $\overline{\theta} : k[y_0, \ldots, y_N]/\mathfrak{a} \to k[x_0, \ldots, x_n]$ induced by θ in Exercise 2.12 as

$$\rho_P^*\left(\frac{f}{g}\right) = \frac{\overline{\theta}(f)}{\overline{\theta}(g)},$$

where f and g are homogenous elements of same degree in $S(Y)$ such that $g(\rho(P)) \neq 0$ (since we are only concerned with homogenous elements of degree 0 in $S(Y)_{\mathfrak{m}_P}$). Clearly ρ_P^* is injective since $\overline{\theta}$ is injective, and ρ_P^* maps $S(Y)_{(\mathfrak{m}_{\rho(P)})}$ into $S(\mathbb{P}^n)_{(\mathfrak{m}_P)}$ since $\theta(y_i)$ is of the same degree for all $0 \leq i \leq N$ and $\mathfrak{a}$ is a homogenous ideal. To show it is surjective, suppose $h/k \in \mathcal{O}_{P,\mathbb{P}^n}$ with $k(P) \neq 0$ and h, k both homogenous of degree e. We can assume e is a multiple d since x_0 is a unit in $\mathcal{O}_{P,\mathbb{P}^n}$, and we can multiply h/k by a sufficent power of x_0/x_0 to obtain the desired degrees in the numerator and denominator. Then, each term in h and k is the product of monomials of degree d, and since $\overline{\theta}$ is surjective onto such elements, ρ_P^* is surjective. Hence, ρ_P^* is an isomorphism. □

6. There are quasi-affine varieties which are not affine. For example, show that $X = \mathbb{A}^2 - \{(0,0)\}$ is not affine.

Proof. The ideal $I(X)$ consists of all polynomials in $k[x, y]$ that vanish at X. If $f \in I(X)$, then viewing f as a regular function on $\mathbb{A}^2$, $f^{-1}(0)$ is a closed and dense subset of $\mathbb{A}^2$, hence it equals to $\mathbb{A}^2$, hence $I(X) = (0)$. Thus, if X is affine, then

$$\mathcal{O}(X) = k[x, y]/I(X) \simeq k[x, y] = \mathcal{O}(\mathbb{A}^2),$$

so by (3.5) and (3.7) $X \cong \mathbb{A}^2$ under the inclusion map, which is ridiculous. □

12. If P is a point on a variety X, then $\dim \mathcal{O}_P = \dim X$.

Proof. We can find an open affine variety Y in X containing P, and since local rings behave well under open subsets, i.e. $\mathcal{O}_{P,Y} \simeq \mathcal{O}_{P,X}$, we can reduce to the affine case. Then, the statement follows from (3.2). □

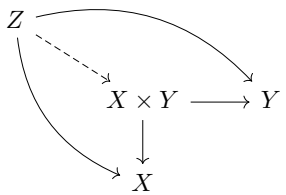
15. *Products of Affine Varieties.* Let $X \subseteq \mathbb{A}^n$ and $Y \subseteq \mathbb{A}^m$ be affine varieties.

(a) Show that $X \times Y \subseteq \mathbb{A}^{n+m}$ with its induced topology is irreducible. The affine variety $X \times Y$ is called the *product* of X and Y. Note that its topology is in general not equal to the product topology.

(b) Show that $A(X \times Y) \simeq A(X) \otimes_k A(Y)$.

(c) Show that $X \times Y$ is a product in the category of varieties.

(d) Show that $\dim X \times Y = \dim X + \dim Y$.



Proof.

(a) Suppose that $X \times Y$ is a union of two closed subsets $Z_1 \cup Z_2$. Let $X_i = \{x \in X \mid x \times Y \subseteq Z_i\}$, $i = 1, 2$ and let $\pi : \mathbb{A}^{n+m} \to \mathbb{A}^m$ be the projection map. For any $x \in X$ since $x \times Y \subseteq Z_1 \cup Z_2$, we can write $x \times Y = (x \times Y \cap Z_1) \cup (x \times Y \cap Z_2)$, so let $C_i = x \times Y \cap Z_i$, then

$$Y = \pi(x \times Y) = \pi(C_1 \cup C_2) = \pi(C_1) \cup \pi(C_2).$$

We can write $Z_i = Z(f_1, \ldots, f_r)$, $i = 1, 2$, for some $f_j \in A(\mathbb{A}^{n+m})$, then we see that

$$\pi(C_i) = \bigcap_{j=1}^{r} Z(f_j(x, y_1, \ldots, y_m)),$$

where we can view $f_j(x, y_1, \ldots, y_m)$ as an element in $A(\mathbb{A}^m)$, which shows $\pi_2(C_i)$ is a closed set in $\mathbb{A}^m$. Since Y is irreducible, we have $\pi(C_2) = \emptyset$, say, hence $Y = \pi(C_1)$, hence $x \times Y = x \times Y \cap Z_1$, hence $x \times Y \subseteq Z_1$. In particular, we have $X = X_1 \cup X_2$, and we have shown $Z_i = X_i \times Y$. We can repeat the argument for $Y_i = \{y \in Y \mid X \times Y \subseteq Z_i\}$, $i = 1, 2$ so that $Z_i = X \times Y_i$. This is only possible if $X = X_1$ and $Y = Y_1$, say, hence $Z_1 = X \times Y$.

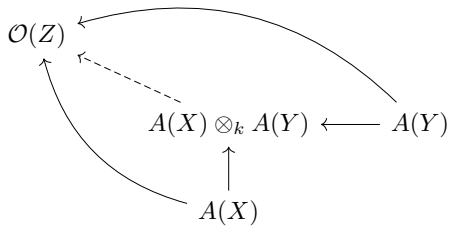
(b) We clearly have $A(\mathbb{A}^{n+m}) \simeq A(\mathbb{A}^n) \otimes_k A(\mathbb{A}^m)$, where $A(\mathbb{A}^n) = k[x_1, \ldots, x_n]$ and $A(\mathbb{A}^m) = k[y_1, \ldots, y_m]$. Then, we have

$$I(X \times Y) = I(X) \otimes_k A(\mathbb{A}^m) + A(\mathbb{A}^n) \otimes_k I(Y),$$

hence we have the isomorphism

$$\begin{aligned} A(X \times Y) &\cong \frac{A(\mathbb{A}^n) \otimes_k A(\mathbb{A}^m)}{I(X \times Y)} \\ &\cong \frac{A(\mathbb{A}^n) \otimes_k A(\mathbb{A}^m)}{I(X) \otimes_k A(\mathbb{A}^m) + A(\mathbb{A}^n) \otimes_k I(Y)} \\ &\cong \frac{A(\mathbb{A}^n)}{I(X)} \otimes_k \frac{A(\mathbb{A}^m)}{I(Y)} \\ &\cong A(X) \otimes_k A(Y). \end{aligned}$$

(c) Suppose we have morphisms $Z \to X$ and $Z \to Y$. This induces k-algebra homomorphisms $A(X) \to \mathcal{O}(Z)$ and $A(Y) \to \mathcal{O}(Z)$, which defines a k-bilinear map $A(X) \times A(Y) \to \mathcal{O}(Z)$. By the universal property of tensor products, we have a unique k-algebra homomorphism $A(X) \otimes_k A(Y) \to \mathcal{O}(W)$, hence we have a unique morphism $W \to X \times Y$.



(d) If $t_1, \ldots, t_n$ and $u_1, \ldots, u_m$ are coordinates for X and Y, respectively, then each t_i, u_j are algebraically independent in $A(X \times Y)$, that is the quotient field of $A(X \times Y)$ has transcendence degree $n + m$, hence $\dim X \times Y = \dim X + \dim Y$.

□

17. *Normal Varieties.* A variety Y is *normal at a point* $P \in Y$ if $\mathcal{O}_P$ is integrally closed ring. Y is *normal* if it is normal at every point.

(a) Show that every conic in $\mathbb{P}^2$ is normal.

(b) Show that the quadric surfaces Q_1, Q_2 in $\mathbb{P}^3$ given by equations $Q_1 : xy = zw$; $Q_2 : xy = z^2$ are normal.

(c) Show that the cuspidal cubic $y^2 = x^3$ in $\mathbb{A}^2$ is not normal.

(d) If Y is affine, then Y is normal $\iff$ $A(Y)$ is integrally closed.

(e) Let Y be an affine variety. Show that there is a normal affine variety $\hat{Y}$, and a morphism $\pi : \hat{Y} \to Y$, with the property that whenever Z is normal variety, and $\varphi : Z \to Y$ is a *dominant* morphism, then there is a unique morphism $\theta : Z \to \hat{Y}$ such that $\varphi = \pi \circ \theta$. $\hat{Y}$ is called the *normalization* of Y.

Proof.

(a) A conic in $\mathbb{P}^2$ can be covered by open affine varieties isomorphic to a conic in $\mathbb{A}^2$. Thus, it suffices to show all irreducible plane algebraic curves of degree 2 are normal, that is we want to show an affine variety X in $\mathbb{A}^2$ defined by the zero set of an irreducible quadratic polynomial f in $k[x, y]$ is normal. Since being integrally closed is a local property, it suffices to show the affine coordinate ring $A(X) = k[x, y]/(f)$ is integrally closed. We first prove (Ex. 1.1c), which states $A(X)$ is isomorphic to either $R = k[x, y]/(y - x^2)$ or $S = k[x, y]/(xy - 1) \simeq k[x, \frac{1}{x}]$. For simplicity assume $\operatorname{char} k \neq 2$. Any quadratic polynomial can be written as

$$f(x, y) = Ax^2 + Bxy + Cy^2 + Dx + Ey + F$$

We claim the following: if $B^2 - 4AC = 0$, then $A(X) \simeq R$, otherwise $A(X) \simeq S$. We proceed by showing that X can be transformed to a variety of the form $y = x^2$ or $xy = 1$ using some affine transformations. Suppose $B^2 - 4AC = 0$, then assume $C \neq 0$ (if $C = 0$, then $B = 0$, so we already have an equation of the desired form up to some translation and stretching), so we have

$$f(x, y) = (\sqrt{A}x + \sqrt{C}y)^2 + Dx + Ey + F,$$

since k is algebraically closed. We can apply the following affine transformation

$$\begin{aligned} x &\mapsto -\sqrt{C}y \\ y &\mapsto -\sqrt{C}x + \sqrt{A}y \end{aligned}$$

which is indeed an affine transformation since it has determinant $C \neq 0$, to obtain

$$f'(x, y) = C^2x^2 - D\sqrt{C}y - E(-\sqrt{C}x + \sqrt{A}y) + F,$$

which is the case when $B = C = 0$.
Now suppose $\Delta^2 = B^2 - 4AC \neq 0$ and assume $A \neq 0$, then we have

$$f(x, y) = \left(\sqrt{A}x + \frac{B + \Delta}{2\sqrt{A}}y\right)\left(\sqrt{A}x + \frac{B - \Delta}{2\sqrt{A}}y\right) + Dx + Ey + F,$$

so we can apply the affine transformation

$$\begin{aligned} \sqrt{A}x + \frac{B + \Delta}{2\sqrt{A}}y &\mapsto x \\ \sqrt{A}x + \frac{B - \Delta}{2\sqrt{A}}y &\mapsto y \end{aligned}$$

which has determinant $1/\Delta \neq 0$, to obtain

$$f'(x, y) = xy + (\text{linear terms}).$$

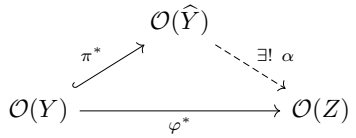
Since affine coordinate rings are invariant up to affine transformations, we have either $A(X) \simeq R$ or $A(X) \simeq S$. The ring $R = k[x, y]/(y - x^2)$ is isomorphic to a polynomial ring over one variable, which is a factorial, hence it is integrally closed. The ring $S = k[x, x^{-1}]$ is a discrete valuation ring, which are integrally closed by (A.M. p. 94).

(b) The quadric surface Q_1 can be covered by open affine varieties isomoprhic to the affine surface $z = xy$. The affine coordinate ring of such surface is isomorphic to $k[x, y]$, which is integrally closed, hence Q_1 is normal. Similarily, Q_2 can be covered by open affine varieties isomoprhic to either $y = x^2$ or $xy = 1$, which was shown to be normal in (a).

(c) The affine coordinate ring $A(Y)$ is isomorphic to $k[t^2, t^3]$. The quotient field is $k(t)$, and the element $t \in k(t)$ is integral over $k[t^2, t^3]$ since $(t)^2 - t^2 = 0$ but $t \notin k[t^2, t^3]$, hence $A(Y)$ is not integrally closed, hence Y is not normal by (d).

(d) Being integrally closed is a local property, that is a ring A is integrally closed if and only if $A_{\mathfrak{m}}$ is integrally closed for all maximal ideals $\mathfrak{m}$ in A by (A.M. p. 63). There is a one-to-one correspondence between maximal ideals of $\mathcal{O}(Y) = A(Y)$ and points of Y, hence Y is normal if and only if $\mathcal{O}_P$ is integrally closed for all $P \in Y$ if and only if $A(Y)_{\mathfrak{m}}$ is integrally closed for all maximal ideals $\mathfrak{m}$ if and only if $A(Y)$ is integrally closed.

(e) We want to show there exists a normal scheme $\widehat{Y}$ with morphism $\pi : \widehat{Y} \to Y$ that is universal amongst all dominant morphisms from normal schemes into Y. Let $A(Y)$ be the affine coordinate ring of Y and let B be the integral closure of $A(Y)$ in its quotient field. Then, B is a finitely generated k-algebra by (3.9A), and it is a subring of a field, so it is an integral domain, which means it defines an affine variety. Let $\widehat{Y}$ be an affine variety with coordinate ring $A(\widehat{Y}) = B$, and let $\pi : \widehat{Y} \to Y$ be the morphism induced by the inclusion map $A(Y) \hookrightarrow B$. Let Z be a normal variety, and $\varphi : Z \to Y$ a dominant morphism. Since Z can be covered by open affine varieties and local rings are perserved under open subsets, we reduce to the case when Z is an affine normal variety. We want to show there exists a ring homomorphism $\alpha : \mathcal{O}(\widehat{Y}) \to \mathcal{O}(Z)$ such that the following diagram commutes:



If $\varphi^* f = 0$ for some $f \in \mathcal{O}(Y)$, then $f = 0$ on a closed dense subset of Y, hence $f = 0$ on Y, so φ^* is injective. Thus, φ^* induces a map between the function fields $\overline{\varphi^*} : K(Y) \to K(Z)$, where we have the inclusions $\mathcal{O}(Y) \hookrightarrow \mathcal{O}(\widehat{Y}) \hookrightarrow K(Y)$ by construction. If $f \in \mathcal{O}(\widehat{Y})$, then there exists an equation of integral dependence of the form

$$f^n + a_1 f^{n-1} + \cdots + a_n = 0, \quad a_i \in \mathcal{O}(Y),$$

and since $\overline{\varphi^*}(a_i) = \varphi^*(a_i) \in \mathcal{O}(Z)$, we can apply $\overline{\varphi^*}$ to the equation above to obtain an equation of integral dependence of $\overline{\varphi^*}(f)$ over $\mathcal{O}(Z)$

$$\overline{\varphi^*}(f)^n + b_1 \overline{\varphi^*}(f)^{n-1} + \cdots + b_n = 0, \quad b_i = \varphi^*(a_i) \in \mathcal{O}(Z),$$

and since $\mathcal{O}(Z)$ is integrally closed, we must have $\overline{\varphi^*}(f) \in \mathcal{O}(Z)$. Thus, we have $\alpha = \overline{\varphi^*}\,|_{\mathcal{O}(\widehat{Y})}$ as the desired map, and it is unique by construction. Hence, we have a unique morphism $\theta : Z \to \hat{Y}$ such that $\varphi = \pi \circ \theta$ and $\theta^* = \alpha : \mathcal{O}(\widehat{Y}) \to \mathcal{O}(Z)$ by (3.5).

□

20. Let Y be a variety of dimension ≥ 2, and let $P \in Y$ be a normal point. Let f be a regular function on $Y - P$.

(a) Show that f extends to a regular function on Y.

(b) Show this would be false for $\dim Y = 1$.

Proof.

(a) Y can be covered by open affine varieties, so we reduce to the case when $Y \subseteq \mathbb{A}^n$ itself is an affine variety. Then by composing with the inclusion morphism $Y \to \mathbb{A}^n$, we futher reduce to the case when $Y = \mathbb{A}^n$, that is we want to show if $f = g/h$ on $\mathbb{A}^n - P$, then we can extend f to a regular function on Y. We show that if a polynomial $h \in k[x_1, \ldots, x_n]$ is nonzero on $\mathbb{A}^n - P$, then f is nonzero on P. We can also assume $P = (0, \ldots, 0)$ and $n = 2$ by induction. Suppose $h(P) = 0$, then h must have zero constant term, so we can write

$$h(x, y) = f_0(x) y^d + f_1(x) y^{d-1} + \cdots + f_d(x), \quad f_i(x) \in k[x] - k,$$

then there exists nonzero $a \in k$ such that $f_i(a) \neq 0$ for some i, we have

$$h(a, y) = a_0 y^d + a_1 y^{d-1} + \cdots a_d \neq 0, \quad a_i = f_i(a),$$

and since k is algebraically closed, $h(a, y)$ must have a solution $y = b$, so (a, b) is a solution to h that is not equal to P, a contradiction.

(b) Consider $Y = \mathbb{A}^1$ and $P = 0$, and let $f = \frac{1}{x}$. Then f is regular on $Y - P$ by definition, but it clearly cannot be extended to the entire affine line.

□

Chapter 1, Section 4

James Lee

October 11, 2026

1. If f and g are regular functions on open subsets U and V of a variety X, and if $f = g$ on $U \cap V$, show that the function which is f on U and g on V is a regular function on $U \cup V$. Conclude that if f is a *rational* function on X, then there is a largest open subset U of X on which f is represented by a regular function. We say that f is *defined* at the points of U.

Proof. Denote the function which is f on U and g on V as F. It suffices to show for all $P \in U \cap V$ that there exists an open neighborhood W of P contained in $U \cap V$ such that F is a rational function on W. Since $U \cap V$ is an open subset of U and V, for all $P \in U \cap V$ there exists open subsets A of U and B of W such that F is rational with $h(Q), h(Q) \neq 0$ for all $Q \in A \cap B$ and $F = f = g$ on $A \cap B$, hence $W = A \cap B \subseteq U \cap V$ is the desired open subset.

If f is a rational function on X, then the set of open subsets of X on which f is represented by regular function is nonempty, and since X is a noetherian space, there is a maximal element in this set. Then, the largest open subset U of X on which f is represented by a regular function is the union of maximal elements in this subset of open subsets. □

2. Same problem for rational maps. If φ is a rational map of X to Y, show there is a largest open set on which φ is represented by a morphism. We say the rational map is *defined* at the points of that open set.

Proof. Let φ and ψ be morphisms on open subsets U and V of a variety X to a variety Y, and suppose $\varphi = \psi$ on $U \cap V$. If f is a regular function on Y, then $\varphi^* f$ and $\psi^* f$ are regular functions on the open subsets U and V which agree on $U \cap V$, so by Exercise 1 the function which is $\varphi^* f$ on U and $\psi^* f$ on V is a regular function $U \cap V$, i.e. the map that is φ on U and ψ on V is indeed a rational map.

If $\varphi : X \to Y$ is a rational map, then the set of open subsets of X on which φ is represented by morphism function is nonempty, and since X is a noetherian space, there is a maximal element in this set. Then, the largest open subset U of X on which φ is represented by a morphism is the union of maximal elements in this subset of open subsets. □

4. A variety Y is *rational* if it is birationally equivalent to $\mathbb{P}^n$ for some n (or, equivalently by (4.5), if $K(Y)$ is a pure transcendental extension of k).

(a) Any conic in $\mathbb{P}^2$ is a rational curve.

(b) The cuspidal cubic $y^2 = x^3$ is a rational curve.

(c) Let Y be the nodal cubic curve $y^2 z = x^2(x + z)$ in $\mathbb{P}^2$. Show that the projection φ from the point $P = (0, 0, 1)$ to the line $z = 0$ induces a birational map from Y to $\mathbb{P}^1$. Thus, Y is a rational curve.

Proof.

(a) A conic in $\mathbb{P}^2$ can be covered by open affine varieties that are either isomorphic to $y = x^2$ or $xy = 1$. The former is isomorphic to $\mathbb{A}^1$, hence it is isomorphic to an open subset of $\mathbb{P}^1$, hence it is birationally equivalent to $\mathbb{P}^1$. The latter has function field isomorphic to $k(x)$, hence it is birationally equivalent to $\mathbb{A}^1$, hence it is also birationally equivalent to $\mathbb{P}^1$.

(b) The cuspidal cubic has coordinate ring $k[t^2, t^3]$, so its function field is $k(t)$, hence it is birationally equivalent to $\mathbb{A}^1$, hence it is birationally equivalent to $\mathbb{P}^1$.

(c) The line $z = 0$ in $\mathbb{P}^2$ corresponds to a hyperplane isomorphic to $\mathbb{P}^1$, so the projection $\varphi : \mathbb{P}^2 - \{P\} \to \mathbb{P}^1$ is a morphism. In coordinates, φ is defined as $(x_0, x_1, x_2) \mapsto (x_0, x_1)$, so φ induces a morphism from $Y - P$ to $\mathbb{P}^1$. Thus, $\varphi(Y - P)$ is the set of all lines in $\mathbb{A}^2$ that pass through the origin and a point in the affine nodal curve $y^2 = x^3 + x^2$. This is an open set in $\mathbb{P}^1$ isomorphic to $\mathbb{A}^1$ since it contains all lines in $\mathbb{A}^2$ besides the one defined by $x = \pm y$. To further elaborate, if $(x, y) \in \mathbb{P}^1$ with $x \neq \pm y$ and $x \neq 0$, say, then write $y = \lambda x$ with $\lambda \neq \pm 1$, so we have

$$\lambda^2 x^2 = x^3 + x^2 \implies x = \lambda^2 - 1.$$

So there exists a line in $\mathbb{A}^2$ passing through a point in $y^2 + x^3 + x^2$ with slope y/x. Rephrasing, we have shown that the map $\varphi : Y - P \to \mathbb{P}^1$ is surjective besides at the two points $(1, 1)$ and $(1, -1)$. It is also injective since the x_2-coordinate can be completely determined by the values of (x_0, x_1), that is if $x_0 \neq 0$, then setting $x_0 = 1$, we have

$$x_1^2 x_2 = 1 + x_2 \implies x_2 = \frac{1}{x_1^2 - 1} \quad \text{or} \quad x_2 = 0,$$

and $x_1 \neq \pm 1$ since $x_0 \neq \pm x_1$, and $x_2 \neq 0$ since the only point on Y with $x_2 = 0$ is P, and φ is restricted $Y - P$. Hence, φ is an isomorphism of the open subset $Y - P$ of Y to the open subset $\mathbb{P}^1 - \{(1, 1), (1, -1)\}$ in $\mathbb{P}^1$, hence Y is birationally equivalent to $\mathbb{P}^1$.

□

5. Show that the quadric surface $Q : xy = zw$ in $\mathbb{P}^3$ is birational to $\mathbb{P}^2$, but not isomorphic to $\mathbb{P}^2$.

Proof. Q is the Segre embedding of $\mathbb{P}^1 \times \mathbb{P}^1$ in $\mathbb{P}^3$, and $\mathbb{P}^1 \times \mathbb{P}^1$ contains a copy of $\mathbb{A}^1 \times \mathbb{A}^1$, so Q contains a copy of $\mathbb{A}^2$, hence Q is birational to $\mathbb{P}^2$. It is an axiom of projective geometry that any two lines intersect in $\mathbb{P}^2$; however, it was shown in Exercise 2.15 that there exists lines in Q that do not intersect, hence Q and $\mathbb{P}^2$ cannot be isomorphic. □

10. Let Y be the cuspidal cubic curve $y^2 = x^3$ in $\mathbb{A}^2$. Blow up the point $O = (0, 0)$, let E be the exceptional curve, and let $\widetilde{Y}$ be the strict transform of Y. Show that E meets $\widetilde{Y}$ in one point, and that $\widetilde{Y} \cong \mathbb{A}^1$. In this case the morphism $\varphi : \widetilde{Y} \to Y$ is bijective and bicontinuous, but it is not an isomorphism.

Proof. Let t, u be homogenous coordinates for $\mathbb{P}^1$. Then X, the blowing-up of $\mathbb{A}^2$ at O, is defined by the equation $xu = ty$ inside $\mathbb{A}^2 \times \mathbb{P}^1$. We obtain the total inverse image of Y in X by considering the equations $y^2 = x^3$ and $xu = ty$ in $\mathbb{A}^2 \times \mathbb{P}^1$. Now $\mathbb{P}^1$ is covered by the open sets $t \neq 0$ and $u \neq 0$, which we consider separately. If $t \neq 0$, we can set $t = 1$, and use u as an affine parameter. Then we have the equations

$$y^2 = x^3, \quad y = xu$$

in $\mathbb{A}^3$ with coordinates x, y, u. Substituting, we get $x^2u^2 - x^3 = 0$, which factors. Thus, we obtain two irreducible components, one defined by $x = 0$, $y = 0$, u arbitrary, which is E, and the other defined by $u = x$ and $y = ux$. This is $\widetilde{Y}$, and $\widetilde{Y}$ meets E at the point $u = 0$. Similarly, if $u \neq 0$, then we can set $u = 1$, and use t as an affine parameter to obtain the equations

$$y^2 = x^3, \quad x = ty$$

in $\mathbb{A}^3$ with coordinates x, y, t. Substituting, we get $y^2 = t^3y^3$, which factors as well. Besides the exceptional curve, we have the component defined by $1 - t^3y = 0$ and $x = ty$, which does not meet E. Hence, E meets $\widetilde{Y}$ at only $(1, 0) \in E$. This also show $\widetilde{Y}$ is contained in the open set defined by $t \neq 0$, so it is isomorphic to the affine variety in $\mathbb{A}^3$ defined by $u = x$ and $y = ux$, which isomorphic to $y = x^2$ in $\mathbb{A}^2$, hence $\widetilde{Y} \cong \mathbb{A}^1$. □

Chapter 1, Section 5

James Lee

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1. Locate the singular points and sketch the following curves in $\mathbb{A}^2$ (assume char $k \neq 2$):

(a) $x^2 = x^4 + y^4$;
(b) $xy = x^6 + y^6$;
(c) $x^3 = y^2 + x^4 + y^4$;
(d) $x^2y + xy^2 = x^4 + y^4$.

f	$\partial f/\partial x$	$\partial f/\partial y$	$\langle \partial f/\partial x, \partial f/\partial y\rangle = \langle 0,0\rangle$	Sing f
$x^4 - x^2 + y^4$	$4x^3 - 2x$	$4y^3$	$(0,0),\ (\pm 1/\sqrt{2}, 0)$	$(0,0)$
$x^6 + y^6 - xy$	$6x^5 - y$	$6y^5 - x$	$(0,0),\ (6^{-1/4}, 6^{-1/4})$	$(0,0)$
$x^4 + y^4 - x^3 + y^2$	$4x^3 - 3x^2$	$4y^3 - 2y$	$\overset{(0,0),(3/4,0),}{(0,\pm 1/\sqrt{2}),(3/4,\pm 1/\sqrt{2})}$	$(0,0)$
$x^4 + y^4 - x^2y - xy^2$	$4x^3 - 2xy - y^2$	$4y^3 - 2xy - x^2$	$(0,0),\ (3/4, 3/4)$	$(0,0)$

(a) is the tacnode, (b) is the node, (c) is the cusp, and (d) is the triple point.

2. Locate the singular points and describe the singularities of the following surfaces in $\mathbb{A}^3$ (assume char $k \neq 2$).

(a) $xy^2 = z^2$;
(b) $x^2 + y^2 = z^2$;
(c) $xy + x^3 + y^3 = 0$.

f	$\partial f/\partial x$	$\partial f/\partial y$	$\partial f/\partial z$	Sing f
$xy^2 - z^2$	y^2	$2xy$	$-2z$	$(t,0,0),\ t \in k$
$x^2 + y^2 - z^2$	$2x$	$2y$	$-2z$	$(0,0,0)$
$x^3 + y^3 + xy$	$3x^2 + y$	$3y^2 + x$	0	$(0,0,t),\ t \in k$

(a) is the pinch point, (b) is the conical double point, and (c) is the double line.

3. *Multiplicities.* Let $Y \subseteq \mathbb{A}^2$ be a curve defined by the equation $f(x,y) = 0$. Let $P = (a,b)$ be a point of $\mathbb{A}^2$. Make a linear change of coordinates so that P becomes the point $(0,0)$. Then write f as a sum $f = f_0 + f_1 + \cdots + f_d$, where f_i is a homogenous polynomial of degree i in x and y. Then we define the *multiplicity* of P on Y, denoted $\mu_P(Y)$, to be the least r such that $f_r \neq 0$. (Note that $P \in Y \iff \mu_P(Y) > 0$.) The linear factors of f_r are called the *tangent directions at* P.

(a) Show that $\mu_P(Y) = 1 \iff P$ is a nonsingular point of Y.
(b) Find the multiplicity of each of the singular points in (Ex. 5.1) above.

Proof.

(a) Let f' be the polynomial $f(x,y)$ after change of coordinates so that P becomes the origin and let $O = (0,0)$ be the origin. Then, P is a nonsingular point of $Y \iff (\partial f/\partial x)(P),\ (\partial f/\partial y)(P) \neq 0 \iff (\partial f'/\partial x)(O),\ (\partial f'/\partial y)(O) \neq 0 \iff \partial f'/\partial x,\ \partial f'/\partial y =$ higher degree terms + constant $\iff f'$ has linear terms $\iff \mu_P(Y) = 1$.

(b) (a), (b), and (c) has multiplicity 2, and (d) has multiplicity 3 at O.

□

6. *Blowing Up Curve Singularities.*

(a) Let Y be the cusp or node of (Ex. 5.1). Show that the curve $\widetilde{Y}$ obtained by blowing up Y at $O = (0, 0)$ is nonsingular.

(b) We define a *node* (also called *ordinary double point*) to be a double point (i.e., a point of multiplicity 2) of a plane curve with distinct tangent directions. If P is a node on a plane curve Y, show that $\varphi^{-1}(P)$ consists of two distinct nonsingular points on the blown-up curve $\widetilde{Y}$. We say that "blowing up P resolves the singularity at P".

(c) Let $P \in Y$ be the tacnode of (Ex. 5.1). If $\varphi : \widetilde{Y} \to Y$ is the blowing-up at P, show that $\varphi^{-1}(P)$ is a node. Using (b) we see that the tacnode can be resolved by two successive blowing-ups.

(d) Let Y be the plane curve $y^3 = x^5$, which has a "high order cusp" at O. Show that O is a triple point; that blowing up O gives rise to a double point (what kind?) and that one further blowing up resolves the singularity.

Proof.

(a) Let Y be the curve defined by the equation $xy = x^6 + y^6$. It is the node of (Ex. 5.1). We consider the equations $xy = x^6 + y^6$ and $xu = ty$ in $\mathbb{A}^2 \times \mathbb{P}^1$, where t, u are homogenous coordinates for $\mathbb{P}^1$. Now $\mathbb{P}^1$ is covered by open sets $t \neq 0$ and $u \neq 0$, which we can consider separately. If $t \neq 0$, we can set $t = 1$, and use u as an affine parameter. Then we have the equations

$$xy = x^6 + y^6$$
$$y = xu$$

in $\mathbb{A}^3$ with coordinates x, y, u. Substituting, we get $x^2u = x^6 + x^6u^6$, which factors. Thus, we obtain two irreducible components, one defined by $x = 0$, $y = 0$, u arbitrary, which is the exceptional curve E, and the other defined by $u = x^4 + x^4u^6$. This is $\widetilde{Y}$, which meets E at the point $u = 0$. This curve is nonsingular since if we set $f(x, y, u) = x^4(1 + u^6) - u$, then

$$\frac{\partial f}{\partial x} = 0, \quad \frac{\partial f}{\partial y} = 0, \quad \frac{\partial f}{\partial u} = -1,$$

so $\widetilde{Y}$ is nonsingular for all points $t \neq 0$. The curve Y is symmetric with respect to $x = y$, so $\widetilde{Y}$ is also nonsingular for all points $u \neq 0$, hence $\widetilde{Y}$ is nonsingular.

(b) We can assume P is the origin so that Y is defined by $f = f_2 + f_3 + \cdots + f_r$. Thus, we can write $f_2 = (ax + by)(cx + dy)$, where $(a, b) \neq (c, d)$ as points in $\mathbb{P}^1$. Take t, u to be homogenous coordinates for $\mathbb{P}^1$. If $t \neq 0$, substituting $y = xu$ we have the equation

$$F_u = (a + bu)(c + du) + f_3' + \cdots + f_r' = 0,$$

where f_i' is obtained by substituting $y = xu$ into f_i and dividing by x^2, which is possible since f_i is homogenous of degree $i > 2$. This equation defines $\widetilde{Y}$, which meets the exceptional curve at $(b, -a)$ and $(d, -c)$. Similarly, if $u \neq 0$, then we obtain the equation

$$F_t = (at + b)(ct + d) + \hat{f}_3 + \cdots + \hat{f}_r = 0,$$

thus $(b, -a)$ and $(d, -c)$ are the only points $\widetilde{Y}$ meets the exceptional curve, that is $\varphi^{-1}(P) = \{(a, b), (c, d)\}$. Also, $\widetilde{Y}$ is nonsingular at both points since

$$\frac{\partial F_u}{\partial u} = b\left(c - \frac{ad}{b}\right) \neq 0, \quad \frac{\partial F_t}{\partial u} = b\left(a - \frac{bc}{d}\right) \neq 0.$$

(c) Let Y be the plane curve defined by $x^2 = x^4 + y^4$ and let t, u, be homogenous coordinates for $\mathbb{P}^1$. If $t \neq 0$, then we have the equations

$$x^2 = x^4 + y^4$$
$$y = xu$$

in $\mathbb{A}^3$ with coordinates x, y, u. Substituting, we get $x^2 = x^4 + x^4u^4$, thus $\widetilde{Y}$ is defined by $1 = x^2 + x^2u^4$, which intersects the exceptional curve at $u = 0$. Similarly, if $u \neq 0$, then $\widetilde{Y}$ is defined by $t^2 = t^2y^2 + y^2$ and intersects the exceptional curve at $t = 0$. These two points correspond to the slopes of the two branches of Y at P, which are the two distinct tangent directions on the node P.

(d) If $t \neq 0$, then we have the equation $x^2 = u^3$, which is a cusp. If $u \neq 0$, then we have the equation $1 = t^5 y^2$, which does not meet the exceptional curve.

□

8. Let $Y \subseteq \mathbb{P}^n$ be a projective variety of dimension r. Let $f_1, \ldots, f_t \in S = k[x_0, \ldots, x_n]$ be homogenous polynomials which generate the ideal of Y. Let $P \in Y$ be a point, with homogenous coordinates $P = (a_0, \ldots, a_n)$. Show that P is nonsingular on Y if and only if the rank of the matrix $\|(\partial f_i/\partial x_j)(a_0, \ldots, a_n)\|$ is $n - r$.

Proof. We follow the hint. The matrix $\|(\partial f_i/\partial x_i)(a_0, \ldots, a_n)\|$ has rank $n - r$ if and only if there exists a $(n-r) \times (n-r)$ submatrix with nonzero determinant. The determinant of a matrix of homogenous polynomials where entries of the same row have same degree is also a homogenous polynomial, hence the rank of $\|(\partial f_i/\partial x_j)(a_0, \ldots, a_n)\|$ is independent of the choice of homogenous coordinates of P. Assuming $a_0 \neq 0$, we can pass to an open affine $U_0 \subseteq \mathbb{P}^n$ containing P and use the affine Jacobian matrix, where U_0 is the open affine subset consisting of all points with nonzero 0th coordinate. In particular, f we set $Y_0 = Y \cap U_0$, then $g_i(y_1, \ldots, y_n) = f_i(1, y_1, \ldots, y_n)$ $(1 \leq i \leq t)$ generate the ideal of Y_0. Thus, P is nonsingular if and only if the rank of the matrix $\|(\partial g_i/\partial y_j)(a_1/a_0, \ldots, a_n/a_0)\|$ is $n - r$. Then, we have

$$\frac{\partial g_i}{\partial y_j} = \frac{\partial f_i(1, y_1, \ldots, y_n)}{\partial y_j} = \frac{\partial f_i(1, x_1, \ldots, x_n)}{\partial x_j}.$$

Since the rank of the Jacobian matrix is independent of the homogenous coordinates chosen for P, we conclude that $||(\partial g_i/\partial y_j)(P)||$ has an invertible $(n-r) \times (n-r)$ minor if and only if $\|(\partial f_i/\partial x_j)(P)\|$ has an invertible $(n-r) \times (n-r)$ minor. □

10. For a point P on a variety X, let $\mathfrak{m}$ be the maximal ideal of the local ring $\mathcal{O}_P$. We define the *Zariski tangent space* $T_P(X)$ of X at P to be the dual k-vector space of $\mathfrak{m}/\mathfrak{m}^2$.

(a) For any point $P \in X$, $\dim T_P(X) \geq \dim X$, with equality if and only if P is nonsingular.

(b) For any morphism $\varphi : X \to Y$, there is a natural induced k-linear map $T_P(\varphi) : T_P(X) \to T_{\varphi(P)}(Y)$.

(c) If φ is the vertical projection of the parabola $x = y^2$ onto the x-axis, show that the induced map $T_0(\varphi)$ of tangent spaces at the origin is the zero map.

Proof.

(a) Since $\mathfrak{m}/\mathfrak{m}^2$ is a finite dimension k-vector space, $\dim_k T_p(X) = \dim_k \mathfrak{m}/\mathfrak{m}^2 \geq \dim X$ by (5.2A), and by (Ex. 3.12),

$$\dim_k \mathfrak{m}/\mathfrak{m}^2 = \dim \mathcal{O}_{P,X} = \dim X.$$

(b) A morphism $\varphi : X \to Y$ induces a map $\varphi_P^* : \mathcal{O}_{\varphi(P),Y} \to \mathcal{O}_{P,X}$ between local rings, where the maximal ideal $\mathfrak{n}$ of $\mathcal{O}_{\varphi(P),Y}$ is mapped into the maximal ideal $\mathfrak{m}$ of $\mathcal{O}_{P,X}$. Thus, we can define $T_P(\varphi)F$ as

$$(T_P(\varphi)F)f = F(d_\varphi f) = F(f \circ \varphi),$$

where $d_\varphi : \mathfrak{n}/\mathfrak{n}^2 \to \mathfrak{m}/\mathfrak{m}^2$ is the map induced by φ_P^*.

(c) By the formula above, it suffices to show d_φ is the zero map at 0. Let X be the parabola $x = y^2$ and let Y by the x-axis. The maximal ideal $\mathfrak{m}$ of the local ring $\mathcal{O}_{0,X}$ is generated by y, thus $\mathfrak{m}/\mathfrak{m}^2$ is a one dimensional k-vector space spanned by y, and similarly $\mathfrak{n}/\mathfrak{n}^2$ of $\mathcal{O}_{0,Y}$ is a one dimensional k-vector space spanned by x. Thus, we have $\varphi^* x = y^2 \in \mathfrak{m}^2$, hence $d_\varphi = 0$, hence $T_0(\varphi) = 0$.

□

Chapter 1, Section 6

James Lee

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2. *An Elliptic Curve.* Let Y be the curve $y^2 = x^3 - x$ in $\mathbb{A}^2$, and assume that the characteristic of the base field k is $\neq 2$. In this exercise we will show that Y is not a rational curve, and hence $K(Y)$ is not a pure transcendental extension of k.

(a) Show that Y is nonsingular, and deduce that $A = A(Y) \simeq k[x,y]/(y^2 - x^3 + x)$ is an integrally closed domain.

(b) Let $k[x]$ be the subring of $K = K(Y)$ generated by the image of x in A. Show that $k[x]$ is a polynomial ring, and that A is the integral closure of $k[x]$ in K.

(c) Show that there is an automorphism $\sigma : A \to A$ which sends y to $-y$ and leaves x fixed. For any $a \in A$, define the *norm* of a to be $N(a) = a\sigma(a)$. Show that $N(a) \in k[x]$, $N(1) = 1$, and $N(ab) = N(a) \cdot N(b)$ for any $a, b \in A$.

(d) Using the norm, show that the units in A are precisely the nonzero elements of k. Show that x and y are irreducible elements of A. Show that A is *not* a unique factorization domain.

(e) Prove that Y is not a rational curve.

Proof.

(a) Let $f(x,y) = y^2 - x^3 + x$. Then, we have

$$\frac{\partial f}{\partial x} = -3x^2 + 1, \quad \frac{\partial f}{\partial y} = 2y.$$

If $(\partial f/\partial y)(P) = 0$ for some $P = (a,b) \in Y$, then $b = 0$, so either $a = 0$ or $a = \pm 1$. In any case, $(\partial f/\partial x)(P) \neq 0$ since char $k \neq 2$. Therefore, both partial derivatives of f do not vanish for any $P \in Y$, hence Y is nonsingular. This means the local ring $A_{\mathfrak{m}}$ for every maximal ideal $\mathfrak{m}$ of A is a regular local ring, and since A has dimension one by Krull's Hauptidealsatz, $A_{\mathfrak{m}}$ also has dimension one, hence $A_{\mathfrak{m}}$ is an integrally closed by (6.2A). Since being integrally closed is a local property by [AM p.63], A is also integrally closed.

(b) Since k is algebraically closed and $x \notin k$, x is transcendental over k, hence $k[x]$ is a polynomial ring. Since $y^2 - x^3 + x = 0$ in K, y is the integral closure of $k[x]$ in K. Thus, the integral closure of $k[x]$ contains A, and A itself is integrally closed, hence A is the integral closure of $k[x]$.

(c) The map σ is clearly bijective. To show it is an automorphism of A, it suffices to show σ as a map from $k[x,y]$ to itself fixes $y^2 - x^3 + x$, which it indeed does. An element of A is of the form $a = f + yg$ for some $f, g \in k[x]$, hence we have

$$N(a) = a\sigma(a) = (f + yg)(f - yg) = f^2 - y^2 g^2 = f^2 - (x^3 - x)g^2 \in k[x].$$

The map σ is an isomorphism, so it fixes k; in particular it fixes 1, hence $N(1) = 1 \cdot 1 = 1$. Lastly, if $a, b \in A$, then

$$N(ab) = (ab)\sigma(ab) = (ab)(\sigma(a)\sigma(b)) = (a\sigma(a))(b\sigma(b)) = N(a) \cdot N(b).$$

(d) Let u be a unit in A and let u^{-1} be its inverses. By (c), we have $N(u)N(u^{-1}) = 1$, so $N(u)$, $N(u^{-1})$ are units in $k[x]$. Since $k[x]$ is a polynomial ring, the units are precisely the nonzero elements of k, that is $N(u) \in k$. Since the norm fixes the degree, we must have $u \in k$.

Suppose $x = ab$ for some $a, b \in A$. Then, $x^2 = N(a)N(b)$ and x is irreducible in $k[x]$, so either $N(a)$, $N(b)$ each are associates with x, or $N(a)$ is an associate of x^2 and $N(b)$ is a unit. It suffices to show

N is not onto $k[x]$, that is there does not exist $c \in A$ such that $N(c) = x$. By the formula above, the degree of f^2 is even while $(x^3 - x)g^2$ has degree 0 or an odd number, so $f^2 - (x^3 - x)g^2$ must have degree greater than 1. Therefore, $N(b)$ is a unit, hence b is a unit. The case for y follows similarly, since $N(y) = x(1 + x)(1 - x)$, so if $y = ab$, then either $N(a)$ is associates with x or $x(1 + x)(1 - x)$, and the former case was shown to be impossible.

A is not a unique factorization domain since $y^2 = x(x + 1)(x - 1)$, and x, y are irreducible and hence prime but are not associates since the only units in A are the nonzero elements of k.

□

3. Show by example that the result of (6.8) is false if either (a) $\dim X \geq 2$, or (b) Y is not projective.

Proof.

(a) Consider the morphism $\mathbb{A}^2 - O \to \mathbb{P}^1$ defined by mapping (x, y) to a point in $\mathbb{P}^1$ with homogenous coordinates (x, y), where O is the origin.

(b) Let X be an abstract nonsingular curve isomorphic to $\mathbb{P}^1$ and write $\mathbb{P}^1 = \mathbb{A}^1 \cup \{\infty\}$. Then, we have an isomorphism $\varphi : \mathbb{P}^1 - \infty \to \mathbb{A}^1$; however, φ clearly cannot be extended to ∞.

□

4. Let Y be a nonsingular projective curve. Show that every nonconstant rational function f on Y defines a surjective morphism $\varphi : Y \to \mathbb{P}^1$, and that for every $P \in \mathbb{P}^1$, $\varphi^{-1}(P)$ is a finite set of points.

Proof. Identifying k with the affine line, f is a rational map between quasi-projective curves. We first show that if f is nonconstant, then, f is a dominant rational map into $\mathbb{A}^1$. Since the closed subsets of $\mathbb{A}^1$ are either finite subsets or the entire line $\mathbb{A}^1$, it suffices to show $\operatorname{im} f$ is an infinite set of points. If $\operatorname{im} f$ is a finite set of points, then it must be of cardinality greater than 1 since f is nonconstant. Every nonsingular projective curve is isomorphic to an abstract nonsingular curve, so the closed subsets of Y are either finite set of points or the entire curve. If $\operatorname{im} f = \{P_1, \ldots, P_n\}$, $n > 1$, then $f^{-1}(P_i)$ cannot be the entire curve Y for any $1 \leq i \leq n$, thus $Y = f^{-1}(\operatorname{im} f) = f^{-1}(P_1) \cup \cdots \cup f^{-1}(P_n)$, which is a finite union of finite sets, implying Y is a finite set of points. This is clearly not true by (Ex. 4.8). Therefore, $\operatorname{im} f$ is infinite set of points, so its closure is the entire affine line, hence f is a dominant rational map.

Then, $\mathbb{A}^1$ is a rational curve, in particular it is birationally equivalent to $\mathbb{P}^1$, so if U is the largest open affine subset of Y such that $f : U \to \mathbb{A}^1$ is defined as a morphism by (Ex. 4.2), we have a dominant morphism $f' : U \to \mathbb{A}^1 \hookrightarrow \mathbb{P}^1$. Since U is open and nonempty, its complement $Y - U$ is closed and proper subset of Y, hence it is a finite set of points, so by (6.8) we can extend f' to be a dominant morphism $\varphi : Y \to \mathbb{P}^1$ between nonsingular projective curves.

It remains to show φ is surjective. A dominant morphism $\varphi : Y \to \mathbb{P}^1$ induces a k-homomorphism $\varphi^* : K(\mathbb{P}^1) \to K(Y)$ between function fields, where $K(\mathbb{P}^1) \simeq k(t)$ for some indeterminant t. Since $k(t)$ and $K(Y)$ both have transcendence degree 1, φ is injective; in particular every valuation ring of $k(t)$ can be extended to one of $K(Y)$ since $K(Y)$ is integrally closed over $k(t)$. The map φ induces a morphism between abstract nonsingular curves $\varphi_\# : C_{K(Y)} \to C_{k(x)}$, where for $P \in C_{K(Y)}$ we have $\varphi_\#(P) = P \cap C_{k(x)}$ (the point P can be identified with a valuation ring in $K(Y)$). Every valuation ring of $k(t)$ is the intersection of some valuation ring of $K(Y)$ and $k(t)$, which implies $\varphi_\#$ is surjective. Since we have the following commutative diagram

$$\begin{array}{ccc} Y & \xrightarrow{\varphi} & \mathbb{P}^1 \\ \downarrow & & \downarrow \\ C_{K(Y)} & \xrightarrow[\varphi_\#]{} & C_{k(t)} \end{array}$$

where the vertical arrows are isomorphisms, φ is also surjective.

Every nonsingular projective curve is isomorphic to an abstract nonsingular curve, where its closed sets are either finite set of points or the entire curve. This implies $\varphi^{-1}(P)$ is either finite set of points or all of Y, and it cannot be Y since φ is surjective onto $\mathbb{P}^1$. □

Chapter 1, Section 7

James Lee

October 11, 2026

1. (a) Find the degree of the d-uple embedding of $\mathbb{P}^n$ in $\mathbb{P}^N$.

(b) Find the degree of the Segre embedding of $\mathbb{P}^r \times \mathbb{P}^s$ in $\mathbb{P}^N$.

Proof.

(a) The homogenous coordinate ring of the d-uple embedding is the $k[y_0, \ldots, y_n]$ where each y_i has degree d as a graded $k[x_0, \ldots, x_n]$-module. The Hilbert polynomial of the d-uple embedding is

$$P(z) = \binom{n+dz}{n} = \frac{(dz+n)(dz+n-1)\cdots(dz)}{n!} = \frac{d^n}{n!}z^n + \cdots,$$

Hence, the degree is d^n.

(b) We have seen in (Ex. 2.14) that the homogenous coordinate ring of the Segre embedding is isomorphic to $k[\{x_i y_j\}]$, which is a subring of $k[x_0, \ldots, x_r, y_0, \ldots, y_s]$. The grading is given by $\bigoplus_{k=0}^{\infty} M_k$ where M_k is the set of polynomials of degree $2k$ in $k[\{x_i y_i\}]$. Since each monomial is made up of half x_i's and half y_j's, the Hilbert polynomial is

$$P(z) = \binom{r+z}{r} + \binom{s+z}{s} = \frac{1}{r!s!}z^{r+s} + \cdots,$$

hence the degree is $\binom{r+s}{r}$.

□

3. *Dual Curve.* Let $Y \subseteq \mathbb{P}^2$ be a curve. We regard the set of lines in $\mathbb{P}^2$ as another projective space, $(\mathbb{P}^2)^*$, by taking (a_0, a_1, a_2) as homogenous coordinates of the line $L : a_0x_0 + a_1x_1 + a_2x_2 = 0$. For each nonsingular point $P \in Y$, show that there is a unique line $T_P(Y)$ whose intersection multiplicity with Y at P is > 1. This is the *tangent line* to Y at P. Show that the mapping $P \mapsto T_P(Y)$ defines a *morphism* of Reg Y (the set of nonsingular points of Y) into $(\mathbb{P}^2)^*$. The closure of the image of this morphism is called the dual curve $Y^* \subseteq (\mathbb{P}^2)^*$ of Y.

Proof. Let x_0, x_1, x_2 be homogenous coordinates for $\mathbb{P}^2$, and let f be an irreducible homogenous polynomial in $S = k[x_0, x_1, x_2]$ that defines Y and let $P = (c_0, c_1, c_2)$ be a nonsingular point on Y. Then $\mathbb{P}^2$ can be covered by U_i where $U_i \cong \mathbb{A}^2$ and is defined by $x_i \neq 0$. Since every P is contained in some U_i, we find an affine line in U_i that is tangent to the affine variety $Y_i := Y \cap U_i$. We claim that the projective closure of this line is the desired unique line $T_P(Y)$.

Without loss of generality, assume $c_0 = 1$ so that $P \in U_0$. Identifying U_0 with $\mathbb{A}^2_k$ with coordinates y_1, y_2, we have $P_0 = (c_1, c_2) \in \mathbb{A}^2_k$ identified with P, and Y_0 is defined by the vanishing of $g(y_1, y_2) = f(1, y_1, y_2)$. Assume $c_1 = c_2 = 0$ for simplicity. Let $\mathfrak{m}_{P_0}$ be the maximal ideal in $S(U_0) \simeq k[y_1, y_2]$ corresponding to the point P_0. Since Y is nonsingular at P, Y_0 is nonsingular at P_0, which means $a_i = (\partial g/\partial y_i)(P_0)$ for $i = 1, 2$ are not both zero. In particular, we have $g \notin \mathfrak{m}^2_{P_0}$, so we can write

$$g = a_1y_1 + a_2y_2 + (\text{higher degree terms}),$$

thus any linear polynomial in $\mathfrak{m}^2_P + (g)$ must be a constant multiple of $t = a_1y_1 + a_2y_2 = 0$. We define $T_P(Y)$ to be the projective closure of t. We show it is the desired unique line by showing if a line $H : h(x_0, x_1, x_2) = b_0x_0 + b_1x_1 + b_2x_2 = 0$ has intersection multiplicity with Y at $P > 1$, then $H \cap U_0 = T$,

which implies $T_P(Y) = H$ since $H \cap U_0 = T$ is dense in both H and T. Setting $S(Y) = S/(f)$ and $\mathfrak{m}_P$ as the homogenous prime ideal in S corresponding to the point $P \in \mathbb{P}^2$, we have $S/(f,h) = S(Y)/(h)$, thus

$$\begin{aligned} i(Y,H;P) > 1 \quad &\iff \quad (S(Y)/(h))_{\mathfrak{m}_P} \text{ has length } k > 1 \text{ as an } S(Y)_{\mathfrak{m}_P}\text{-module} \\ &\iff \quad \mathfrak{m}_P/(h) \text{ has length } k \text{ as an } S(Y)\text{-module} \\ &\iff \quad (h) = \mathfrak{m}_P^k \text{ as an extended ideal in } S(Y)_{\mathfrak{m}_P} \end{aligned}$$

since $S(Y)_{\mathfrak{m}_P}$ is a discrete valuation ring by non-singularity of Y at P. In terms of affine coordinates, let $\ell = b_0 + b_1 y_1 + b_2 y_2$ so that $H \cap U_0$ can be identified with the affine variety defined by the vanishing of ℓ, where $b_0 = 0$ necessarily since $P_0 = (0,0)$ is a root of ℓ. Then, by the statements above, we have $\ell \in \mathfrak{m}_{P_0}^k + (g) \subseteq \mathfrak{m}_{P_0}^2 + (g)$, which is possible if and only if $\ell = \lambda t$ for some $\lambda \neq 0$, hence $H \cap U_0 = T$. □

4. Given a curve Y of degree d in $\mathbb{P}^2$, show that there is a nonempty open subset U of $(\mathbb{P}^2)^*$ in its Zariski topology such that for each $L \in U$, L meets Y in exactly d points. This result shows that we could have defined the degree of Y to be the number d such that almost all lines in $\mathbb{P}^2$ meet Y in d points, where "almost all" refers to a nonempty open set of the set of lines, when this set is identified with the dual projective space $(\mathbb{P}^2)^*$.

Proof. Following the hint, we show that the set of lines in $(\mathbb{P}^2)^*$ which are either tangent to Y or pass through a singular point of Y is contained in a property closed subset. By the previous exercise, a line can meet Y at exactly d points if and only if it has intersection multiplicity equal to 1 at every point of intersection, thus the set of lines which are either tangent to Y or pass through a singular point of Y is contained in the dual curve Y^* in $(\mathbb{P}^2)^*$, hence the set of lines that meet Y at exactly d points is contained in the open subset $(\mathbb{P}^2)^* - Y^*$. □

Chapter 2

Schemes

Chapter 2, Section 1

James Lee

October 11, 2026

1. Let A be an Abelian group, and define the *constant presheaf* associated to A on the topological space X to be the presheaf $U \mapsto A$ for all $U \neq \emptyset$, with restriction maps the identity. Show that the constant sheaf $\mathcal{A}$ defined in the text is the sheaf associated to this presheaf.

Proof. Let $\mathcal{C}^+$ be the sheaf associated to the constant presheaf $\mathcal{C}$ defined above. It suffices to show $\mathcal{C}^+$ and $\mathcal{A}$ are isomorphic at the level of stalks. Fix $P \in X$, then the stalk of $\mathcal{C}^+$ at P is the same as the stalk of $\mathcal{C}$ at P, and since the restriction morphisms of $\mathcal{C}$ are identity maps, we must have $\mathcal{C}^+_P = \mathcal{C}_P = A$. The constant sheaf $\mathcal{A}$ is defined as $U \mapsto \{\text{continuous maps } U \to A\}$, where A has the discrete topology. Then, we can take the stalk of $\mathcal{A}$ at x to be the direct limit of $\mathcal{A}(U)$ where U is a connected open neighborhood of x, and if U is connected, $\mathcal{A}(U) = A$ since the image of a connected set must be connected, i.e. $U \to A$ is continuous if and only if it is constant. Therefore, we have $\mathcal{A}_P = A$. Hence, $\mathcal{A}^+$ and $\mathcal{A}$ are isomorphic. □

2. (a) For any morphism of sheaves $\varphi : \mathcal{F} \to \mathcal{G}$, show that for each point P, $(\ker \varphi)_P = \ker(\varphi_P)$ and $(\operatorname{im} \varphi)_P = \operatorname{im}(\varphi_P)$.

(b) Show that φ is injective (respectively, surjective) if and only if the induced maps on the stalks φ_P is injective (respectively, surjective) for all P.

(c) Show that a sequence $\cdots \to \mathcal{F}^{i-1} \xrightarrow{\varphi^{i-1}} \mathcal{F}^i \xrightarrow{\varphi^i} \mathcal{F}^{i+1} \to \dots$ of sheaves and morphisms is exact if and only if for each $P \in X$ the corresponding sequence of stalks is exact as a sequence of abelian groups.

Proof.

(a) It is true in general that kernels commute with limits and cokernels commmute with colimits in any abelian category, since kernels and cokernels can be realized as limits and colimits, respectively.

(b) φ is injective $\iff \ker \varphi = 0 \iff (\ker \varphi)_P = 0 \iff \ker \varphi_P = 0$.
φ is surjective $\iff \operatorname{im} \varphi = \mathcal{G} \iff \operatorname{im}(\varphi_P) = (\operatorname{im} \varphi)_P = \mathcal{G}_P$.

(c) The sequence $\cdots \to \mathcal{F}^{i-1} \xrightarrow{\varphi^{i-1}} \mathcal{F}^i \xrightarrow{\varphi^i} \mathcal{F}^{i+1} \to \dots$ is exact $\iff \operatorname{im} \varphi^{i-1} = \ker \varphi^i$ for all $i \iff (\operatorname{im} \varphi^{i-1})_P = (\ker \varphi^i)_P$ for all i and $P \in X \iff \operatorname{im}(\varphi^{i-1}_P) = \ker(\varphi^i_P)$ for all i and $P \in X$ by part (a) above.

□

3. (a) Let $\varphi : \mathcal{F} \to \mathcal{G}$ be a morphism of sheaves on X. Show that φ is surjective if and only if the following condition holds: for every open set $U \subseteq X$, and for every $s \in \mathcal{G}(U)$, there is a covering $\{U_i\}$ of U, and there are elements $t_i \in \mathcal{F}(U_i)$, such that $\varphi(t_i) = s\big|_{U_i}$ for all i.

(b) Give an example of a surjective morphism of sheaves $\varphi : \mathcal{F} \to \mathcal{G}$, and an open set U such that $\varphi(U) : \mathcal{F}(U) \to \mathcal{G}(U)$ is not surjective.

Proof.

(a) Suppose $\varphi : \mathcal{F} \to \mathcal{G}$ is a surjective morphism of sheaves. Then for any $P \in U$, the induced morphism of stalks $\varphi_P : \mathcal{F}_P \to \mathcal{G}_P$ is surjective. The elements of $\mathcal{G}_P$ (and similarly $\mathcal{F}_P$) are an equivalence class of ordered pairs (U, s) with U an open neighborhood of P and $s \in \mathcal{G}(U)$ and the equivalence relation $(U, s) \sim (V, t)$ if and only if there exists some open $W \subseteq U \cap V$ containing P such that $s\big|_W = t\big|_W$. If φ_P is surjective, then for every $(U, s) \in \mathcal{G}_P$ there exists some $(V, f) \in \mathcal{G}_P$ such that

$$\varphi_P(V, f) = (V, \varphi f), \quad \varphi : \mathcal{F}(V) \to \mathcal{G}(V),$$

so there exists some open subset $W \subseteq U \cap V$ such that $\varphi f\big|_W = s\big|_W$. We can repeat this process for all $x \in U$ to obtain an open cover with the desired properties, that is $\{W_x\}_{x\in U}$ of U where W_x is defined as above with an associated $f_x \in \mathcal{F}(W_x)$ such that by definition $\varphi f_x = s\big|_{W_x}$.
Conversely, it suffices to show $\varphi_P : \mathcal{F}_P \to \mathcal{G}_P$ is surjective for all $P \in X$. Fix $P \in X$ and consider an arbitrary element $(U, s) \in \mathcal{F}_P$. We want to show there exists some $(V, f) \in \mathcal{G}_P$ such that $(V, \varphi_V(f)) \sim (U, s)$. To that end, by assumption there exists some open subset $W \subseteq U$ and $g \in \mathcal{F}(W)$ such that $\varphi_W(g) = s\big|_W$, hence $\varphi_P(W, g) = (U, s)$.

(b) We provide an example of a sheaf from open sets to sets that satisfies the condition. Let $X, Y = S^1$. Let $\pi_X : X \to S^1$ be the identity map, and let $\tau : Y \to X$ be defined by $z \mapsto z^2$ where X, Y is identified with the unit circle in the complex plane, and let $\pi_Y := \tau \circ \pi_X$. Define a sheaf $\mathcal{F}$ on S^1 for a nonempty open subset U of S^1 as

$$\begin{aligned}\mathcal{F}(U) &= \{\text{sections } s : U \to X \text{ with respect to } \pi_X\} \\ &= \{\text{continuous maps } s : U \to X \text{ such that } \pi_X \circ s = \mathrm{id}_U\},\end{aligned}$$

and similarly define $\mathcal{G}$ as the map that maps U to the set of sections from U to Y with respect to π_Y. If $s \in \mathcal{G}(U)$ so that $\pi_Y \circ s = \mathrm{id}_U$, then we can compose s with τ so that

$$\pi_X \circ (\tau \circ s) = (\pi_X \circ \tau) \circ s = \pi_Y \circ s = \mathrm{id}_U,$$

which shows $\tau \circ s$ is a section from to X, so we can define a morphism $\tau_\# : \mathcal{G} \to \mathcal{F}$ induced by τ as

$$\begin{aligned}\tau_\#(U) : \mathcal{G}(U) &\to \mathcal{F}(U) \\ s &\mapsto \tau \circ s.\end{aligned}$$

Let U be the entire space S^1, then $\mathcal{G}(S^1) = \emptyset$ and $\mathcal{F}(S^1) = \{\mathrm{id} : S^1 \to X\}$, so $\mathcal{G}(S^1) \to \mathcal{F}(S^1)$ cannot be surjective. However, for any proper open subset U of S^1, the map $\tau_\# : \mathcal{G}(U) \to \mathcal{F}(U)$ is surjective, thus by part (a) above $\tau_\#$ is a surjective morphism.

□

5. Show that a morphism of sheaves is an isomorphism if and only if it is both injective and surjective.

Proof. $\varphi : \mathcal{F} \to \mathcal{G}$ is an isomorphism $\iff$ $\varphi_P : \mathcal{F}_P \to \mathcal{G}_P$ is an isomorphism for all $P \in X$ $\iff$ φ is injective and surjective by Exercise 2. □

6. (a) Let $\mathcal{F}'$ be a subsheaf of a sheaf $\mathcal{F}$. Show that the natural map of $\mathcal{F}$ to the quotient sheaf $\mathcal{F}/\mathcal{F}'$ is surjective, and has kernel $\mathcal{F}'$. Thus, there is an exact sequence

$$0 \longrightarrow \mathcal{F}' \longrightarrow \mathcal{F} \longrightarrow \mathcal{F}/\mathcal{F}' \longrightarrow 0.$$

(b) Conversely, if $0 \to \mathcal{F}' \to \mathcal{F} \to \mathcal{F}'' \to 0$ is an exact sequence, show that $\mathcal{F}'$ is isomorphic to a subsheaf of $\mathcal{F}$, and that $\mathcal{F}''$ is isomorphic to the quotient $\mathcal{F}$ by this subsheaf.

Proof.

(a) The map $\mathcal{F}(U) \to \mathcal{F}(U)/\mathcal{F}'(U)$ is surjective for all open $U \subseteq X$, so $\pi : \mathcal{F} \to \mathcal{F}/\mathcal{F}'$ is surjective. To show π has kernel $\mathcal{F}'$, by Exercise 2 it suffices to show $(\ker \pi)_P = \mathcal{F}'_P$ for all $P \in X$. Each map $\mathcal{F}(U) \to \mathcal{F}(U)/\mathcal{F}'(U)$ has kernel $\mathcal{F}'(U)$, hence we have $(\ker \pi)_P = \varinjlim \mathcal{F}'(U) = \mathcal{F}'_P$.

(b) By Exercise 2, We have the following equivalent statements:

$$\begin{aligned}&0 \to \mathcal{F}' \to \mathcal{F} \to \mathcal{F}'' \to 0 \text{ is exact.} \\ &\iff 0 \to \mathcal{F}'_P \to \mathcal{F}_P \to \mathcal{F}''_P \to 0 \text{ is exact.} \\ &\iff \mathcal{F}'_P \subseteq \mathcal{F}_P \text{ and } \mathcal{F}''_P \simeq \mathcal{F}_P/\mathcal{F}'_P \simeq (\mathcal{F}/\mathcal{F}')_P \\ &\iff \mathcal{F}' \text{ is isomorphic to a subsheaf of } \mathcal{F} \text{ and } \mathcal{F}'' \cong \mathcal{F}/\mathcal{F}'.\end{aligned}$$

□

7. Let $\varphi : \mathcal{F} \to \mathcal{G}$ be a morphism of sheaves.

(a) Show that $\operatorname{im}\varphi \cong \mathcal{F}/\ker\varphi$.

(b) Show that $\operatorname{coker}\varphi \cong \mathcal{G}/\operatorname{im}\varphi$.

Proof.

(a) It suffices to show $(\operatorname{im}\varphi)_P \simeq \mathcal{F}_P/(\ker\varphi)_P$, which again follows from the fact that for all open $U \subseteq X$, $\operatorname{im}(\varphi(U)) \simeq \mathcal{F}(U)/\ker(\varphi(U))$.

(b) In the same vain, $\operatorname{coker}\varphi \simeq \mathcal{G}/\operatorname{im}\varphi$ follows from the fact that $\operatorname{coker}(\varphi(U)) \simeq \mathcal{G}(U)/\operatorname{im}(\varphi(U))$.

□

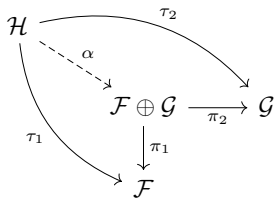
8. For any open subset $U \subseteq X$, show that the functor $\Gamma(U, \cdot)$ from sheaves on X to abelian groups is a left exact functor, i.e. if $0 \to \mathcal{F}' \xrightarrow{f} \mathcal{F} \xrightarrow{g} \mathcal{F}''$ is an exact sequence of sheaves, then $0 \to \Gamma(U, \mathcal{F}') \xrightarrow{f_U} \Gamma(U, \mathcal{F}) \xrightarrow{g_U} \Gamma(U, \mathcal{F}'')$ is an exact sequence of groups.

Proof. Exactness at $\Gamma(U, \mathcal{F}')$ follows from the fact that $\mathcal{F}' \to \mathcal{F}$ is an injective morphism of sheaves if and only if $\Gamma(U, \mathcal{F}') \to \Gamma(U, \mathcal{F})$ is injective for all U. To show exactness at $\Gamma(U, \mathcal{F})$, the sequence $\mathcal{F}' \to \mathcal{F} \to \mathcal{F}''$ is exact if and only if $\mathcal{F}'_P \to \mathcal{F}_P \to \mathcal{F}''_P$ is an exact sequence for all $P \in X$

If $s \in \operatorname{im} f_U$ then $(U, s) \in \ker g_P$ for all $P \in X$, so $(U, g(s))$ equals to 0 everywhere locally, hence $s \in g_U$. Conversely, if $s \in \ker g_U$, then by Exercise 3 there exists an open cover $\{U_i\}$ and elements $t_i \in \Gamma(U_i, \mathcal{F}')$ such that $f_U(t_i) = s\big|_{U_i}$. For any i, j, consider $t_i\big|_{U_i \cap U_j}$, $t_j\big|_{U_i \cap U_j}$. Since $f_{U_i \cap U_j} : \Gamma(U_i \cap U_j, \mathcal{F}') \to \Gamma(U_i \cap U_j, \mathcal{F})$ is injective and $f_{U_i \cap U_j}(t_i\big|_{U_i \cap U_j}) = f_{U_i \cap U_j}(t_j\big|_{U_i \cap U_j}) = s\big|_{U_i \cap U_j}$, we have must $t_i\big|_{U_i \cap U_j} = t_j\big|_{U_i \cap U_j}$, so by the sheaf property there exists $t \in \Gamma(U, \mathcal{F}')$ such that $t\big|_{U_i} = t_i$, so $f_U(t) = s$, hence $s \in \operatorname{im} f_U$. □

9. *Direct Sum.* Let $\mathcal{F}$ and $\mathcal{G}$ be sheaves on X. Show that the presheaf $U \mapsto \mathcal{F}(U) \oplus \mathcal{G}(U)$ is a sheaf. It is called the *direct sum* of $\mathcal{F}$ and $\mathcal{G}$, and is denotes by $\mathcal{F} \oplus \mathcal{G}$. Show that it plays the role of direct sum and of direct product in the category of sheaves of abelian groups.

Proof. Since the category of sheaves on X is a full subcategory of presheaves on X, it suffices to show it satisfies the universal property of a direct sum and of direct product in the category of presheaves of abelian groups on X, then show it satisfies the sheaf property. Let $\pi_1 : \mathcal{F} \oplus \mathcal{G} \to \mathcal{F}$, $\pi_2 : \mathcal{F} \oplus \mathcal{G}_{\to}\mathcal{G}$ be the canonical projection morphisms, and let $\mathcal{H} \in \mathfrak{PreSh}_X$ with morphisms $\tau_1 : \mathcal{H} \to \mathcal{F}$ and $\tau_2 : \mathcal{H} \to \mathcal{G}$ such that the following diagram commutes:

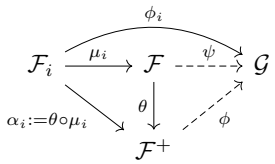


we find the find a unique morphism $\alpha : \mathcal{H} \to \mathcal{F} \oplus \mathcal{G}$ such that the diagram above commmutes. For any open $U \subseteq X$, define $\alpha(U) : \mathcal{H}(U) \to (\mathcal{F} \oplus \mathcal{G})(U)$ as $\alpha(h) = (\tau_1(h), \tau_2(h))$. Since $\alpha(U)$ is unique by the same universal property for abelian groups, α must be the desired unique morphism, and since the direct sum and direct product are essentially the same for finite collections, we have shown $\mathcal{F} \oplus \mathcal{G}$ is a direct sum and product in the category of presheaves.

Now we show $\mathcal{F} \oplus \mathcal{G}$ is a sheaf. Let U be an open subset of X, and let $\{V_i\}$ be a cover of U by open sets. If $(s, t) \in \Gamma(U, \mathcal{F} \oplus \mathcal{G})$, then $(s\big|_{U_i}, t\big|_{U_i}) = (s, t)\big|_{U_i} = 0$ by universal property nonsense, so $s = 0$ and $t = 0$. Also, if we have elements $(s_i, t_i) \in \Gamma(V_i, \mathcal{F} \oplus \mathcal{G})$ such that for all i, j, $(s_i, t_i)\big|_{V_i \cap V_j} = (s_j, t_j)\big|_{V_i \cap V_j}$, then again $(s_i\big|_{U_i \cap U_j}, t_i\big|_{U_i \cap U_j}) = (s_j\big|_{U_i \cap U_j}, t_j\big|_{U_i \cap U_j})$ if and only if $s_i\big|_{U_i \cap U_j} = s_j\big|_{U_i \cap U_j}$ and $t_i\big|_{U_i \cap U_j} = t_j\big|_{U_i \cap U_j}$, so by the sheaf properties of $\mathcal{F}$ and $\mathcal{G}$, there exists $s \in \Gamma(U, \mathcal{F})$ and $t \in \Gamma(U, \mathcal{G})$ such that $(s, t)\big|_{U_i} = (s_i, t_i)$. □

10. *Direct Limit.* Let $\{\mathcal{F}_i\}$ be a direct system of sheaves and morphisms on X. We define the *direct limit* of the system $\{\mathcal{F}_i\}$, denoted $\varinjlim \mathcal{F}_i$, to be the sheaf associated to the presheaf $U \mapsto \varinjlim \mathcal{F}_i(U)$. Show that this is a direct limit in the category of sheaves on X.

Proof. Let $\mathcal{F}$ denote the presheaf defined $U \mapsto \varinjlim \mathcal{F}_i$, and let $\mathcal{F}^+$ denote the sheaf associated to $\mathcal{F}$ with associated morphism $\theta : \mathcal{F} \to \mathcal{F}^+$. From the direct system, we have morphisms $\mu_{ij} : \mathcal{F}_i \to \mathcal{F}_j$ and $\mu_i : \mathcal{F}_i \to \mathcal{F}$ such that for all i, j, $\mu_i = \mu_j \circ \mu_{ij}$. To show $\mathcal{F}^+$ is a direct limit in the category-theoretic sense, we must provide the data of morphisms (between sheaves) $\alpha_i : \mathcal{F}_i \to \mathcal{F}^+$ such that $\alpha_i = \alpha_j \circ \mu_{ij}$ for all i, j satisfying the following universal property: if $\mathcal{G}$ is any sheaf on X with the data of morphisms $\{\phi_i : \mathcal{F}_i \to \mathcal{G}\}$ satisfying for all i, j, $\phi_i = \phi_j \circ \mu_{ij}$, then there exists a unique morphism $\phi : \mathcal{F}^+ \to \mathcal{G}$ such that $\phi_i = \phi \circ \alpha_i$. Observe the following commutative diagram:



We claim $\{\alpha_i := \theta \circ \mu_i\}$ are the desired morphisms. By the universal property of the direct limit which defines $\mathcal{F}$ and the direct system of morphisms $\{\psi_i\}$, there exists a unique morphism between presheaves $\phi : \mathcal{F} \to \mathcal{G}$ such that $\phi_i = \psi \circ \mu_i$. Then, by the universal property of the sheaf $\mathcal{F}^+$ associated to the presheaf $\mathcal{F}$, there exists a unique morphism between sheaves $\phi : \mathcal{F}^+ \to \mathcal{G}$ such that $\psi = \phi \circ \theta$. It remains to check $\phi_i = \phi \circ \alpha_i$. Indeed, we have

$$\phi_i = \psi \circ \mu_i = (\phi \circ \theta) \circ \mu_i = \phi \circ (\theta \circ \mu_i) = \phi \circ \alpha_i,$$

and ϕ is unique by the universal properties above. □

11. Let $\{\mathcal{F}_i\}$ be a direct system of sheaves on a noetherian topological space X. In this case show that the presheaf $U \mapsto \varinjlim \mathcal{F}_i(U)$ is already a sheaf. In particular, $\Gamma(X, \varinjlim \mathcal{F}_i) = \varinjlim \Gamma(X, \mathcal{F}_i)$.

Proof. We directly show the presheaf $\mathcal{F}$ defined by $\mathcal{F}(U) = \varinjlim \mathcal{F}_i(U)$ is a sheaf. Let U be an open subset of X, and let $\{V_i\}$ be open cover of U. By the noetherian hypothesis, we can assume $\{V_j\}$ to be a finite set $(1 \leq j \leq n)$, then we have the following exact sequence of abelian groups

$$0 \longrightarrow \Gamma(U, \mathcal{F}_i) \longrightarrow \prod_{j=1}^n \Gamma(V_j, \mathcal{F}_i) \longrightarrow \prod_{j,k} \Gamma(V_j \cap V_k, \mathcal{F}_i),$$

and since both products $\prod_{j=1}^n \Gamma(V_j, \mathcal{F}_i)$ and $\prod_{j \neq k} \Gamma(V_j \cap V_k, \mathcal{F}_i)$ are finite, it is equivalent to the coproduct; in particular, colimits commute with colimits, so we have the exact sequence

$$0 \longrightarrow \varinjlim \Gamma(U, \mathcal{F}_i) \longrightarrow \prod_{j=1}^n \varinjlim \Gamma(V_j, \mathcal{F}_i) \longrightarrow \prod_{j,k} \varinjlim \Gamma(V_j \cap V_k, \mathcal{F}_i)$$

so the sheaf properties immediately follow. □

12. *Inverse Limit.* Let $\{\mathcal{F}_i\}$ be an inverse system of sheaves on X. Show that the presheaf $U \mapsto \varprojlim \mathcal{F}_i(U)$ is a sheaf. It is called the *inverse limit* of the system $\{\mathcal{F}_i\}$, and is denoted by $\varprojlim \mathcal{F}_i$. Show that this it has the universal property of an inverse limit in the category of sheaves.

Proof. Let $\mathcal{F}$ be the presheaf $U \mapsto \varprojlim \mathcal{F}_i(U)$. We have morphisms $\pi_{ij} : \mathcal{F}_i \to \mathcal{F}_j$ and $\pi_i : \mathcal{F} \to \mathcal{F}_i$ such that for all i, j, $\pi_j = \pi_{ij} \circ \pi_i$. If U is an open set, if $\{V_\alpha\}$ is an open covering of U, then we have the exact sequence of abelian groups

$$0 \longrightarrow \Gamma(U, \mathcal{F}_i) \longrightarrow \prod_\alpha \Gamma(V_\alpha, \mathcal{F}_i) \longrightarrow \prod_{\alpha,\beta} \Gamma(V_\alpha \cap V_\beta, \mathcal{F}_i)$$

and direct limits commute with direct products, so the fact that $\mathcal{F}$ is a sheaf follows from the exact sequence

$$0 \longrightarrow \varprojlim \Gamma(U, \mathcal{F}_i) \longrightarrow \prod_\alpha \varprojlim \Gamma(V_\alpha, \mathcal{F}_i) \longrightarrow \prod_{\alpha,\beta} \varprojlim \Gamma(V_\alpha \cap V_\beta, \mathcal{F}_i).$$

To show $\mathcal{F}$ is an inverse limit in a category-theoretic sense, if $\mathcal{G}$ is some sheaf on X with a collection of morphisms $\tau_i : \mathcal{G} \to \mathcal{F}_i$ such that $\tau_j = \pi_{ij} \circ \tau_i$ for all i, j, then we want to show there exists a unique $\theta : \mathcal{G} \to \mathcal{F}$ such that $\tau_i = \pi_i \circ \theta$. Since for each open subset U in X we have a direct system $\tau_i(U) : \Gamma(U, \mathcal{G}) \to \Gamma(U, \mathcal{F}_i)$, by the universal property of the inverse limit $\Gamma(U, \mathcal{F}) = \varprojlim \Gamma(U, \mathcal{F}_i)$, there exists a unique morphism $\theta(U) : \Gamma(U, \mathcal{G}) \to \varprojlim \Gamma(U, \mathcal{F}_i)$ such that $\tau_i(U) = \pi_i(U) \circ \theta(U)$, so we can define θ as such, then it is unique by construction. □

14. *Support.* Let $\mathcal{F}$ be a sheaf on X, and let $s \in \mathcal{F}(U)$ be a section over an open set U. The *support* of s, denoted Supp s, is defined to be $\{P \in U \mid s_P \neq 0\}$, where s_P denotes the germ of s in the stalk $\mathcal{F}_P$. Show that Supp s is a closed subset of U. We define the *support* of $\mathcal{F}$, Supp $\mathcal{F}$, to be $\{P \in X \mid \mathcal{F}_P \neq 0\}$. It need not be a closed subset.

Proof. We show $U - \text{Supp } s$ is an open set. If $Q \in U$ such that $s_Q = 0$, then by definition of the direct limit, there exists an open neighborhood V of Q in U such that $0 = (V, s\big|_V) \in \mathcal{F}_Q$, hence $s_R = 0$ for all $R \in V$. □

15. *Sheaf Hom.* Let $\mathcal{F}$, $\mathcal{G}$ be sheaves of abelian groups on X. For any open subset $U \subseteq X$, show that the set $\text{Hom}\,(\mathcal{F}\big|_U, \mathcal{G}\big|_U)$ of morphisms of the restricted sheaves has a natural structure of an abelian group. Show that the presheaf $U \mapsto \text{Hom}(\mathcal{F}\big|_U, \mathcal{G}\big|_U)$ is a sheaf. It is called the *sheaf of local morphisms* of $\mathcal{F}$ into $\mathcal{G}$, "sheaf hom" for short, and is denoted $\mathcal{H}om(\mathcal{F}, \mathcal{G})$.

Proof. Morphisms $f, g \in \text{Hom}\,(\mathcal{F}\big|_U, \mathcal{G}\big|_U)$ define for all open $V \subseteq U$ homomorphisms between abelian groups $f(V), g(V) : \mathcal{F}(V) \to \mathcal{G}(V)$, so we can define $f + g$ as $(f + g)(V) = f(V) + g(V)$. It is obviously an abelian group with identity 0 as the zero morphism defined by $0(V) \equiv 0$.

Let U be an open subset of X, and let $\{V_i\}$ be an open cover of U. We make some clarifying remarks. If $s \in \Gamma(U, \mathcal{H}om\,(\mathcal{F}, \mathcal{G})) = \text{Hom}\,(\mathcal{F}\big|_U, \mathcal{G}\big|_U)$, then s is a natural transformation of functors $\mathcal{F}\big|_U \to \mathcal{G}\big|_U$, so $s\big|_{V_i}$ refers to the induced natural transformation of functors $\mathcal{F}\big|_{V_i} \to \mathcal{G}\big|_{V_i}$ by only considering open subsets $W \subseteq V_i$. Thus, we have the following commutative diagram:

$$\begin{array}{ccccccc} 0 & \longrightarrow & \Gamma(W, \mathcal{F}) & \longrightarrow & \prod \Gamma(W \cap V_i, \mathcal{F}) & \longrightarrow & \prod \Gamma(W \cap V_i \cap V_j, \mathcal{F}) \\ & & \downarrow{\scriptstyle s(W)} & & \downarrow{\scriptstyle \prod s(W \cap V_i)} & & \downarrow{\scriptstyle \prod s(W \cap V_i \cap V_j)} \\ 0 & \longrightarrow & \Gamma(W, \mathcal{G}) & \longrightarrow & \prod \Gamma(W \cap V_i, \mathcal{G}) & \longrightarrow & \prod \Gamma(W \cap V_i \cap V_j, \mathcal{G}) \end{array}$$

hence $\mathcal{H}om(\mathcal{F}, \mathcal{G})$ is a sheaf. □

17. *Skyscraper Sheaves.* Let X be a topological space, let P be a point, and let A be an abelian group. Define a sheaf $i_P(A)$ on X as follows: $i_P(A)(U) = A$ if $P \in U$, 0 otherwise. Verify that the stalk of $i_P(A)$ is A at every point $Q \in \overline{\{P\}}$, and 0 elsewhere. Hence, the name "Skyscraper sheaf." Show that this sheaf could also be described as $i_*(A)$, where A denotes the constant sheaf A on the closed subspace $\overline{\{P\}}$, and $i : \overline{\{P\}} \to X$ is the inclusion.

Proof. We first identify the restriction maps of $i_P(A)$. Let $U \subseteq V$ be open subsets of X, then $i_P(A)(V) \to i_P(A)(U)$ is the zero map if either V or U (hence V as well) does not contain P, and is the identity map if both V and U contain P. If $Q \in \overline{\{P\}}$, then every open neighborhood of Q contains P, so the direct system of open neighborhoods of Q induces the direct system consisting of a copy of A for every $U \ni Q$ with the identity map as transition maps, hence $(i_P(A))_Q = A$. If $Q \notin \overline{\{P\}}$, then there exists an open neighborhood W of Q not containing P, which means $i_P(A)(W) = 0$, hence $(i_P(A))_Q = 0$. The last statement is obvious. □

18. *Adjoint Property of f^{-1}.* Let $f : X \to Y$ be a continuous map of topological spaces. Show that for any sheaf $\mathcal{F}$ on X there is a natural map $f^{-1}f_*\mathcal{F} \to \mathcal{F}$, and for any sheaf $\mathcal{G}$ on Y there is a natural map $\mathcal{G} \to f_*f^{-1}\mathcal{G}$. Use these maps to show that there is a natural bijection of sets, for any sheaves $\mathcal{F}$ on X and $\mathcal{G}$ on Y,

$$\text{Hom}_X(f^{-1}\mathcal{G}, \mathcal{F}) = \text{Hom}_Y(\mathcal{G}, f_*\mathcal{F}).$$

Proof. Let U, V be open in X, Y, respectively. We unpack the definitions:

$$(f^{-1}f_*\mathcal{F})(U) = \varinjlim_{W \supseteq f(U)} (f_*\mathcal{F})(W) = \varinjlim_{W \supseteq f(U)} \mathcal{F}(f^{-1}(W)),$$
$$(f_*f^{-1}\mathcal{G})(V) = (f^{-1}\mathcal{G})(f^{-1}(V)) = \varinjlim_{W \supseteq f(f^{-1}(V))} \mathcal{G}(W).$$

We also have

$$(f_*f^{-1}f_*\mathcal{F})(V) = (f^{-1}f_*\mathcal{F})(f^{-1}(V))$$

$$
\begin{aligned}
&= \varinjlim_{W \supseteq f(f^{-1}(V))} \mathcal{F}(f^{-1}(W)) \\
&= \mathcal{F}(f^{-1}(V)) \\
&= (f_*\mathcal{F})(V) \\
(f^{-1}f_*f^{-1}\mathcal{G})(U) &= \varinjlim_{W \supseteq f(U)} (f_*f^{-1}\mathcal{G})(W) \\
&= \varinjlim_{W \supseteq f(U)} \varinjlim_{W' \supseteq f(f^{-1}(W))} \mathcal{G}(W') \\
&= \varinjlim_{W \supseteq f(U)} \mathcal{G}(W) \\
&= (f^{-1}\mathcal{G})(U)
\end{aligned}
$$

Note that $V' := f(f^{-1}(V)) = V \cap f(X)$. An element of $(f^{-1}f_*\mathcal{F})(U)$ is of the form (W, s) where $W \supseteq f(U)$ and $s \in \mathcal{F}(f^{-1}(W))$. Define $\alpha : f^{-1}f_*\mathcal{F} \to \mathcal{F}$ as

$$
\begin{aligned}
\alpha(U) : (f^{-1}f_*\mathcal{F})(U) &\to \mathcal{F}(U) \\
(W, s) &\mapsto s\big|_U.
\end{aligned}
$$

An element of $(f_*f^{-1}\mathcal{G})(V)$ is of the form (W, t) where $W \supseteq V'$ and $t \in \mathcal{G}(W)$. Define $\beta : \mathcal{G} \to f_*f^{-1}\mathcal{G}$ as

$$
\begin{aligned}
\beta(V) : \mathcal{G}(V) &\to (f_*f^{-1}\mathcal{G})(V) \\
t &\mapsto (V, t).
\end{aligned}
$$

Let $\varphi : \mathcal{F} \to \mathcal{F}'$, $\psi : \mathcal{G} \to \mathcal{G}'$, where $\mathcal{F}, \mathcal{F} \in \mathfrak{Sh}_X$, $\mathcal{G}, \mathcal{G}' \in \mathfrak{Sh}_Y$, $\varphi \in \mathrm{Hom}_X(\mathcal{F}, \mathcal{F}')$, and $\psi \in \mathrm{Hom}_Y(\mathcal{G}, \mathcal{G}')$. We explicitly describe the induced lifts with respect to f, namely $f_*\varphi : f_*\mathcal{F} \to f_*\mathcal{F}'$ and $f^{-1}\psi : f^{-1}\mathcal{G} \to f^{-1}\mathcal{G}'$. Let U, V be open sets in X, Y, respectively. Define

$$
\begin{aligned}
f_*\varphi(V) : f_*\mathcal{F}(V) &\to f_*\mathcal{F}'(V) \\
s &\mapsto \varphi(f^{-1}(V))(s)
\end{aligned}
$$

where $s \in \Gamma(f^{-1}(V), \mathcal{F})$, and define

$$
\begin{aligned}
f^{-1}\psi(U) : f^{-1}\mathcal{G}(U) &\to f^{-1}\mathcal{G}'(U) \\
(W, t) &\mapsto (W, \psi(W)(t)),
\end{aligned}
$$

where $(W, t) \in \varinjlim_{W \supseteq f(U)} \mathcal{G}(W)$. For every $\varphi : f^{-1}\mathcal{G} \to \mathcal{F}$, we have the morphism $f_*\varphi : f_*f^{-1}\mathcal{G} \to f_*\mathcal{F}$, so we define $\beta^* : \mathrm{Hom}_X(f^{-1}\mathcal{G}, \mathcal{F}) \to \mathrm{Hom}_Y(\mathcal{G}, f_*\mathcal{F})$ as

$$
\beta^*\varphi = (f_*\varphi) \circ \beta.
$$

Similarly, for every $\psi : \mathcal{G} \to f_*\mathcal{F}$, we have the morphism $f^{-1}\psi : f^{-1}\mathcal{G} \to f^{-1}f_*\mathcal{F}$, so we define $\alpha_* : \mathrm{Hom}_Y(\mathcal{G}, f_*\mathcal{F}) \to \mathrm{Hom}_X(f^{-1}\mathcal{G}, \mathcal{F})$ as

$$
\alpha_*\psi = \alpha \circ (f^{-1}\psi).
$$

To show α_*, β^* are bijections with inverses to each other, it suffices to show $\alpha_* \circ \beta^* = \mathrm{id}_{\mathrm{Hom}_X(f^{-1}\mathcal{G}, \mathcal{F})}$ and $\beta^* \circ \alpha_* = \mathrm{id}_{\mathrm{Hom}_Y(\mathcal{G} \to f_*\mathcal{F})}$:

$$
\begin{aligned}
(\alpha_* \circ \beta^*)(\varphi) &= \alpha_*((f_*\varphi) \circ \beta) = \alpha \circ f^{-1}((f_*\varphi) \circ \beta) = \alpha \circ (f^{-1}f_*\varphi) \circ (f^{-1}\beta) \\
(\beta^* \circ \alpha^*)(\psi) &= \beta^*(\alpha \circ (f^{-1}\psi)) = f_*(\alpha \circ (f^{-1}\psi)) \circ \beta = (f_*\alpha) \circ (f_*f^{-1}\psi) \circ \beta.
\end{aligned}
$$

Again, we unpack the definitions. Let $(W, s) \in f^{-1}\mathcal{G}(U)$, where $f(U) \subseteq W \subseteq Y$ with $t \in \Gamma(W, \mathcal{G})$. We want to show

$$
(\alpha_*\beta^*\varphi)(U)((W, t)) = \varphi(U)((W, t)).
$$

We have

$$
\begin{aligned}
(\alpha_*\beta^*\varphi)(U)((W, t)) &= (\alpha(U) \circ (f^{-1}f_*\varphi))(U) \circ (f^{-1}\beta)(U))((W, t)) \\
&= (\alpha(U) \circ (f^{-1}f_*\varphi)(U))((W, t)))
\end{aligned}
$$

$$
\begin{aligned}
&= \alpha(U)((f^{-1}(W), \varphi(f^{-1}(W)(t)))) \\
&= \varphi(f^{-1}(W)((W,t)))\big|_U \\
&= \varphi(U)((W,t)).
\end{aligned}
$$

Here, we used the fact that the restriction map commutes with $\varphi(U)$ by definition of morphisms between sheaves. Let $t \in \Gamma(V, \mathcal{G})$. We similarly have

$$
\begin{aligned}
(\beta * \alpha^* \psi)(V)(t) &= ((f_*\alpha)(V) \circ (f_* f^{-1}\psi)(V) \circ \beta(V))(t) \\
&= ((f_*\alpha)(V) \circ (f_* f^{-1}\psi(V)))((V,t)) \\
&= (f_*\alpha)(V)(\psi(V)(t)) \\
&= \psi(V)(t).
\end{aligned}
$$

Hence, α_* and β^* are bijections with inverses to each other. □

22. *Glueing Sheaves.* Let X be a topological space, let $\mathfrak{U} = \{U_i\}$ be an open cover of X, and suppose we are given for each i a sheaf $\mathcal{F}_i$ on U_i, and for each i, j an isomorphism $\varphi_{ij} : \mathcal{F}_i\big|_{U_i \cap U_j} \xrightarrow{\sim} \mathcal{F}_j\big|_{U_i \cap U_j}$ such that

(1) for each i, $\varphi_{ii} = \mathrm{id}$,

(2) and for each i, j, k, $\varphi_{ik} = \varphi_{jk} \circ \varphi_{ij}$ on $U_i \cap U_j \cap U_k$.

Then there exists a unique sheaf $\mathcal{F}$ on X together with isomorphisms $\psi_i : \mathcal{F}\big|_{U_i} \xrightarrow{\sim} \mathcal{F}_i$ such that for each i, j, $\psi_j = \varphi_{ij} \circ \psi_i$ on $U_i \cap U_j$. We say loosely that $\mathcal{F}$ is obtained by *glueing* the sheaves $\mathcal{F}_i$ via the isomorphisms φ_{ij}.

Shame on James for skipping this "standard exercise."

Chapter 2, Section 2

James Lee

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1. Let A be a ring, let $X = \operatorname{Spec} A$, let $f \in A$ and let $D(f) \subseteq X$ be the open complement of $V((f))$. Show that the locally ringed space $(D(f), \mathcal{O}_X\big|_{D(f)})$ is isomorphic to $\operatorname{Spec} A_f$.

Proof. The natural map $A \to A_f$ defined by $a \mapsto a/1$ induces an inclusion of schemes $\varphi^* : \operatorname{Spec} A_f \to D(f) \subseteq X$ with a morphism of sheaves which is a homeomorphism onto its image. This is by the one-to-one correspondence between prime ideals of A_f and prime ideals of A not containing f. It is an isomorphism at the local level. Let $x = \mathfrak{p}A_f$ be a point in $\operatorname{Spec} A_f$ where $\mathfrak{p}$ is a prime ideal of A not containing f, let $y = \varphi^*(x) = \mathfrak{p} \in \operatorname{Spec} A$, and let $\bar{f}$ denote the image of f in $A_\mathfrak{p}$. Since $\mathfrak{p}$ does not contain f, $\bar{f}$ is a unit, which means $(A_\mathfrak{p})_{\bar{f}} \cong A_\mathfrak{p}$, so we have

$$\mathcal{O}_{X,\varphi(x)} = A_\mathfrak{p} \cong (A_\mathfrak{p})_{\bar{f}} \cong (A_f)_{\mathfrak{p}A_f} = \mathcal{O}_{\operatorname{Spec} A_f, x}$$

(in particular, localization commutes, see (A.M. Ex. 3.3). Hence, $(D(f), \mathcal{O}_X\big|_{D(f)}))$ is isomorphic to $\operatorname{Spec} A_f$. □

2. Let $(X, \mathcal{O}_X)$ be a scheme, and let $U \subseteq X$ be any open subset. Show that $(U, \mathcal{O}_X\big|_U)$ is a scheme. We call this the *induced scheme structure* on the open set U, and we refer to $(U, \mathcal{O}_X\big|_U)$ as an *open subscheme* of X.

Proof. We want to show U can be covered by open affine subsets. Let $\{X_i = \operatorname{Spec} A_i\}$ be an affine open cover of X. The open subsets $U \cap A_i$ cover U, so we reduce to the case when $X = \operatorname{Spec} A$ for some commutative ring A. Open sets of the form $D(f) = X - V((f))$ are a basis for X, so for any $\mathfrak{p} \in U$, we can find $f \in A$ such that $\mathfrak{p} \in D(f) \subseteq U$. By Problem 1, $(D(f), \mathcal{O}_X\big|_{D(f)})$ is isomorphic to the affine scheme $\operatorname{Spec} A_f$, hence U is a scheme. □

3. *Reduced Schemes.* A scheme $(X, \mathcal{O}_X)$ is *reduced* if for every open set $U \subseteq X$, the ring $\mathcal{O}_X(U)$ has no nilpotent elements.

(a) Show that $(X, \mathcal{O}_X)$ is reduced if and only if for every $P \in X$, the local ring $\mathcal{O}_{X,P}$ has no nilpotent elements.

(b) Let $(X, \mathcal{O}_X)$ be a scheme. Let $(\mathcal{O}_X)_{\text{red}}$ be the sheaf associated to the presheaf $U \mapsto \mathcal{O}_X(U)_{\text{red}}$, where for any ring A, we denote by A_{red} the quotient of A by its ideal of nilpotent elements. Show that $(X, (\mathcal{O}_X)_{\text{red}})$ is a scheme. We call it the *reduced scheme* associated to X, and denote it by X_{red}. Show that there is a morphism of schemes $X_{\text{red}} \to X$, which is a homeomorphism on the underlying topological spaces.

(c) Let $f : X \to Y$ be a morphism of sheaves, and assume that X is reduced. Show that there is a unique morphism $g : X \to Y_{\text{red}}$ such that f is obtained by composing g with the natural map $Y_{\text{red}} \to Y$.

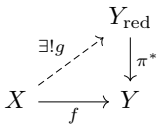
Proof.

(a) We show a commutative ring A has no nilpotent elements if and only if $A_\mathfrak{p}$ has no nilpotent elements for all prime ideals $\mathfrak{p}$ of A, that is *being reduced is a local property.* In general, if M is any A-module, then $M = 0$ if and only if $M_\mathfrak{p} = 0$ for all prime ideals $\mathfrak{p}$ in A. In particular, if M is the nilradical of A, then A has no nilpotent elements if and only if $M = 0$ if and only if $M_\mathfrak{p} = 0$ for all $\mathfrak{p}$ if and only if $A_\mathfrak{p}$ has no nilpotent elements.

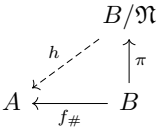
(b) We proceed with the case when $X = \operatorname{Spec} A$ is affine. Let $\mathfrak{N}$ be the nilradical of A. It is the ideal of all nilpotent elements of A and can be realized as intersection of all prime ideals of A. We show the sheaf of rings $(X, (\mathcal{O}_X)_{\text{red}})$ is isomorphic to the affine scheme $\operatorname{Spec} A/\mathfrak{N}$. By definition $\operatorname{Spec} A \cong \operatorname{Spec} A/\mathfrak{N}$ as topological spaces since the prime ideal structures of the rings A, $A/\mathfrak{N}$ are the same. Also, if $f \in A$, then $\Gamma(D(f), (\mathcal{O}_X)_{\text{red}} = A_f/\mathfrak{N}A_f$ since the nilradical of A_f is $\mathfrak{N}A_f$. Thus, we see that $\mathcal{O}_{\operatorname{Spec} A/\mathfrak{N}}$ and $(\mathcal{O}_X)_{\text{red}}$ agree on basic open sets $D(f)$, hence they must be isomorphic.

In the general case, let $U_i = \operatorname{Spec} A_i$ be some affine cover of X. Denote $\tilde{U}_i = \operatorname{Spec} A_i/\mathfrak{N}_i$ the reduced scheme associated to U_i, where $\mathfrak{N}_i$ is the nilradical of A_i. The scheme obtained by glueing together $\tilde{U}_i$ is exactly $(X, (\mathcal{O}_X)_{\text{red}})$. From the canonical projection maps $A_i \to A_i/\mathfrak{N}_i$ we obtain a morphism of schemes $X_{\text{red}} \to X$ which is a homeomorphism on the underlying topological spaces.

(c) Again, we proceed with the affine case. Let $X = \operatorname{Spec} A$, $Y = \operatorname{Spec} B$, and $Y_{\text{red}} = \operatorname{Spec} B/\mathfrak{N}$, where A is a reduced ring and $\mathfrak{N}$ is the nilradical of the ring B. The category of affine schemes is anti-equivalent to the category of commutative rings, so we can completely reduce this problem to the level of rings. We have the canonical projection map $\pi : B \to B/\mathfrak{N}$ which induces the morphism of schemes $\pi^* : Y_{\text{red}} \to Y$, and giving a morphism $f : X \to Y$ is equivalent to giving a ring homomorphism $f_\# : B \to A$. We want to show there exists a unique $g : X \to Y_{\text{red}}$ such that the diagram commutes



Correspondingly, we want to show there exists a unique ring homomorphism $h : B/\mathfrak{N} \to A$ such that the diagram commutes



Since the image of $f_\#$ must be a reduced subring of A, the kernel of $f_\#$ must contain the nilradical $\mathfrak{N}$ of B, so we can take $b + \mathfrak{N} \mapsto f_\#(b)$ to be h, and it is unique by construction. In the general case, we can cover X by open affine subsets $U_i = \operatorname{Spec} A_i$ such that A_i is a reduced ring, then glue together the obtained morphisms $g_i : U_i \to Y_{\text{red}}$.

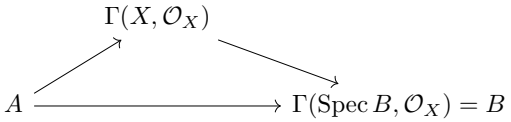
□

4. Let A be a ring and let $(X, \mathcal{O}_X)$ be a scheme. Given a morphism $f : X \to \operatorname{Spec} A$, we have an associated map on sheaves $f^\# : \mathcal{O}_{\operatorname{Spec} A} \to f_*\mathcal{O}_X$. Taking global sections we obtain a homomorphism $A \to \Gamma(X, \mathcal{O}_X)$. Thus, there is a natural map

$$\alpha : \operatorname{Hom}_{\mathfrak{Sch}}(X, \operatorname{Spec} A) \to \operatorname{Hom}_{\mathfrak{Rings}}(A, \Gamma(X, \mathcal{O}_X)).$$

Show that α is bijective.

Proof. We want to show the homomorphism between global sections $A \to \Gamma(X, \mathcal{O}_X)$ completely determines the morphism $f : X \to \operatorname{Spec} B$. We can check this locally on X. Let $V = \operatorname{Spec} B$ be an open affine subset of X. We also have a homomorphism of global sections $A \to \Gamma(V, \mathcal{O}_X) = B$ by composing with the inclusion morphism, which fits into the following commutative diagram



where $\Gamma(X, \mathcal{O}_X) \to B$ is the natural restriction homomorphism. By equivalence of the category of rings and affine schemes, we conclude that α is injective.

It remains to show α is surjective. Let $\varphi : A \to \Gamma(X, \mathcal{O}_X)$ be a homomorphism of global sections. For each $x \in X$, let $\mathfrak{m}_x$ be the maximal ideal of $\mathcal{O}_{x,X}$, and let $i_x : \Gamma(X, \mathcal{O}_X) \to \mathcal{O}_{x,X}$ be the natural

map. Then define $\tilde{\varphi} : X \to \operatorname{Spec} A$ by $x \mapsto (i_x \circ \varphi)^{-1}(\mathfrak{m}_x)$. This coincides with the case when X is affine. Indeed, suppose $X = \operatorname{Spec} B$ for some ring B, and let $\mathfrak{p} \in \operatorname{Spec} B$ corresponding to x. The inclusion homeomorphism $i_x : \Gamma(X, \mathcal{O}_X) \to \mathcal{O}_{x,X}$ is precisely the natural map $i_{\mathfrak{p}} : B \to B_{\mathfrak{p}}$ associated to localization. Thus, $i_x(\mathfrak{m}_x) = \mathfrak{p} \in B$, and $\tilde{\varphi}(x) = \varphi^{-1}(\mathfrak{p})$. □

5. Describe $\operatorname{Spec} \mathbb{Z}$, and show that is a final object for the category of schemes, i.e., each scheme X admits a unique morphism to $\operatorname{Spec} \mathbb{Z}$.

Proof. The ring of integers $\mathbb{Z}$ is a principal ideal domain, so all of its prime ideals are of the form (p) for $p = 0$ or prime. If $p \neq 0$, then (p) is a closed point in $\operatorname{Spec} \mathbb{Z}$ since it is maximal as an ideal in $\mathbb{Z}$. It is also a reduced scheme since $\mathbb{Z}$ is a domain. To show $\operatorname{Spec} \mathbb{Z}$ is final in the category of schemes, it suffices to show $\mathbb{Z}$ is initial in the category of commutative rings, because any morphism of schemes $X \to \operatorname{Spec} Z$ is determined by the ring homomorphism $\mathbb{Z} \to \Gamma(X, \mathcal{O}_X)$ by Exercise 4. Indeed, if A is any commutative ring, the multiplicative unit $1 \in \mathbb{Z}$ additively generates $\mathbb{Z}$, so the homomorphism $f : \mathbb{Z} \to A$ is completely determined by $f(1)$, and since $f(1) = 1$, f must be unique. □

7. Let X be a scheme. For any $x \in X$, let $\mathcal{O}_x$ be the local ring at x, and $\mathfrak{m}_x$ its maximal ideal. We define the *residue field* of x on X to be the field $k(x) = \mathcal{O}_x/\mathfrak{m}_x$. Now let K be any field. Show that to give a morphism of $\operatorname{Spec} K$ to X it is equivalent to give a point $x \in X$ and an inclusion map $k(x) \hookrightarrow K$.

Proof. There are two associated maps to any morphism of schemes: a continuous map $f : X \to Y$ between topological spaces, and a morphism between sheaves $f_{\#} : \mathcal{O}_Y \to f_*\mathcal{O}_X$. Any field has only the zero ideal (0) as a prime ideal, so $\operatorname{Spec} K$ as a topological space consists of only one point. To give a continuous map $\iota : \operatorname{Spec} K \to X$ is equivalent to giving a point $x \in X$ to map (0) to. Also, any morphism $\mathcal{O}_X \to \iota_*\mathcal{O}_{\operatorname{Spec} K}$ induces a morphism of local rings $\mathcal{O}_x \to K_{(0)} = K$, which is equivalent to giving an inclusion map of fields $k(x) \hookrightarrow K$. Giving any inclusion $k(x) \hookrightarrow K$, we have a morphism between spectra $\operatorname{Spec} K \to \operatorname{Spec} k(x)$, which can be naturally extended to $\operatorname{Spec} K \to \operatorname{Spec} k(x) \to X$ by mapping the unique point of $\operatorname{Spec} k(x)$ to $x \in X$. □

8. Let X be a scheme. For any point $x \in X$, we define the *Zariski tangent space* T_x to X at x to be the dual of the $k(x)$-vector space $\mathfrak{m}_x/\mathfrak{m}_x^2$. Now assume that X is a scheme over a field k, and let $k[\varepsilon]/(\varepsilon^2)$ be the *ring of dual numbers over* k. Show that to give a k-morphism of $\operatorname{Spec} k[\varepsilon]/(\varepsilon^2)$ to X is equivalent to giving a point $x \in X$, *rational over* k (i.e., such that $k(x) = k$), and an element of T_x.

Proof. We first identify the prime ideals of $k[\varepsilon]/(\varepsilon^2)$ and determine the topology of $\operatorname{Spec} k[\varepsilon]/(\varepsilon^2)$. The only prime ideal is (ε), in particular it is a local ring with nilpotent elements, so $\operatorname{Spec} k[\varepsilon]/(\varepsilon^2)$ has only one point. We have the canonical projection $k[\varepsilon]/(\varepsilon^2) \to k$, which given a morphism of schemes $d : \operatorname{Spec} k[\varepsilon]/(\varepsilon^2) \to X$ induces

$$\operatorname{Spec} k \to \operatorname{Spec} k[\varepsilon]/(\varepsilon^2) \to X.$$

By Exercise 7, this is equivalent to giving a point $x \in X$ and an inclusion of fields $k(x) \hookrightarrow k$. Also, any d induces a morphism between tangent spaces

$$d^* : T_{(\varepsilon)}(\operatorname{Spec} k[\varepsilon]/(\varepsilon^2)) \to T_x(X),$$

and since $T_{(\varepsilon)}(\operatorname{Spec} k[\varepsilon]/(\varepsilon^2))$ is a one-dimensional k-vector space, a linear map such as d^* is equivalent to giving an element of $T_x(X)$. □

9. If X is a topological space, and Z an irreducible closed subset of X, a *generic point* for Z is a point ζ such that $Z = \overline{\{\zeta\}}$. If X is a scheme, show that every (nonempty) irreducible closed subset has a unique generic point.

Proof. The case when X is an affine scheme is easy. If $X = \operatorname{Spec} A$, then any irreducible closed subset Z of X is of the form $Z = V(\mathfrak{p})$ for some prime ideal $\mathfrak{p}$ of A, so $\mathfrak{p}$ as a point in X is by definition the unique generic point of Z (uniqueness follows from $\mathfrak{p}$ being the smallest, hence unique, prime ideal contained in all elements of $V(\mathfrak{p})$). Note that any proper open subset of an irreducible closed subset of an affine scheme must contain the unique generic point, otherwise this implies the unique generic point is contained in a closed set, hence is the entire space. Generally, suppose X is a scheme, and let Z be a nonempty irreducible closed subset of X. Cover X with open affine subsets $U_i = \operatorname{Spec} A_i$, $i \in I$. The closed subset Z can be obtained by glueing $Z_i = Z \cap U_i$ along $Z_i \cap Z_j$ for all i, j. Since Z_i is a closed irreducible subset of U_i it has a unique generic point, say ζ_i. The set $Z_i \cap Z_j$ is an open in Z_i and Z_j, so it must contain ζ_i and ζ_j, thus they must be identified by the glueing process. □

10. Describe $\operatorname{Spec}\mathbb{R}[x]$. How does its topological space compare to the set $\mathbb{R}$? To $\mathbb{C}$?

Proof. The maximal ideals of $\mathbb{R}[x]$ are of the form $(x-\alpha)$ and $(x^2+\beta x+\gamma)$, where $\alpha,\beta,\gamma\in\mathbb{R}$ with $\beta^2-4\gamma<0$, and the only non-maximal prime ideal of $\mathbb{R}[x]$ is the zero ideal (0). There is a bijection between quadratic polynomials $x^2+\beta x+\gamma$ and pairs of conjugate complex numbers $a\pm ib$ with $b\neq 0$, namely

$$x^2+\beta x+\gamma \mapsto \frac{-\beta\pm\sqrt{\beta^2-4\gamma}}{2}$$

so besides the generic point (0), we can identify $\operatorname{Spec}\mathbb{R}[x]$ with the complex plane $\mathbb{C}$ "folded in half along the real axis." □

12. *Glueing Lemma.* Generalize the glueing procedure described in the text (2.3.5) as follows. Let $\{X_i\}$ be a family of schemes (possible infinite). For each $i\neq j$, suppose given an open subset $U_{ij}\subseteq X_i$, and let it have the induced scheme structure. Suppose also given for each $i\neq j$ an isomorphism of schemes $\varphi_{ij}: U_{ij}\to U_{ji}$ such that

(1) for each i,j, $\varphi_{ji}=\varphi_{ij}^{-1}$,
(2) and for each i,j,k, $\varphi_{ij}(U_{ij}\cap U_{ik})=U_{ji}\cap U_{jk}$, and $\varphi_{ik}=\varphi_{jk}\circ\varphi_{ij}$ on $U_{ij}\cap U_{ik}$.

Then, show that there is a scheme X, together with morphisms $\psi_i: X_i\to X$ for each i, such that

(1) ψ_i is an isomorphism of X_i onto an open subscheme of X,
(2) the $\psi_i(X_i)$ cover X,
(3) $\psi_i(U_{ij})=\psi_i(X_i)\cap\psi_j(X_j)$,
(4) and $\psi_i=\psi_j\circ\varphi_{ij}$ on U_{ij}.

We say that X is obtained by *glueing* the schemes X_i along the isomorphisms φ_{ij}. An interesting special case is when the family X_i is arbitrary, but the U_{ij} and φ_{ij} are all empty. Then the scheme X is called the *disjoint union* of the X_i, and is denoted $\coprod X_i$.

Proof. Let $X=\coprod X_i/\sim$, where the equivalence relation identifies U_{ij} with U_{ji} for all $i\neq j$. It is a scheme topologically, that is there is an open cover of X such that each element of the cover is homeomorphic to the spectra of some ring. We have already have a scheme struture on each open subset X_i of X, so by Exercise 1.22 we can obtain a unique sheaf on X satisfies the properties above. In particular, the structure sheaf of X is given by

$$\mathcal{O}_X(V)=\{\langle s_i\rangle_{i\in I}\mid s_i\in\mathcal{O}_X(\iota_i^{-1}(V)) \text{ and } \varphi_{ij}(s_i\big|_{\iota_i^{-1}(V)\cap U_{ij}})=s_j\big|_{\iota_j^{-1}(V)\cap U_j}\}.$$

□

13. A topological space is *quasi-compact* if every open cover has a finite subcover.

(a) Show that a topological space is noetherian if and only if every open subset is quasi-compact.
(b) If X is an affine scheme, show that $\operatorname{sp}(X)$ is quasi-compact, but not in general noetherian. We say a scheme X is *quasi-compact* if $\operatorname{sp}(X)$ is.
(c) If A is a noetherian ring, show that $\operatorname{sp}(\operatorname{Spec}A)$ is a noetherian topological space.
(d) Give an example to show that $\operatorname{sp}(\operatorname{Spec}A)$ can be noetherian even when A is not.

Proof.

(a) Any open subset of a noetherian space if noetherian, and any noetherian space is quasi-compact. Conversely, let X be a topological space such that every open subset is quasi-compact. If $U_1\subseteq U_2\subseteq\cdots$ is an ascending chain of open subsets of X, then $U=\bigcup U_i$ is an open subset of X, so it is quasi-compact with open cover $\{U_i\}_{i=1}^{\infty}$. Hence, there exists a finite subcover $U_{i_1},\ldots,U_{i_n}$, so the ascending chain must terminate at $\max\{i_j\}$.

(b) Let $\{U_i\}_{i\in I}$ be an open cover of $\operatorname{Spec} A$. We can assume U_i to be basic open sets of the form $D(f_i)$ for some $f_i \in A$. If $\operatorname{Spec} A = \bigcup D(f_i)$, then $V((f_i)_{i\in I}) = \emptyset$, which implies the ideal generated by f_i is the entire ring A. That means there exists a finite number of elements $f_1, \dots f_n$ such that $\sum_{i=1}^n a_i f_i = 1$ for some $a_i \in A$, hence $D(f_1), \dots, D(f_n)$ is a finite subcover of $\operatorname{Spec} A$.

(c) The descending chain of irreducible closed subsets in $\operatorname{sp}(\operatorname{Spec} A)$ corresponds to an ascending chain of prime ideals in A. If A is noetherian any ascending chain of ideals must terminate, hence the corresponding descending chain of irreducible closed subsets must terminate.

(d) Let $A = \prod_{i=1}^{\infty} k$, where k is any field. The prime ideals of A are of the form $\mathfrak{p}_i = \cdots \times k \times 0 \times k \cdots$, so $\operatorname{Spec} A$ has countably infinite number of points, and closed subsets of $\operatorname{Spec} A$ are finite set of points. Hence, $\operatorname{Spec} A$ is a noetherian topological space. However, A itself is not noetherian, e.g. $0 \times 0 \times \cdots \subset k \times 0 \times 0 \times \cdots \subset k \times k \times 0 \times \cdots \subset \cdots$.

□

14. (a) Let S be a graded ring. Show that $\operatorname{Proj} S = \emptyset$ if and only if every element of S_+ is nilpotent.

(b) Let $\varphi : S \to T$ be a graded homomorphism of graded rings (preserving degrees). Let $U = \{\mathfrak{p} \in \operatorname{Proj} T \mid \mathfrak{p} \not\supseteq \varphi(S_+)\}$. Show that U is an open subset of $\operatorname{Proj} T$, and show that φ determines a natural morphism $f : U \to \operatorname{Proj} S$.

(c) The morphism f can be an isomorphism even when φ is not. For example, suppose that $\varphi_d : S_d \to T_d$ is an isomorphism for all $d \geq d_0$, where d_0 is an integer. Then show that $U = \operatorname{Proj} T$ and the morphism $f : \operatorname{Proj} T \to \operatorname{Proj} S$ is an isomorphism. We can define $f : U \to \operatorname{Proj} S$ to be $f(\mathfrak{p}) = \varphi^{-1}(\mathfrak{p})$.

(d) Let V be a projective variety with homogenous coordinate ring S. Show that $t(V) \cong \operatorname{Proj} S$.

Proof.

(a) By (Ex. 2.2), $Z((0)) = \operatorname{Proj} S = \emptyset \iff \sqrt{(0)} \supseteq S_+ \iff S_+$ is nilpotent.

(b) Notice that $U = \operatorname{Proj} T - V(\varphi(S_+))$, so U is indeed open. We can define $f : U \to \operatorname{Proj} S$ to be $f(\mathfrak{p}) = \varphi^{-1}(\mathfrak{p})$. Since φ does not contain $\varphi(S_+)$, $\varphi^{-1}(\mathfrak{p})$ does not contain S_+, so it is an element of $\operatorname{Proj} S$. Also, φ preserves degrees; in particular the preimage of a homogenous element is homogenous, so if $\mathfrak{p}$ is a homogenous prime ideal, then so is $f(\mathfrak{p})$.

(c) We first show $U = \operatorname{Proj} T$. Assume $d_0 > 0$ and suppose $\mathfrak{p}$ is a homogenous prime ideal in T and $\mathfrak{p} \supseteq \varphi(S_+)$. We want to show $\mathfrak{p} \supseteq T_+$. Towards that end, φ_d is an isomorphism for all $d \geq d_0$, which means $\bigoplus_{d\geq d_0} T_d \subseteq \varphi(S_+) \subseteq \mathfrak{p}$. Thus, for any $x \in T_+$, there exists a sufficiently large $n > 0$ such that the lowest degree term of x^n is greater than d_0, i.e., $x^n \in \bigoplus_{d\geq d_0} T_d \subseteq T_+$ for $n \mathfrak{g} 0$. Hence, $T_+ \subseteq \sqrt{\mathfrak{p}} = \mathfrak{p}$. By the same reasoning as above, it is bijective.

(d) A point in $t(V)$ corresponds to an irreducible closed subset of V, which corresponds to a homogenous prime ideal of S, so $t(V)$ and $\operatorname{Proj} S$ are homeomorphic as topological spaces. Let $\alpha : V \to t(V)$ be the map in (2.6), so that $(t(V), \alpha_* \mathcal{O}_V)$ is sheaf of rings on $t(V)$. There is a one-to-one correspondence between the open subsets of V and open subsets of $t(V)$, hence $t(V)$ and $\operatorname{Proj} S$ are isomorphic at the level of stalks.

□

15. (a) Let V be a variety over an algebraically closed field k. Show that a point $P \in t(V)$ is a closed point if and only if its residue field is k.

(b) If $f : X \to Y$ is a morphism of schemes over k, and if $P \in X$ is a point with residue field k, then $f(P) \in Y$ also has residue field k.

(c) Now show that if V, W are any two varieties over k, then the natural map

$$\operatorname{Hom}_{\mathfrak{Var}}(V, W) \to \operatorname{Hom}_{\mathfrak{Sch}/k}(t(V), t(W))$$

is bijective.

Proof.

(a) $P \in t(V)$ is a closed point $\iff t(Y) = \{P\}$ for some closed subset $Y \subseteq V \iff Y$ is a closed point in $V \iff$ its residue field is k.

(b) We can reduce to the case when X and Y are affine over k by taking open affine neighborhoods of P and $f(P)$. Let $X = \operatorname{Spec} A$ and $Y = \operatorname{Spec} B$ where A, B are k-algebras. A morphism of affine schemes $f : X \to Y$ correspond to a ring homomorphism $\varphi : B \to A$. The point $P \in X$ corresponds to a prime ideal $\mathfrak{p}$ in A, and $f(P) \in Y$ corresponds to the prime ideal $\mathfrak{P} = \varphi^{-1}(\mathfrak{p})$ in B. If P has residue field k, then φ induces a map $B/\mathfrak{P} \to k$ since $A/\mathfrak{p} = k$, so the residue field of $f(P)$ is a subfield of k. Since the residue field of $f(P)$ is also a k-algebra and k is algebraically closed, $f(P)$ must have k as its residue field.

(c) Injectivity is easy by the fact that any two continuous maps $V \to W$ induces the same continuous map between $t(V) \to t(W)$, which defines the same morphism of sheaves. We show surjectivity. Let $f : t(V) \to t(W)$ be a morphism of schemes, and let α_U, α_V be the maps as in (2.6) for V, respectively. By part (a) and (b), the map $f\big|_{\operatorname{im} \alpha_U} : \operatorname{im} \alpha_U \to \operatorname{im} \alpha_V$ is the desired map such that $t(f\big|_{\operatorname{im} \alpha_U}) = f$.

□

16. Let X be a scheme, let $f \in \Gamma(X, \mathcal{O}_X)$, and define X_f to be the subset of points $x \in X$ such that the stalk f_x of f at x is not contained in the maximal ideal $\mathfrak{m}_x$ of the local ring $\mathcal{O}_x$.

(a) If $U = \operatorname{Spec} B$ is an open *affine* subscheme of X, and if $\bar{f} \in B = \Gamma(U, \mathcal{O}_X)$ is the restriction of f, show that $U \cap X_f = D(\bar{f})$. Conclude that X_f is an open subset of X.

(b) Assume that X is quasi-compact. Let $A = \Gamma(X, \mathcal{O}_X)$, and let $a \in A$ be an element whose restriction to X_f is 0. Show that for some $n > 0$, $f^n a = 0$.

(c) Now assume that X has a finite cover by open affines U_i such that each intersection $U_i \cap U_j$ is quasi-compact. Let $b \in \Gamma(X_f, \mathcal{O}_X)$. Show that for some $n > 0$, $f^n b$ is the restriction of an element of A.

(d) With the hypothesis of (c), conclude that $\Gamma(X_f, \mathcal{O}_X) = A_f$.

Proof.

(a) The set $D(\bar{f})$ can be identified as the set of all prime ideals of B that contain $\bar{f}$. For each $\mathfrak{p} \in U$, the stalk of $\mathcal{O}_X$ at $\mathfrak{p}$ is the local ring $B_{\mathfrak{p}}$, hence $U \cap X_f \subseteq D(\bar{f})$. Conversely, the image of $\bar{f}$ in $B_{\mathfrak{p}}$ is precisely the stalk f_x of f at x, so $\bar{f} \in \mathfrak{p}$ implies $f_x \in \mathfrak{p}B_{\mathfrak{p}}$, where x and $\mathfrak{p}$ correspond to the same point in X, and since $\mathfrak{m}_x = \mathfrak{p}B_{\mathfrak{p}}$, we have inclusion in the other direction. We can cover X_f by affine open subschemes so that X_f is an open subset of an open subset, which implies X_f is open.

(b) First assume X is affine so that $X = \operatorname{Spec} A$. If $a \in A$ restricts to 0 in X_f, then since $X_f \simeq \operatorname{Spec} A_f$ and $\Gamma(X_f, \mathcal{O}_X) = A_f$, which means the image of a in A_f is 0, which is true if and only if $f^n a = 0$ for some $n > 0$. The general case (with the quasi-compactness assumption) follows from the fact that we can let $X = \bigcup_{i=1}^m \operatorname{Spec} A_i$, and for each i, we have $f^{n_i} a = 0 \in \Gamma(\operatorname{Spec} A_i, \mathcal{O}_X)$ for some $n_i > 0$, and we can take $n = \max\{n_i\}$.

(c) Let $X = \bigcup_{i=1}^n U_i$, where $U_i = \operatorname{Spec} A_i$, and let $X_{f,i} = X_f \cap U_i$. Assume $X_{f,i} \neq \emptyset$. We first show that for some $n_i > 0$, $f^{n_i} b_i$ is the restriction of an element of $\Gamma(U_i, \mathcal{O}_X) = A_i$, where b_i is the image of b in $\Gamma(X_{f,i}, \mathcal{O}_X)$. It is not hard to see that $\Gamma(X_{f,i}, \mathcal{O}_X) = (A_i)_f$, so $b_i = b_i'/f^{n_i}$ for some $b_i' \in A_i$ and $n_i > 0$ (the case is trivial when $n_i = 0$), thus $f^{n_i} b_i$ is an element of A_i. Let $n = \max\{n_i\}$, then $f^n b_i$ is an element of A_i for all i, and for all $i \neq j$, $f^n b_i$ and $f^n b_j$ restrict to the same element on $U_i \cap U_j$, so by part (b), there exists $n_{i,j} > 0$ such that $f^{n_{i,j}}(f^n b_i - f^n b_j) = 0$. Again, let $m = \max\{n_{i,j}\}$, then $f^{m+n} b$ is the restriction of an element of A.

(d) The element f is clearly invertible in $\Gamma(X_f, \mathcal{O}_X)$, so by part (c), every element of $\Gamma(X_f, \mathcal{O}_X)$ is of the form a/f^n for some $a \in A$ and $n \geq 0$.

□

17. *A Criterion for Affineness.*

(a) Let $f : X \to Y$ be a morphism of schemes, and suppose that Y can be covered by open subsets U_i, such that for each i, the induced map $f^{-1}(U_i) \to U_i$ is an isomorphism. Then f is an isomorphism.

(b) A scheme X is affine if and only if there is a finite set of elements $f_1, \ldots, f_r \in A = \Gamma(X, \mathcal{O}_X)$, such that the open subsets X_{f_i} are affine, and $f_1, \ldots, f_r$ generate the unit ideal in A.

Proof.

(a) Localize.

(b) One direction is obvious. Let X be a scheme satisfying the hypothesis above. If X_{f_i} are affine, then by Exercise 2.6 and 2.16d, $X_{f_i} \simeq \operatorname{Spec} A_{f_i}$. Also, $X = \bigcup X_{f_i}$, since if there exists $P \in X$ such that $f_i \in \mathfrak{m}_P$ for all $i = 1, \ldots, r$, then $1 \in \mathfrak{m}_P$, which is not possible. Hence, by (a), X and $\operatorname{Spec} A$ are isomorphic.

□

19. Let A be a ring. Show that the following condition are equivalent:

(i) $\operatorname{Spec} A$ is disconnected;

(ii) there exist nonzero elements $e_1, e_2 \in A$ such that $e_1 e_2 = 0$, $e_1^2 = e_1$, $e_2^2 = e_2$, $e_1 + e_2 = 1$ (these elements are called *orthogonal idempotents*);

(iii) A is isomorphic to a direct product $A_1 \times A_2$ of two nonzero rings.

Proof. (i) $\implies$ (iii) Write $\operatorname{Spec} A = \operatorname{Spec} A/\mathfrak{a}_1 \amalg \operatorname{Spec} A/\mathfrak{a}_2$ where $\mathfrak{a}_i$ are the ideals corresponding to the open and closed sets $U_i = \operatorname{Spec} A/\mathfrak{a}_i$ for $i = 1, 2$. We have

$$V(\mathfrak{a}_1) \cup V(\mathfrak{a}_2) = V(\mathfrak{a}_1 \mathfrak{a}_2) = V((0)) = \operatorname{Spec} A$$

and

$$V(\mathfrak{a}_1) \cap V(\mathfrak{a}_2) = V(\mathfrak{a}_1 + \mathfrak{a}_2) = V((1)) = \emptyset.$$

By (A, M Prop 1.10), $A \simeq A/\mathfrak{a}_1 \times A/\mathfrak{a}_2$.

(iii) $\implies$ (ii) $e_1 = (1, 0), e_2 = (0, 1)$

(ii) $\implies$ (i) $\operatorname{Spec} A = V((e_1)) \amalg V((e_2))$ because any prime ideal of A must contain either e_i since $e_1 e_2 = 0$, but cannot contain both. □

Chapter 2, Section 3

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1. Show that a morphism $f : X \to Y$ is locally of finite type if and only if for every open affine subset $V = \operatorname{Spec} B$ of Y, $f^{-1}(V)$ can be covered by open affine subsets $U_j = \operatorname{Spec} A_j$, where each A_j is a finitely generated B-algebra.

Proof. We reduce to proving the following statement: let $f : X \to Y$ be a morphism of schemes with $Y = \operatorname{Spec} B$, which can be covered by open affine subsets $V_i = \operatorname{Spec} B_i$, such that for each i, $f^{-1}(V_i)$ can be covered by open affine subsets $U_{ij} = \operatorname{Spec} A_{ij}$, where each A_{ij} is a finitely generated B_i-algebra. Then X can be covered by open affine subsets $U_k = \operatorname{Spec} A_k$, where each A_k is a finitely generated B-algebra.

We show each B_i can be chosen to be a finitely generated B-algebra, then by restriction of scalars each A_{ij} is a finitely generated B-algebra, which proves the statement. For each i, there exists $g_i \in B$ such that $D(g_i) \subseteq V_i$, where $D(g_i) \cong \operatorname{Spec} B_{g_i}$. Then $f^{-1}(D(g_i)) \subseteq f^{-1}(V_i)$, so we can cover $f^{-1}(D(g_i))$ by open affines of the form $\operatorname{Spec}((A_{ij})_{h_k})$ with $h_k \in A_{ij}$. Since each $(A_{ij})_{h_k}$ is a finitely generated A_{ij}-algebra, and each $(A_{ij})_{h_k}$ is a finitely generated $B_{g_i} \cong (B_i)_{\overline{g}_i}$, where each $\overline{g}_i$ is the image of g_i in B_i, what we have shown is each $f^{-1}(D(g_i))$ can be covered by open affine subsets $U'_{ij} = \operatorname{Spec} A'_{ij}$, where each A'_{ij} is a finitely generated B_{g_i}-algebra. Each B_{g_i} is a finitely generated B-algebra, generated by 1 and $1/g_i$, which is what we wanted to show. □

2. A morphism $f : X \to Y$ of schemes is *quasi-compact* if there is a cover of Y by open affines V_i such that $f^{-1}(V_i)$ is quasi-compact for each i. Show that f is quasi-compact if and only if for *every* open affine subset $V \subseteq Y$, $f^{-1}(V)$ is quasi-compact.

Proof. We reduce to proving the following statement: let $f : X \to Y$ be a morphism of schemes with Y affine, which can be covered by open subsets V_i such that $f^{-1}(V_i)$ is quasi-compact in X. Then X is quasi-compact. Indeed, Y is quasi-compact, so a finite number of i will do, and a finite union of quasi-compact sets is quasi-compact, and $X = f^{-1}(Y) = f^{-1}(\bigcup_{i=1}^{n} V_i) = \bigcup_{i=1}^{n} f^{-1}(V_i)$, hence X is quasi-compact. □

3. (a) Show that a morphism $f : X \to Y$ is of finite type if and only if it is locally of finite type and quasi-compact.

(b) Conclude from this that f is of finite type if and only if for *every* open affine subset $V = \operatorname{Spec} B$ of Y, $f^{-1}(V)$ can be covered by a finite number of open affines $U_j = \operatorname{Spec} A_j$, where each A_j is a finitely generated B-algebra.

(c) Show also if f is of finite type, then for *every* open affine subset $V = \operatorname{Spec} B \subseteq Y$, and for *every* open affine subset $U = \operatorname{Spec} A \subseteq f^{-1}(V)$, A is a finitely generated B-algebra.

Proof.

(a) Suppose $f : X \to Y$ is of finite type. Then by definition it is also locally of finite type, so there exists a covering of Y by open affine subsets $V_i = \operatorname{Spec} B_i$, such that each i, $f^{-1}(V_i)$ can be covered by a finite number of open affine subsets $U_{ij} = \operatorname{Spec} A_{ij}$, where each A_{ij} is a finitely generated B_i-algebra. Each $f^{-1}(V_i)$ is quasi-compact since every affine scheme is quasi-compact and $f^{-1}(V_i)$ is a finite union of quasi-compact sets. Conversely, if $f : X \to Y$ is locally of finite type and quasi-compact, then each $f^{-1}(V_i)$ is quasi-compact, so it can be covered by a finite number of open affines $U_{ij} = \operatorname{Spec} A_{ij}$, where each A_{ij} is a finitely generated B_i-algebra.

(b) A morphism $f : X \to Y$ is of finite type if and only if it is locally of finite type and quasi-compact if and only if for every open affine subset $V = \operatorname{Spec} B$ of Y, $f^{-1}(V)$ is quasi-compact and can be covered by open affines $U_j = \operatorname{Spec} A_j$, where each A_j is a finitely generated B-algebra if and only if for every open affine subset $V = \operatorname{Spec} B$ of Y, $f^{-1}(V)$ can be covered by a finite number of open affines $U_j = \operatorname{Spec} A_j$, where each A_j is a finitely generated B-algebra.

(c) By restricting the domain of f, it suffices to prove the following: if X is a scheme over a ring B that can be covered by a finite number of open subsets that are the spectra of finitely generated B-algebras, then for every open affine $U = \operatorname{Spec} A \subseteq X$, A is a finitely generated B-algebra. By Nike's trick, we can cover U by the distinguished basic sets $\operatorname{Spec} A_f$, where each A_f is a finitely generated B-algebra. Since U is quasi-compact, a finite number will do.

So now, we have reduced to the following purely algebraic problem: A is a B-algebra, $f_1, \ldots, f_r$ are a finite number of elements of A, which generate the unit ideal, and each localization A_{f_i} is a finitely generated B-algebra. Then A is finitely generated over B. Write $\sum_{i=1}^{r} g_i f_i$. Let S be the union of $\{g_1, \ldots, g_r, f_1, \ldots, f_r\}$ and a finite subset of A such that the image of it under the natural map $A \to A_{f_i}$ generates A_{f_i} as a B-algebra for all i, and let $B[S]$ be the B-subalgebra of A generated by S. Localization is left-exact, so $B[S]_{f_i} \to A_{f_i}$ is injective for all i, and it is surjective as well by construction of S. Thus, the inclusion $B[S] \to A$ viewed as a $B[S]$-module homomorphism is an isomorphism by the following statement from commutative algebra: let R be any ring, let M by any R-module, and let $h_1, \ldots, h_n$ be elements of R that generate the unit ideal. If $M_{h_i} = 0$ for all i, then $M = 0$. Indeed, if $\mathfrak{p}$ is any prime ideal of R, then $h_i \notin \mathfrak{p}$ for some i. Denote $\bar{h}_i$ the image of h_i in $A_{\mathfrak{p}}$. Then, $M_{\mathfrak{p}} = (M_{\mathfrak{p}})_{\bar{h}_i} = (M_{h_i})_{\mathfrak{p}R_{h_i}} = 0$ (A.M. §3).

□

Theorem 0.1 (Nike's Trick)**.** *Let X be a scheme, and let $U_i = \operatorname{Spec} A_i$, $i = 1, 2$, be open affine subsets of X. Then there is an open cover of $U_1 \cap U_2$ consisting of open sets that are distinguished basic open sets in both U_i.*

Lemma 0.2. *If $(f, f^{\#}) : (X, \mathcal{O}_X) \to (Y, \mathcal{O}_Y)$ is a morphism of locally ringed spaces, then $f^{-1}(Y_g) = X_{f^{\#}(g)}$ for any $g \in \Gamma(Y, \mathcal{O}_Y)$.*

Proof. Pick $g \in A_1$ such that $\operatorname{Spec}(A_1)_g = (U_1)_g \subseteq U_1 \cap U_2$, and set $U_1 = (U_1)_g$, so we have an open immersion $\iota : U_1 \hookrightarrow U_2$, which induces a ring homomorphism $\varphi : A_2 \to (A_1)_g$. Pick $h \in A_2$ such that $(U_2)_h \subseteq U_1$. By lemma 1, $\iota^{-1}((U_2)_h) = (U_1)_{\varphi(h)}$, and since ι is an open immersion, $(U_1)_{\varphi(h)} \cong (U_2)_h$ (we can assume $\varphi(h) \in A$). □

4. Show that a morphism $f : X \to Y$ is finite if and only if for *every* open affine subset $V = \operatorname{Spec} B$ of Y, $f^{-1}(V)$ is affine, equal to $\operatorname{Spec} A$, where A is a finite B-module.

Proof. Let $f : X \to Y$ be a finite morphism of schemes. Then there exists an open covering of Y by sets $V_i = \operatorname{Spec} B_i$, such that for each i, $f^{-1}(V_i)$ is affine, equal to $\operatorname{Spec} A_i$, where A_i is a finite B_i-module. Let $V = \operatorname{Spec} B$ be an open subset of Y. By Nike's trick, we can cover V by open affines that are distinguished open set in V and some V_i, i.e., open sets of the form $\operatorname{Spec} B_g = \operatorname{Spec}(B_i)_h$ for some $g \in B, h \in B_i$. Since V is quasi-compact, a finite number will do. By lemma 1, $f^{-1}(\operatorname{Spec}(B_i)_h) = \operatorname{Spec}(A_i)_h \subseteq f^{-1}(V)$. Since $(A_i)_h$ is a finite $(B_i)_h$-module and $B_g \cong (B_i)_h$, we deduce that there exists a finite cover of V by basic open sets $\operatorname{Spec} B_{g_j}$ for some $g_j \in B$ such that $f^{-1}(\operatorname{Spec} B_{g_j}) = \operatorname{Spec} C_j$, where each C_j is a finite B_{g_j}-module. We show $X' = f^{-1}(V)$ is affine using the criterion in (Ex. 2.17b). The restriction of f to $X' \to \operatorname{Spec} B$ induces a ring homomorphism $B \to A = \Gamma(X', \mathcal{O}_X)$. Denote $\bar{g}_i$ the image of g_i in A. Since g_i generate the unit ideal in B, its image in A also generate the unit ideal. Also, by lemma 1 $X'_{\bar{g}_j} = f^{-1}(\operatorname{Spec} B_{g_j}) = \operatorname{Spec} C_j$, hence $X' = \operatorname{Spec} A$. It remains to show A is a finite B-module, which we reduce to the following algebraic problem: let A be a B-algebra, let $g_i \in B$ $(1 \leq i \leq n)$ generate the unit ideal, and suppose A_{g_i} is a finite B_{g_i}-module. Then A is a finite B-module. The proof is identical to (Ex. 3.3c), so we omit this part. □

6. Let X be an integral scheme. Show that the local ring $\mathcal{O}_\xi$ of the generic point ξ of X is a field. It is called the *function field* of X, and is denoted by $K(X)$. Show also that if $U = \operatorname{Spec} A$ is any open affine subset of X, then $K(X)$ is isomorphic to the quotient field of A.

Proof. Any nonempty open set U of X must contain ξ since $X - U$ is a proper closed subset of X. In particular, any element of $\mathcal{O}_\xi$ can be represented as a pair $(\operatorname{Spec} A, f)$ where $\operatorname{Spec} A$ is an open affine set in X and $f \in A$. We further assume $\operatorname{Spec} A$ is connected, so A is an integral domain. If f is a nonzero element of A, then $(\operatorname{Spec} A, f)$ is a nonzero element of $\mathcal{O}_\xi$ with inverse $(\operatorname{Spec} A, f)^{-1} = (\operatorname{Spec} A_f, 1/f)$, hence $\mathcal{O}_\xi$ is a field. Lastly, the distinguished open sets $\operatorname{Spec} A_f$ form a neighborhood basis of ξ, so any element of $\mathcal{O}_\xi$ can be written as $(\operatorname{Spec} A_f, a/f^n)$, which is the quotient field of A. □

8. *Normalization.* A scheme is *normal* if all of its local rings are integrally closed domains. Let X be an integral scheme. For each open affine subset $U = \operatorname{Spec} A$ of X, let $\tilde{A}$ be the integral closure of A in its quotient field, and let $\tilde{U} = \operatorname{Spec} \tilde{A}$. Show that one can glue the schemes $\tilde{U}$ to obtain a normal integral scheme $\tilde{X}$, called the *normalization* of X. Show also that there is a morphism $\tilde{X} \to X$, having the following universal property: for every normal integral scheme Z, and for every dominant morphisms $f : Z \to X$, f factors uniquely through $\tilde{X}$. If X is of finite type over a field k, then the morphism $\tilde{X} \to X$ is a finite morphism.

Lemma 0.3. *If $f : Z \to X$ is a dominant morphism of schemes with Z reduced, then $f^\# : \mathcal{O}_X \to f_*\mathcal{O}_Z$ is injective.*

Proof. Let U be any open subset of X. We want to show if $g \in \Gamma(U, \mathcal{O}_X)$ such that $f^\#(g) = 0 \in \Gamma(f^{-1}(U), \mathcal{O}_Z)$, then $g = 0$. By lemma 1, we have $f^{-1}(U_g) = (f^{-1}(U))_{f^\#(g)}$, and if $f^\#(g) = 0$, then $(f^{-1}(U))_{f^\#(g)} = \emptyset$, so U_g must not meet the $f(Z)$. But $f(Z)$ is dense in X, and U_g is an open set in X by (Ex. 2.16a), so $U_g = \emptyset$, which implies g is nilpotent (this result corresponds to the algebraic fact that the intersection of all prime ideals of a ring is the nilradical of the ring), hence $g = 0$. □

Proof. Let $X = \bigcup \operatorname{Spec} A_i$ be an open affine covering of X where each A_i is an integral domain. For each $i \neq j$, we have an identification $\varphi_{ij} : U_i \to U_j$, which is an isomorphism of open subschemes $U_i \subseteq \operatorname{Spec} A_i, U_j \subseteq \operatorname{Spec} A_j$. By Nike's trick, there exists an open covering of U_i by basic open sets $\operatorname{Spec}(A_i)_{f_k}$ with $f_k \in A_i$ such that $\varphi_{ij}(\operatorname{Spec}(A_i)_{f_k}) = \operatorname{Spec}(A_j)_{g_k}$ with $g_k \in A_j$. For each i, let $\tilde{A}_i$ be the integral closure of A_i in its quotient field (note by (Ex. 6), every A_i has the same quotient field), and let $\pi_i : \operatorname{Spec} \tilde{A}_i \to \operatorname{Spec} A_i$ be the morphism induced by the inclusion $A_i \hookrightarrow \tilde{A}_i$. We have $\pi_i^{-1}(\operatorname{Spec}(A_i)_{f_k}) = \operatorname{Spec}(\tilde{A}_i)_{f_k}$ and $(\tilde{A}_i)_{f_k} \cong (\tilde{A}_j)_{g_k}$, so we can naturally glue open subsets of X' that are of the form $\pi_i^{-1}(U_i)$ using φ_{ij} to obtain $\tilde{X}$. The morphism $\pi : \tilde{X} \to X$ is obtained by glueing π_i accordingly.

Now suppose Z is a normal integral scheme, and let $f : Z \to X$ be a dominant morphism of schemes. It is clear we can assume $X = \operatorname{Spec} A$, where A is an integral domain. Then f induces a ring homomorphism $\varphi : A \to B = \Gamma(Z, \mathcal{O}_Z)$. Let $\tilde{X} = \operatorname{Spec} \tilde{A}$ be the normalization of X with associated morphism $\pi : \tilde{X} \to X$ induced by the inclusion homomorphism $\iota : A \to \tilde{A}$, where $\tilde{A}$ is the integral closure of A in its quotient field. We want to show there exists a unique morphism $\tilde{f} : Z \to \tilde{X}$ such that $f = \pi \circ \tilde{f}$. Being integrally closed is a local property (A.M. 5.13), so B is integrally closed (it is automatically an integral domain by definition of an integral scheme). Also, φ is injective by lemma 2. Since $\tilde{X}$ is affine, by the bijection in (Ex. 24), we have reduced to proving the following universal property for the integral closure of a domain $A \xrightarrow{\iota} \tilde{A}$: for any injective homomorphism $\varphi : A \to B$ where B is an integrally closed domain, there exists a unique homomorphism $\psi : \tilde{A} \to B$ such that $\varphi = \psi \circ \iota$. Any injective homomorphism between integral domains induces an inclusion of fraction fields, so let $\Phi : \operatorname{Frac}(A) \to \operatorname{Frac}(B)$ be induced by φ, where $\Phi\big|_A = \varphi$. Note that we have inclusions $A \subseteq \tilde{A} \subseteq \operatorname{Frac}(A)$, thus we claim $\psi = \Phi\big|_{\tilde{A}} : \tilde{A} \to \operatorname{Frac}(B)$ is the desired ring homomorphism. It suffices to show the image of ψ is contained in B. If $f \in \tilde{A}$, then there exists an equation of integral dependence $f^n + a_1 f^{n-1} + \cdots + a_n = 0$ where $a_i \in A$. Then $\Phi(f)$ has an equation of integral dependence $\Phi(f)^n + \varphi(a_1)\Phi(f)^{n-1} + \cdots + \varphi(a_n) = 0$, and since B is integrally closed, $\Phi(f)$ must be an element of B. Since any other $\psi' : \tilde{A} \to B$ such that $\varphi = \psi' \circ \iota$ must agree with Φ on $\tilde{A}$, ψ is unique by construction. □

10. *Fibers of a Morphism.*

(a) If $f : X \to Y$ is a morphism, and $y \in Y$ a point, show that $\operatorname{sp}(X_y)$ is homeomorphic to $f^{-1}(y)$ with the induced topology.

(b) Let $X = \operatorname{Spec} k[s,t]/(s - t^2)$, let $Y = \operatorname{Spec} k[s]$, and let $f : X \to Y$ be the morphism defined by sending $s \to s$. If $y \in Y$ is the point $a \in k$ with $a \neq 0$, show that the fiber X_y, consists of two points, with residue

field k. If $y \in Y$ corresponds to $0 \in k$, show that the fiber X_y is a nonreduced one-point scheme. If η is the generic point of Y, show that X_η is a one-point scheme, whose residue field is an extension of degree two of the residue field of η. (Assume k is algebraically closed.)

Proof.

(a) We can assume Y to be affine by taking any open affine neighborhood of $y \in Y$ and restricting f to its preimage in X. Also, if $U_i = \operatorname{Spec} A_i$ is an open affine cover of X, then $f^{-1}(y) = \bigcup_i (\operatorname{Spec} A_i \cap f^{-1}(y))$, and $\operatorname{sp}(X_y) = \bigcup_i \operatorname{sp}(U_i \times_Y k(y))$, so we can also assume X to be affine. Let $X = \operatorname{Spec} A$ and $Y = \operatorname{Spec} B$, then f induces a ring homomorphism $f^\# : B \to A$. A point $y \in Y$ corresponds to a prime ideal $\mathfrak{q}$ in B, where $k(y) = B_\mathfrak{q}/\mathfrak{q}B_\mathfrak{q}$. Thus, $f^{-1}(y)$ is the set of all prime ideals $\mathfrak{p}$ in A such that $f^{\#-1}(\mathfrak{p}) = \mathfrak{q}$. Next, we look at $\operatorname{sp}(X_y)$. Notice that $X_y = \operatorname{Spec} A \otimes_B k(y)$, and $A \otimes_B k(y) = A \otimes_B B_\mathfrak{q}/\mathfrak{q}B_\mathfrak{q} = A_\mathfrak{q}/\mathfrak{q}A_\mathfrak{q}$. The prime ideals of $A_\mathfrak{q}/\mathfrak{q}A_\mathfrak{q}$ correspond to prime ideals of A that contain the image of $\mathfrak{q}$ and does not contain meet the image of $B - \mathfrak{q}$, which are precisely the prime ideals of A such that $f^{\#-1}(\mathfrak{p}) = \mathfrak{q}$.

(b) $X_y = \operatorname{Spec} k[t]/(a - t^2)$, $X_\eta = \operatorname{Spec} k(s)[\sqrt{s}]$. Any element of $k(s)[\sqrt{s}]$ can be written as $F + G\sqrt{s}$, where $F, G \in k(s)$. It is a field, since $(F + G\sqrt{s})^{-1} = (F - G\sqrt{s})/(F^2 - sG^2)$.

□

11. *Closed Subschemes.*

(a) Closed immersions are stable under base extension: if $f : Y \to X$ is a closed immersion, and if $X' \to X$ is any morphism, then $f' : Y \times_X X' \to X'$ is also a closed immersion.

(b) If Y is a closed subscheme of an affine scheme $X = \operatorname{Spec} A$, then Y is also affine, and in fact Y is the closed subscheme determined by a suitable ideal $\mathfrak{a} \subseteq A$ as the image of the closed immersion $\operatorname{Spec} A/\mathfrak{a} \to \operatorname{Spec} A$.

(c) Let Y be a closed subset of a scheme X, and give Y the reduced induced subscheme structure. If Y' is any other closed subscheme of X with the same underlying topological space, show that the closed immersion $Y \to X$ factors through Y'. We express this property by saying that the reduced induced structure is the smallest subscheme structure on a closed subset.

(d) Let $f : Z \to X$ be a morphism. Then there is a unique closed subscheme Y of X with the following property: the morphism f factors through Y, and if Y' is any other closed subscheme of X through which f factors, then $Y \to X$ factors through Y' also. We call Y the *scheme-theoretic image* of f. If Z is a reduced scheme, then Y is just the reduced induced structure on the closure of the image $f(Z)$.

Proof.

(a) Consider the special case when $X = \operatorname{Spec} A, Y = \operatorname{Spec} A/\mathfrak{a}$, and $X' = \operatorname{Spec} B$ where $\mathfrak{a}$ is an ideal of A and B is any A-algebra. The natural map $A \to A/\mathfrak{a}$ induces a closed immersion $Y \to X$, and the structure homomorphism $A \to B$ induces a morphism of schemes $X' \to X$. The fiber product $Y \times_X X'$ is equal to the spectra of the tensor product $A/\mathfrak{a} \otimes_A B$, which is isomorphic to $B/\mathfrak{a}B$. The induced structure morphism $Y \times_X X' \to X'$ corresponds to the canonical homomorphism $B \to B/\mathfrak{a}B$, thus $Y \times_X X' \to X'$ is a closed immersion. In other words, this property of closed immersions corresponds to the algebraic fact that the tensor operation is a right exact functor. In the general case, we can still assume X to be affine by taking an open affine cover of X. Then Y is an affine scheme by part (b), and if U is any open subset of X', $f^{'-1}(U) = Y \times_X U$, so by taking an open affine cover of X', we can reduce to the case when X' is affine, which is just the special case as in above.

(b) Let Y be a closed subscheme of an affine scheme $X = \operatorname{Spec} A$, and let $\varphi : A \to B = \Gamma(Y, \mathcal{O}_Y)$ be the induced ring homomorphism of global sections. Fix $y \in Y$, let $V = \operatorname{Spec} C$ be an open affine neighborhood of y as a subspace of Y, and let $\rho : B \to C$ be the restriction homomorphism. By definition of the subspace topology, there exists an open set U of X such that $V = U \cap Y$. We can cover U by distinguished open sets, and at least one of them must contain y, so let X_f be such set with $f \in A$. By lemma 1, $X_f \cap Y = Y_{\varphi(f)} = \operatorname{Spec} C_{\rho(\varphi(f))}$. Thus, we can cover Y by open affines of the form $X_{f_i} \cap Y$. Since Y is homeomorphic to a closed subset of a quasi-compact set, it is also quasi-compact, so a finite number will do. By adding some more f_i with $D(f_i) \cap Y = \emptyset$ by taking an open cover of $X - Y$, we assume X_{f_i} cover X. Such collection of X_{f_i} cover X if and only if f_i generate the unit ideal in A, so $\varphi(f_i)$ must generate the unit ideal of B. Hence, Y is affine by (Ex. 2.17b). The quotient ring $A/\ker\varphi$ is a subring of B, so the affine scheme $X' = \operatorname{Spec} A/\ker\varphi$ contains Y as a dense subset. X' is also homeomorphic to a closed subset of X, hence $Y = \operatorname{Spec} A/\ker\varphi$.

(c) By taking an affine cover of X, we reduce to the case when $X = \operatorname{Spec} A$ is affine. Let $\mathfrak{a}$ be the ideal in A that corresponds to the reduced induced structured of Y. By part (b), there exists an ideal $\mathfrak{b}$ in A such that $Y' = \operatorname{Spec} A/\mathfrak{b}$. A morphism of schemes $Y \to Y'$ corresponds to a ring homomorphism $A/\mathfrak{b} \to A/\mathfrak{a}$. Recall that $\mathfrak{a}$ is the largest ideal of A such that $V(\mathfrak{a}) = \operatorname{sp}(Y) = \operatorname{sp}(Y')$; in particular, $\mathfrak{b} \subseteq \mathfrak{a}$, so the natural projection map $A/\mathfrak{b} \to A/\mathfrak{a}$ is well-defined and is the desired ring homomorphism.

(d) Again, reduce to the affine case. A morphism of affine schemes $f : \operatorname{Spec} B \to \operatorname{Spec} A$ correspond to a ring homomorphism $\varphi : A \to B$. Then $Y = \operatorname{Spec} A/\ker\varphi$ is the desired closed subscheme. If Z is reduced, then B is a reduced ring. Then $\ker\varphi$ contains the nilradical of A, so $A/\ker\varphi$ is reduced, hence the intersection of all prime ideals of $A/\ker\varphi$ is the zero ideal.

□

13. *Properties of Morphisms of Finite Type.*

(a) A closed immersion is a morphism of finite type.

(b) A quasi-compact open immersion is of finite type.

(c) A composition of two morphisms of finite type is of finite type.

(d) Morphisms of finite type are stable under base extension.

(e) If X and Y are schemes of finite type over S, then $X \times_S Y$ is of finite type over S.

(f) If $X \xrightarrow{f} Y \xrightarrow{g} Z$ are two morphisms, and if f is quasi-compact, and $g \circ f$ is of finite type, then f is of finite type.

(g) If $f : X \to Y$ is a morphism of finite type, and if Y is noetherian, then X is noetherian.

Proof.

(a) Let $f : Y \to X$ be a closed immersion. By abuse of notation, also denote Y as a closed subset of $\operatorname{sp}(X)$. If U is quasi-compact in X, then $f^{-1}(U) = U \cap Y$, which is quasi-compact since any closed subset of a quasi-compact set is also quasi-compact, so f is a quasi-compact morphism. Also, if $U = \operatorname{Spec} A$ is an open affine subset of X, then $f^{-1}(U) = U \cap Y$ is a closed subscheme of U equal to the spectra of $A/\mathfrak{a}$ for some ideal $\mathfrak{a}$ of A by (Ex. 3.11b), which is a finitely generated A-algebra, so f is also locally of finite type. Hence, f is of finite type.

(b) Let $f : Y \to X$ be an open immersion. If $U = \operatorname{Spec} A$ is any open affine set in X, then $f^{-1}(U) = U \cap Y$ can be covered by distinguished open sets of U, which are spectra of A_f for some $f \in A$. Any such A_f is a finitely generated A-algebra. Hence, an open immersion is locally of finite type, so a quasi-compact open immersion is of finite type by (Ex. 3.3a).

(c) Let $f : X \to Y$ and $g : Y \to Z$ be morphisms of finite type. Let $W = \operatorname{Spec} C$ be an open affine subset of Z. Since g is of finite type, $g^{-1}(W)$ can be covered by a finite number of open affines $V_i = \operatorname{Spec} B_i$, where each B_i is a finitely generated C-algebra, and similarly each $f^{-1}(V_i)$ can be covered by a finite number of open affines $U_{ij} = \operatorname{Spec} A_{ij}$, where each A_{ij} is a finitely generated B_i-algebra. Each A_{ij} is also a finitely-generated A-algebra since B_i is a finitely generated A-algebra, and there are a finite number of U_{ij} with $(g \circ f)^{-1}(W) = f^{-1}(g^{-1}(W)) = \bigcup U_{ij}$, $g \circ f$ is a morphism of finite type.

(d) This corresponds to the algebraic fact that if A is a finitely generated B-algebra and C is any other B-algebra, then $A \otimes_A C$ is a finitely generated C-algebra.

(e) This corresponds to the algebraic fact that the tensor product of two finitely generated algebras is also finitely generated.

(f) By (Ex. 3.3a), it suffices to show f is locally of finite type. Let $W = \operatorname{Spec} C$ be an open affine subset of Z, and let $V_i = \operatorname{Spec} B_i$ be an open affine cover of $g^{-1}(W)$. By (Ex. 3.3c), for each i, we can cover $f^{-1}(V_i)$ by open affines $U_{ij} = \operatorname{Spec} A_{ij}$ where each A_{ij} is a finitely generated C-algebra. We have a composition of structure morphisms $C \to B_i \to A_{ij}$, where A_{ij} is finitely generated over the image of C in A_{ij}. Since the image of C in A_{ij} is contained in the image of B_i in A_{ij}, A_{ij} is also a finitely generated B_i-algebra.

(g) This corresponds to the fact that any finitely generated algebra over a noetherian ring is also noetherian as a ring.

□

16. *Noetherian Induction.* Let X be a noetherian topological space, and let $\mathscr{P}$ be a property of closed subsets of X. Assume that for any closed subset Y of X, if $\mathscr{P}$ holds for every proper closed subset of Y, then $\mathscr{P}$ holds for Y. (In particular, $\mathscr{P}$ must hold for the empty set.) Then $\mathscr{P}$ holds for X.

Proof. Suppose $\mathscr{P}$ does not hold for X, and set $X_1 = X$. Recursively define X_{n+1} to be any proper closed subset of X_n such that $\mathscr{P}$ does not hold. This is a nonterminating decreasing sequence of closed subsets of X by construction, a contradiction. □

20. *Dimension.* Let X be an integral scheme of finite type over a field k (not necessarily algebraically closed). Use appropriate results from (I,§1) to prove the following.

(a) For any closed point $P \in X$, $\dim X = \dim \mathcal{O}_P$, where for rings, we always mean the Krull dimension.

(b) Let $K(X)$ be the function field of X. Then $\dim X = \text{tr.d.}K(X)/k$.

(c) If Y is a closed subset of X, then $\operatorname{codim}(Y, X) = \inf\{\dim \mathcal{O}_{P,X} \mid P \in Y\}$.

(d) If Y is a closed subset of X, then $\dim Y + \operatorname{codim}(Y, X) = \dim X$.

(e) If U is a nonempty open subset of X, then $\dim U = \dim X$.

(f) If $k \subseteq k'$ is a field extension, then every irreducible component of $X' = X \times_k k'$ has dimension $= \dim X$.

Proof.

(a) Let V be a variety over k. The bijection between the open sets of V and open sets of $t(V)$ induced by the map α in the proof of (2.6) implies $\dim V = \dim t(V)$. If X is an integral scheme of finite type over a field k, then we can cover X by a finite number of open affines $U_i = \operatorname{Spec} A_i$, where each A_i is of the form $k[x_1, \ldots, x_{r_i}]/\mathfrak{p}_i$ for some $r_i > 0$ and prime ideal $\mathfrak{p}_i$ in $k[x_1, \ldots, x_{r_i}]$. Then $P \in U_i$ for some i, and if P is closed, then it corresponds to a maximal ideal $\mathfrak{m}_P$ in A_i, so by (A.M. 11.25) and (I, 1.7), $\dim \mathcal{O}_P = \dim A_i = \dim U_i$. Also, $\dim U_i = \dim X$ for all i by (I, Ex. 1.10b) and from the fact that $\dim U_i = \dim U_j$. Indeed, X itself is irreducible since it is an integral scheme and therefore has a unique generic point. Then $U_i \cap U_j \neq \emptyset$ for any i, j, so for all $P \in U_i \cap U_j$, $\dim U_i = \dim \mathcal{O}_{U_i,P} = \dim \mathcal{O}_{U_j,P} = \dim U_j$. Hence, $\dim X = \dim \mathcal{O}_P$ for all closed $P \in X$.

(b) By part (a), (II, Ex. 3.6), (I, 1.8A), $\text{tr. d. } K(X)/k = \dim U_i = \dim X$.

(c) Let $X = \operatorname{Spec} A$, where A is an integral domain that is finitely generated over a field k, and let Y be an irreducible, closed subset of X. The closed subscheme Y corresponds to a prime ideal $\mathfrak{p}$ in A, and any irreducible closed set containing Y corresponds to a prime ideal contained in $\mathfrak{p}$, so $\operatorname{codim} Y X$ equals to the height of $\mathfrak{p}$. Points in Y correspond to prime ideals containing $\mathfrak{p}$, so the infimum of the dimension of all local rings $\mathcal{O}_{P,X}$ over $P \in Y$ equals to the infimum of the height of all prime ideals of A which contain $\mathfrak{p}$, which is just the height of $\mathfrak{p}$. If Y is any closed subset, then Y corresponds to an ideal $\mathfrak{a}$ in A, and any irreducible closed set of Y contained in Y corresponds to a prime ideal of A that contains $\mathfrak{a}$. From the case when Y is irreducible, the infimum of the codimension of all irreducible closed sets contained in Y equals to the infimum of the height of prime ideals containing $\mathfrak{a}$. The height of such prime ideals is precisely the dimension of the local ring of points $P \in Y$. If X is any integral scheme of finite type over a field k, then we can cover X by a finite number of open affines $U_i = \operatorname{Spec} A_i$ where $\dim A_i = \dim A_j$. If $P \in U_i \cap U_j \cap Y$, then $\dim \mathcal{O}_{P,X} = \dim \mathcal{O}_{P,U_i} = \dim \mathcal{O}_{P,U_j}$; in particular, the prime ideals that P corresponds to in A_i, A_j have the same height. Thus, we can just reduce to the affine case.

(d) By the same reason as part (c), we can reduce to the affine case, which follows from (§1, 1.7) and (§1, 1.8Ab).

(e) By (§1, Ex. 1.10b), $\dim U = \sup \dim U \cap U_i = \dim U_i = \dim X$.

(f) If $U_i = \operatorname{Spec} A_i$ is an open affine cover of X, then $U_i' = U_i \times_k k' = \operatorname{Spec} A_i \otimes_k k'$ is an open affine cover of X, so by (§1, Ex. 1.10b) and part (a), we reduce to the case when $X = \operatorname{Spec} A$ is affine. Suppose $A = k[x_1, \ldots, x_n]/\mathfrak{p}$ for some prime ideal $\mathfrak{p}$. Then X' is the spectra of the ring $k'[x_1, \ldots, x_n]/\mathfrak{p}'$, where $\mathfrak{p}'$ is the extension of $\mathfrak{p}$ in $k'[x_1, \ldots, x_n]$. The irreducible components of X' correspond to minimal prime ideals of $\mathfrak{p}'$. Let $\mathfrak{q}$ be a minimal prime ideal of $\mathfrak{p}'$ so that $\mathfrak{p} = \mathfrak{q} \cap k[x_1, \ldots, x_n]$. We want to show $\dim k[x_1, \ldots, x_n]/\mathfrak{p} = \dim k'[x_1, \ldots, x_n]/\mathfrak{q}$. The case when k' is an algebraic extension of k is immediate from the going-up going-down theorems, so assume k and k' are algebraically closed. Let K, K' be the fraction fields of $k[x_1, \ldots, x_n]/\mathfrak{p}$, $k'[x_1, \ldots, x_n]/\mathfrak{q}$, then by part (b) or (§1, 1.8Aa) it suffices to show

$\mathrm{tr.d.}K/k = \mathrm{tr.d.}K'/k'$. Since k, k' are algebraically closed, K, K' are purely transcendental extensions, so we can write $K = k(y_1, \ldots, y_r), K' = k'(y'_1, \ldots, y'_{r'})$, where $r = \mathrm{tr.d.}K/k, r' = \mathrm{tr.d.}K'/k'$. By Noether's normalization lemma, we can choose y_i to be linear combinations of $x_1, \ldots, x_n$, so after a suitable k-algebra automorphism, we can assume $\mathfrak{p} = (x_1, \ldots, x_{n-r})$ so that $y_i = x_i$ for $i = n-r+1, \ldots, n$. Then the extension of $\mathfrak{p}$ in $k'[x_1, \ldots, x_n]$ is also a prime ideal generated by $x_1, \ldots, x_{n-r}$, which implies $r = r'$. Since $n = \dim k[x_1, \ldots, x_n] = \dim k'[x_1, \ldots, x_n]$, by (§1, 1.8Ab) we have the following corollary:

Corollary 0.4. *Let k'/k be any field extension. If $\mathfrak{p}$ is a prime ideal in $k[x_1, \ldots, x_n]$, then any minimal prime ideal of the extension of $\mathfrak{p}$ in $k'[x_1, \ldots, x_n]$ has the same height as $\mathfrak{p}$.*

□

Chapter 2, Section 4

James Lee

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1. Show that a finite morphism is proper.

Lemma 0.1. *If $\varphi : A \to B$ is an integral homomorphism of rings, then the corresponding morphism between affine schemes $\varphi^* : \operatorname{Spec} B \to \operatorname{Spec} A$ is a closed mapping.*

Proof. We can decompose φ into $A \to A/\ker\varphi \hookrightarrow B$, and $\operatorname{Spec}(A/\ker\varphi)$ is homeomorphic to a closed subspace of $\operatorname{Spec} A$, so we reduce to the case when φ is injective. Let $\mathfrak{b}$ be any ideal of B, then we want to show $\varphi^*(V(\mathfrak{b})) = V(\mathfrak{b} \cap A)$ (note that in general, we only have $\overline{\varphi(V(\mathfrak{b}))} = V(\varphi^{-1}(\mathfrak{b}))$). This follows from (A.M. 5.10), which states if $A \subseteq B$ are rings, B integral over A, and $\mathfrak{p}$ prime ideal of A, then there exists a prime ideal $\mathfrak{q}$ of B such that $\mathfrak{q} \cap A = \mathfrak{p}$. □

Lemma 0.2. *Let $f : B \to B'$ be a homomorphism of A-algebras, and let C be an A-algebra. If f is integral, prove that $f \otimes 1 : B \otimes_A C \to B' \otimes_A C$ is integral.*

Proof. It suffices to show all pure tensors $b' \otimes c$ in $B' \otimes_A C$ have an equation of integral dependence over $B \otimes_A C$. Since B' is an integral B-algebra, we have

$$b'^n + d_1 b'^{n-1} + \cdots + d_n = 0$$

for some $d_i \in B$, $n > 0$, then

$$(b' \otimes c)^n + (d_1 \otimes 1)(b' \otimes c)^{n-1} + \cdots + d_n \otimes c = 0.$$

□

Proof. It follows from these lemmas that if $f : A \to B$ is integral and C is any A-algebra, then the mapping $(f \otimes 1)^* : \operatorname{Spec}(B \otimes_A C) \to \operatorname{Spec} C$ is a closed map. Let $f : X \to Y$ be a finite morphism of schemes. A finite morphism is an affine morphism (Ex. 3.4), so by (4.6f) we reduce to the case when $X = \operatorname{Spec} B$ and $Y = \operatorname{Spec} A$, where B is a finite A-module (hence integral over A), and f is induced by a ring homomorphism $A \to B$. Morphisms between affine schemes are separated (4.1), and finite morphisms are of finite type, so it remains to show f is universally closed. If $Y' \to Y$ is any morphism, then we want to show $X \times_Y Y' \to Y'$ is a closed mapping. There is an open cover of Y' by spectra of A-algebras C_i so that the fiber product $X \times_Y Y'$ is covered by spectra of $B \otimes_A C_i$. By the above remarks, the morphisms $\operatorname{Spec}(B \otimes_A C_i) \to \operatorname{Spec} C_i$ are closed, hence $X \times_Y Y' \to Y'$ is closed. □

2. Let S be a scheme, let X be a reduced scheme over S, and let Y be a separated scheme over S. Let f and g be two S-morphisms of X to Y which agree on an open dense subset of X. Show that $f = g$. Give examples to show that this result fails if either (a) X is nonreduced, or (b) Y is nonseparated.

Proof. Let U be an open dense subset of X such that f and g agree on U, let $h : X \to Y \times_S Y$ be the map obtained from f and g, and let $\Delta : Y \to Y \times_S Y$ be the diagonal morphism. By hypothesis, $h(U) \subseteq \Delta(Y)$. But U is dense in X, and $\Delta(Y)$ is closed since Y is separated over S, so $h(X) \subseteq \Delta(Y)$. This says that f and g agree topologically, so $f_*\mathcal{O}_X = g_*\mathcal{O}_X$. Set $\mathcal{O} = f_*\mathcal{O}_X = g_*\mathcal{O}_X$. Let V be open subset of Y, and let $t \in \Gamma(V, \mathcal{O}_Y)$. We want to show $f^\#(t) = g^\#(t) \in \mathcal{O}(V)$. By hypothesis, we have $W = (f^{-1}(V))_{f^\#(t) - g^\#(t)} \subseteq X - U$, but W is an open subset and U is dense, which implies $W = \emptyset$. This implies $f^\#(t) - g^\#(t)$ is nilpotent, but X is reduced, hence $f^\#(t) - g^\#(t) = 0$.

(a) Let k be a field, let $X = \operatorname{Spec} k[\varepsilon]/(\varepsilon^2)$ (Ex. 2.8), and let Y be any scheme over k. Giving a k-morphism $X \to Y$ is equivalent to giving a point in $y \in Y$ rational over k, and an element of $\mathfrak{m}_y/\mathfrak{m}_y^2$.

(b) Let X be the affine line, and let Y be the affine line with the origin doubled. We have two possible open immersions of X into Y with each one having either origin in its image, and the open immersions agree on the complement of the origin of X, which is an open dense subset of X.

□

6. Let $f : X \to Y$ be a proper morphism of affine varieties over k. Then f is a finite morphism.

Proof. Let $X = \operatorname{Spec} B, Y = \operatorname{Spec} A$, where A and B are finitely generated k-algebras that are integral domains. Let $\varphi : A \to B$ be a k-algebra homomorphism such that B is a finitely generated A-algebra, and f is induced by φ. Closed immersions are proper, so we reduce to the case when φ is injective. We want to show B is a finite A-module, which is equivalent to B being finitely generated and integral over A, so it suffices to show B is integral over A. Let K be the field of fractions of B so that A and B are subrings of K. By (4.11A), the integral closure of A in K is the intersection of all valuation rings of K which contains A, so it suffices to show B is contained in all such subrings. This is an easy consequence of the valuative criterion of properness: given any valuation ring R containing A, we have inclusions $A \to R$ and $B \to K$ forming a commutative diagram

$$\begin{array}{ccc} A & \longrightarrow & R \\ \downarrow & \nearrow & \downarrow \\ B & \longrightarrow & K \end{array}$$

and the valuation criterion of properness implies there exists a unique homomorphism $B \to R$ making the whole diagram commute. All homomorphisms are inclusions, so $B \to R$ is an inclusion, which is what we wanted to show. (See an alternative proof of this result that uses the universally closed property instead of the valuative criterion in (A.M. Ex. 5.35).) □

8. Let $\mathscr{P}$ be a property of morphisms of schemes such that:

(a) a closed immersion has $\mathscr{P}$;

(b) a composition of two morphisms having $\mathscr{P}$ has $\mathscr{P}$;

(c) $\mathscr{P}$ is stable under base extension.

Then show that:

(d) a product of morphisms having $\mathscr{P}$ has $\mathscr{P}$;

(e) if $f : X \to Y$ and $g : Y \to Z$ are two morphisms, and if $g \circ f$ has $\mathscr{P}$ and g is separated, then f has $\mathscr{P}$.

(f) if $f : X \to Y$ has $\mathscr{P}$, then $f_{\text{red}} : X_{\text{red}} \to Y_{\text{red}}$ has $\mathscr{P}$.

Proof.

(d) Let $X \to Y$ and $X' \to Y'$ be two morphisms having $\mathscr{P}$. By (c) $X \times_Y (Y \times Y') \to Y \times Y'$ and $X' \times'_Y (Y \times Y') \to Y \times Y'$ have $\mathscr{P}$. Hence, $X \times X' = (X \times_Y (Y \times Y')) \times_{Y \times Y'} (X' \times_{Y'} (Y \times Y')) \to Y \times Y'$ has $\mathscr{P}$.

(e) We can base extend $g \circ f : X \to Z$ by $g : Y \to Z$ so that $h : X \times_Z Y \to Y$ has $\mathscr{P}$. Then f factors through h, so by (b) it suffices to show $\Gamma_f : X \to X \times_Z Y$ has $\mathscr{P}$. By hypothesis, the diagonal morphism $\Delta : Y \to Y \times_Z Y$ has $\mathscr{P}$. We can obtain Γ_f by base extension of Δ by $(f, 1_Y) : X \times_Z Y \to Y \times_Z Y$ since $(X \times_Z Y) \times_{Y \times_Z Y} Y \cong X \times_{Y \times_Z Y} (Y \times_Z Y) \cong X$. Hence, by (b) Γ_f has $\mathscr{P}$.

(f) By the universal property of the reduced scheme associated to Y, f_{red} is the unique morphism that makes the diagram

$$\begin{array}{ccccc} & & & & Y_{\text{red}} \\ & & \overset{f_{\text{red}}}{\nearrow} & & \downarrow \\ X_{\text{red}} & \longrightarrow & X & \xrightarrow[f]{} & Y \end{array}$$

commute. The associated morphisms $X_{\text{red}} \to X, Y_{\text{red}} \to Y$ are closed immersions; in particular, $X_{\text{red}} \to Y$ has $\mathscr{P}$ and $Y_{\text{red}} \to Y$ is separated. Hence, by (e) f_{red} has $\mathscr{P}$.

Remark. In the affine case, we can translate the above to statements about rings. Let $\mathscr{Q}$ be a property of homomorphisms of rings such that

(a′) a surjective homomorphism has $\mathscr{Q}$;

(b′) a composition of two homomorphisms having $\mathscr{Q}$ has $\mathscr{Q}$;

(c′) if $A \to B$ has $\mathscr{Q}$ and C is any A-algebra, then $C \to B \otimes_A C$ has $\mathscr{Q}$.

Then

(d′) a product of homomorphisms having $\mathscr{Q}$ has $\mathscr{Q}$;

(e′) if $\varphi : A \to B$ and $\psi : B \to C$ are two homomorphisms, and if $\psi \circ \varphi$ has $\mathscr{Q}$, then ψ has $\mathscr{Q}$;

(f′) if $\varphi : A \to B$ has $\mathscr{Q}$, then $\varphi_{\text{red}} : A_{\text{red}} \to B_{\text{red}}$ has $\mathscr{Q}$.

Note that we can ignore the condition of g separated in (e′) since any morphism between affine schemes is separable. Indeed, (e') can be proved from the sequence

$$A \xrightarrow{f} B \xrightarrow{g} C$$

by tensoring with B and extending the sequence as follows

$$B \cong A \otimes_A B \xrightarrow{f\otimes 1_B} B \otimes_A B \xrightarrow{g\otimes 1_B} C \otimes_A B \longrightarrow C$$

where $C \otimes_A B \to C$ is defined by $c \otimes b \mapsto g(b)c$. By (c′), $B \to C \otimes_A B$ has $\mathscr{Q}$, and $C \otimes_A B \to C$ is surjective, so it has $\mathscr{Q}$ by (a′). Notice that composing the homomorphisms give g, hence by (b′) g has $\mathscr{Q}$. □

9. Show that a composition of projective morphisms is projective. Conclude that projective morphisms have properties (a)-(f) of (Ex. 4.8) above.

Proof. By the results of (Ex. 3.13), (4.6), and (4.8), it suffices to show if $f : X \to \mathbb{P}^r$ is a projective morphism, then X is projective over $\operatorname{Spec}\mathbb{Z}$. If f is projective, then there exists a closed embedding $i : X \to \mathbb{P}^r \times \mathbb{P}^s$ such that f factors through $\mathbb{P}^r \to \mathbb{P}^s$. The Segre embedding (§1, 2.14) $\psi : \mathbb{P}^r \times \mathbb{P}^s \to \mathbb{P}^{rs+r+s}$ is a closed embedding, so $\psi \circ i : X \to \mathbb{P}^{rs+r+s}$ is a closed embedding such that $X \to \operatorname{Spec}\mathbb{Z}$ factors through $\mathbb{P}^{rs+r+s} \to \operatorname{Spec}\mathbb{Z}$, which is what we wanted to show. □

Chapter 2, Section 5

James Lee

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1. Let $(X, \mathcal{O}_X)$ be a ringed space, and let $\mathcal{E}$ be a locally free $\mathcal{O}_X$-module of finite rank. We define the *dual* of $\mathcal{E}$, denoted $\mathcal{E}^\vee$ to be the sheaf $\mathcal{H}om_{\mathcal{O}_X}(\mathcal{E}, \mathcal{O}_X)$.

(a) Show that $(\mathcal{E}^\vee)^\vee \cong \mathcal{E}$.

(b) For any $\mathcal{O}_X$-module $\mathcal{F}$, $\mathcal{H}om_{\mathcal{O}_X}(\mathcal{E}, \mathcal{F}) \cong \mathcal{E}^\vee \otimes_{\mathcal{O}_X} \mathcal{F}$.

(c) For any $\mathcal{O}_X$-modules $\mathcal{F}$, $\mathcal{G}$, $\mathrm{Hom}_{\mathcal{O}_X}(\mathcal{E} \otimes \mathcal{F}, \mathcal{G}) \cong \mathrm{Hom}_{\mathcal{O}_X}(\mathcal{F}, \mathcal{H}om_{\mathcal{O}_X}(\mathcal{E}, \mathcal{G}))$.

(d) (*Projection Formula*). If $f : (X, \mathcal{O}_X) \to (Y, \mathcal{O}_Y)$ is a morphism of ringed spaces, if $\mathcal{F}$ is an $\mathcal{O}_X$-module, and if $\mathcal{E}$ is a locally free $\mathcal{O}_Y$-module of finite rank, then there is a natural isomorphism $f_*(\mathcal{F} \otimes_{\mathcal{O}_X} f^*\mathcal{E}) \cong f_*(\mathcal{F}) \otimes_{\mathcal{O}_Y} \mathcal{E}$.

Lemma 0.1. *Let $(X, \mathcal{O})$ be a ringed space, let $\mathcal{F}, \mathcal{G}$ be $\mathcal{O}$-modules, and let $\mathcal{F}$ be locally free of finite rank. For any $x \in X$, $\mathcal{H}om(\mathcal{F}, \mathcal{G})_x \cong \mathrm{Hom}_{\mathcal{O}_{x,X}}(\mathcal{F}_x, \mathcal{G}_x)$.*

Proof. An element of $\mathcal{H}om(\mathcal{F}, \mathcal{G})_x$ can be represented as a pair $\langle U, f\rangle$, where U is an open neighborhood of x and $f : \mathcal{F}\big|_U \to \mathcal{G}\big|_U$ is a morphism of sheaves on U. Since $(\mathcal{F}\big|_U)_x \cong \mathcal{F}_x, (\mathcal{G}\big|_U)_x \cong \mathcal{G}_x$, there is a natural map

$$\begin{aligned} \alpha : \mathcal{H}om(\mathcal{F}, \mathcal{G})_x &\to \mathrm{Hom}_{\mathcal{O}_{x,X}}(\mathcal{F}_x, \mathcal{G}_x) \\ \langle U, f\rangle &\mapsto f_x. \end{aligned}$$

We show α is a bijection. If $\mathcal{F}$ has rank n, then $\mathcal{F}_x \cong \mathcal{O}_x^{\oplus n}$ since colimits commute and the image of f_x is a finitely generated $\mathcal{O}_x$-module. Thus, we can represent f_x as an $n \times n$-matrix, and there are only a finite number of entries, so we can further assume the entries lie in $\Gamma(W, \mathcal{O})$ for a smaller open neighborhood $W \subseteq U$. It is immediate that α is injective. Also, by this representation any $\varphi \in \mathrm{Hom}_{\mathcal{O}_x}(\mathcal{F}_x, \mathcal{G}_x)$ defines a $\mathcal{O}_X(W)$-module homomorphism $\mathcal{F}(W) = \mathcal{O}_X(W)^{\oplus n} \to \mathcal{G}(W)$, and by sheafification it ascends to a morphism of sheaves $\mathcal{F}\big|_W \to \mathcal{G}\big|_W$. Hence, α is surjective. □

Lemma 0.2. *Let $(X, \mathcal{O}_X)$ be a ringed space, let $\mathcal{F}, \mathcal{G}$ be $\mathcal{O}_X$-modules, and let $x \in X$. Then $(\mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{G})_x \cong \mathcal{F}_x \otimes_{\mathcal{O}_{x,X}} \mathcal{G}_x$.*

Proof. (A.M. Ex. 2.20). □

Lemma 0.3. *If $(f, f^\#) : (X, \mathcal{O}_X) \to (Y, \mathcal{O}_Y)$ is a morphism of ringed spaces, and $\mathcal{F}$ and $\mathcal{G}$ are $\mathcal{O}_Y$-modules, then $f^*(\mathcal{F} \otimes_{\mathcal{O}_Y} \mathcal{G}) \cong f^*(\mathcal{F}) \otimes_{\mathcal{O}_X} f^*(\mathcal{G})$.*

Proof. Check locally at each point and use Lemma 2. □

Proof.

(a) The question is local, so assume $\mathcal{E}$ is free of rank n. Then $\mathcal{E}_x = \mathcal{O}_{x,X}^{\oplus n}$, and by lemma 1, we have

$$(\mathcal{E}^\vee)_x^\vee = \mathrm{Hom}_{\mathcal{O}_{x,X}}(\mathrm{Hom}_{\mathcal{O}_{x,X}}(\mathcal{O}_{x,X}^{\oplus n}, \mathcal{O}_{x,X}), \mathcal{O}_{x,X}) = (\mathcal{O}_{x,X}^{\oplus n\vee})^\vee \cong \mathcal{O}_{x,X}^{\oplus n} = \mathcal{E}_x,$$

where the last isomorphism is a basic fact of finite free modules. The sheaves $(\mathcal{E}^\vee)^\vee$ and $\mathcal{E}$ are isomorphic at the level of stalks, hence they are isomorphic.

(b) The question is local, so assume $\mathcal{E}$ is free of rank n. For any open set $U \subseteq X$, we have

$$\mathcal{H}om_X(\mathcal{E},\mathcal{F})(U) = \mathrm{Hom}_{\mathcal{O}_U}(\mathcal{E}\big|_U, \mathcal{F}\big|_U),$$
$$(\mathcal{E}^\vee \otimes_{\mathcal{O}_X} \mathcal{F})(U) = \mathcal{E}^\vee(U) \otimes_{\mathcal{O}_U} \mathcal{F}(U) = \mathrm{Hom}_{\mathcal{O}_U}(\mathcal{E}\big|_U, \mathcal{O}_X\big|_U) \otimes_{\mathcal{O}_U} \mathcal{F}(U),$$

and since $\mathcal{E}$ is free, we have $\mathcal{E}(U) = \mathcal{O}_X(U)^{\oplus n}$. Let $f_i : \mathcal{E} \to \mathcal{O}_X$ be the projection morphisms onto the ith component, let $e_i\big|_U$ be the standard basis of $\mathcal{E}(U)$, and let $f : \mathcal{H}om_{\mathcal{O}_X}(\mathcal{E},\mathcal{O}_X) \to \mathcal{E}^\vee \otimes_{\mathcal{O}_X} \mathcal{F}$ be the morphism of sheaves defined by

$$f(U) : \mathcal{H}om_X(\mathcal{E},\mathcal{F})(U) \to (\mathcal{E}^\vee \otimes_{\mathcal{O}_X} \mathcal{F})(U)$$
$$g \mapsto \sum_{i=1}^n f_i\big|_U \otimes g(e_i\big|_U)$$

It is isomorphism of sheaves from the fact that $\mathcal{E}$ is free and g is $\mathcal{O}_X$-linear.

(c) Let $f \in \mathrm{Hom}_{\mathcal{O}_X}(\mathcal{E}\otimes\mathcal{F},\mathcal{G})$, which for every open $U \subseteq X$ defines a homomorphism of abelian groups

$$f(U) : (\mathcal{E}\otimes\mathcal{F})(U) \to \mathcal{G}(U).$$

We want to show f naturally defines an element of $\mathrm{Hom}_{\mathcal{O}_X}(\mathcal{F}, \mathcal{H}om_{\mathcal{O}_X}(\mathcal{E},\mathcal{G}))$. Define $\tilde{f} : \mathcal{F} \to \mathcal{H}om_{\mathcal{O}_X}(\mathcal{E},\mathcal{G})$ as the following: for each $s \in \mathcal{F}(U)$, let $\tilde{f}(s) : \mathcal{E}\big|_U \to \mathcal{G}\big|_U$ be the morphism defined by

$$\tilde{f}(s) : \mathcal{E}\big|_U \to \mathcal{G}\big|_U$$
$$t \in \Gamma(V, \mathcal{E}\big|_U) \mapsto f(t \otimes s\big|_V)$$

where V is any open subset of U. The fact that this defines an isomorphism of abelian groups follows the corresponding algebraic fact about finite free modules over rings.

(d) The question is local on Y, so assume $\mathcal{E}$ is free of finite rank. Also, if the statement is true for two sheaves of $\mathcal{O}_Y$-modules, then it true for their direct sum, because left-adjoint functors preserve colimits, so we can assume $\mathcal{E} = \mathcal{O}_Y$. Thus, we have the following composite of canonical isomorphisms

$$f_*\mathcal{F} \otimes_{\mathcal{O}_Y} \mathcal{O}_Y \cong f_*\mathcal{F} \cong f_*(\mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{O}_X) \cong f_*(\mathcal{F} \otimes_{\mathcal{O}_X} f^*\mathcal{O}_Y).$$

The last isomorphism is because structure sheaves pullback to structure sheaves, i.e.,

$$f^*\mathcal{O}_Y = f^{-1}\mathcal{O}_Y \otimes_{f^{-1}\mathcal{O}_Y} \mathcal{O}_X \cong \mathcal{O}_X.$$

□

7. Let X be a noetherian scheme, and let $\mathcal{F}$ be a coherent sheaf.

(a) If the stalk $\mathcal{F}_x$ is a free $\mathcal{O}_x$-module for some point $x \in X$, then there is a neighborhood U of x such that $\mathcal{F}\big|_U$ is free.

(b) $\mathcal{F}$ is locally free if and only if its stalks $\mathcal{F}_x$ are free $\mathcal{O}_x$-modules for all $x \in X$.

(c) $\mathcal{F}$ is invertible (i.e. locally free of rank 1) if and only if there is a coherent sheaf $\mathcal{G}$ such that $\mathcal{F}\otimes\mathcal{G} \cong \mathcal{O}_X$.

Proof.

(a) This is a local question, so assume $X = \operatorname{Spec} A$ and $\mathcal{F} = \tilde{M}$ for a noetherian ring A and a finitely generated A-module M. We reduce to the following algebraic problem: if there exists a prime ideal $\mathfrak{p}$ such that $M_\mathfrak{p} \cong M \otimes_A A_\mathfrak{p}$ is a finite free $A_\mathfrak{p}$-module, then there exists $f \in A - \mathfrak{p}$ such that $M_f \cong M \otimes_A A_f$ is a finite free A_f-module. Let $m_1, \ldots, m_r \in M$ such that the image of m_i form a basis in $M_\mathfrak{p}$, and let $\phi : A^{\oplus r} \to M$ be an A-module homomorphism defined by $e_i \mapsto m_i$, where e_i is the standard basis of $A^{\oplus r}$. We have an exact sequence

$$0 \to \ker\phi \to A^{\oplus r} \to M \to M/\operatorname{im}\phi \to 0,$$

and localization is an exact functor, so we have an induced exact sequence of $A_\mathfrak{p}$-modules

$$0 \to (\ker\phi)_\mathfrak{p} \to A_\mathfrak{p}^{\oplus r} \to M_\mathfrak{p} \to (M/\operatorname{im}\phi)_\mathfrak{p} \to 0.$$

Since $A_\mathfrak{p}^{\oplus r} \to M_\mathfrak{p}$ is an isomorphism, we have $(\ker\phi)_\mathfrak{p}, (M/\operatorname{im}\phi)_\mathfrak{p} = 0$. Submodules and quotients of noetherian modules are noetherian; in particular they are finitely generated, so by (A.M. Ex. 2.1) we can find $f \in A - \mathfrak{p}$ such that $(\ker\phi)_f, (M/\operatorname{im}\phi)_f = 0$. Hence, $M_f \cong A_f^{\oplus r}$.

(b) The if direction follows from (a). The converse direction follows from the following facts: we can realize finite free modules as a colimit, the stalk of a sheaf is defined as a colimit, and colimits commute.

(c) By the previous parts and lemma 2, we reduce to the following algebraic problem: let A be a noetherian local ring with maximal ideal $\mathfrak{m}$, and let M be a finitely generated A-module. Then $M \cong A$ if and only if there exists a finitely generated A-module N such that $M \otimes_A N \cong A$. One direction is clear. Conversely, suppose M, N are finitely generated A-modules such that $M \otimes_A N \cong A$. Let $k = A/\mathfrak{m}$ be the residue field of A. Tensoring with k gives

$$(M \otimes_A k) \otimes_k (N \otimes_A k) \cong (M \otimes_A N) \otimes_A k \cong A \otimes_A k \cong k,$$

which implies $M \otimes_A k \cong k$. By Nakayama's lemma, M is generated by a single element, which implies $M \cong A/\mathfrak{a}$ for some ideal $\mathfrak{a}$ of A, and similarly $N = A/\mathfrak{b}$. We have

$$A \cong M \otimes_A N \cong A/\mathfrak{a} \otimes_A A/\mathfrak{b} \cong A/(\mathfrak{a} + \mathfrak{b}),$$

which implies $\mathfrak{a} + \mathfrak{b} = (0)$. Hence, $M, N \cong A$.

□

8. Again let X be a noetherian scheme, and $\mathcal{F}$ a coherent sheaf on X. We will consider the function

$$\varphi(x) = \dim_{k(x)} \mathcal{F}_x \otimes_{\mathcal{O}_x} k(x),$$

where $k(x) = \mathcal{O}_x/\mathfrak{m}_x$ is the residue field at the point x. Use Nakayama's lemma to prove the following results:

(a) The function φ is *upper semi-continuous*, i.e. for any $n \in \mathbb{Z}$, the set $\{x \in X \mid \varphi(x) \geq n\}$ is closed.

(b) If $\mathcal{F}$ is locally free, and X is connected, then φ is a constant function.

(c) Conversely, if X is reduced, and φ is constant, then $\mathcal{F}$ is locally free.

Lemma 0.4 (Nakayama)**.** *Let A be a local ring with residue field k, and let M be a finitely generated A-module. Then any k-basis of M/mM lifts to a minimal set of generators of M.*

Proof.

(a) We show the set $\{x \in X : \varphi(x) < n\}$ is open. By (5.4), we reduce to the following algebraic problem: let A be a noetherian ring, let M be a finitely generated A-module, and let $\mathfrak{p}$ be a prime ideal of A with residue field $k(\mathfrak{p})$. If $\dim_{k(\mathfrak{p})} M_\mathfrak{p} \otimes_{A_\mathfrak{p}} k(\mathfrak{p}) < n$ for some $n \in \mathbb{Z}$, there exists a basic open neighborhood $\operatorname{Spec} A_s$ of $\mathfrak{p}$ such that $\dim_{k(\mathfrak{q})} M_\mathfrak{q} \otimes_{A_\mathfrak{q}} k(\mathfrak{q}) < n$ for all $\mathfrak{q} \in \operatorname{Spec} A_s$, where $s \in A - \mathfrak{p}$. By Nakayama's lemma, there exists $m_1, \ldots, m_r \in M$ with $r < n$ such that their image in $M_\mathfrak{p} \otimes_{A_\mathfrak{p}} k(\mathfrak{p})$ is a $k(\mathfrak{p})$-basis. Let $f : A^{\oplus r} \to M$ be the A-module homomorphism defined by $e_i \mapsto m_i$. We have an exact sequence

$$A^{\oplus r} \to M \to M/\operatorname{im} f.$$

Localizing gives

$$A_\mathfrak{p}^{\oplus r} \to M_\mathfrak{p} \to (M/\operatorname{im} f)_\mathfrak{p},$$

where $(M/\operatorname{im} f)_\mathfrak{p} = 0$. Quotients of noetherian modules are noetherian, so by (A.M. Ex. 2.1), we can find $s \in A - \mathfrak{p}$ such that $(M/\operatorname{im} f)_s = 0$. Thus, we have an exact sequence

$$A_s^{\oplus r} \to M_s \to 0,$$

which implies $M_\mathfrak{q} \otimes_{A_\mathfrak{q}} k(\mathfrak{q})$ has rank at most r for any $\mathfrak{q} \in \operatorname{Spec} A_s$.

(b) Let U_i for $i = 1, \ldots, n$ be a finite open cover of X such that $\mathcal{F}\big|_{U_i}$ is a free $\mathcal{O}_X\big|_{U_i}$-module. If X is connected, then for all U_i, there exists U_j with $j \neq i$ such that $U_i \cap U_j$. Clearly φ is constant on each U_i, so let r_i be the rank of $\mathcal{F}\big|_{U_i}$. Choose any $x \in U_i \cap U_j$, then $r_i = r_j = \varphi(x)$.

(c) Let $x \in X$, let $\operatorname{Spec} A$ be an open affine neighborhood of x for some noetherian reduced ring A, and let $\mathfrak{p}$ be the prime ideal corresponding to $x \in \operatorname{Spec} A$. Set $r = \varphi(x)$. By (5.4), there exists a finitely generated A-module M such that $\mathcal{F}\big|_{\operatorname{Spec} A} = \tilde{M}$. Let $m_1, \ldots, m_r \in \mathcal{F}_x \cong M_\mathfrak{p}$ such that their image in $\mathcal{F}_x \otimes_{\mathcal{O}_x} k(x)$ form a $k(x)$-basis. By Nakayama's lemma, the m_i's generate $M_\mathfrak{p}$ over $A_\mathfrak{p}$, so they generate $M_\mathfrak{q}$ for all prime ideals $\mathfrak{q} \subseteq \mathfrak{p}$. Suppose $\sum_{i=1}^r a_i m_i = 0$ for $a_i \in A_\mathfrak{p}$. Since the image of the m_i in $M_\mathfrak{q} \otimes_{A_\mathfrak{q}} k(\mathfrak{q})$ for all $i = 1, \ldots, r$ form a $k(\mathfrak{q})$-basis, the images of a_i in $k(\mathfrak{q})$ must be zero for all i. This implies a_i is contained in all prime ideals of $A_\mathfrak{p}$. However, $A_\mathfrak{p}$ is reduced, so $a_i = 0$. Hence, m_i are linearly independent.

□

9. Let S be a graded ring, generated by S_1 as an S_0-algebra, let M be a graded S-module, and let $X = \operatorname{Proj} S$.

(a) Show that there is a natural homomorphism $\alpha : M \to \Gamma_*(\tilde{M})$.

(b) Assume now that $S_0 = A$ is a finitely generated k-algebra for some field k, that S_1 is a finitely generated A-module, and that M is a finitely generated S-module. Show that the map α is an isomorphism in all large enough degrees, i.e. there is a $d_0 \in \mathbb{Z}$ such that for all $d \geq d_0$, $\alpha_d : M_d \to \Gamma(X, \tilde{M}(d))$ is an isomorphism.

(c) With the same hypotheses, we define an equivalence relation $\approx$ on graded S-modules by saying $M \approx M'$ is there is an integer d such that $M_{\geq d} \cong M'_{\geq d}$. Here $M_{\geq d} = \bigoplus_{n \geq d} M_n$. We will say that a graded S-module M is *quasi-finitely generated* if it is equivalent to a finitely generated module. Now show that the functors $\tilde{}$ and Γ_* induce an equivalence of categories between the category of quasi-finitely generated graded S-modules modulo the equivalence relation $\approx$, and the category of coherent $\mathcal{O}_X$-modules.

Proof.

(a) Write $M = \bigoplus_{d=0}^{\infty} M_d$. If $s \in M_d$, then s determines in a natural way a global section $s \in \Gamma(X, \tilde{M}(d))$, so define $\alpha_d : M_d \to \Gamma(X, \tilde{M}(d))$ in this way. Define β by extending this map linearly for all d.

(b) By (I, 7.4), there is a finite filtration

$$0 = M^0 \subseteq M^1 \subseteq \cdots \subseteq M^r = M$$

of M by graded submodules, where for each i, $M^i/M^{i-1} \cong (S/\mathfrak{p}_i)(n_i)$ for some homogenous prime ideal $\mathfrak{p}_i \subseteq S$, and some integer n_i. This filtration gives a filtration of M and short exact sequences

$$0 \to \tilde{M}^{i-1} \to \tilde{M}^i \to \widetilde{M^i/M^{i-1}} \to 0.$$

Twisting by d and taking global sections, all maps are natural, so we have the following commutative diagram with exact rows

$$\begin{array}{ccccccc} 0 & \longrightarrow & M_d^{i-1} & \longrightarrow & M_d^i & \longrightarrow & (M^i/M^{i-1})_d \\ & & \downarrow & & \downarrow & & \downarrow \\ 0 & \longrightarrow & \Gamma(X, \tilde{M}^{i-1}(d)) & \longrightarrow & \Gamma(X, \tilde{M}^i(d)) & \longrightarrow & \Gamma(X, \widetilde{M^i/M^{i-1}}(d)) \end{array}$$

where the vertical arrows are maps defined in (a). By the five lemma, to show that $M_d^i \to \Gamma(X, \tilde{M}^i(d))$ is surjective for large enough d, it will be sufficient to show that $(S/\mathfrak{p})_d \to \Gamma(X, \widetilde{S/\mathfrak{p}}(d))$ is surjective for large enough d, for each $\mathfrak{p}$ and n. Thus, we have reduced to the following special case: Let S be a graded integral domain, finitely generated by S_1 as an S_0-algebra, where $S_0 = A$ is a finitely generated domain over k, and let $X = \operatorname{Proj} S$. Then the map $\alpha : S \to S' = \bigoplus_{d \geq 0} \Gamma(X, \mathcal{O}_X(d))$ is an isomorphism in all large enough degrees.

Let $x_0, \ldots, x_r \in S_1$ be a set of generators of S_1 as an A-module. Following the argument in (5.13), S' is a ring, containing S, and contained in the intersection $\bigcap S_{x_i}$ of the localizations of S at the elements $x_0, \ldots, x_r$. By the proof of (5.19), since S'_d is a finitely generated A-module for every d, there exists $d_0 \geq 0$ such that $S_n S'_d \subseteq S_{n+d} \subseteq S'_{n+d}$ for some large enough n and all $d \geq d_0$. It follows by $S_n S'_d = S'_{n+d}$ that $S_{n+d} = S'_{n+d}$ for all $d \geq d_0$.

(c) The natural homomorphism α viewed as a morphism between equivalence classes of quasi-finitely generated graded S-modules is an isomorphism by (b). Hence, the two categories are equivalent with β from (5.15).

□

12. (a) Let X be a scheme over a scheme Y, and let $\mathcal{L}$, $\mathcal{M}$ be two very ample invertible sheaves on X. Show that $\mathcal{L} \otimes \mathcal{M}$ is also very ample.

(b) Let $f : X \to Y$ and $g : Y \to Z$ be two morphisms of schemes. Let $\mathcal{L}$ be a very ample invertible sheaf on X relative to Y, and let $\mathcal{M}$ be a very ample invertible sheaf on Y relative to Z. Show that $\mathcal{L} \otimes f^*\mathcal{M}$ is a very ample invertible sheaf on X relative to Z.

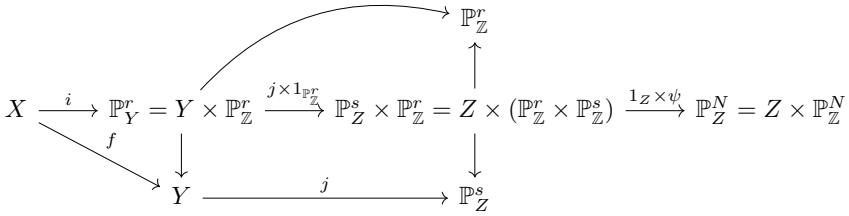
Proof.

(a) Let $g : X \to \mathbb{P}^r_Y, h : X \to \mathbb{P}^s_Y$ be immersions such that $\mathcal{L} \cong g^*(\mathcal{O}_{\mathbb{P}^r_Y}(1)), \mathcal{M} \cong h^*(\mathcal{O}_{\mathbb{P}^s_Y}(1))$ respectively, let $\psi : \mathbb{P}^r_Y \times \mathbb{P}^s_Y \to \mathbb{P}^N_Y$ be a Segre embedding with $N = rs + r + s$, let $\pi_r, \pi_s : \mathbb{P}^r_Y \times \mathbb{P}^s_Y \to \mathbb{P}^s_Y, \mathbb{P}^s_Y$ be the natural projection morphisms, and let $f : X \to \mathbb{P}^r_Y \times \mathbb{P}^s_Y$ be the unique morphism such that $g = \pi_r \circ f, h = \pi_s \circ f$. A Segre embedding is an immersion, and compositions of immersions are immersions. In particular, $\psi \circ f$ is an immersion, and we have the following isomorphisms

$$\begin{aligned}
\mathcal{L} \otimes \mathcal{M} &\cong g^*(\mathcal{O}_{\mathbb{P}^r_Y}(1)) \otimes h^*(\mathcal{O}_{\mathbb{P}^s_Y}(1)) \\
&\cong (p_r \circ f)^*(\mathcal{O}_{\mathbb{P}^r_Y}(1)) \otimes (p_s \circ f)^*(\mathcal{O}_{\mathbb{P}^s_Y}(1) \\
&\cong f^*(p_r^*(\mathcal{O}_{\mathbb{P}^r_Y}(1)) \otimes p_s^*(\mathcal{O}_{\mathbb{P}^s_Y}(1))) \\
&\cong f^*(\mathcal{O}_{\mathbb{P}^r_Y \times \mathbb{P}^s_Y}(1)) \\
&\cong f^*(\psi^*(\mathcal{O}_{\mathbb{P}^N_Y}(1))) \\
&\cong (\psi \circ f)^*(\mathcal{O}_{\mathbb{P}^N_Y}(1)).
\end{aligned}$$

Hence, $\mathcal{L} \otimes \mathcal{M}$ is very ample.

(b) Let $i : X \to \mathbb{P}^r_Y$ be an immersion such that $\mathcal{L} \cong i^*\mathcal{O}(1)$, let $j : Y \to \mathbb{P}^s_Z$ be an immersion such that $\mathcal{M} \cong j^*\mathcal{O}(1)$, and let $\psi : \mathbb{P}^r_\mathbb{Z} \times \mathbb{P}^s_\mathbb{Z} \to \mathbb{P}^N_Z$ be a Segre embedding, where $N = rs + r + s$. We have the following commutative diagram



where $p : Y \times \mathbb{P}^r_\mathbb{Z} \to Y, q : \mathbb{P}^s_Z \times \mathbb{P}^r_Z \to \mathbb{P}^s_Z$ are the natural projection maps. A composition of immersions is an immersion, and a product of immersions is an immersion, so $\phi = (1_Z \times \psi) \circ (j \times 1_{\mathbb{P}^r_\mathbb{Z}}) \circ i$ is an immersion. Hence, $\mathcal{L} \otimes f^*\mathcal{M} \cong \phi^*\mathcal{O}(1)$.

□

13. Let S be a graded ring, generated by S_1 as an S_0-algebra. For any integer $d > 0$, let $S^{(d)}$ be the graded ring $\bigoplus_{n \geq 0} S^{(d)}_n$ where $S^{(d)}_n = S_{nd}$. Let $X = \operatorname{Proj} S$. Show that $\operatorname{Proj} S^{(d)} \cong X$, and that the sheaf $\mathcal{O}(1)$ on $\operatorname{Proj} S^{(d)}$ corresponds via this isomorphism to $\mathcal{O}_X(d)$.

Proof. The inclusion $i : S^{(d)} \hookrightarrow S$ induces a morphism of schemes $\phi : X \to X' = \operatorname{Proj} S^{(d)}$ via $\mathfrak{p} \mapsto \mathfrak{p} \cap S^{(d)}$. If $\mathfrak{p} \in X$ so that $\mathfrak{p} \not\supseteq S_+ = \bigoplus_{n>0} S_n$, then $\mathfrak{p} \not\supseteq S^{(d)}_+$. Otherwise, $s^d \in \mathfrak{p}$ for all $s \in S_+$, which implies $S_+ \supseteq \mathfrak{p}$, a contradiction. Thus, ϕ is well-defined. Since $S^{(d)}$ is generated by $S^{(d)}_1$ as an S_0-algbera, we can cover X' by open affines $\operatorname{Spec} S^{(d)}_{(f)}$, where $f \in S^{(d)}_1 = S_d$ and $S^{(d)}_{(f)}$ consists of all degree zero elements of the form s/f^n for some $s \in S^{(d)}$. Also, $\phi^{-1}(\operatorname{Spec} S^{(d)}_{(f)}) = \operatorname{Spec} S_{(f)}$, and clearly $S_{(f)} \cong S^{(d)}_{(f)}$ as rings. Hence, ϕ is an isomorphism of schemes. Lastly, we also have an isomorphism of twisted sheaves $\mathcal{O}_{X'}(1) \cong (i_*\mathcal{O}_X)(1) = i_*(\mathcal{O}_X(d))$. □

16. Let $(X, \mathcal{O}_X)$ be a ringed spaces, and let $\mathcal{F}$ be a sheaf of $\mathcal{O}_X$-modules. We define the *tensor algebra*, *symmetric algebra*, and *exterior algebra* of $\mathcal{F}$ by taking the sheaves associated to the presheaf, which to each open set U assigns the corresponding tensor operation applied to $\mathcal{F}(U)$ as an $\mathcal{O}_X(U)$-module. The results are $\mathcal{O}_X$-algebras, and their components in each degree are $\mathcal{O}_X$-modules.

(a) Suppose that $\mathcal{F}$ is locally free of rank n. Then $T^r(\mathcal{F}), S^r(\mathcal{F})$, and $\bigwedge^r(\mathcal{F})$ are also locally free, of ranks n^r, $\binom{n+r-1}{n-1}$, and $\binom{n}{r}$ respectively.

(b) Again let $\mathcal{F}$ be locally free of rank n. Then the multiplication map $\bigwedge^r \mathcal{F} \otimes \bigwedge^{n-r} \mathcal{F} \to \bigwedge^n \mathcal{F}$ is a perfect pairing for any r, i.e. it induces an isomorphism of $\bigwedge^r \mathcal{F}$ with $(\bigwedge^{n-r} \mathcal{F})^\vee \otimes \bigwedge^n \mathcal{F}$. As a special case, note if $\mathcal{F}$ has rank 2, then $\mathcal{F} \cong \mathcal{F}^\vee \otimes \bigwedge^2 \mathcal{F}$.

(c) Let $0 \to \mathcal{F}' \to \mathcal{F} \to \mathcal{F}'' \to$ be an exact sequence of locally free sheaves. Then for any r there is a finite filtration of $S^r(\mathcal{F})$,

$$S^r(\mathcal{F}) = F^0 \supseteq F^1 \supseteq \cdots \supseteq F^r \supseteq F^{r+1} = 0$$

with quotients

$$F^p/F^{p+1} \cong S^p(\mathcal{F}') \otimes S^{r-p}(\mathcal{F}'')$$

for each p.

(d) Same statement as (c), with exterior powers isntead of symmetric powers. In particular, if $\mathcal{F}', \mathcal{F}, \mathcal{F}''$ have ranks n', n, n'' respectively, there is an isomorphism $\bigwedge^n \mathcal{F} \cong \bigwedge^{n'} \mathcal{F} \otimes \bigwedge^{n''} \mathcal{F}''$.

(e) Let $f : X \to Y$ be a morphism of ringed spaces, and let $\mathcal{F}$ be an $\mathcal{O}_Y$-module. Then f^* commutes with all the tensor operations on $\mathcal{F}$, i.e. $f^*(S^n(\mathcal{F})) = S^n(f^*\mathcal{F})$.

Proof.

(a) The question is local, so the statements follow from the case for rings. For example, let U be an open set in X such that $\mathcal{F}\big|_U \cong \mathcal{O}_U^{\oplus n}$. Then $T^r(\mathcal{F})\big|_U \cong T^r(\mathcal{O}_U^{\oplus n}) \cong \mathcal{O}_U^{\oplus n^r}$. We remark that locally, the symmetric algebra $S(\mathcal{F})$ is isomorphic to the polynomial ring $\mathcal{O}[T_1, \ldots, T_n]$, and $S^r(\mathcal{F})$ correspond to the homogenous elements of degree r in $\mathcal{O}[T_1, \ldots, T_n]$. Thus, $S^r(\mathcal{F})$ has rank $\binom{n+r-1}{n-1}$. Let $x_1, \ldots, x_n \in \Gamma(U, \mathcal{O}_X)$ be a $\mathcal{O}_U$-basis for $\mathcal{F}\big|_U$. Then $\bigwedge^r(\mathcal{F})$ is spanned by $x_{i_1} \wedge \cdots \wedge x_{i_r}$, where $1 \leq i_1 < \cdots < i_r \leq n$. Hence, $\bigwedge^r(\mathcal{F})$ has rank $\binom{n}{r}$.

□

Chapter 2, Section 6

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1. Let X be a scheme satisfying $(*)$. Then $X \times \mathbb{P}^n$ also satisfies $(*)$, and $\mathrm{Cl}(X \times \mathbb{P}^n) \cong (\mathrm{Cl}\, X) \times \mathbb{Z}$.

Proof. Since $\mathbb{P}^n$ is a union of $n+1$ copies of $\mathbb{A}^n$, it is not hard to see from (6.6) that $X \times \mathbb{P}^n$ is Noetherian and integral. It's left to show it is separated. Indeed, $X \times \mathbb{P}^n \to X$ is a projective morphism of Noetherian schemes, hence it is proper by (4.9). Composition of separated morphisms is separated. Hence, $X \times \mathbb{P}^n$ satisfies $(*)$.

The projective n-space is the union of a hyperplane $\mathbb{P}^{n-1}$ and a copy of $\mathbb{A}^n$, and $\mathbb{P}^{n-1}$ is a prime divisor of $\mathbb{P}^n$. Thus, $X \times \mathbb{P}^{n-1}$ is a prime divisor of $X \times \mathbb{P}^n$. Also, $X \times \mathbb{P}^n - X \times \mathbb{P}^{n-1} = X \times \mathbb{A}^n$, and $\mathrm{Cl}(X \times \mathbb{A}^n) \cong \mathrm{Cl}\, X$. Hence, we have the exact sequence

$$0 \to \mathbb{Z} \xrightarrow{1 \mapsto X \times \mathbb{P}^{n-1}} \mathrm{Cl}(X \times \mathbb{P}^n) \to \mathrm{Cl}\, X \to 0.$$

This exact sequence splits via $Y \in \mathrm{Cl}\, X \mapsto Y \times \mathbb{P}^n \in \mathrm{Cl}(X \times \mathbb{P}^n)$. Injectivity of this map follows from $\mathbb{P}^n$ being birational to $\mathbb{A}^n$ and (6.6). Hence, $\mathrm{Cl}(X \times \mathbb{P}^n) \cong \mathrm{Cl}(X) \times \mathbb{Z}$. □

2. *Varieties in Projective Space.* Let k be an algebraically closed field, and let X be a closed subvariety of $\mathbb{P}^n_k$ which is nonsingular in codimension one (hence satisfies $(*)$). For any divisor $D = \sum n_i Y_i$ on X, we define the *degree* of D to be $\sum n_i \deg Y_i$, where $\deg Y_i$ is the degree of Y_i, considered as a projective variety itself.

(a) Let V be an irreducible hypersurface in $\mathbb{P}^n$ which does not contain X, and let Y_i be the irreducible components of $V \cap X$. They all have codimension 1 by (I, Ex. 1.8). For each i, let f_i be a local equation for V on some open set U_i of $\mathbb{P}^n$ for which $Y_i \cap U_i \neq \emptyset$, and let $n_i = v_{Y_i}(\bar{f}_i)$, where $\bar{f}_i$ is the restriction of f_i to $U_i \cap X$. Then we define the *divisor* $V.X$ to be $\sum n_i Y_i$. Extend by linearity and show that this gives a well-defined homomorphism from the subgroup of $\mathrm{Div}\, \mathbb{P}^n$ consisting of divisors, none of whose components contain X, to $\mathrm{Div}\, X$.

(b) If D is a principal divisor on $\mathbb{P}^n$, for which $D.X$ is defined as in (a), show that $D.X$ is principal on X. Thus, we get a homomorphism $\mathrm{Cl}\, \mathbb{P}^n \to \mathrm{Cl}\, X$.

(c) Show that the integer n_i defined in (a) is the same as the intersection multiplicity $i(X, V; Y_i)$ defined in (I, §7). Then use the generalized Bézout's theorem (I, 7.7) to show that for any divisor D on $\mathbb{P}^n$, none of whose components contain X,

$$\deg(D.X) = (\deg D) \cdot (\deg X).$$

(d) If D is a principal divisor on X, show that there is a rational function f on $\mathbb{P}^n$ such that $D = (f).X$. Conclude that $\deg D = 0$. Thus, the degree function defines a homomorphism $\deg : \mathrm{Cl}\, X \to \mathbb{Z}$. Finally, there is a commutative diagram

$$\begin{array}{ccc} \mathrm{Cl}\, \mathbb{P}^n & \longrightarrow & \mathrm{Cl}\, X \\ \cong \downarrow \deg & & \downarrow \deg \\ \mathbb{Z} & \xrightarrow[\cdot(\deg X)]{} & \mathbb{Z} \end{array}$$

and in particular, we see that the map $\mathrm{Cl}\, \mathbb{P}^n \to \mathrm{Cl}\, X$ is injective.

Proof.

(a) Let f_i' be another local equation for V on some open set U_i' of $\mathbb{P}^n$ for which $Y_i \cap U_i' \neq \emptyset$. Since X is irreducible, $U_i \cap U_i' \cap X \neq \emptyset$, and $f_i/f_i' \in \Gamma(U_i \cap U_i', \mathcal{O}^*_{\mathbb{P}^n})$. Hence, $\bar{f}_i/\bar{f}_i' \in \Gamma(U_i \cap U_i' \cap X, \mathcal{O}^*_X)$. Hence, n_i is independent of the choice of U_i.

(b) Suppose $D = (f)$, and let $i : X \hookrightarrow \mathbb{P}^n$ be a closed immersion. Then $D.X = (i^{-1}f)$.

(c) We recall the definition of the intersection multiplicity:

$$i(V, X; Y_i) = \text{length}_{S_{\mathfrak{p}_i}}(S/(I_X + I_V))_{\mathfrak{p}_i},$$

where $S = k[x_0, \ldots, x_n]$, and I_X, I_V are the homogeneous ideals defining X and V, respectively. The homogeneous prime ideal $\mathfrak{p}_i$ defines Y_i. Note that $\text{length}_{S_\mathfrak{p}} M_\mathfrak{p} = \text{length}_{S_{(\mathfrak{p})}} M_{(\mathfrak{p})}$. Let $\mathfrak{P}_i$ be the image of $\mathfrak{p}_i$ in S/I_X, and let $A = (S/I_X)_{(\mathfrak{P}_i)}$. Then A is precisely the valuation ring of Y_i as a prime divisor of X. If we denote I_V' the extension of the ideal I_V in S, it is not hard to see that $\text{length}_A A/I_V' = v_{Y_i}(\overline{f})$, where f is any local equation defining V.
Let $D = \sum n_j V_j$ be any $\mathbb{P}^n$ with each V_j an irreducible hypersurface not containing X. By the generalized Bézout's theorem,

$$\begin{aligned}
\deg(D.X) &= \sum n_j \deg(V_j.X) \\
&= \sum n_j \sum i(V_j, X; V_{jk}) \deg V_{jk} \\
&= \sum n_j (\deg V_j \cdot \deg X) \\
&= (\deg D) \cdot (\deg X).
\end{aligned}$$

(d) Let $D = (f)$ for some $f \in K(X)$, where $K(X)$ is the function field of X. If X is locally defined by an ideal $I \subset k[y_1, \ldots, y_n]$ on some open subset $\text{Spec}\, k[y_1, \ldots, y_n]$ of $\mathbb{P}^n$ where f is regular, then we can write $f = g/h$ for some $g, h \in k[y_1, \ldots, y_n]$ such that h is nowhere zero on some open set in X, which is what we wanted to show. Since any principal divisor has degree zero, by (c), $\deg(D) = \deg((f).X) = 0 \cdot \deg X = 0$.

□

4. Let k be a field of characteristic $\neq 2$. Let $f \in k[x_1, \ldots, x_n]$ be a *square-free* nonconstant polynomial, i.e., in the unique factorization of f into irreducible polynomials, there are no repeated factors. Let $A = k[x_1, \ldots, x_n, z]/(z^2 - f)$. Show that A is an integrally closed ring.

Proof. The quotient field K of A is just $k(x_1, \ldots, x_n)[\sqrt{f}]$. It is a Galois extension of $k[x_1, \ldots, x_n]$ with Galois group $\mathbb{Z}/2\mathbb{Z}$ generated by $\sqrt{f} \mapsto -\sqrt{f}$. If $\alpha = g + h\sqrt{f} \in K$, where $g, h \in k(x_1, \ldots, x_n)$, then the minimal polynomial of α is $X^2 - 2gX + (g^2 - h^2 f)$. Suppose α is integral over $k[x_1, \ldots, x_n]$. Since $k[x_1, \ldots, x_n]$ is integrally closed, the coefficients of p_α lie in $k[x_1, \ldots, x_n]$ (A.M. 5.15). It follows $g, h \in k[x_1, \ldots, x_n]$. Hence, the integral closure of $k[x_1, \ldots, x_n]$ in K lies in A. The converse is immediate by the formula for the minimal polynomial of any $\alpha \in K$. Hence, A is an integrally closed ring. □

7. Let X be the nodal cubic curve $y^2 z = x^3 + x^2 z$ in $\mathbb{P}^2$. Imitate (6.11.4) and show that the group of Cartier divisors of degree 0, $\text{CaCl}^\circ X$, is naturally isomorphic to the multiplicative group $\mathbb{G}_m$.

Proof. Take $Z = (0 : 0 : 1)$ and $P_0 = (0 : 1 : 0)$. There is a bijection with the closed points of $X - Z$, the set of non-singular points of X, and $\text{CaCl}^\circ X$. It remains to show $X - Z \cong \mathbb{G}_m$, where $\mathbb{G}_m = \mathbb{A}^1 - \{0\}$ is the multiplicative group defined to be the spectrum of the ring $k[t, t^{-1}]$. Looking at the affine subset $z \neq 1$, we see that any line through the singular point Z, which has affine coordinates $(0, 0)$, intersects the curve one more time. Plugging in $y = \lambda x$ into the defining equation of X for some $\lambda \in k^*$, we have $x = \lambda^2 - 1$. Thus, we have a map $k^* - \{\pm 1\} \to X - Z$ defined by $\lambda \mapsto (\lambda^2 - 1, (\lambda^2 - 1)^3 + (\lambda^2 - 1)^2)$. Extend this map to all of k^* via $1 \mapsto P_0$ and $-1 \mapsto (-1, 0)$, and we are done. □

8. (a) Let $f : X \to Y$ be a morphism of schemes. Show that $\mathcal{L} \mapsto f^*\mathcal{L}$ induces a homomorphism of Picard groups, $f^* : \text{Pic}\, Y \to \text{Pic}\, X$.

(b) If f is a finite morphism of nonsingular curves, show that this homomorphism corresponds to the homomorphism $f^* : \text{Cl}\, Y \to \text{Cl}\, X$ defined in the text, via the isomorphisms of (6.16).

(c) If X is a locally factorial integral closed subscheme of $\mathbb{P}^n_k$, and if $f : X \to \mathbb{P}^n$ is the inclusion map, then f^* on Pic agrees with the homomorphism on divisor class groups defined in (Ex. 6.2) via the isomorphisms of (6.16).

Proof.

(a) If V is an open set in Y such that $\mathcal{L}\big|_V \cong \mathcal{O}_V$, then $f^*\mathcal{L}\big|_{f^{-1}V} \cong \mathcal{O}_{f^{-1}V}$.

(b) Let D be a Weil divisor on Y. We want to show $f^*(\mathcal{O}_Y(D)) \cong \mathcal{O}_X(f^*D)$. By the isomorphism $\operatorname{CaCl} X \cong \operatorname{Cl} X$, there exists a representation $\{(U_i, s_i)\}$ of D as a Cartier divisor. A finite morphism of nonsingular curves induces an inclusion of function fields $K(Y) \to K(X)$. Since the s_i's are elements of $K(Y)$, by (a), $f^*(\mathcal{O}_X(D))$ is the Cartier divisor defined by $\{(f^{-1}U_i, f^{-1}s_i)\}$. It is clear that the Weil divisor associated to $\{(f^{-1}U_i, f^{-1}s_i)\}$ is f^*D.

(c) The map defined in (Ex. 6.2) is precisely $f^* : \operatorname{Cl} Y \to \operatorname{Cl} X$ as defined in the text.

□

10. *The Grothendiek Group $K(X)$.* Let X be a noetherian scheme. We define $K(X)$ to be the quotient of the free abelian group generated by all the coherent sheaves on X, by the subgroup generated by all expressions $\mathcal{F} - \mathcal{F}' - \mathcal{F}''$, whenever there is an exact sequence $0 \to \mathcal{F}' \to \mathcal{F} \to \mathcal{F}'' \to 0$ of coherent sheaves on X. If $\mathcal{F}$ is a coherent sheaf, we denote by $\gamma(\mathcal{F})$ its image in $K(X)$.

(a) If $X = \mathbb{A}^1_k$, then $K(X) \cong \mathbb{Z}$.

(b) If X is any integral scheme, and $\mathcal{F}$ a coherent sheaf, we define the *rank* of $\mathcal{F}$ to be $\dim_K \mathcal{F}_\xi$, where ξ is the generic point of X, and $K = \mathcal{O}_\xi$ is the function field of X. Show that the rank function defines a surjective homomorphism $\operatorname{rank} : K(X) \to \mathbb{Z}$.

(c) If Y is a closed subcheme of X, there is an exact sequence

$$K(Y) \longrightarrow K(X) \longrightarrow K(X - Y) \longrightarrow 0,$$

where the first map is extension by zero, and the second map is restriction.

Proof.

(a) Suppose $X = \operatorname{Spec} A$ for some principle ideal domain A. Coherent sheaves on X correspond to finitely generated A-modules. Any such module can be decomposed as a direct sum of a free and torsion submodule. Any torsion module is a direct sum of $A/\mathfrak{p}$, where $\mathfrak{p}$ is a prime ideal of A. Since A is a principle ideal domain, all prime ideals are rank one A-modules, so $\gamma(A) = \gamma(\mathfrak{p})$. We have the exact sequence

$$0 \to \mathfrak{p} \to A \to A/\mathfrak{p} \to 0,$$

which implies $\gamma(A/\mathfrak{p}) = 0$. Hence, we can identify elements of $K(X)$ to the rank of their free parts.

(b) Take $\mathcal{F} = \mathcal{O}_X^{\oplus n}$ for each n, then $\operatorname{rank} \mathcal{F} = n$.

□

11. *The Grothendiek Group of a Nonsingular Curve.* Let X be a nonsingular curve over an algebraically closed field k. We will show that $K(X) \cong \operatorname{Pic} X \oplus \mathbb{Z}$, in several steps.

(a) For any divisor $D = \sum n_i P_i$ on X, let $\psi(D) = \sum n_i \gamma(k(P_i)) \in K(X)$, where $k(P_i)$ is the skyscraper sheaf k at P_i and 0 elsewhere. If D is an effective divisor, let $\mathcal{O}_D$ be the structure sheaf of the associated subscheme of codimension 1, and show that $\psi(D) = \gamma(\mathcal{O}_D)$. Then use (6.18) to show that for any D, $\psi(D)$ depends only on the linear equivalence class of D, so ψ defines a homomorphism $\psi : \operatorname{Cl} X \to K(X)$.

(b) For any coherent sheaf $\mathcal{F}$ on X, show that there exist locally free sheaves $\mathcal{E}_0$ and $\mathcal{E}_1$ and an exact sequence $0 \to \mathcal{E}_1 \to \mathcal{E}_0 \to \mathcal{F} \to 0$. Let $r_0 = \operatorname{rank} \mathcal{E}_0$, $r_1 = \operatorname{rank} \mathcal{E}_1$, and define $\det \mathcal{F} = (\bigwedge^{r_0} \mathcal{E}_0) \otimes (\bigwedge^{r_1} \mathcal{E}_1)^{-1} \in \operatorname{Pic} X$. Show that $\det \mathcal{F}$ is independent of the resolution chosen, and that it gives a homomorphism $\det : K(X) \to \operatorname{Pic} X$. Finally, show that if D is a divisor, then $\det(\psi(D)) = \mathcal{L}(D)$.

(c) if $\mathcal{F}$ is any coherent sheaf of rank r, show that there is a divisor D on X and an exact sequence $0 \to \mathcal{L}^{\oplus r}(D) \to \mathcal{F} \to \mathcal{T} \to 0$, where $\mathcal{T}$ is a torsion sheaf. Conclude that if $\mathcal{F}$ is a sheaf of rank r, then $\gamma(\mathcal{F}) - r\gamma(\mathcal{O}_X) \in \operatorname{im} \psi$.

(d) Using the maps $\psi, \det, \operatorname{rank}$, and $1 \mapsto \gamma(\mathcal{O}_X)$ from $\mathbb{Z} \to K(X)$, show that $K(X) \cong \operatorname{Pic} X \oplus \mathbb{Z}$.

Proof.

(a) Let $D = \sum n_i P_i$ be an effective divisor, and let $\{(U_i, f_i)\}$ be the effective Cartier divisor associated to D, where $f_i \in \Gamma(U_i, \mathcal{O}_{U_i})$ and $v_{P_i}(f_i) = n_i$ for all i. Thus, if t_i is an uniformizing element of $\mathcal{O}_{P_i,X}$ for each i, then $\mathcal{O}_D$ be can be identified with the direct sum of skyscraper sheaves $\bigoplus \mathcal{O}_{P_i,X}/(t_i^{n_i})$ on each P_i. Note that $\mathcal{O}_{P_i,X}/(t_i^{n_i})$ is non-zero for only finitely many P_i. Since k is algebraically closed, the residue field of $\mathcal{O}_{P_i,X}$ is k, and since each $\mathcal{O}_{P_i,X}$ is a discrete valuation ring, $\mathcal{O}_{P_i,X}/(t_i^{n_i})$ is isomorphic to $k^{\oplus n_i}$ as a k-vector space, so $\gamma(\mathcal{O}_{P_i,X}/(t_i^{n_i})) = \gamma(k(P_i)^{\oplus n_i}) = n_i\gamma(k(P_i)) = \psi(n_i P_i)$, which is what we wanted to show.

We need to show for any principal divisor $D = (f)$, where $f \in K^*$, the function field of X, $\psi(D) = 0$. Any principal divisor can be written as a difference of two effective principal divisors, so we assume D is effective. Let Y be the associated closed subscheme of D. Then the ideal sheaf $\mathcal{I}_Y$ is generated by f, so we have an isomorphism of $\mathcal{O}_X$-modules $\mathcal{O}_X \to \mathcal{I}_Y$ defined by multiplication by f. Hence, $\psi(D) = \gamma(\mathcal{O}_D) = \gamma(\mathcal{O}_X/\mathcal{I}_Y) = \gamma(\mathcal{O}_X) - \gamma(\mathcal{I}_Y) = 0$ by (6.18).

(b) Let $U_i = \operatorname{Spec} A_i$ be an open affine cover of X such that $\mathcal{F}\big|_{U_i} \cong \widetilde{M_i}$ for some finitely generated A_i-module. Since X is noetherian, a finite number will do, and let $N > 0$ such that the minimal number of generators for M_i is at most N for all i.

□

12. Let X be a complete nonsingular curve. Show that there is a unique way to define the *degree* of any coherent sheaf on X, $\deg \mathcal{F} \in \mathbb{Z}$, such that:

(1) If D is a divisor, $\deg \mathcal{L}(D) = \deg D$.

(2) If $\mathcal{F}$ is a *torsion sheaf* (meaning a sheaf whose stalk at the generic point is zero), then

$$\deg \mathcal{F} = \sum_{P \in X} \operatorname{length}(\mathcal{F}_P).$$

(3) If $0 \to \mathcal{F}' \to \mathcal{F} \to \mathcal{F}'' \to 0$ is an exact sequence, then $\deg \mathcal{F} = \deg \mathcal{F}' + \deg \mathcal{F}''$.

Proof. Define the degree map to be the composition of the determinant map $K(X) \to \operatorname{Pic} X$ and the degree map $\operatorname{Pic} X \to \mathbb{Z}$, where $\deg : \operatorname{Pic} X \to \mathbb{Z}$ is defined as in (6.10) by identifying $\operatorname{Pic} X$ with $\operatorname{Cl} X$ (6.16). □

Chapter 2, Section 7

James Lee

October 11, 2026

1. Let $(X, \mathcal{O}_X)$ be a locally ringed space, and let $f : \mathcal{L} \to \mathcal{M}$ be a surjective map of invertible sheaves on X. Show that f is an isomorphism.

Proof. We reduce to the following algebraic problem: any A-linear surjective map $\phi : A \to A$ is an isomorphism. Indeed, ϕ is determined by $r = \phi(1)$, and ϕ surjective implies there exists $a \in A$ such that $ar = \phi(a) = 1$. Thus, r is a unit, so ϕ is injective. □

2. Let X be a scheme over a field k. Let $\mathcal{L}$ be an invertible sheaf on X, and let $\{s_0, \dots, s_n\}$ and $\{t_0, \dots, t_m\}$ be two sets of sections of $\mathcal{L}$, which generate the same subspace $V \subseteq \Gamma(X, \mathcal{L})$, and which generate the sheaf $\mathcal{L}$ at every point. Suppose $n \leq m$. Show that the corresponding morphisms $\varphi : X \to \mathbb{P}^n_k$ and $\psi : X \to \mathbb{P}^m_k$ differ by a suitable linear projection $\mathbb{P}^m - L \to \mathbb{P}^n$ and an automorphism of $\mathbb{P}^n$, where L is a linear subspace of $\mathbb{P}^m$ of dimension $m - n - 1$.

3. Let $\varphi : \mathbb{P}^n_k \to \mathbb{P}^m_k$ be a morphism. Then:

(a) either $\varphi(\mathbb{P}^n) = pt$ or $m \geq n$ and $\dim \varphi(\mathbb{P}^n) = n$;

(b) in the second case, φ can be obtained as the composition of (1) a d-uple embedding $\mathbb{P}^n \to \mathbb{P}^N$ for a uniquely determined $d \geq 1$, (2) a linear projection $\mathbb{P}^N - L \to \mathbb{P}^m$, and an automorphism of $\mathbb{P}^m$. Also, φ has finite fibers.

Proof.

(a) Let $\mathcal{L}$ be the invertible sheaf on $\mathbb{P}^n$ associated to the morphism $\varphi : \mathbb{P}^n \to \mathbb{P}^m$, which is generated by global sections $s_i = \varphi^*(x_i)$, $i = 0, 1, \dots, m$. Pick $P \in \varphi^{-1}((1, 0, \dots, 0))$ (we can assume such P always exists by applying an automorphism of $\mathbb{P}^m$). Identifying $\mathcal{L}_P \cong \mathcal{O}_{P,\mathbb{P}^n}$, we see that $(s_i)_P$ for $i \neq 0$ is not a unit, and they generate the local ring of $\mathcal{O}_{P,\mathbb{P}^n}$. Since $\mathbb{P}^n$ is smooth, the minimal number of generators of the maximal ideal of $\mathcal{O}_{P,\mathbb{P}^n}$ is equal to the dimension n. The sections s_i generate $\mathcal{O}_{P,\mathbb{P}^n}$. Hence, $m \geq n$. Our proof also shows the induced homomorphism of local rings $\varphi_P : \mathcal{O}_{\varphi(P),\mathbb{P}^m} \to \mathcal{O}_{P,\mathbb{P}^n}$ is surjective, which implies $\mathcal{O}_{\varphi(P),\varphi(\mathbb{P}^n)} = \mathcal{O}_{\varphi(P),\mathbb{P}^m} / \ker \varphi_P \cong \mathcal{O}_{P,\mathbb{P}^n}$. Hence, $\varphi(P)$ has dimension n.

□

5. Establish the following properties of ample and very ample invertible sheaves on a noetherian scheme X. $\mathcal{L}, \mathcal{M}$ will denote invertible sheaves, and for (d), (e) we assume furthermore that X is of finite type over a noetherian ring A.

(a) If $\mathcal{L}$ is ample and $\mathcal{M}$ is generated by global sections, then $\mathcal{L} \otimes \mathcal{M}$ is ample.

(b) If $\mathcal{L}$ is ample and $\mathcal{M}$ is arbitrary, then $\mathcal{M} \otimes \mathcal{L}^n$ is ample for sufficiently large n.

(c) If $\mathcal{L}, \mathcal{M}$ are both ample, so is $\mathcal{L} \otimes \mathcal{M}$.

(d) If $\mathcal{L}$ is very ample and $\mathcal{M}$ is generated by global sections, then $\mathcal{L} \otimes \mathcal{M}$ is very ample.

(e) If $\mathcal{L}$ is ample, then there is an $n_0 > 0$ such that $\mathcal{L}^n$ is very ample for all $n \geq n_0$.

Proof. Fix a coherent sheaf $\mathcal{F}$ on X.

(a) Let $n \gg 0$ such that $\mathcal{F} \otimes \mathcal{L}^n$ is generated by global sections. If $\mathcal{M}$ is generated by global sections, then so is $\mathcal{M}^n$ for any $n > 0$. More generally, the tensor product of two invertible sheaves generated by global sections is generated by global sections. Indeed, if t_i and r_j are generating global sections of $\mathcal{L}$ and $\mathcal{M}$, then $\mathcal{L} \otimes \mathcal{M}$ is generated by $t_i \otimes r_j$ for all pairs (i, j) (tensor products commutes with colimits). Thus, $\mathcal{F} \otimes \mathcal{L}^n \otimes \mathcal{M}^n \cong \mathcal{F} \otimes (\mathcal{L} \otimes \mathcal{M})^n$ is generated by global sections. Hence, $\mathcal{L} \otimes \mathcal{M}$ is ample.

(b) Let $n > 0$ such that $\mathcal{M} \otimes \mathcal{L}^{n-1}$ is generated by global sections. By (a), $\mathcal{L} \otimes (\mathcal{M} \otimes \mathcal{L}^{n-1}) = \mathcal{M} \otimes \mathcal{L}^n$ is ample.

(c) If $\mathcal{M}$ is ample, then $\mathcal{M}^n$ is generated by global sections for large enough n. Also, $\mathcal{L}^n$ is ample by (II, 7.5), so $\mathcal{L}^n \otimes \mathcal{M}^n$ is ample. Hence, $\mathcal{L} \otimes \mathcal{M}$ is ample by (II, 7.5) again.

(d) Let $i : X \to \mathbb{P}^n_A$ be an immersion such that $\mathcal{L} \cong i^*\mathcal{O}_{\mathbb{P}^n_A}(1)$, and let $\varphi : X \to \mathbb{P}^m_A$ be the unique A-morphism corresponding to $\mathcal{M}$ (II, 7.1). By the universal property of the fiber project $\mathbb{P}^n_A \times_A \mathbb{P}^m_A$, there exists a unique A-morphism $\phi : X \to \mathbb{P}^n_A \times_A \mathbb{P}^m_A$ such that $i = p_n \circ \phi$ and $\varphi = p_m \circ \phi$, where $p_n : \mathbb{P}^n_A \times_A \mathbb{P}^m_A \to \mathbb{P}^n_A$ and $p_m : \mathbb{P}^n_A \times_A \mathbb{P}^m_A \to \mathbb{P}^m_A$ are the natural projection maps. It is not hard to see ϕ is an immersion, and composing it with a Segre embedding (Ex. 5.12), we obtain an immersion $\phi' : X \to \mathbb{P}^N_A$, where $N = nm + n + m$, such that $\mathcal{L} \otimes \mathcal{M} \cong \phi'^*\mathcal{O}(1)$.

(e) If $\mathcal{L}$ is ample, then there exists $n_0 > 0$ such that $\mathcal{L}^{n-1}$ is generated by global sections for all $n \geq n_0$. By (d), $\mathcal{L}^n = \mathcal{L} \otimes \mathcal{L}^{n-1}$ is very ample for all $n \geq n_0$.

□

Chapter 2, Section 8

James Lee

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1. Here we will strengthen the results of the text to include information about the sheaf of differentials at a not necessarily closed point of a scheme X.

(a) Generalize (8.7) as follows. Let B be a local ring containing a field k, and assume that the residue field $k(B) = B/\mathfrak{m}$ of B is a separably generated extension of k. Then the exact sequence of (8.4A),

$$0 \longrightarrow \mathfrak{m}/\mathfrak{m}^2 \xrightarrow{\delta} \Omega_{B/k} \otimes k(B) \longrightarrow \Omega_{k(B)/k} \longrightarrow 0$$

is exact on the left also.

(b) Generalize (8.8) as follows. With B, k as above, assume furthermore that k is perfect, and that B is a localization of an algebra of finite type over k. Then show that B is a regular local ring if and only if $\Omega_{B/k}$ is free of rank $= \dim B + \text{tr. d. } k(B)/k$.

(c) Strengthen (8.15) as follows. Let X be an irreducible scheme of finite type over a perfect field k, and let $\dim X = n$. For any point $x \in X$, not necessarily closed, show that the local ring $\mathcal{O}_{x,X}$ is a regular local ring if and only if the stalk $(\Omega_{X/k})_x$ of the sheaf of differentials at x is free of rank n.

(d) Strengthen (8.16) as follows. If X is a variety over an algebraically closed field k, then $U = \{x \in X \mid \mathcal{O}_x \text{ is regular}\}$ is an open dense subset of X.

Proof.

(a) In copying the proof of (8.7), we want to show the map

$$\delta^\vee : \text{Der}_{k(B)}(B, k(B)) \to \text{Hom}_{k(B)}(\mathfrak{m}/\mathfrak{m}^2, k(B))$$

of dual vector spaces is surjective. If $d : B \to k(B)$ is a derivation, then the $\delta^\vee(d)$ is obtained by restricting to $\mathfrak{m}$. This is well-defined, since $\mathfrak{m} = 0$ in $k(B)$, so $d\mathfrak{m}^2 = \mathfrak{m}d\mathfrak{m} = 0 \subset k(B)$. Now to show $\delta^\vee$ is surjective, let $h \in \text{Hom}_{k(B)}(\mathfrak{m}/\mathfrak{m}^2, k(B))$. Since $B/\mathfrak{m}^2$ is a complete local ring with residue field $k(B)$, there exists a field of representatives $K \subseteq B$ for B (8.25A). Thus, for any $b \in B$, $\bar{b} \in B/\mathfrak{m}^2$, the image of b, can be written as $\bar{b} = \lambda + \bar{c}$, $\lambda \in K, \bar{c} \in \mathfrak{m}/\mathfrak{m}^2$, uniquely. Define $db = h(\bar{c})$. Let $b, b' \in B$, and write $\bar{b} = \lambda + \bar{c}, \bar{b}' = \lambda' + \bar{c}'$ for some $\lambda, \lambda' \in K, \bar{c}', \bar{c}' \in \mathfrak{m}/\mathfrak{m}^2$. Note that $\bar{b} = \lambda, \bar{b}' = \lambda'$ and $d\bar{b} = d\bar{c}, d\bar{b}' = d\bar{c}'$ in $k(B)$, $bb' = \lambda\bar{c}' + \lambda'\bar{c} \in \mathfrak{m}/\mathfrak{m}^2$. Hence, $dbb' = d(\lambda'\bar{c} + \lambda\bar{c}') = \lambda' d\bar{c} + \lambda d\bar{c}' = b'db + bdb'$, so d is a well-defined $k(B)$-derivation.

(b) Immediate by the exact sequence of (a), (8.6A), and (8.8).

(c) If $x \in X$ is any point, then the local ring $B = \mathcal{O}_{x,X}$ has dimension n, residue field some finitely generated, hence separable, extension $k(B)$ (since k is perfect), and is a localization of a k-algebra of finite type. Furthermore, the module $\Omega_{B/k}$ of differentials of B over k is equal to the stalk $(\Omega_{X/k})_x$ of the sheaf $\Omega_{X/k}$. Thus, we can apply (b) and we see that $(\Omega_{X/k})_x$ is free of rank n if and only if B is a regular local ring.

(d) Follows from (c) and (Ex. 5.7a).

□

2. Let X be a variety of dimension n over k. Let $\mathcal{E}$ be a locally free sheaf of rank $> n$ on X, and let $V \subseteq \Gamma(X, \mathcal{E})$ be a vector space of global sections which generate $\mathcal{E}$. Then show that there is an element $s \in V$, such that for each $x \in X$, we have $s_x \notin \mathfrak{m}_x \mathcal{E}_x$. Conclude that there is a morphism $\mathcal{O}_X \to \mathcal{E}$ giving rise to an exact sequence

$$0 \longrightarrow \mathcal{O}_X \longrightarrow \mathcal{E} \longrightarrow \mathcal{E}' \longrightarrow 0$$

where $\mathcal{E}'$ is also locally free.

Proof. Let m be the rank of $\mathcal{E}$, and let $r = \dim_k V$. For any closed point $x \in X$, we can define a map of k-vector spaces $\varphi_x : V \to \mathcal{E}_x/\mathfrak{m}_x\mathcal{E}_x$ in the obvious way. It is surjective by hypothesis, and $\dim_k \mathcal{E}_x/\mathfrak{m}_x\mathcal{E}_x = m$, which shows $r \geq m$. Now considering the vector space V as an affine space over k, consider the subset $B \subseteq X \times V$ consisting of all pairs (x, s) such that $x \in X$ is a closed point and $s \in \ker \varphi_x$. Clearly B is the set of closed points of a closed subset of $X \times V$, which we denote by B, and which we give a reduced induced structure. Consider the first projection $p_1 : B \to X$. It is surjective, with fiber an affine space of dimension $r - m$ (in particular, each fiber is a linear subspace of V). Hence, B is irreducible, and has dimension $r - m + n$. By hypothesis $n < m$, so $\dim B \leq r - 1$. Therefore, considering the second projection $p_2 : B \to V$, we have $\dim p_2(B) \leq r - 1$. Since $\dim V = r$, we conclude that $p_2(B) \subset V$. Pick any $s \in V - p_2(B)$, then $X \times \{s\} \subset X \times V - B$, which is what we wanted to show. For the conclusion, the morphism $\mathcal{O}_X \to \mathcal{E}$ defined by $1 \mapsto s$ gives the desired exact sequence. □

6. *The Infinitesimal Lifting Property.* The following result is very important in studying deformations of nonsingular varieties. Let k be an algebraically closed field, let A be a finitely generated k-algebra such that $\operatorname{Spec} A$ is a nonsingular variety over k. Let $0 \to I \to B' \to B \to 0$ be an exact sequence, where B' is a k-algebra, and I is an ideal with $I^2 = 0$. Finally suppose given a k-algebra homomorphism $f : A \to B$. Then there exists a k-algebra homomorphism $g : A \to B'$ making a commutative diagram

$$\begin{array}{ccc} & & 0 \\ & & \downarrow \\ & & I \\ & & \downarrow \\ & & B' \\ & \overset{g}{\nearrow} & \downarrow \\ A & \xrightarrow[f]{} & B \\ & & \downarrow \\ & & 0 \end{array}$$

We call this result the *Infinitesimal lifting property* for A. We prove this result in several steps.

(a) First suppose that $g : A \to B'$ is a given homomorphism lifting f. If $g' : A \to B'$ is another such homomorphism, show that $\theta = g - g'$ is a k-derivation of A into I, which we can consider as an element of $\operatorname{Hom}_A(\Omega_{A/k}, I)$. Note that since $I^2 = 0$, I has a natural structure of B-module and hence also of A-module. Conversely, for any $\theta \in \operatorname{Hom}_A(\Omega_{A/k}, I)$, $g' = g + \theta$ is another homomorphism lifting f. (For this step, you do not need the hypothesis about $\operatorname{Spec} A$ being nonsingular.)

(b) Now let $P = k[x_1, \ldots, x_n]$ be a polynomial ring over k of which A is a quotient, and let J be the kernel. Show that there does exist a homomorphism $h : P \to B'$ making a commutative diagram,

$$\begin{array}{ccc} 0 & & 0 \\ \downarrow & & \downarrow \\ J & & I \\ \downarrow & & \downarrow \\ P & \xrightarrow{h} & B' \\ \downarrow & & \downarrow \\ A & \xrightarrow[f]{} & B \\ \downarrow & & \downarrow \\ 0 & & 0 \end{array}$$

and show that h induces an A-linear map $\overline{h} : J/J^2 \to I$.

(c) Now use the hypothesis $\operatorname{Spec} A$ nonsingular and (8.17) to obtain an exact sequence

$$0 \longrightarrow J/J^2 \longrightarrow \Omega_{P/k} \otimes A \longrightarrow \Omega_{A/k} \longrightarrow 0 \ .$$

Show furthermore that applying the functor $\mathrm{Hom}_A(\cdot, I)$ gives an exact sequence

$$0 \longrightarrow \mathrm{Hom}_A(\Omega_{A/k}, I) \longrightarrow \mathrm{Hom}_P(\Omega_{P/k}, I) \longrightarrow \mathrm{Hom}_A(J/J^2, I) \longrightarrow 0.$$

Let $\theta \in \mathrm{Hom}_P(\Omega_{P/k}, I)$ be an element whose image gives $\overline{h} \in \mathrm{Hom}_A(J/J^2, I)$. Consider θ as a derivation of P to B'. Then let $h' = h - \theta$, and show that h' is a homomorphism of $P \to B'$ such that $h'(J) = 0$. Thus, h' induces the desired homomorphism $g : A \to B'$.

Proof.

(a) Let $\pi : B' \to B$ be the natural projection homomorphism. If $\pi \circ g = \pi \circ g'$, then $\pi \circ \theta = 0$. Hence, $\theta(A) \subseteq \ker \pi = I$. Let a, a' be elements of A. We have $g(a) = a, g'(a') = a' \in B$, so the natural A-module structure of I gives

$$\begin{aligned}\theta(aa') &= g(a)g(a') - g'(a)g'(a') \\ &= g(a)g(a') - g(a)g'(a') + g(a)g'(a') - g'(a)g'(a') \\ &= g(a)(g(a') - g'(a')) + g'(a')(g(a) - g'(a)) \\ &= a\theta(a') - a'\theta(a).\end{aligned}$$

Also, g, g' are k-linear, so $\theta(\lambda) = 0$ for all $\lambda \in k$. Hence, θ is a k-derivation.

In the converse direction, since $\mathrm{im}\,\theta \subseteq I$, $\pi \circ g = \pi \circ g'$, it is enough to check that g' is a well-defined homomorphism. It is clear it is additive. For any $a, a' \in A$, we have $\theta(a)\theta(a') \in I^2 = 0$ and $g(a)\theta(a') = a\theta(a'), g(a')\theta(a) = a'\theta(a)$. It follows that

$$\begin{aligned}g'(aa') &= g(aa') + \theta(aa') \\ &= g(a)g(a') + a\theta(a') + a'\theta(a) \\ &= g(a)g(a') + a\theta(a') + a'\theta(a) + \theta(a)\theta(a') \\ &= (g(a) + \theta(a))(g(a') + \theta(a')) \\ &= g'(a)g'(a').\end{aligned}$$

(b) For each $i = 1, \ldots, n$, let $y_i \in B$ be the image of $x_i \in P$. Then $f(A) = k[y_i, \ldots, y_n]$, and $\pi : B' \to B$ is surjective, so there exists $z_i \in B'$ such that $\pi(z_i) = y_i$. Let $A' = k[z_1, \ldots, z_n] \subseteq B'$. We have $\pi(A') = f(A)$, and there is a natural map $h : P \to A' \hookrightarrow B'$ defined by $x_i \mapsto z_i$. It satisfies the desired conditions by construction.

To show h induces an A-lienar map $\bar{h} : J/J^2 \to I$, we need to show $h(J) \subseteq I$ and $h(J^2) = 0$. Indeed, the diagram above commutes with h, and J gets mapped to 0 in B. Hence, $h(J) \subseteq \ker \pi = I$. Let $cc' \in J^2$ for some $c, c' \in J$. Then $h(cc') = h(c)h(c') \in I^2 = 0$ since $h(c) \in I$. Hence, we can obtain $\bar{h}$ by restricting h to J and passing to the quotient J/J^2.

(c) Let $X = \mathbb{A}^n = \mathrm{Spec}\, P$, and let $X = \mathrm{Spec}\, A$. By (8.17), (5.5), and (5.10), taking global sections of the exact sequence in (8.17) gives the desired exact sequence.

In general, $\mathrm{Hom}_A(\cdot, I)$ is only left exact. In particular, the sequence

$$0 \longrightarrow \mathrm{Hom}_A(\Omega_{A/k}, I) \longrightarrow \mathrm{Hom}_A(\Omega_{P/k} \otimes A, I) \longrightarrow \mathrm{Hom}_A(J/J^2, I) \longrightarrow 0$$

is not necessarily exact on the right. However, the middle term is isomorphic to $\mathrm{Hom}_P(\Omega_{P/k}, I)$, which by definition can be identified with $\mathrm{Der}_k(P, I)$, the set of all k-derivations of P to I. Noting that P/J^2 has dimension $1 + \dim A$ as a k-vector space by non-singularity of X and k being algebraically closed, we can uniquely write any element of P/J^2 as a sum $\lambda + c$, where $\lambda \in k, c \in J/J^2$. We conclude that the sequence above is exact on the right as well. Imitating the proof of (a), it remains to show $h'(J) = 0$. For any θ, $\theta(J^2) = 0$, passing to the quotient P/J^2 gives $h'(J) = (\bar{h} - \bar{h})(J + J^2) = 0$.

□

Chapter 2, Section 9

James Lee

October 11, 2026

1. Let X be a Noetherian scheme, Y a closed subscheme, and $\hat{X}$ the completion of X along Y. We call the ring $\Gamma(\hat{X}, \mathcal{O}_{\hat{X}})$ the ring of *formal-regular functions* on X along Y. In this exercise we show that if Y is a connected, nonsingular, positive dimensional subvariety of $X = \mathbb{P}^n_k$ over an algebraically closed field k, then $\Gamma(\hat{X}, \mathcal{O}_{\hat{X}}) = k$.

(a) Let $\mathcal{I}$ be the ideal sheaf of Y. Use (8.13) and (8.17) to show that here is an inclusion of sheaves on $Y, \mathcal{I}/\mathcal{I}^2 \hookrightarrow \mathcal{O}_Y(-1)^{n+1}$.

(b) Show that for any $r \geq 1$, $\Gamma(Y, \mathcal{I}^r/\mathcal{I}^{r+1}) = 0$.

(c) Use the exact sequences

$$0 \to \mathcal{I}^r/\mathcal{I}^{r+1} \to \mathcal{O}_X/\mathcal{I}^{r+1} \to \mathcal{O}_X/\mathcal{I}^r \to 0$$

and induction on r to show that $\Gamma(Y, \mathcal{O}_X/\mathcal{I}^r) = k$ for all $r \geq 1$.

(d) Conclude that $\Gamma(\hat{X}, \mathcal{O}_{\hat{X}}) = k$.

2. Use the result of (Ex. 9.1) to prove the following geometric result. Let $Y \subseteq X = \mathbb{P}^n_k$ be as above, and let $f : X \to Z$ be a morphism of k-varieties. Suppose that $f(Y)$ is a single closed point $P \in Z$. Then $f(X) = P$ also.

3. Prove the analogue of (5.6) for formal schemes, which says, if $\mathfrak{X}$ is an affine formal scheme, and if

$$0 \to \mathfrak{F}' \to \mathfrak{F} \to \mathfrak{F}'' \to 0$$

is an exact sequence of $\mathcal{O}_{\mathfrak{X}}$-modules, and if $\mathfrak{F}'$ is coherent, then the sequence of global sections

$$0 \to \Gamma(\mathfrak{X}, \mathfrak{F}') \to \Gamma(\mathfrak{X}, \mathfrak{F}) \to \Gamma(\mathfrak{X}, \mathfrak{F}'') \to 0$$

is exact. For the proof, proceed in the following steps.

(a) Let $\mathfrak{J}$ be an ideal of definition for $\mathfrak{X}$, and for each $n > 0$ consider the exact sequence

$$0 \to \mathfrak{F}'/\mathfrak{J}^n\mathfrak{F}' \to \mathfrak{F}/\mathfrak{J}^n\mathfrak{F} \to \mathfrak{F}'' \to 0.$$

Use (5.6), slightly modified, to show that for every open affine subset $\mathfrak{U} \subseteq \mathfrak{X}$, the sequence

$$0 \to \Gamma(\mathfrak{U}, \mathfrak{F}'/\mathfrak{J}^n\mathfrak{F}') \to \Gamma(\mathfrak{U}, \mathfrak{F}/\mathfrak{J}^n\mathfrak{J}) \to \Gamma(\mathfrak{U}, \mathfrak{F}'') \to 0$$

is exact.

(b) Now pass to the limit, using (9.1), (9.2), and (9.6). Conclude that $\mathfrak{F} \cong \varprojlim \mathfrak{F}/\mathfrak{J}\mathcal{F}$ and that the sequence of global sections above is exact.

4. Use (Ex. 9.3) to prove that if

$$0 \to \mathfrak{F}' \to \mathfrak{F} \to \mathfrak{F}'' \to 0$$

is an exact sequence of $\mathcal{O}_{\mathfrak{X}}$-modules on a Noetherian formal scheme $\mathscr{X}$, and if $\mathfrak{F}', \mathfrak{F}''$ are coherent, then $\mathcal{F}$ is coherent.

5. If $\mathcal{F}$ is a coherent sheaf on a Noetherian formal scheme $\mathscr{X}$, which can be generated by global sections, show in fact that it can be generated by a finite number of its global sections.

6. Let $\mathfrak{X}$ be a Noetherian formal scheme, let $\mathfrak{J}$ be an ideal of definition, and for each n, let Y_n be the scheme $(\mathfrak{X}, \mathcal{O}_{\mathfrak{X}}/\mathfrak{J}^n)$. Assume that the inverse system of groups $(\Gamma(Y_n, \mathcal{O}_{Y_n}))$ satisfies the Mittag-Leffler condition. Then prove that $\operatorname{Pic}\mathfrak{X} = \varprojlim \operatorname{Pic} Y_n$. As in the case of a scheme, we define $\operatorname{Pic}\mathfrak{X}$ to be the group of locally free $\mathcal{O}_{\mathfrak{X}}$-modules of rank 1 under the operation $\otimes$. Proceed in the following steps.

(a) Use the fact that $\ker(\Gamma(Y_{n+1}, \mathcal{O}_{Y_{n+1}})) \to \Gamma(Y_n, \mathcal{O}_{Y_n})$ is a nilpotent ideal to show that the inverse system $(\Gamma(Y_n, \mathcal{O}_{Y_n}^*))$ of units in the respective rings also satisfies (ML).

(b) Let $\mathfrak{F}$ be a coherent sheaf of $\mathcal{O}_{\mathfrak{X}}$-modules, and assume that for each n, there is some isomorphism $\varphi_n : \mathfrak{F}/\mathfrak{J}^n\mathfrak{F} \cong \mathcal{O}_{Y_n}$. Be careful, because the φ_n may not be compatible with the maps in the two inverse systems $(\mathfrak{F}/\mathfrak{J}^n\mathfrak{F})$ and $(\mathcal{O}_{Y_n})$! Then show that there is an isomorphism $\mathfrak{F} \cong \mathcal{O}_{\mathfrak{X}}$. Conclude that the natural map $\operatorname{Pic}\mathfrak{X} \to \varprojlim \operatorname{Pic} Y_n$ is injective.

(c) Given an invertible sheaf $\mathscr{L}_n$ on Y_n for each n, and given isomorphism $\mathscr{L}_{n+1} \otimes \mathcal{O}_{Y_n} \cong \mathscr{L}_n$, construct maps $\mathscr{L}_{n'} \to \mathscr{L}_n$ for each $n' \geq n$ so as to make an inverse system, and show that $\mathfrak{L} = \varprojlim \mathscr{L}_n$ is a coherent sheaf on $\mathfrak{X}$. Then show that $\mathfrak{L}$ is locally free of rank 1, and thus conclude that the map $\operatorname{Pic}\mathfrak{X} \to \varprojlim \operatorname{Pic} Y_n$ is surjective. Again be careful, because even though each $\mathscr{L}_n$ is locally free of rank 1, the open sets needed to make them free might get smaller and smaller with n.

(d) Show that the hypothesis "$(\Gamma(Y_n, \mathcal{O}_{Y_n}))$ satisfies (ML)" is satisfied if either $\mathfrak{X}$ is affine, or each Y_n is projective over a field k.

Chapter 3

Cohomology

Chapter 3, Section 2

James Lee

October 11, 2026

4. *Mayer-Vietoris Sequence.* Let Y_1, Y_2 be two closed subsets of X. Then there is a long exact sequence of cohomology with supports

$$\cdots \longrightarrow \mathrm{H}^i_{Y_1\cap Y_2}(X,\mathcal{F}) \longrightarrow \mathrm{H}^i_{Y_1}(X,\mathcal{F}) \oplus \mathrm{H}^i_{Y_2}(X,\mathcal{F}) \longrightarrow \mathrm{H}^i_{Y_1\cup Y_2}(X,\mathcal{F}) \longrightarrow$$

$$\longrightarrow \mathrm{H}^{i+1}_{Y_1\cap Y_2}(X,\mathcal{F}) \longrightarrow \cdots .$$

Proof. There is an exact sequence of sheaves

$$\begin{array}{ccccccccc}
 & & 0 & & 0 & & 0 & & \\
 & & \downarrow & & \downarrow & & \downarrow & & \\
0 & \longrightarrow & \mathcal{H}^0_{Y_1\cap Y_2}(\mathcal{F}) & \longrightarrow & \mathcal{F} & \longrightarrow & \mathcal{F}_{X-Y_1\cap Y_2} & \longrightarrow & 0 \\
 & & \downarrow & & \downarrow & & \downarrow & & \\
0 & \longrightarrow & \mathcal{H}^0_{Y_1}(\mathcal{F}) \oplus \mathcal{H}^0_{Y_2}(\mathcal{F}) & \longrightarrow & \mathcal{F}\oplus\mathcal{F} & \longrightarrow & \mathcal{F}_{X-Y_1} \oplus \mathcal{F}_{X-Y_2} & \longrightarrow & 0 \\
 & & \downarrow & & \downarrow & & \downarrow & & \\
0 & \longrightarrow & \mathcal{H}^0_{Y_1\cup Y_2}(\mathcal{F}) & \longrightarrow & \mathcal{F} & \longrightarrow & \mathcal{F}_{X-Y_1\cup Y_2} & \longrightarrow & 0 \\
 & & \downarrow & & \downarrow & & \downarrow & & \\
 & & 0 & & 0 & & 0 & &
\end{array}$$

which induces the desired long sequence of cohomology with supports by (1.1A). □

6. Let X be a Noetherian topological space, and let $\{\mathcal{I}_\alpha\}_{\alpha\in A}$ be a direct system of injective sheaves of Abelian groups on X. Then $\varinjlim \mathcal{I}_\alpha$ is also injective.

Proof. We follow the hint. One direction is clear. Conversely, let $i : \mathcal{N} \to \mathcal{M}$ be an injective morphism of sheaves. By the proof of (2.7) we can write $\mathcal{N} = \varinjlim \mathcal{N}_\beta$ where $\mathcal{N}_\beta$ is generated by the sections on some open U_β, and similarly for $\mathcal{M} = \varinjlim \mathcal{M}_\beta$. Notice that we can assume $\mathcal{N}$ and $\mathcal{M}$ are defined over the same direct system so that they belong to the same Abelian category. Thus, the inclusion map $i : \mathcal{N} \to \mathcal{M}$ can be broken down into inclusion maps $i_\beta : \mathcal{N}_\beta \to \mathcal{M}_\beta$. A direct system of morphisms $\mathcal{N}_\beta \to \mathcal{M}$ induces the same inclusion morphism $\mathcal{N} = \varinjlim \mathcal{N}_\beta \to \mathcal{M}$, so we reduce to the case when $\mathcal{N}$ and $\mathcal{M}$ are generated by a single section over some open set U. We have an exact sequence

$$\begin{array}{ccccccccc}
0 & & 0 & & & & & & \\
 & \searrow & \downarrow & & & & & & \\
 & & \mathcal{K} & & & & 0 & & \\
 & & \downarrow & \searrow & & & \downarrow & & \\
0 & \longrightarrow & \mathcal{R} & \longrightarrow & \mathbb{Z}_U & \longrightarrow & \mathcal{N} & \longrightarrow & 0 \\
 & & & & & \searrow & \downarrow & & \\
 & & & & & & \mathcal{M} & & \\
 & & & & & & & \searrow & \\
 & & & & & & & & 0
\end{array}$$

where all the maps are natural, and $\mathcal{R}, \mathcal{K}$ are kernels of the quotients $\mathcal{N}, \mathcal{M}$, respectively. It is not hard to see from above that any $f : \mathcal{N} \to \mathcal{I}$ naturally extends to $\mathcal{M}$, which is what we wanted to show.

Next, we show any subsheaf $\mathcal{R} \subseteq \mathbb{Z}_U$ such that $\mathbb{Z}_U/\mathcal{R}$ is generated by a single section must be finitely generated. Indeed, fix some $x \in X$. Following the proof of (2.7), there exists some open neighborhood $x \in V \subseteq U$ such that $\mathcal{R}\big|_V \cong d \cdot \mathbb{Z}\big|_V$ for some positive integer d. Since X is noetherian, we can cover U by finite number of such V, say V_i for $i = 1, \ldots, n$. Therefore, there is an exact sequence

$$0 \longrightarrow \mathcal{R} \longrightarrow \bigoplus_{i=1}^{n} d_i \cdot \mathbb{Z}_{V_i} \longrightarrow \bigoplus_{i,j,k} d_{ijk} \cdot \mathbb{Z}_{V_i \cap V_j \cap V_k}$$

where d_{ijk} is the minimum of d_i, d_j, d_k. The terms on the right are finitely generated. Thus, $\mathcal{R}$ is finitely generated, and any $\mathcal{R} \to \varinjlim \mathcal{I}_\alpha$ must factor through one of the $\mathcal{I}_\alpha$ (each generator s_i of $\mathcal{R}$ factors through one of the $\mathcal{I}_{\alpha_i}$, so take any $\beta > \alpha_i$, which exists by definition of a direct system). □

7. Let S^1 be the circle (with its usual topology), and let $\mathbb{Z}$ be the constant sheaf $\mathbb{Z}$.

(a) Show that $H^1(S^1, \mathbb{Z}) = \mathbb{Z}$, using our definition of cohomology.

(b) Now let $\mathcal{R}$ be the sheaf of germs of continuous real-valued functions on S^1. Show that $H^1(S^1, \mathcal{R}) = 0$.

Proof.

(a) We remark that cohomology commutes with colimits on paracompact Hausdorff spaces. In particular, the statements of (II, Ex. 1.11), (2.9) hold for S^1. Let A, B be closed subsets of S^1 homeomorphic to the unit interval such that $A \cup B = S^1$ and $A \cap B = \{P, Q\}$ for two distinct points P, Q in S^1. From now on, for any closed subset C of S^1, denote $\mathbb{Z}_C = i_*\mathbb{Z}$, where $i : C \hookrightarrow S^1$ is the inclusion map and $\mathbb{Z}$ is the constant sheaf on C. Without ambiguity $\mathbb{Z}$ will denote the constant sheaf on the ambient space. We claim the following sequence of sheaves

$$0 \longrightarrow \mathbb{Z} \xrightarrow{\Delta} \mathbb{Z}_A \oplus \mathbb{Z}_B \xrightarrow{\tau} \mathbb{Z}_{A \cap B} \longrightarrow 0$$

defined by $\Delta(a) = (a, a)$ and $\tau(a, b) = b - a$ is exact. In particular, there exists natural maps $i_A^\#, i_B^\# : \mathbb{Z} \to \mathbb{Z}_A, \mathbb{Z}_B$ associated to the inclusion maps $i_A, i_B : A, B \hookrightarrow S^1$, so that $\Delta = (i_A^\#, i_B^\#)$. In the same way, the associated morphism of sheaves of the inclusion maps $j_A, j_B : A \cap B \to A, B$ ascends to naturally defined maps $j_A^\#, j_B^\# : \mathbb{Z}_A, \mathbb{Z}_B \to \mathbb{Z}_{A \cap B}$ in the form of a restriction morphism. Thus, $\tau = j_B^\# - j_A^\#$. Exactness can be checked at the level of stalks. Suppose $R \notin A \cap B$. Then either $R \in A$ or $R \in B$, say $R \in A$. Then the stalks are $(\mathbb{Z})_R = \mathbb{Z}$, $(\mathbb{Z}_A)_R = \mathbb{Z}$, $(\mathbb{Z}_B)_R = 0$, which is an exact sequence. If $R \in A \cap B$, then the stalks are $(\mathbb{Z})_R = \mathbb{Z}$, $(\mathbb{Z}_A)_R = \mathbb{Z}$, $(\mathbb{Z}_B)_R = \mathbb{Z}$, $\mathbb{Z}_{A \cap B} = \mathbb{Z}$ defined by Δ and τ, which is clearly exact. Hence, the sequence is exact at all points, so the sequence is exact.

Taking cohomology, we get a long exact sequence of cohomology groups

$$0 \longrightarrow \mathrm{H}^0(S^1, \mathbb{Z}) \longrightarrow \mathrm{H}^0(S^1, \mathbb{Z}_A) \oplus \mathrm{H}^0(S^1, \mathbb{Z}_B) \xrightarrow{\tau_0} \mathrm{H}^0(S^1, \mathbb{Z}_{A \cap B}) \longrightarrow$$

$$\longrightarrow \mathrm{H}^1(S^1, \mathbb{Z}) \longrightarrow \mathrm{H}^1(S^1, \mathbb{Z}_A) \oplus \mathrm{H}^1(S^1, \mathbb{Z}_B) \longrightarrow \mathrm{H}^1(S^1, \mathbb{Z}_{A \cap B}) \longrightarrow \cdots$$

where $\mathrm{H}^i(\mathbb{Z}_A \oplus \mathbb{Z}_B) \cong H^i(\mathbb{Z}_A) \oplus H^i(\mathbb{Z}_B)$ by (2.9). By (2.10), we have

$$\begin{aligned} \mathrm{H}^0(S^1, \mathbb{Z}),\ \mathrm{H}^0(S^1, \mathbb{Z}_A),\ \mathrm{H}^0(S^1, \mathbb{Z}_B) &= \mathbb{Z}, \\ \mathrm{H}^0(\mathbb{Z}_{A \cap B}) &= \mathbb{Z} \oplus \mathbb{Z}, \\ \mathrm{H}^1(\mathbb{Z}_{A \cap B}) &= 0 \end{aligned}$$

The first line follows from the fact that A, B, S^1 are all connected and locally connected. The intersection $A \cap B$ is a noetherian space of dimension zero with two irreducible components, namely the points P and Q, so its space of global sections is free of rank two. Lastly, $H^1(S^1, \mathbb{Z}_{A \cap B}) = H^1(A \cap B, \mathbb{Z}) = 0$ by (2.7). By exactness, we reduce to the following exact sequence

$$0 \longrightarrow \mathbb{Z} \oplus \mathbb{Z} / \operatorname{im} \tau_0 \longrightarrow \mathrm{H}^1(S^1, \mathbb{Z}) \longrightarrow \mathrm{H}^1(S^1, \mathbb{Z}_A) \oplus H^1(S^1, \mathbb{Z}_B).$$

The homomorphism τ_0 is defined by $\tau_0(a,b) = (b-a, b-a)$, which is the diagonal map. Thus, the term on the left is free of rank one. It remains to show $H^1(S^1, \mathbb{Z}_A) = H^1(S^1, \mathbb{Z}_B) = 0$. By (2.10), it suffices to show $H^1(A, \mathbb{Z}) = 0$.

From here, $\mathbb{Z}$ will denote the constant sheaf on A. Identifying A with the closed unit interval $[0,1]$, we repeat the procedure above for A. Pick any $t \in (0,1)$, say $t = 2^{-1}$. Then $X = [0,t]$ and $Y = [t,1]$ cover A, so taking cohomology groups, we get a long exact sequence

$$0 \longrightarrow \mathrm{H}^0(A, \mathbb{Z}) \longrightarrow \mathrm{H}^0(A, \mathbb{Z}_X) \oplus \mathrm{H}^0(A, \mathbb{Z}_Y) \longrightarrow \mathrm{H}^0(A, \mathbb{Z}_{X\cap Y}) \longrightarrow$$

$$\longrightarrow \mathrm{H}^1(A, \mathbb{Z}) \longrightarrow \mathrm{H}^1(A, \mathbb{Z}_X) \oplus \mathrm{H}^1(A, \mathbb{Z}_Y) \longrightarrow \mathrm{H}^1(A, \mathbb{Z}_{X\cap Y}) \longrightarrow \cdots$$

Imitating the previous calculation, the first row is exact and $H^1(A, \mathbb{Z}_{X\cap Y}) = 0$ by (2.7) and (2.10). Thus, we reduce to the following exact sequence

$$0 \longrightarrow \mathrm{H}^1(A, \mathbb{Z}) \longrightarrow \mathrm{H}^1(A, \mathbb{Z}_X) \oplus \mathrm{H}^1(A, \mathbb{Z}_Y) \longrightarrow 0.$$

Since $X \cong A, B$, $\mathrm{H}^1(A, \mathbb{Z}_X), H^1(A, \mathbb{Z}_Y) \cong H^1(A, \mathbb{Z})$ by (2.10), which is possible if and only if $\mathrm{H}^1(A, \mathbb{Z}) = 0$.

(b) Let $\mathcal{M}$ be the sheaf of germs of measurable real-valued functions on S^1 modulo equivalence almost everywhere. It is clearly flasque, since for any measurable $f : V \to \mathbb{R}$ where $V \subseteq U \subseteq \mathbb{R}$ are open sets, the extension of f by zero on U is a measurable function. Thus, we have an exact sequence of sheaves

$$0 \longrightarrow \mathcal{R} \longrightarrow \mathcal{M} \longrightarrow \mathcal{M}/\mathcal{R} \longrightarrow 0.$$

Any measurable function on S^1 is continuous except for on a set of measure zero in S^1, so any global section of $\mathcal{M}/\mathcal{R}$ is locally zero almost everywhere. Since we are considering equivalence classes of functions, where two functions are considered equal if they are equal except for on a measure zero set, we conclude that $\mathcal{M}/\mathcal{R}$ has no non-zero global section. Taking cohomology groups, we have $\mathrm{H}^1(S^1, \mathcal{R}) = 0$.

□

Chapter 3, Section 3

James Lee

October 11, 2026

1. Let X be a Noetherian scheme. Show that X is affine if and only if X_{red} (II, Ex. 2.3) is affine.

   Proof. One direction is clear. Suppose $X_{\text{red}} = \operatorname{Spec} A$ where A is a Noetherian ring with no nilpotent elements, let $f : X_{\text{red}} \to X$ be the natural map, and let $\mathcal{F}$ be any quasi-coherent sheaf on X. Following the hint, consider the filtration
   $$\mathcal{F} \supseteq \mathcal{N} \cdot \mathcal{F} \supseteq \mathcal{N}^2 \cdot \mathcal{F} \supseteq \cdots,$$
   where $\mathcal{N}$ is the sheaf of nilpotent elements on X. Note that $X \cong X_{\text{red}}$ as topological space, and the associated morphism of sheaves $\mathcal{O}_X \to f_*\mathcal{O}_{X_{\text{red}}}$ is surjective with kernel $\mathcal{N}$. Thus, each of the quotients of this filtration can be naturally viewed as A-modules. In particular, we have a natural isomorphism (2.10)
   $$\mathrm{H}^i(X, \mathcal{N}^r \cdot \mathcal{F}/\mathcal{N}^{r+1}\mathcal{F}) \cong \mathrm{H}^i(X_{\text{red}}, f^*(\mathcal{N}^r \cdot \mathcal{F}/\mathcal{N}^{r+1} \cdot \mathcal{F})).$$
   Also, the nilradical of a Noetherian ring is nilpotent, so there exists a positive integer $r > 0$ such that $\mathcal{N}^r = 0$ (A.M. 7.15). Using our hypothesis and (3.7), we climb up the filtration and deduce that $H^1(X, \mathcal{F}) = 0$. Hence, X is affine by (3.7). □

2. Let X be a reduced Noetherian scheme. Show that X is affine if and only if each irreducible component is affine.

   Proof. Suppose $X = \operatorname{Spec} A$ is affine for some reduced Noetherian ring A. The irreducible components of X correspond to the minimal prime ideals $\mathfrak{p}$ of A (A.M. Ex. 1.20). In particular, the irreducible components of X are precisely $\operatorname{Spec} A/\mathfrak{p}$. Conversely, let X_i be the irreducible components of X, and let $\phi : \mathcal{F} \to \bigoplus_i j_*\mathcal{F}\big|_{X_i}$ be the natural map of $\mathcal{O}_X$-modules, where $j : X_i \hookrightarrow X$ is the inclusion. Since X is Noetherian, $X_i \cap X_j$ is quasi-compact, so we can cover it with a finite number of open affine subsets X_{ijk}. Because X is reduced, ϕ is injective, so we can extend ϕ by the following exact sequence
   $$0 \longrightarrow \mathcal{F} \longrightarrow \bigoplus_i j_*\mathcal{F}\big|_{X_i} \longrightarrow \bigoplus_{i,j} j_*\mathcal{F}\big|_{X_{ijk}}.$$
   Each $j_*\mathcal{F}\big|_{X_i}, j_*\mathcal{F}\big|_{X_{ijk}}$ has vanishing cohomology for $i > 0$ by (2.10), (3.5), and (3.7). While the sequence above is not surjective on the right, the image is still a quasi-coherent sheaf, so using the long exact sequence of cohomology, we deduce that $H^i(X, \mathcal{F}) = 0$ for $i > 0$. Hence, X is affine by (3.7). □

6. Let X be a Noetherian scheme.

   (a) Show that the sheaf $\mathcal{G}$ constructed in the proof of (3.6) is an injective object in the category $\mathfrak{Qco}(X)$ of quasi-coherent sheaves on X. Thus, $\mathfrak{Qco}(X)$ has enough injectives.

   (b) Show that any injective object of $\mathfrak{Qco}(X)$ is flasque.

   (c) Conclude that one can compute cohomology as the derived functors of $\Gamma(X, \cdot)$, considered as a functor $\mathfrak{Qco}(X)$ to $\mathfrak{Ab}$.

   Proof.

(a) The Hom functor commutes with finite direct sums in the second argument, so we can assume $\mathcal{G} = j_*\tilde{I}$, where $j : U = \operatorname{Spec} A \to X$ is the inclusion, and I is an injective A-module. Suppose $\mathcal{N} \to \mathcal{M}$ is an injective map of $\mathcal{O}_X$-modules, and we are given any $f : \mathcal{N} \to j_*\tilde{I}$. Since j^* is left exact when j is an open immersion, the induced map of A-modules $j^*\mathcal{N} \to j^*\mathcal{M}$ is also injective. For any such f there is an associated morphism of A-modules $g : j^*\mathcal{N} \to \tilde{I}$ by adjointness of j_*, so there exists an extension of g to $j^*\mathcal{M}$ by injectivity of $\tilde{I}$. By adjointness of j^* again, we obtain a morphism $\mathcal{M} \to j_*\tilde{I}$ that naturally extends f, which is what we wanted to show.

(b) Essentially imitating (a) but replacing i^* with i_* and vice versa, we deduce that $\mathcal{I}\big|_U$ is an injective object of $\mathfrak{Qco}(U)$. Covering X with finite number of open affines $U_i = \operatorname{Spec} A_i$, we have $\mathcal{I}\big|_{U_i} \cong \tilde{I}_i$ for some injective A_i-module I_i for each i by (II, 5.5). Each $\tilde{I}_i$ is flasque by (3.4), so $\mathcal{I}$ is flasque on a local basis. Hence, $\mathcal{I}$ is flasque.

(c) Considering $\Gamma(X, \cdot)$ as a functor from $\mathfrak{Qco}(X)$ to $\mathfrak{Ab}$, we calculate its derived funcotrs by taking injective resolutions in the category $\mathfrak{Qco}(X)$. But any injective is flasque (b), and flasques are acyclic (2.5), so this resolution gives the usual cohomology functors (1.2A).

□

7. Let A be a Noetherian ring, let $X = \operatorname{Spec} A$, let $\mathfrak{a} \subseteq A$ be an ideal, and let $U \subseteq X$ be the open set $X - V(\mathfrak{a})$.

   (a) For any A-module M, establish the following formula of Deligne:

   $$\Gamma(U, \widetilde{M}) \cong \varinjlim_n \operatorname{Hom}_A(\mathfrak{a}^n, M).$$

   (b) Apply this in the case of an injective A-module I, to give another proof of (3.4).

Proof.

(a) To define a map $\phi : \varinjlim_n \operatorname{Hom}_A(\mathfrak{a}^n, M) \to \Gamma(U, \widetilde{M})$, it suffices to define A-homomoprhisms

$$\phi_n : \operatorname{Hom}_A(\mathfrak{a}^n, M) \to \Gamma(U, \widetilde{M})$$

that respect the direct system

$$M \cong \operatorname{Hom}_A(A, M) \xrightarrow{\mu_0} \operatorname{Hom}_A(\mathfrak{a}, M) \xrightarrow{\mu_1} \operatorname{Hom}_A(\mathfrak{a}^2, M) \xrightarrow{\mu_2} \cdots,$$

i.e., $\phi_n = \phi_{n+1} \circ \mu_n$ for all n. By the Noetherian hypothesis, $\mathfrak{a}$ is generated by finitely many elements, say $f_1, \ldots, f_r$. Consider the localization of any $\alpha \in \operatorname{Hom}_A(\mathfrak{a}^n, M)$ with respect to f_i for any i, that is $\alpha_i : (\mathfrak{a}^n)_{f_i} \to M_{f_i}$. Then $(\mathfrak{a}^n)_{f_i} \cong A_{f_i}$ since $f_i^n \in \mathfrak{a}^n$ and f_i is a unit in A_{f_i} (we exclude the case when f_i is nilpotent, since $A_{f_i} = 0$). Let $\phi_n(\alpha)$ be the section equal to $\alpha_i(1)$ on U_i, where U_i is the distinguished open set associated to f_i. We remark that $U = \bigcup_{i=1}^r U_i$, so this definition is well-defined, and it respects the direct system of above since the localization of the inclusion $\mathfrak{a}^n \hookrightarrow \mathfrak{a}^{n-1}$ is the identity map on A_{f_i}. Thus, to show $\phi_n(\alpha)$ is a well-defined section, it is sufficient to show $\alpha_i(1)$ and $\alpha_j(1)$ agree on $U_i \cap U_j \cong \operatorname{Spec} A_{f_i f_j}$ for all i, j. Indeed, $\alpha(f_i^n) = \alpha_i(f_i^n)$ where we naturally view $\alpha(f_i^n)$ as an element of M_{f_i}, α_i is A_{f_i}-linear, and the restriction map $U_i \to U_i \cap U_j$ is given by $m/f_i^n \mapsto f_j^n/(f_i f_j)^n$. Thus, we have

$$\begin{aligned}
\alpha_i(1)\big|_{U_i \cap U_j} &= \alpha_i(f_i^{-n} f_i^n)\big|_{U_i \cap U_j} \\
&= \left.\frac{\alpha(f_i^n)}{f_i^n}\right|_{U_i \cap U_j} \\
&= \frac{f_j^n \alpha(f_i^n)}{(f_i f_j)^n} \\
&= \frac{f_i^n \alpha(f_j^n)}{(f_i f_j)^n} \\
&= \alpha_j(1)\big|_{U_i \cap U_j}.
\end{aligned}$$

Also, each ϕ_n is injective, so the induced map $\phi : \varinjlim_n \operatorname{Hom}_A(\mathfrak{a}^n, M) \to \Gamma(U, \widetilde{M})$ is injective. It remains to show ϕ is surjective. Let $s \in \Gamma(U, \widetilde{M})$ be any section. For each i, we can express $s\big|_{U_i}$ as $m_i/f_i^{n_i}$ for some $n_i > 0$ and $m_i \in M$ such that $m_i/f_i^{n_i}\big|_{U_i \cap U_j} = m_j/f_j^{n_j}\big|_{U_i \cap U_j}$, where $U_i \cap U_j = \operatorname{Spec} A_{f_i f_j}$ for all i, j. Choose $n = n_i$ that works for all i, so we have

$$\frac{f_j^n m_i}{(f_i f_j)^n} = \frac{f_i^n m_j}{(f_i f_j)^n}.$$

We want to show there exists an A-homomoprhism $\alpha : \mathfrak{a}^N \to M$ for some $N \geq n$ such that $\alpha_i(1) = f_i^{N-n} m_i / f_i^N$. Imitating the proof of (3.3), let $\mathfrak{b}_1 \subseteq \mathfrak{b}_2 \subseteq \cdots$ be the sequence of annihilators of $\cdots \subseteq \mathfrak{a}^2 \subseteq \mathfrak{a}$. Since A is Noetherian, there is an r such that $\mathfrak{b}_r = \mathfrak{b}_{r+1} = \cdots$. Define $\sigma : \mathfrak{a}^{n+r} \to M$ by sending f_i^{n+r} to $f_i^r m_i$ for all i and extending by zero. This is a well-defined homomorphism because the annihilator of f_i^{n+r} is $\mathfrak{b}_{n+r} = \mathfrak{b}_r$, and $\mathfrak{b}_r$ annihilates $f_i^r m_i$. Hence, $\phi_{n+r}(\sigma) = s$.

(b) Let I be an injective A-module. It will be sufficient to show for any open set $U \subseteq X$, where $U = X - V(\mathfrak{a})$ for some ideal $\mathfrak{a}$ of A, that $\Gamma(X, \tilde{I}) \to \Gamma(U, \tilde{I})$ is surjective. The map $\Gamma(X, \tilde{I}) \to \Gamma(U, \tilde{I})$ is induced by the natural maps $\mathrm{Hom}_A(A, I) \to \mathrm{Hom}_A(\mathfrak{a}^n, I)$, which is surjective since I is injective. Hence, $\Gamma(X, I) \to \Gamma(U, I)$ is surjective.

□

Chapter 3, Section 4

James Lee

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1. Let $f : X \to Y$ be an affine morphism of Noetherian separated schemes (II, Ex. 5.17). Show that for any quasi-coherent sheaf $\mathcal{F}$ on X, there are natural isomorphisms for all $i \geq 0$,

$$\mathrm{H}^i(X, \mathcal{F}) \cong \mathrm{H}^i(Y, f_*\mathcal{F}).$$

Proof. Let $\mathfrak{V}$ be an open affine cover of Y so that $\mathfrak{U} = (f^{-1}(V))_{V \in \mathfrak{V}}$ is an open affine cover of X. We can compute $H^i(X, \mathcal{F})$ using the Čech complex defined by the open covering $\mathfrak{U}$. Also, $f_*\mathcal{F}$ is a quasi-coherent $\mathcal{O}_Y$-module by (II, 5.8), so the cohomology of $f_*\mathcal{F}$ can be computed via the Čech complex defined by $\mathfrak{V}$ (4.5). Lastly, $f_*\mathcal{F}(V) = \mathcal{F}(U)$ for all $V \in \mathfrak{V}$ and corresponding $U = f^{-1}(V) \in \mathfrak{U}$, so the Čech cohomology of $\mathcal{F}$ and $f_*\mathcal{F}$ with respect to $\mathfrak{U}$ and $\mathfrak{V}$, respectively, are isomorphic. Hence, $\mathrm{H}^i(X, \mathcal{F}) \cong \mathrm{H}^i(Y, f_*\mathcal{F})$ for all $i \geq 0$ by (4.5). □

2. Prove Chevalley's theorem: Let $f : X \to Y$ be a finite surjective morphism of Noetherian separated schemes, with X affine. Then Y is affine.

(a) Let $f : X \to Y$ be a finite surjective morphism of integral Noetherian schemes. Show that there is a coherent sheaf $\mathcal{M}$ on X, and a morphism of sheaves $\alpha : \mathcal{O}_Y^r \to f_*\mathcal{M}$ for some $r > 0$, such that α is an isomorphism at the generic point of Y.

(b) For any coherent sheaf $\mathcal{F}$ on Y, show that there is a coherent sheaf $\mathcal{G}$ on X, and a morphism $\beta : f_*\mathcal{G} \to \mathcal{F}^r$ which is an isomorphism at the generic point of Y.

(c) Now prove Chevalley's theorem.

Proof.

(a) Let $\mathcal{L}$ and $\mathcal{K}$ be the sheaf of total quotient rings of X and Y, respectively. Since the function field of X is a finite field extension of that of Y, there exists an isomorphism $\phi : \mathcal{K}^r \to f_*\mathcal{L}$ for some $r > 0$. Let $\alpha : \mathcal{O}_Y^r \to f_*\mathcal{L}$ be the compositon of the natural map $\mathcal{O}_Y^r \to \mathcal{K}^r$ with ϕ. This is an isomorphism at the generic point of Y. Replacing $\mathcal{L}$ by the sub-$\mathcal{O}_X$-module spanned by the image of $\mathcal{K}^r$, we obtain the desired $\mathcal{M}$.

(b) Let $\mathcal{F}$ be any coherent sheaf $\mathcal{F}$ on Y, $\mathcal{A} = f_*\mathcal{O}_X$, and $\alpha : \mathcal{O}_Y^r \to f_*\mathcal{M}$ as in (a) for some coherent sheaf $\mathcal{M}$ on X. Applying $\mathcal{H}om(\cdot, \mathcal{F})$ to α gives $\beta : \mathcal{H}om(f_*\mathcal{M}, \mathcal{F}) \to \mathcal{F}^r$, which is an isomorphism at the generic point of Y. The $\mathcal{O}_Y$-module $\mathcal{H}om(f_*\mathcal{M}, \mathcal{F})$ is naturally a coherent $\mathcal{A}$-module, and finite morphisms are also affine, so by the equivalence of categories of coherent $\mathcal{O}_X$-modules and coherent $\mathcal{A}$-modules (II, 5.17e), there exists a coherent sheaf $\mathcal{G}$ on X such that $f_*\mathcal{G} \cong \mathcal{H}om(f_*\mathcal{M}, \mathcal{F})$.

(c) Let X_{red} and Y_{red} be the reduced schemes of X and Y, respectively (II, Ex. 2.3). By the universal property of $Y_{\mathrm{red}} \to Y$, there exists a unique morphism $X_{\mathrm{red}} \to Y_{\mathrm{red}}$ making the following diagram commute

$$\begin{array}{ccc} X_{\mathrm{red}} & \longrightarrow & Y_{\mathrm{red}} \\ \downarrow & & \downarrow \\ X & \longrightarrow & Y \end{array}$$

A scheme is affine if and only if its reduced scheme is affine (Ex. 3.1) so assume X and Y are reduced by replacing $X \to Y$ with $X_{\mathrm{red}} \to Y_{\mathrm{red}}$. Because f is surjective, every irreducible component of Y has a nonempty irreducible preimage and has a structure of a closed subscheme of X. All closed subschemes of X are affine (II, Ex. 3.11b), so we reduce to the case X and Y integral.

Let $\mathcal{F}$ be a coherent sheaf on Y. We will show $H^i(Y, \mathcal{F}^r) = 0$ for all $i > 0$ (2.9) (3.7). Let $\beta : f_*\mathcal{G} \to \mathcal{F}^r$ as in (b). Suppose the set Σ of proper closed subsets of Y which are not affine is nonempty, and let

$Z \subset Y$ be a minimal element in Σ. Thus, all proper closed subsets of Z are affine. By replacing X and Y with $f^{-1}Z$ and Z, and $\mathcal{F}$ with any coherent sheaf $\mathcal{F}$ with support in Z, we show Z must be affine, i.e., assume every proper closed subset of Y is affine, and by noetherian induction, this implies Y itself is affine.

By (II, 5.7), $\ker\beta$ and $\operatorname{coker}\beta$ are coherent sheaves on Y, so $K = \operatorname{Supp}(\ker\beta)$ and $C = \operatorname{Supp}(\operatorname{coker}\beta)$ are closed subsets of Y. Also, K and C are both proper subsets because β is an isomorphism at the generic point, so by hypothesis, K and C are affine. For any closed subset $Z \subseteq Y$, let $j : Z \to Y$ denote the inclusion map. We have $\beta \cong j_* j^* \ker\beta$ and similarly for $\operatorname{coker}\beta$, so by (3.7) and (Ex. 4.1),

$$\mathrm{H}^i(Y, \ker\beta) \cong \mathrm{H}^i(K, j^*\ker\beta) = 0, \quad \mathrm{H}^i(Y, \operatorname{coker}\beta) \cong \mathrm{H}^i(C, j^*\operatorname{coker}\beta) = 0.$$

The exact sequence

$$0 \longrightarrow \ker\beta \longrightarrow f_*\mathcal{G} \longrightarrow \mathcal{F}^r \longrightarrow \operatorname{coker}\beta \longrightarrow 0$$

induces two long exact sequences in cohomology

$$0 \longrightarrow \mathrm{H}^0(Y, \ker\beta) \longrightarrow \mathrm{H}^0(Y, f_*\mathcal{G}) \longrightarrow \mathrm{H}^0(Y, \operatorname{im}\beta) \longrightarrow \cdots,$$

$$0 \longrightarrow \mathrm{H}^0(Y, \operatorname{im}\beta) \longrightarrow \mathrm{H}^0(Y, \mathcal{F}^r) \longrightarrow \mathrm{H}^0(Y, \operatorname{coker}\beta) \longrightarrow \cdots.$$

Again by (Ex. 4.1), $\mathrm{H}^i(Y, f_*\mathcal{G}) \cong \mathrm{H}^i(X, \mathcal{G}) = 0$. By exactness of above, we conclude that $\mathrm{H}^i(Y, \mathcal{F}^r) = 0$.

□

3. Let $X = \mathbb{A}^2_k = \operatorname{Spec} k[x, y]$, and let $U = X - \{(0, 0)\}$. Using a suitable cover of U by open affine subsets, show that $H^1(U, \mathcal{O}_U)$ is isomorphic to the k-vector space spanned by $\{x^i y^j \mid i, j < 0\}$. In particular, it is infinite-dimensional.

Proof. Let $\mathfrak{U}$ be the open covering by the two open sets $V = X - \{x = 0\}$ and $U = X - \{y = 0\}$, with affine coordinates obtained by restricting the ones from X. Then the Čech complex has only two terms:

$$C^0 = \Gamma(V, \mathcal{O}_V) \times \Gamma(W, \mathcal{O}_W),$$
$$C^1 = \Gamma(V \cap W, \mathcal{O}_{V \cap W}).$$

Now

$$\Gamma(V, \Omega) = k\left[x, \frac{1}{x}, y\right]$$
$$\Gamma(W, \Omega) = k\left[x, y, \frac{1}{y}\right]$$
$$\Gamma(V, W, \Omega) = k\left[x, y, \frac{1}{x}, \frac{1}{y}\right]$$

and the map $d : C^0 \to C^1$ is given by $(f, g) \mapsto f - g$. To compute H^1, the image of d is the set of all expressions $x^i y^j$ where at least one of i, j is non-negative. Hence, $\mathrm{H}^1(U, \mathcal{O}_U)$ is spanned by $\{x^i y^j \mid i, j < 0\}$. □

4. On an arbitrary topological space X with an arbitrary Abelian sheaf $\mathcal{F}$, Čech cohomology may not give the same result as the derived functor cohomology. But here we show that for H^1, there is an isomorphism if one takes the limit over all coverings.

(a) Let $\mathfrak{U} = (U_i)_{i \in I}$ be an open covering of the topological space X. A *refinement* of $\mathfrak{U}$ is a covering $\mathfrak{V} = (V_j)_{j \in J}$, together with a map $\lambda : J \to I$ of the index sets, such that for each $j \in J$, $V_j \subseteq U_{\lambda(j)}$. If $\mathfrak{V}$ is a refinement of $\mathfrak{U}$, show that there is a natural induced map on Čech cohomology for any Abelian sheaf $\mathcal{F}$, and for each i,

$$\lambda^i : \check{\mathrm{H}}^i(\mathfrak{U}, \mathcal{F}) \to \check{\mathrm{H}}^i(\mathfrak{V}, \mathcal{F}).$$

The coverings of X form a partially ordered set under refinement, so we can consider the Čech cohomology in the limit

$$\varinjlim_{\mathfrak{U}} \check{\mathrm{H}}^i(\mathfrak{U}, \mathcal{F}).$$

(b) For any Abelian sheaf $\mathcal{F}$ on X, show that the natural maps (4.4) for each covering

$$\check{\mathrm{H}}^i(\mathfrak{U}, \mathcal{F}) \to \mathrm{H}^i(X, \mathcal{F})$$

are compatible with the refinement maps above.

(c) Now prove the following theorem. Let X be a topological space, $\mathcal{F}$ a sheaf of Abelian groups. Then the natural map

$$\varinjlim_{\mathfrak{U}} \check{\mathrm{H}}^1(\mathfrak{U}, \mathcal{F}) \to \mathrm{H}^1(X, \mathcal{F})$$

is an isomorphism.

Proof.

(a) An element $\alpha \in C^p(\mathfrak{U}, \mathcal{F})$ is determined by giving an element

$$\alpha_{i_0,\ldots,i_p} \in \mathcal{F}(U_{i_0,\ldots,i_p}),$$

for each $(p+1)$-tuple $i_0 < \cdots < i_p$ of elements of I. We define the map $\Lambda^p : C^p(\mathfrak{U}) \to C^p(\mathfrak{V})$ by setting

$$(\Lambda\alpha)_{j_0,\ldots,j_p} = \alpha_{\lambda(j_0),\ldots,\lambda(j_p)}\big|_{V_{j_0,\ldots,j_p}}$$

if $\lambda(j_i)$ are all distinct, and 0 otherwise. One has $d_{\mathfrak{V}}\Lambda = \Lambda d_{\mathfrak{U}}.$, so Λ induces a map of cohomology groups.

(b) We want to show the following diagram commutes

$$\begin{array}{ccc} & \mathcal{I}^\bullet & \\ \nearrow & & \nwarrow \\ \mathcal{C}^\bullet(\mathfrak{U}, \mathcal{F}) & \longrightarrow & \mathcal{C}^\bullet(\mathfrak{V}, \mathcal{F}) \end{array}$$

where $0 \to \mathcal{F} \to \mathcal{I}^\bullet$ is any injective resolution of $\mathcal{F}$ in $\mathfrak{Ab}(X)$. This follows from the universal property of $\mathcal{I}^\bullet$. In particular, there is only one extension of $\mathcal{F} \to \mathcal{I}^\bullet$ to $\mathcal{C}^\bullet(\mathfrak{U}, \mathcal{F})$.

(c) Embed $\mathcal{F}$ in a flasque sheaf $\mathcal{G}$, and let $\mathcal{R} = \mathcal{G}/\mathcal{F}$, so that we have an exact sequence

$$0 \longrightarrow \mathcal{F} \longrightarrow \mathcal{G} \longrightarrow \mathcal{R} \longrightarrow 0.$$

Define a complex $D^\bullet$ by

$$0 \longrightarrow C^\bullet(\mathfrak{U}, \mathcal{F}) \longrightarrow C^\bullet(\mathfrak{U}, \mathcal{F}) \longrightarrow D^\bullet(\mathfrak{U}) \longrightarrow 0.$$

For any refinement $\mathfrak{V}$ of $\mathfrak{U}$, we have a commutative diagram

$$\begin{array}{ccccccccc} 0 & \longrightarrow & C^\bullet(\mathfrak{U}, \mathcal{F}) & \longrightarrow & C^\bullet(\mathfrak{U}, \mathcal{G}) & \longrightarrow & D^\bullet(\mathfrak{U}) & \longrightarrow & 0 \\ & & \downarrow & & \downarrow & & \downarrow & & \\ 0 & \longrightarrow & C^\bullet(\mathfrak{V}, \mathcal{F}) & \longrightarrow & C^\bullet(\mathfrak{V}, \mathcal{G}) & \longrightarrow & D^\bullet(\mathfrak{V}) & \longrightarrow & 0 \end{array}$$

Let $\mathrm{H}^i_D(\mathfrak{U})$ be the cohomology of the complex $D^\bullet(\mathfrak{U})$. The natural map of complex

$$D^\bullet(\mathfrak{U}) \to C^\bullet(\mathfrak{U}, \mathcal{R})$$

induces a natural map of cohomology groups $H^i_D(\mathfrak{U}) \to \check{H}^i(\mathfrak{U}, \mathcal{R})$. Noting that the Čech cohomology of $\mathcal{G}$ vanishes for $i > 0$, consider the following commutative diagram

$$\begin{array}{ccccccccccc} 0 & \longrightarrow & \varinjlim_{\mathfrak{U}} \check{\mathrm{H}}^0(\mathfrak{U}, \mathcal{F}) & \longrightarrow & \varinjlim_{\mathfrak{U}} \check{\mathrm{H}}^0(\mathfrak{U}, \mathcal{G}) & \longrightarrow & \varinjlim_{\mathfrak{U}} \mathrm{H}^0_D(\mathfrak{U}) & \longrightarrow & \varinjlim_{\mathfrak{U}} \check{\mathrm{H}}^1(\mathfrak{U}, \mathcal{F}) & \longrightarrow & 0 \\ & & \downarrow\wr & & \downarrow\wr & & \downarrow & & \downarrow & & \\ 0 & \longrightarrow & \mathrm{H}^0(X, \mathcal{F}) & \longrightarrow & \mathrm{H}^0(X, \mathcal{G}) & \longrightarrow & \mathrm{H}^0(X, \mathcal{R}) & \longrightarrow & \mathrm{H}^1(X, \mathcal{F}) & \longrightarrow & 0 \end{array}$$

By the Snake lemma, it suffices to show

$$\varinjlim_{\mathfrak{U}} \mathrm{H}^0_D(\mathfrak{U}) \cong \mathrm{H}^0(X, \mathcal{R}).$$

□

5. For any ringed space $(X, \mathcal{O}_X)$, let $\operatorname{Pic} X$ be the group of isomorphism classes of invertible sheaves (II, §6). Show that $\operatorname{Pic} X \cong H^1(X, \mathcal{O}_X^*)$ where $\mathcal{O}_X^*$ denotes the sheaf whose sections over an open set U are the units in the ring $\Gamma(U, \mathcal{O}_X)$, with multiplication as the group operation.

Proof. For any invertible sheaf $\mathcal{L}$ on X, let $\mathfrak{U} = (U_i)_{i \in I}$ be an open cover of X such that $\mathcal{L}\big|_{U_i}$ is free for each i. Fix isomorphisms $\varphi_i : \mathcal{O}_{U_i} \to \mathcal{L}\big|_{U_i}$. Then on $U_i \cap U_j$, we get an isomorphism $\varphi_i^{-1} \circ \varphi_j$ of $\mathcal{O}_{U_i \cap U_j}$ with itself. In particular, $(\varphi_i^{-1} \circ \varphi_j)(1)$ is a unit in $\mathcal{O}_{U_i \cap U_j}$. For another $U_k \in \mathfrak{U}$, one has

$$(\varphi_j^{-1} \circ \varphi_k)(1)\big|_{U_i \cap U_j \cap U_k} \cdot (\varphi_i^{-1} \circ \varphi_j)(1)\big|_{U_i \cap U_j \cap U_k} = (\varphi_i^{-1} \circ \varphi_k)(1)\big|_{U_i \cap U_j \cap U_k}$$

by (I, Ex. 1.22), so the isomorphisms $\varphi_i^{-1} \circ \varphi_j$ give an element of $\check{H}^1(\mathfrak{U}, \mathcal{O}_X^*)$. Define a map

$$L : \varinjlim_{\mathfrak{U}} \check{\mathrm{H}}^1(\mathfrak{U}, \mathcal{O}_X^*) \to \operatorname{Pic} X$$

as follows: for every $\alpha \in \check{\mathrm{H}}^1(\mathfrak{U}, \mathcal{O}_X^*)$, define $L(\alpha)$ to be the invertible sheaf on X obtained by glueing $\mathcal{O}_{U_i}$ via the transition functions defined by $\varphi_{ij}(1) = \alpha_{ij}$. This is an isomorphism by the prevoius remarks, hence $\operatorname{Pic} X \cong \mathrm{H}^1(X, \mathcal{O}_X^*)$ by (Ex. 4.4). □

Chapter 3, Section 5

James Lee

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1. Let X be a projective scheme over a field k, and let $\mathcal{F}$ be a coherent sheaf on X. We define the *Euler characteristic* of $\mathcal{F}$ by

$$\chi(\mathcal{F}) = \sum (-1)^i \dim_k \mathrm{H}^i(X, \mathcal{F}).$$

If

$$0 \longrightarrow \mathcal{F}' \longrightarrow \mathcal{F} \longrightarrow \mathcal{F}'' \longrightarrow 0$$

is a short exact sequence of coherent sheaves on X, show that $\chi(\mathcal{F}) = \chi(\mathcal{F}') + \chi(\mathcal{F}'')$.

Proof. We get an exact sequence of finite dimensional k-vector spaces (5.2)

$$\cdots \xrightarrow{\delta_{i-1}} \mathrm{H}^i(X, \mathcal{F}') \longrightarrow \mathrm{H}^i(X, \mathcal{F}) \longrightarrow \mathrm{H}^i(X, \mathcal{F}'') \xrightarrow{\delta_i} \cdots,$$

where the δ_i's are connecting homomorphisms. By exactness, we have

$$\begin{aligned}
\chi(\mathcal{F}) &= \sum (-1)^i \dim_k \mathrm{H}^i(X, \mathcal{F}) \\
&= \sum (-1)^i (\dim_k \operatorname{coker} \delta_{i-1} + \dim_k \ker \delta_i) \\
&= \sum (-1)^i (\dim_k \mathrm{H}^i(X, \mathcal{F}') - \dim_k \operatorname{im} \delta_{i-1} + \dim_k \mathrm{H}^i(X, \mathcal{F}'') - \dim_k \operatorname{im} \delta_i) \\
&= \sum (-1)^i \dim_k \mathrm{H}^i(X, \mathcal{F}') + \sum (-1)^i \dim_k \mathrm{H}^i(X, \mathcal{F}'') \\
&= \chi(\mathcal{F}') + \chi(\mathcal{F}'').
\end{aligned}$$

□

2. (a) Let X be a projective scheme over a field k, let $\mathcal{O}_X(1)$ be a very ample invertible sheaf on X over k, and let $\mathcal{F}$ be a coherent sheaf on X. Show that there is a polynomial $P(z) \in \mathbb{Q}[z]$, such that $\chi(\mathcal{F}(n)) = P(n)$ for all $n \in \mathbb{Z}$. We call P the *Hilbert polynomial* of $\mathcal{F}$ with respect to the sheaf $\mathcal{O}_X(1)$.

(b) Now let $X = \mathbb{P}^r_k$, and let $M = \Gamma_*(\mathcal{F})$, considered as a graded $S = k[x_0, \ldots, x_r]$-module. Use (5.2) to show that the Hilbert polynomial of $\mathcal{F}$ just defined is the same as the Hilbert polynomial of M defined in (I, §7).

Proof.

(a) Since $\mathcal{O}_X(1)$ is a very ample sheaf on X over k, there is a closed immersion $i : X \to \mathbb{P}^r_k$ of schemes over k, for some r, such that $\mathcal{O}_X(1) = i^*\mathcal{O}_{\mathbb{P}^r}(1)$. If $\mathcal{F}$ is coherent on X, then $i_*\mathcal{F}$ is coherent on $\mathbb{P}^r_k$ (II, Ex. 5.5), and the cohomology is the same (2.10). The Euler characteristic is defined in terms of cohomology, so we reduce to the case $X = \mathbb{P}^r_k$.

Cover X by a finite number of open affines $U_i \cong \operatorname{Spec} k[y_1, \ldots, y_r]$. Then there exist finitely generated $k[y_1, \ldots, y_r]$-modules M_i such that $\mathcal{F}\big|_{U_i} \cong \tilde{M}_i$. Since $\operatorname{Spec} A_i \cap \operatorname{Supp} \mathcal{F}$ is a closed subset of $\operatorname{Spec} A_i$; in fact, $\operatorname{Spec} A_i \cap \operatorname{Supp} \mathcal{F} = V(\operatorname{Ann} M_i)$ (AM, Ex. 3.19), $\operatorname{Supp} \mathcal{F}$ is a closed subset of X. We proceed by induction on $\dim \operatorname{Supp} \mathcal{F}$. If $\operatorname{Supp} \mathcal{F}$ has dimension zero, then $\operatorname{Supp} \mathcal{F} \cap U_i$ is affine (II, Ex. 3.11b) equal to spectra of a Noetherian ring A of dimension zero, which are precisely Artinian rings (AM, 8.5). So $U_i \cap \operatorname{Supp} \mathcal{F}$ is a finite set (AM, Ex. 8.2), and since X can be covered by a finite number of U_i, $\operatorname{Supp} \mathcal{F}$ is a finite set of closed points. Thus, $\mathcal{F}$ is isomorphic to a direct sum of skyscraper sheaves. More precisely, $\mathcal{F} = \bigoplus_{P \in \operatorname{Supp} \mathcal{F}} \mathcal{F}_P$, where $\mathcal{F}_P$ is the stalk of $\mathcal{F}$ at P, and cohomology commutes with direct sums, so by

(Ex. 1), we reduce to the case when $\operatorname{Supp}\mathcal{F}$ consists of a single point. Then, $\chi(\mathcal{F}(n))$ is just the Hilbert polynomial of $\mathcal{F}_P$ (I, 7.5) as a graded $A[y_1, \ldots, y_r]$-module.

Assume the statement to be true for all coherent sheaves with suPort dimension less than n. Consider the exact sequence

$$0 \longrightarrow \mathcal{O}_X(-1) \xrightarrow{x_r} \mathcal{O}_X \longrightarrow \mathcal{O}_H \longrightarrow 0,$$

where H is the hyperplane $x_r = 0$. Taking the tensor product of this sequence with $\mathcal{F}$ gives us

$$0 \longrightarrow \mathcal{R} \longrightarrow \mathcal{F}(-1) \longrightarrow \mathcal{F} \longrightarrow \mathcal{F}_H \longrightarrow 0,$$

where $\mathcal{R}$ is the kernel of the map $\mathcal{F}(-1) \to \mathcal{F}$. Twisting by $\mathcal{O}_X(n)$ is exact, so we get

$$\chi(\mathcal{F}(n)) - \chi(\mathcal{F}(n-1)) = \chi(\mathcal{F}_H(n)) - \chi(\mathcal{R}(n))$$

by (III, Ex. 5.1). Consider the coherent $\mathcal{O}_X$-module $\mathcal{R}$. It vanishes under the multiplication map x_r, where x_r locally defines one of the coordinates such that $x_r = 0$ is the defining equation for H, and such that there is a inclusion $\operatorname{Supp}\mathcal{R} \subseteq H$, which exists by Bertini's theorem. Since $H \cong \mathbb{P}_k^{r-1}$, we can apply the induction hypothesis to $\mathcal{R}$ and $\mathcal{F}_H$, so let $P_{\mathcal{R}}(z), P_{\mathcal{F}_H}(z) \in \mathbb{Q}[z]$ be polynomials such that

$$\chi(\mathcal{R}(n)) = P_{\mathcal{R}}(n), \quad \chi(\mathcal{F}_H(n)) = P_{\mathcal{F}_H}(n).$$

The difference function of $\chi(\mathcal{F}(n))$ is equal to $P_{\mathcal{F}_H}(z) - P_{\mathcal{R}}(n) \in \mathbb{Q}[z]$, so by (§1, 7.3), we conclude that $\chi(\mathcal{F}(n)) = P(n)$ for some $P(z) \in \mathbb{Q}[z]$.

(b) By (5.2), there exists an integer n_0 such that for each $i > 0$ and each $n \geq n_0$, $H^i(X, \mathcal{F}(n)) = 0$. Let $P(n)$ be the Hilbert polynomial just defined, and let $P'(n)$ be the polynomial from associated to the graded S-module M from (I, §7). Then $P(n) = \dim_k H^0(X, \mathcal{F}(n))$ for each $n \geq n_0$, but $M_n = \Gamma(X, \mathcal{F}(n)) = H^0(X, \mathcal{F}(n))$ and $P'(n) = \dim_k M_n$; in particular, $P(n)$ and $P'(n)$ agree at infinitely many values. Hence, they must be equal.

□

3. *Arithmetic Genus.* Let X be a projective scheme of dimension r over a field k. We define the *arithmetic genus* p_a of X by

$$p_a(X) = (-1)^r(\chi(\mathcal{O}_X) - 1).$$

Note that it depends only on X, not on any projective embedding.

(a) If X is integral, and k algebraically closed, show that $\mathrm{H}^0(X, \mathcal{O}_X) \cong k$, so that

$$p_a(X) = \sum_{i=0}^{r-1} (-1)^i \dim_k \mathrm{H}^{r-i}(X, \mathcal{O}_X).$$

In particular, if X is a curve, we have

$$p_a(X) = \dim_k \mathrm{H}^1(X, \mathcal{O}_X).$$

(b) If X is a closed subvariety of $\mathbb{P}_k^r$, show that this $p_a(X)$ coincides with the one defined in (I, Ex. 7.2), which aParently depended on the projective embedding.

(c) If X is a nonsingular projective curve over an algebraically closed field k, show that $p_a(X)$ is in fact a *birational* invariant. Conclude that a nonsingular plane curve of degree $d \geq 3$ is not rational.

Proof.

(a) By (II, Ex. 2.4), there exists a natural bijection

$$\operatorname{Hom}_{\mathbf{Sch}(k)}(X, \mathbb{A}^1) \cong \operatorname{Hom}_{\mathbf{Alg}(k)}(k[t], \Gamma(X, \mathcal{O}_X)).$$

The space on the right is naturally isomorphic to $\mathrm{H}^0(X, \mathcal{O}_X)$ as vector spaces over k, so it is enough to show any morphism $f : X \to \mathbb{A}^1$ over k must be constant. Since X is projective over k, and $\mathbb{A}^1$ is a separated scheme over k, by (II, Ex. 4.8) f is a proper morphism. In particular, f is a closed map, so the image of X in $\mathbb{A}^1$ is a closed subscheme. The proper closed subschemes of $\mathbb{A}^1$ are finite sets of points. But the image of a proper scheme is proper (II, Ex. 4.4), and $\mathbb{A}^1$ is not a proper scheme over k. Hence, f must be constant.

(b) Let $\mathcal{I}_X$ be the ideal sheaf of X on $\mathbb{P}^r_k$. The homogenous coordinate ring $S(X)$ of X for the given embedding is $k[x_0, \ldots, x_r]/I$, where I is the ideal $\Gamma_*(\mathcal{I}_X)$ (II, Ex. 5.14). The result follows by (Ex. 1) and (Ex. 2b).

(c) Two nonsingular projective curves are birational if and only if they are isomorphic (II, 6.7), so they must have the same cohomology. But by the degree-genus formula, any curve of degree ≥ 3 has non-zero arithematic genus, while $\mathbb{P}^1$ has genus zero.

□

6. *Curves on a Nonsingular Quadric Surface.* Let Q be the nonsingular quadric surface $xy = zw$ in $X = \mathbb{P}^3_k$ over a field k. We will consider locally principal closed subschemes Y of Q. These correspond to Cartier divisors on Q by (II, 6.17.1). On the other hand, we know that $\operatorname{Pic} Q \cong \mathbb{Z} \oplus \mathbb{Z}$, so we can talk about the *type* (a, b) of Y (II, 6.16) and (II, 6.6.1). Let us denote the invertible sheaf $\mathcal{L}(Y)$ by $\mathcal{O}_Q(a, b)$. Thus, for any $n \in \mathbb{Z}$, $\mathcal{O}_Q(n) = \mathcal{O}_Q(n, n)$.

(a) Use the special cases $(q, 0)$ and $(0, q)$, with $q > 0$, when Y is a disjoint union of q lines $\mathbb{P}^1$ in Q, to show:
   (1) if $|a - b| \leq 1$, then $\mathrm{H}^1(Q, \mathcal{O}_Q(a, b)) = 0$;
   (2) if $a, b < 0$, then $\mathrm{H}^1(Q, \mathcal{O}_Q(a, b)) = 0$;
   (3) if $a \leq -2$, then $\mathrm{H}^1(Q, \mathcal{O}_Q(a, 0)) \neq 0$.

(b) Now use these results to show:
   (1) if Y is a locally principal closed subscheme of type (a, b) with $a, b > 0$, the Y is connected;
   (2) now assume k is algebraically closed. Then for any $a, b > 0$, there exists an irreducible nonsingular curve Y of type (a, b). Use (II, 7.6.2) and (II, 8.18).
   (3) an irreducible nonsingular curve Y of type (a, b), $a, b > 0$ on Q is projectively normal (II, Ex. 5.14) if and only if $|a - b| \leq 1$. In particular, this gives lots of examples of nonsingular, but not projectively normal curves in $\mathbb{P}^3$. The simplest is the one of type $(1, 3)$, which is just the rational quartic curve (I, Ex. 3.18).

(c) If Y is a locally principal subscheme of type (a, b) in Q, show that $p_a(Y) = ab - a - b + 1 = (a-1)(b-1)$.

Proof.

(a)

(b)

(c)

□

7. Let X (respectively, Y) be proper schemes over a Noetherian ring A. We note by $\mathcal{L}$ an invertible sheaf.

(a) If $\mathcal{L}$ is ample on X, and Y is any closed subscheme of X, then $i^*\mathcal{L}$ is ample on Y, where $i : Y \to X$ is the inclusion.

(b) $\mathcal{L}$ is ample on X if and only if $\mathcal{L}_{\mathrm{red}} = \mathcal{L} \otimes \mathcal{O}_{X_{\mathrm{red}}}$ is ample on X_{red}.

(c) SuPose X is reduced. Then $\mathcal{L}$ is ample on X if and only if $\mathcal{L} \otimes \mathcal{O}_{X_i}$ is ample on X_i, for each irreducible component X_i of X.

(d) Let $f : X \to Y$ be a finite surjective morphism, and let $\mathcal{L}$ be an invertible sheaf on Y. Then $\mathcal{L}$ is ample on Y if and only if $f^*\mathcal{L}$ is ample on X.

Proof.

(a)

(b)

(c)

(d)

□

10. Let X be a projective scheme over a Noetherian ring A, and let $\mathcal{F}^1 \to \mathcal{F}^2 \to \cdots \to \mathcal{F}^r$ be an exact sequence of coherent sheaves on X. Show that there is an integer n_0, such that for all $n \geq n_0$, the sequence of global sections

$$\Gamma(X, \mathcal{F}^1(n)) \longrightarrow \Gamma(X, \mathcal{F}^2(n)) \longrightarrow \cdots \longrightarrow \Gamma(X, \mathcal{F}^r(n))$$

is exact.

Proof. Denote $f_i : \mathcal{F}^i :\to \mathcal{F}^{i+1}$ for each $i = 1, \ldots, n-1$. For every $i = 2, \ldots, n-1$, we have the following three short exact sequence

$$0 \longrightarrow \operatorname{im} f_{i-1} \longrightarrow \mathcal{F}^i \longrightarrow \operatorname{im} f_i \longrightarrow 0,$$

$$0 \longrightarrow \ker f_{i-1} \longrightarrow \mathcal{F}^{i-1} \longrightarrow \operatorname{im} f_{i-1} \longrightarrow 0,$$

$$0 \longrightarrow \operatorname{im} f_i \longrightarrow \mathcal{F}^{i+1} \longrightarrow \operatorname{coker} f_i \longrightarrow 0.$$

Every sheaf above is coherent (II, 5.7). By (5.2), there exists an integer n_i such that for each $i > 0$ and each $n \geq n_i$, the cohomology vanishes for every sheaf, i.e., there is an exact sequence of global sections

$$0 \longrightarrow \Gamma(X, \operatorname{im} f_{i-1}(n)) \longrightarrow \Gamma(X, \mathcal{F}^i(n)) \longrightarrow \Gamma(X, \operatorname{im} f_i(n)) \longrightarrow 0$$

$$0 \longrightarrow \Gamma(X, \ker f_{i-1}(n)) \longrightarrow \Gamma(X, \mathcal{F}^{i-1}(n)) \longrightarrow \Gamma(X, \operatorname{im} f_{i-1}(n)) \longrightarrow 0$$

$$0 \longrightarrow \Gamma(X, \operatorname{im} f_i(n)) \longrightarrow \Gamma(X, \mathcal{F}^{i+1}(n)) \longrightarrow \Gamma(X, \operatorname{coker} f_i(n)) \longrightarrow 0$$

which implies the sequence

$$\Gamma(X, \mathcal{F}^{i-1}(n)) \longrightarrow \Gamma(X, \mathcal{F}^i(n)) \longrightarrow \Gamma(X, \mathcal{F}^{i+1}(n))$$

is exact. Take $n_0 = \max n_i$ to obtain the desired integer. □

Chapter 3, Section 6

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1. Let $(X, \mathcal{O}_X)$ be a ringed space, and let $\mathcal{F}', \mathcal{F}'' \in \mathfrak{Mod}(X)$. An *extension* of $\mathcal{F}''$ by $\mathcal{F}'$ is a short exact sequence

$$0 \longrightarrow \mathcal{F}' \longrightarrow \mathcal{F} \longrightarrow \mathcal{F}'' \longrightarrow 0$$

in $\mathfrak{Mod}(X)$. Two extensions are *isomorphic* if there is an isomorphism of the short exact sequences, inducing the identity maps on $\mathcal{F}'$ and $\mathcal{F}''$. Given an extension as above consider the long exact sequence arising from $\mathrm{Hom}(\mathcal{F}'', \cdot)$, in particular the map

$$\delta : \mathrm{Hom}(\mathcal{F}'', \mathcal{F}'') \to \mathrm{Ext}^1(\mathcal{F}'', \mathcal{F}'),$$

and let $\xi \in \mathrm{Ext}^1(\mathcal{F}'', \mathcal{F}')$ by $\delta(1_{\mathcal{F}''})$. Show that this process gives a one-to-one correspondence between isomorphism classes of extensions of $\mathcal{F}''$ by $\mathcal{F}'$, and elements of the group $\mathrm{Ext}^1(\mathcal{F}'', \mathcal{F}')$.

Proof. Let $0 \to \mathcal{F}' \to \mathcal{I}^\bullet$ be an injective resolution of $\mathcal{F}'$. An element of the group $\xi \in \mathrm{Ext}^1(\mathcal{F}'', \mathcal{F}')$ gives a morphism $\xi : \mathcal{F}'' \to \mathcal{I}^1$ such that

$$\begin{array}{ccc} \mathcal{F}'' & & \\ {\scriptstyle\xi}\downarrow & \searrow^{0} & \\ \mathcal{I}^1 & \xrightarrow{\psi} & \mathcal{I}^2 \end{array}$$

commutes, i.e., $\xi \in \mathrm{Hom}(\mathcal{F}'', \mathcal{I}^0/\mathcal{F}')$. From now, denote this subsheaf of $\mathcal{I}^1$ as $\mathcal{J} = \mathcal{I}^0/\mathcal{F}'$. The pull-back diagram of ξ and ψ gives the following commutative diagram with exact rows

$$\begin{array}{ccccccccc} 0 & \longrightarrow & \mathcal{F}' & \longrightarrow & \mathcal{I}^0 \times_{\mathcal{I}^1} \mathcal{F}'' & \dashrightarrow & \mathcal{F}'' & \longrightarrow & 0 \\ & & \| & & \downarrow & & {\scriptstyle\xi}\downarrow & & \\ 0 & \longrightarrow & \mathcal{F}' & \longrightarrow & \mathcal{I}^0 & \xrightarrow{\psi} & \mathcal{J} & \longrightarrow & 0 \end{array},$$

so the top row is an extension of $\mathcal{F}''$ by $\mathcal{F}'$. If we are given another representative $\bar{\xi}$ of ξ, then $\bar{\xi} - \xi = j \circ \psi$ for some $\psi \in \mathrm{Hom}(\mathcal{F}'', \mathcal{I}^0)$, and repeating the construction above for $j \circ \psi$ gives the trivial extension $\mathcal{F}' \times \mathcal{F}''$ of $\mathcal{F}''$ by $\mathcal{F}'$, so this construction is well-defined. Because $\mathrm{Hom}(\mathcal{F}'', \cdot)$ is a covariant δ-functor, there is a commutative diagram

$$\begin{array}{ccc} \mathrm{Hom}(\mathcal{F}'', \mathcal{F}'') & \xrightarrow{\delta} & \mathrm{Ext}^1(\mathcal{F}'', \mathcal{F}') \\ \downarrow & & \| \\ \mathrm{Hom}(\mathcal{F}'', \mathcal{J}) & \longrightarrow & \mathrm{Ext}^1(\mathcal{F}'', \mathcal{F}') \end{array}$$

which ensures $\delta(1_{\mathcal{F}''}) = \xi$. Hence, this process is surjective.

To prove injectivity, we show $\mathcal{F} \cong \mathcal{I}^0 \times_{\mathcal{I}^1} \mathcal{F}''$. Let $\alpha : \mathcal{F} \to \mathcal{F}''$ be the epimorphism from the extension. By injectivity of $\mathcal{I}^0$, there exists a morphism $\beta : \mathcal{F} \to \mathcal{I}^0$ that extends $\mathcal{F}' \to \mathcal{I}^0$, which induces a morphism $\gamma : \mathcal{F}'' \to \mathcal{I}^1$ making the following diagram commute

$$\begin{array}{ccccccccc} 0 & \longrightarrow & \mathcal{F}' & \longrightarrow & \mathcal{F} & \longrightarrow & \mathcal{F}'' & \longrightarrow & 0 \\ & & \| & & {\scriptstyle\beta}\downarrow & & {\scriptstyle\gamma}\downarrow & & \\ 0 & \longrightarrow & \mathcal{F}' & \longrightarrow & \mathcal{I}^0 & \longrightarrow & \mathcal{J} & \longrightarrow & 0 \end{array}.$$

The pull-back $\mathcal{I}^0 \times_{\mathcal{I}^1} \mathcal{F}''$ is unique up to unique isomorphism, so it remains to show γ is a representative of ξ. Again, $\mathrm{Hom}(\mathcal{F}'', \cdot)$ is a covariant δ-functor with $\mathrm{Hom}(\mathcal{F}'', \mathcal{F}'') \to \mathrm{Hom}(\mathcal{F}'', \mathcal{J})$ given by composition with γ, so γ is a representative of ξ.

□

3. Let X be a Noetherian scheme, and let $\mathcal{F}, \mathcal{G} \in \mathfrak{Mod}(X)$.

(a) If $\mathcal{F}, \mathcal{G}$ are both coherent, then $\mathcal{E}xt^i(\mathcal{F}, \mathcal{G})$ is coherent, for all $i \geq 0$.

(b) If $\mathcal{F}$ is coherent and $\mathcal{G}$ is quasi-coherent, then $\mathcal{E}xt^i(\mathcal{F}, \mathcal{G})$ is quasi-coherent, for all $i \geq 0$.

Proof.

(a) The question is local (6.2), so assume $X = \operatorname{Spec} A$ for some Noetherian ring A, and $\mathcal{F} \cong \tilde{M}, \mathcal{G} \cong \tilde{N}$ for some A-modules M, N. Because A is Noetherian, there exists a finite free resolution of M

$$F_\bullet \longrightarrow M \longrightarrow 0$$

where each $F_\bullet$ is a free A-module of finite rank. Applying the $\tilde{}$-functor and setting $\mathcal{L}_\bullet = \tilde{F}_\bullet$, we obtain a free resolution of $\mathcal{F}$ of finite rank

$$\mathcal{L}_\bullet \longrightarrow \mathcal{F} \longrightarrow 0,$$

which we can use to compute $\mathcal{E}xt^i(\mathcal{F}, \mathcal{G})$ (6.5)

$$\mathcal{E}xt^i(\mathcal{F}, \mathcal{G}) \cong h^i(\mathcal{H}om(\mathcal{L}_\bullet, \mathcal{G})).$$

By equivalence of categories between coherent $\mathcal{O}_X$-modules and finitely generated A-modules (§2, 5.5), one has

$$\mathcal{H}om(\mathcal{L}_\bullet, \mathcal{G}) \cong \widetilde{\mathrm{Hom}_A(F_\bullet, N)}.$$

Each $\mathcal{H}om(\mathcal{L}_\bullet, \mathcal{G})$ is coherent because each $\mathrm{Hom}(F_\bullet, N)$ is finitely generated, and A is Noetherian, so any sub-$\mathcal{O}_X$-module of $\mathcal{H}om(\mathcal{L}_\bullet, \mathcal{G})$ is also coherent (§2, 5.7). Hence, $\mathcal{E}xt^i(\mathcal{F}, \mathcal{G})$ is coherent for every $i \geq 0$.

(b) The proof is essentially the same as (a) except $\mathcal{G} \cong \tilde{N}$ for some A-module N, and the result still follows by (§2, 5.7).

□

6. Let A be a regular local ring, and let M be a finitely generated A-module. In this case, strengthen the result (6.10A) as follows.

(a) M is projective if and only if $\mathrm{Ext}^i(M, A) = 0$ for all $i > 0$.

(b) Use (a) to show that for any n, $\operatorname{pd} M \leq n$ if and only if $\mathrm{Ext}^i(M, A) = 0$ for all $i > n$.

Proof.

(a) Noetherian local rings have finite Krull dimension, so $n = \operatorname{pd}(M) < \infty$ (6.11A), and $\mathrm{Ext}^i_A(M, N) = 0$ for all $i > n$ and all A-modules N (6.10A). Suppose $\mathrm{Ext}^i_A(M, A) = 0$ for all $i > 0$. Let N be any finitely generated A-module. There exists an exact sequence

$$0 \longrightarrow K \longrightarrow A^{\oplus m} \longrightarrow N \longrightarrow 0,$$

where K is some finitely generated A-module. Applying $\mathrm{Hom}_A(M, \cdot)$ gives a long exact sequence in cohomology

$$0 \longrightarrow \mathrm{Hom}_A(M, K) \longrightarrow \mathrm{Hom}_A(M, A^{\oplus m}) \longrightarrow \mathrm{Hom}_A(M, N) \longrightarrow$$

$$\longrightarrow \mathrm{Ext}^1_A(M, K) \longrightarrow \mathrm{Ext}^1_A(M, A^{\oplus m}) \longrightarrow \mathrm{Ext}^1_A(M, N) \longrightarrow \cdots$$

$$\cdots \longrightarrow \mathrm{Ext}^n_A(M, K) \longrightarrow \mathrm{Ext}^n_A(M, A^{\oplus m}) \longrightarrow \mathrm{Ext}^n_A(M, N) \longrightarrow 0.$$

The $\mathrm{Hom}_A(M,\cdot)$ functor commute with direct sums, so $\mathrm{Ext}^i_A(M,A^{\oplus m}) \cong \mathrm{Ext}^i_A(M,A)^{\oplus m} = 0$ for all $i > 0$. By the long exact sequence of above, $\mathrm{Ext}^{i-1}_A(M,N) \cong \mathrm{Ext}^i_A(M,K)$ and $\mathrm{Ext}^n_A(M,N) = 0$. Thus, $\mathrm{Ext}^n_A(M,N) = 0$ for all finitely generated A-modules N, and descending induction on i shows $\mathrm{Ext}^i_A(M,N) = 0$ for all $i > 0$ and all finitely generated A-modules N.

By assumption, M is finitely generated, so there exists an exact sequence

$$0 \longrightarrow Q \longrightarrow A^{\oplus r} \longrightarrow M \longrightarrow 0$$

where Q is a finitely generated A-module. Since $\mathrm{Ext}^1(M,Q) = 0$, which classifies all extensions of M by Q, $A^{\oplus r}$ must be the trivial extension of M by Q, i.e., $A^{\oplus r} \cong M \oplus Q$.

Let $\pi : N' \to N$ be a surjective A-module homomorphism and $\phi : M \to N$ any A-module homomorphism. There is a natural extension of ϕ to $A^{\oplus r}$ via taking the pull-back by the projection $A^{\oplus r} \to M$. Any free module is projective, so there exists unique A-module homomorphism $h : A^{\oplus r} \to N'$ such that $\phi' = \pi \circ h$, so by restriction one has $\phi = \pi \circ h\big|_M$. Hence, M is projective. In particular, a module is projective if and only if it is a direct summand of a free module.

(b) We prove the following general result from (Weibel, 1994):

Lemma 0.1. *Let $F : \mathcal{C} \to \mathcal{D}$ be a left exact contravariant functor between two Abelian categories. Assume $\mathcal{A}$ has enough projectives. If*

$$0 \longrightarrow N \longrightarrow P_m \longrightarrow \cdots \longrightarrow P_0 \longrightarrow M \longrightarrow 0$$

is an exact sequence in $\mathcal{C}$ with the P_i projective (or F-cyclic), then $\mathrm{R}^i F(M) \cong \mathrm{R}^{i-m-1} F(N)$ for $i \geq m+2$.

For each $0 \leq j \leq m+1$, label the morphisms $f_j : P_j \to P_{j-1}$, where P_{-1}, P_{m+1} understood as M, N, respectively. We have an exact sequence

$$0 \longrightarrow \mathrm{im}\, f_{j+1} \longrightarrow P_j \longrightarrow \mathrm{im}\, f_j \longrightarrow 0,$$

which induces the natural isomorphism

$$\mathrm{R}^i F(\mathrm{im}\, f_j) \cong \mathrm{R}^{i-1} F(\mathrm{im}\, f_{j+1})$$

for all $i \geq 2$. Thus,

$$\mathrm{R}^i F(\mathrm{im}\, f_j) \cong \mathrm{R}^{i-k} F(\mathrm{im}\, f_{j+k})$$

for all $i \geq k+1$. Since $\mathrm{im}\, f_0 = M$ and $\mathrm{im}\, f_{m+1} = N$, setting $j = 0$ and $k = m+1$, we obtain our desired result.

Continuing on to our main problem, let M be a finitely generated A-module, and suppose $\mathrm{Ext}^i(M,A) = 0$ for all $i > n$. Let $P_\bullet \to M \to 0$ be any projective resolution of M, and consider the exact sequence

$$0 \longrightarrow N \longrightarrow P_{n-1} \longrightarrow \cdots \longrightarrow P_0 \longrightarrow M \longrightarrow 0$$

where N is the image of $P_{n+1} \to P_n$. Applying Lemma 1 with $F = \mathrm{Hom}(\cdot, A)$ and $i = n+1$ so that $\mathrm{R}^i F = \mathrm{Ext}^i(\cdot, A)$, one has an isomorphism

$$\mathrm{Ext}^{n+1}(M,A) \cong \mathrm{Ext}^1(N,A).$$

But $\mathrm{Ext}^{n+1}(M,A) = 0$, so N is projective by (a). Hence, $\mathrm{pd}\, M \leq n$.

□

7. Let $X = \mathrm{Spec}\, A$ be an affine Noetherian scheme. Let M, N be A-modules, with M finitely generated. Then

$$\mathrm{Ext}^i_X(\tilde{M}, \tilde{N}) \cong \mathrm{Ext}^i_A(M,N)$$

and

$$\mathcal{E}\mathcal{X}\mathcal{t}^i_X(\tilde{M}, \tilde{N}) \cong \widetilde{\mathrm{Ext}^i_A(M,N)}.$$

Proof. Both sides are δ-functors in N from $\mathfrak{Mod}(A)$ to $\mathfrak{Ab}$, and the functors $\mathrm{Hom}_X(\tilde{M}, \tilde{\cdot})$ and $\mathrm{Hom}_A(M, \cdot)$ are equal (§2, 5.5), so their derived functors (as functors from $\mathfrak{Mod}(A)$ to $\mathfrak{Ab}$) are the same.

For the next isomorphism, both sides are δ-functors in N from $\mathfrak{Mod}(A)$ to $\mathfrak{Mod}(X)$ because the $\tilde{}$-functor is an exact functor. Both sides vanish for $i > 0$ and N injective A-module because localization is exact, and $\tilde{N}$ is flasque if N is injective (3.4). Thus, to invoke (1.3A), we need to show the functors are equal for $i = 0$. We check locally. If $x \in X$ is a point, let $\mathfrak{p}$ be the prime ideal in A corresponding to x. By (§2, 5.1) and (6.8), we have

$$\mathcal{E}\mathcal{X}\mathcal{T}^i_X(\tilde{M}, \tilde{N})_x \cong \mathrm{Ext}^i_{A_\mathfrak{p}}(M_\mathfrak{p}, N_\mathfrak{p}), \quad \widetilde{\mathrm{Ext}^i_A(M, N)}_x \cong \mathrm{Ext}^i(M, N)_\mathfrak{p},$$

and there is a natural map of $\mathcal{O}_X$-modules

$$\begin{aligned} \widetilde{\mathrm{Hom}_A(M, N)} &\to \mathcal{H}om_X(\tilde{M}, \tilde{N}) \\ f/s &\mapsto (m \mapsto f(m)/s). \end{aligned}$$

Thus, we reduce to the following algebraic problem: let A be a Noetherian ring, M, N A-modules with M finitely generated. Then the natural map of $A_\mathfrak{p}$-modules

$$\mathrm{Hom}_A(M, N)_\mathfrak{p} \to \mathrm{Hom}_{A_\mathfrak{p}}(M_\mathfrak{p}, N_\mathfrak{p})$$

is an isomorphism. If M is finitely generated, there exists an exact sequence

$$A^{\oplus n} \longrightarrow A^{\oplus r} \longrightarrow M \longrightarrow 0.$$

Applying $\mathrm{Hom}_A(\cdot, N)$ and localizing the first exact sequence gives

$$0 \longrightarrow \mathrm{Hom}_A(M, N)_\mathfrak{p} \longrightarrow \mathrm{Hom}_A(A^{\oplus r}, N)_\mathfrak{p} \longrightarrow \mathrm{Hom}_A(A^{\oplus n}, N)_\mathfrak{p}$$

Switching the order, i.e., localizing first then applying $\mathrm{Hom}_{A_\mathfrak{p}}(\cdot, N_\mathfrak{p})$, gives

$$0 \longrightarrow \mathrm{Hom}_{A_\mathfrak{p}}(M_\mathfrak{p}, N_\mathfrak{p}) \longrightarrow \mathrm{Hom}_{A_\mathfrak{p}}(A_\mathfrak{p}^{\oplus r}, N_\mathfrak{p}) \longrightarrow \mathrm{Hom}_{A_\mathfrak{p}}(A_\mathfrak{p}^{\oplus n}, N_\mathfrak{p}).$$

The statement obvious holds For finite free modules

$$\mathrm{Hom}_A(A^{\oplus r}, N)_\mathfrak{p} \cong N_\mathfrak{p}^{\oplus r} \cong \mathrm{Hom}_{A_\mathfrak{p}}(A_\mathfrak{p}^{\oplus r}, N_\mathfrak{p})$$

We obtain our desired isomorphism by applying the five lemma to the following commutative diagram

$$\begin{array}{ccccccc} 0 & \longrightarrow & \mathrm{Hom}_A(M, N)_\mathfrak{p} & \longrightarrow & \mathrm{Hom}_A(A^{\oplus r}, N)_\mathfrak{p} & \longrightarrow & \mathrm{Hom}_A(A^{\oplus n}, N)_\mathfrak{p} \\ & & \downarrow & & \downarrow & & \downarrow \\ 0 & \longrightarrow & \mathrm{Hom}_{A_\mathfrak{p}}(M_\mathfrak{p}, N_\mathfrak{p}) & \longrightarrow & \mathrm{Hom}_{A_\mathfrak{p}}(A_\mathfrak{p}^{\oplus r}, N_\mathfrak{p}) & \longrightarrow & \mathrm{Hom}_{A_\mathfrak{p}}(A_\mathfrak{p}^{\oplus n}, N_\mathfrak{p}) \end{array}.$$

□

8. Prove the following theorem of Kleiman: if X is a Noetherian, integral, separated, locally factorial scheme, then every coherent sheaf on X is a quotient of a locally free sheaf (of finite rank).

(a) First show that open sets of the form X_s, for various $s \in \Gamma(X, \mathcal{L})$, and various invertible sheaves $\mathcal{L}$ on X, form a base for the topology of X.

(b) Now use (II, 5.14) to show that any coherent sheaf is a quotient of a direct sum $\bigoplus \mathcal{L}_i^{n_i}$ for various invertible sheaves $\mathcal{L}_i$ and various integers n_i.

10. *Duality for a Finite Flat Morphism.*

(a) Let $f : X \to Y$ be a finite morphism of Noetherian schemes. For any quasi-coherent $\mathcal{O}_Y$-module $\mathcal{G}$, $\mathcal{H}om_Y(f_*\mathcal{O}_X, \mathcal{G})$ is a quasi-coherent $f_*\mathcal{O}_X$-module, hence corresponds to a quasi-coherent $\mathcal{O}_X$-module, which we call $f^!\mathcal{G}$ (II, Ex. 5.17e).

(b) Show that for any coherent $\mathcal{F}$ on X and any quasi-coherent $\mathcal{G}$ on Y, there is a natural isomorphism

$$f_* \mathcal{H}om_X(\mathcal{F}, f^!\mathcal{G}) \xrightarrow{\sim} \mathcal{H}om_Y(f_*\mathcal{F}, \mathcal{G}).$$

(c) For each $i \geq 0$, there is a natural map

$$\varphi_i : \mathrm{Ext}^i_X(\mathcal{F}, f^!\mathcal{G}) \to \mathrm{Ext}^i_Y(f_*\mathcal{F}, \mathcal{G}).$$

(d) Now assume that X and Y are separated, $\mathfrak{Coh}(X)$ has enough locally frees, and assume that $f_*\mathcal{O}_X$ is locally free on Y (this is equivalent to saying f flat - see §9). Show that φ_i is an isomorphism for all i, all $\mathcal{F}$ coherent on X, and all $\mathcal{G}$ quasi-coherent on Y.

Proof.

(b) The question is local on Y, so we can assume $Y = \operatorname{Spec} A$ for some Noetherian ring A. Then $X = \operatorname{Spec} B$ for some finite A-algebra B so that

$$f_*\mathcal{O}_X \cong \tilde{B}, \quad \mathcal{F} \cong \tilde{M}, \quad \mathcal{G} \cong \tilde{N}$$

for some finitely generated B-module M, and A-module N. Now, the statement just says there is a natural isomorphism of A-modules

$$\mathrm{Hom}_B(M, \mathrm{Hom}_A(B, N)) \xrightarrow{\sim} \mathrm{Hom}_A(M, N).$$

(c) If $\mathcal{H}$ is any quasi-coherent $\mathcal{O}_X$-module, there is a natural map

$$\mathrm{Hom}_X(\mathcal{F}, \mathcal{H}) \to \mathrm{Hom}_Y(f_*\mathcal{F}, f_*\mathcal{H})$$

by functoriality of f_*. Note that f_* is an exact functor when f is an a finite morphism (8.1), so both $\mathrm{Ext}^i_X(\mathcal{F}, \cdot)$ and $\mathrm{Ext}^i_Y(f_*\mathcal{F}, f_*\cdot)$ is a δ-functor in $\mathcal{H}$ from $\mathfrak{Mod}(X)$ to $\mathfrak{Ab}$. Since $\mathrm{Ext}^i_X(\mathcal{F}, \cdot)$ is a universal δ-functor (1.4), there is an induced natural map by definition

$$\psi_i : \mathrm{Ext}^i_X(\mathcal{F}, \mathcal{H}) \to \mathrm{Ext}^i_Y(f_*\mathcal{F}, f_*\mathcal{H}).$$

Now, consider the case $\mathcal{H} = f^!\mathcal{G}$. Taking the global sections of (b) shows

$$\mathrm{Hom}_X(f^!\mathcal{G}, f^!\mathcal{G}) \cong \mathrm{Hom}_Y(f_*f^!\mathcal{G}, \mathcal{G}),$$

so let $\xi : f_*f^!\mathcal{G} \to \mathcal{G}$ be the morphism corresponding to the identitiy $1_{f^!\mathcal{G}}$. Then ξ induces a map of derived functors

$$\xi_i : \mathrm{Ext}^i_Y(f_*\mathcal{F}, f_*f^!\mathcal{G}) \to \mathrm{Ext}_Y(f_*\mathcal{F}, \mathcal{G}).$$

Hence, $\varphi_i = \xi_i \circ \psi_i$ is our desired natural map.

□

Chapter 3, Section 7

James Lee

October 11, 2026

1. Let X be an integral projective scheme of dimension ≥ 1 over a field k, and let $\mathcal{L}$ be an ample invertible sheaf on X. Then $\mathrm{H}^0(X, \mathcal{L}^{-1}) = 0$.

Proof. A global section of $\mathcal{L}^{-1}$ defines an $\mathcal{O}_X$-module morphism

$$\varphi : \mathcal{L} \to \mathcal{O}_X,$$

and hence induces morphisms

$$\varphi^{\otimes n} : \mathcal{L}^{\otimes n} \to \mathcal{O}_X$$

for all $n > 0$. Since $\mathcal{L}$ is locally free of rank one, φ is locally given by multiplication by a section of $\mathcal{O}_X$. If for some $x \in X$ the stalk map $\varphi_x^{\otimes n}$ is not injectve, it must be the zero map. As X is integral, this forces φ to vanish at the generic point, hence $\varphi = 0$. Assume therefore $\varphi \neq 0$. Since the global section functor is left exact, $\varphi^{\otimes n}$ induces an injective map

$$\Gamma(\varphi^{\otimes n}) : \mathrm{H}^0(X, \mathcal{L}^{\otimes n}) \to \mathrm{H}^0(X, \mathcal{O}_X).$$

However, $\mathrm{H}^0(X, \mathcal{O}_X)$ is finite dimensional over k (5.2), while for $n \geqslant 0$, we have

$$\dim_k \mathrm{H}^0(X, \mathcal{L}^{\otimes n}) = \chi(\mathcal{L}^{\otimes n}),$$

which is a polynomial in n (5.3), and hence eventually exceeds $\dim_k \mathrm{H}^0(X, \mathcal{O}_X)$. This is a contradiction, and therefore $\varphi = 0$. □

2. Let $f : X \to Y$ be a finite morphism of projective schemes of the same dimension over a field k, and let ω_Y° be a dualizing sheaf for Y.

(a) Show that $f^!\omega_Y^\circ$ is a dualizing sheaf for X, where $f^!$ is defined as in (Ex. 6.10).

(b) If X and Y are both nonsingular, and k algebraically closed, conclude that there is a natural trace map $t : f_*\omega_X \to \omega_Y$.

Proof.

(a) We must first define a trace map $t_X : \mathrm{H}^n(X, f^!\omega_Y^\circ) \to k$. By (Ex. 4.1), we have a natural isomorphism

$$\mathrm{H}^n(X, f^!\omega_Y^\circ) \xrightarrow{\sim} \mathrm{H}^n(Y, f_*f^!\omega_Y^\circ).$$

There is a natural morphism of $\mathcal{O}_Y$-modules $f_*f^!\omega_Y \to \omega_Y$, which induces a map between cohomology

$$\mathrm{H}^n(Y, f_*f^!\omega_Y^\circ) \to \mathrm{H}^n(Y, \omega_Y^\circ).$$

Composing with the trace map of ω_Y° gives us a trace map for $f^!\omega_Y^\circ$. It remains to show t_X satisfies the universal property, so let $\mathcal{F}$ be a coherent sheaf on X. Then $f_*\mathcal{F}$ is also a coherent sheaf on Y, so by (Ex. 6.10), we have

$$\mathrm{Hom}_X(\mathcal{F}, f^!\omega_Y) \cong \mathrm{Hom}_Y(f_*\mathcal{F}, \omega_Y) \cong \mathrm{H}^n(Y, f_*\mathcal{F})^\vee \cong \mathrm{H}^n(X, \mathcal{F})^\vee.$$

(b) By (Ex. 6.10), defining a morphim $t : f_*\omega_X \to \omega_Y$ is the same as defining a morphism $\omega_X \to f^!\omega_Y$, so we can let t be the morphism corresponding to a fixed (unique) isomorphism $\omega_X \to f^!\omega_Y$.

□

3. Let $X = \mathbb{P}^n_k$. Show that $\mathrm{H}^q(X, \Omega^p_X) = 0$ for $p \neq q$, k for $p = q$, $0 \leq p, q \leq n$.

Proof. Consider the Euler sequence of (II, 8.13)

$$0 \longrightarrow \Omega_X \longrightarrow \mathcal{O}_X(-1)^{\oplus n+1} \longrightarrow \mathcal{O}_X \longrightarrow 0.$$

For each $0 \leq p \leq n$, let $\mathcal{G}^{p+1} = \bigwedge^{p+1} \mathcal{O}_X(-1)^{\oplus n+1}$, which is isomorphic to the direct sum of $\binom{n+1}{p+1}$ copies of $\mathcal{O}_X(-1)$. By (II, Ex. 5.16c), for each $0 \leq p \leq n$, there exists a filtration $\{\mathcal{F}^q\}$ of $\mathcal{G}^{p+1}$ such that

$$\mathcal{F}^q/\mathcal{F}^{q+1} \cong \Omega^q_X \otimes \bigwedge^{p+1-q} \mathcal{O}_X = \begin{cases} \Omega^q_X & q = p, p+1 \\ 0 & \text{otherwise.} \end{cases}.$$

Since $\mathcal{O}_X$ has rank one,

$$\mathcal{G}^{p+1} = \mathcal{F}^0 = \mathcal{F}^1 = \cdots = \mathcal{F}^p, \quad \mathcal{F}^p/\mathcal{F}^{p+1} \cong \Omega^p_X, \quad \mathcal{F}^{p+1} \cong \Omega^{p+1}_X,$$

which can be put into the short exact sequence

$$0 \longrightarrow \Omega^{p+1}_X \longrightarrow \mathcal{G}^{p+1} \longrightarrow \Omega^p_X \longrightarrow 0.$$

Taking the long exact sequence of cohomology gives

$$\cdots \longrightarrow \mathrm{H}^i(X, \mathcal{G}^{p+1}) \longrightarrow \mathrm{H}^i(X, \Omega^p_X) \longrightarrow \mathrm{H}^{i+1}(X, \Omega^{p+1}_X) \longrightarrow \mathrm{H}^{i+1}(X, \mathcal{G}^{p+1}) \longrightarrow \cdots$$

Direct sums commute with cohomology (2.9), and $\mathrm{H}^i(X, \mathcal{O}_X(-1)) = 0$ for all i, which means $\mathrm{H}^i(X, \mathcal{G}^{p+1}) = 0$, so from the long exact sequence, we obtain the isomorphisms

$$\mathrm{H}^i(X, \Omega^p_X) \cong \mathrm{H}^{i+1}(X, \Omega^{p+1}_X).$$

Hence, if $q \geq p$,

$$\mathrm{H}^q(X, \Omega^p_X) \cong \mathrm{H}^{q-p}(X, \mathcal{O}_X) = \begin{cases} k & p = q, \\ 0 & \text{otherwise.} \end{cases}$$

The other Hodge numbers are obtained via Serre duality. □

4. *The Cohomology Class of a Subvariety.* Let X be a nonsingular projective variety of dimension n over an algebraically closed field k. Let Y be a nonsingular subvariety of codimension p (hence dimension $n-p$). From the natural map $\Omega_X \otimes \mathcal{O}_Y \to \Omega_Y$ of (II, 8.12) we deduce a map $\Omega^{n-p}_X \to \Omega^{n-p}_Y$. This induces a map on cohomology $\mathrm{H}^{n-p}(X, \Omega^{n-p}_X) \to \mathrm{H}^{n-p}(Y, \Omega^{n-p}_Y)$. Now $\Omega^{n-p}_Y = \omega_Y$ is a dualizing sheaf for Y, so we have the trace map $t_Y : \mathrm{H}^{n-p}(Y, \Omega^{n-p}_Y) \to k$. Composing, we obtain a linear map $\mathrm{H}^{n-p}(X, \Omega^{n-p}_X) \to k$. By (7.13) this corresponds to an element $\eta(Y) \in \mathrm{H}^p(X, \Omega^p_X)$, which we call the *cohomology class* of Y.

(a) If $P \in X$ is a closed point, show that $t_X(\eta(P)) = 1$, where $\eta(P) \in \mathrm{H}^n(X, \Omega^n)$ and t_X is the trace map.

(b) If $X = \mathbb{P}^n$, identify $\mathrm{H}^p(X, \Omega^p)$ with k by (Ex. 7.3), and show that $\eta(Y) = (\deg Y) \cdot 1$, where $\deg Y$ is its *degree* as a projective variety (I, §7).

(c) For any scheme X of finite type over k, we define a homomorphism of sheaves of Abelian groups $\mathrm{d}\log : \mathcal{O}^*_X \to \Omega_X$ by $\mathrm{d}\log(f) = f^{-1}df$. Here $\mathcal{O}^*_X$ is a group under multiplication, and Ω_X is a group under addition. This induces a map on cohomology $\operatorname{Pic} X = \mathrm{H}^1(X, \mathcal{O}^*_X) \to \mathrm{H}^1(X, \Omega_X)$ which we denote by c.

(d) Returning to the hypotheses above, suppose $p = 1$. Show that $\eta(Y) = c(\mathcal{L}(Y))$, where $\mathcal{L}(Y)$ is the invertible sheaf corresponding to the divisor Y.

Chapter 3, Section 8

James Lee

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1. Let $f : X \to Y$ be a continuous map of topological spaces. Let $\mathcal{F}$ be a sheaf of Abelian groups on X, and assume that $\mathrm{R}^i f_*(\mathcal{F}) = 0$ for all $i > 0$. Show that there are natural isomorphisms, for each $i \geq 0$,

$$\mathrm{H}^i(X, \mathcal{F}) \cong \mathrm{H}^i(Y, f_*\mathcal{F}).$$

Proof. Let $0 \to \mathcal{F} \to \mathcal{I}^\bullet$ be an injective resolution of $\mathcal{F}$. By hypothesis, $0 \to f_*\mathcal{F} \to f_*\mathcal{I}^\bullet$ is also exact. The pushforward of an injective object in $\mathfrak{Ab}(X)$ is also injective in $\mathfrak{Ab}(Y)$ because f_* is a right adjoint functor, so $0 \to f_*\mathcal{F} \to f_*\mathcal{I}^\bullet$ is an injective resolution of $f_*\mathcal{F}$ in $\mathfrak{Ab}(Y)$. Suppose $\mathcal{G} \to \mathcal{G}'$ is a monomorphism of sheaves on Y, and $\mathcal{G} \to f_*\mathcal{I}$ is any morphism of sheaves. By adjunction, there is a morphism $f^{-1}\mathcal{G} \to \mathcal{I}$, and because f^{-1} is an exact functor, $f^{-1}\mathcal{G} \to f^{-1}\mathcal{G}'$ is also a monomorphism (note that in general, f^* is only right exact). By injectivity of $\mathcal{I}$, there is an extension of $f^{-1}\mathcal{G} \to \mathcal{I}$ making the following diagram commute

$$\begin{array}{ccc} & & \mathcal{I} \\ & \nearrow & \uparrow \\ f^{-1}\mathcal{G} & \longrightarrow & f^{-1}\mathcal{G}' \end{array} .$$

Hence, by adjunction again, there is an extension of $\mathcal{G} \to f_*\mathcal{I}$ making the following diagram commute

$$\begin{array}{ccc} & & f_*\mathcal{I} \\ & \nearrow & \uparrow \\ \mathcal{G} & \longrightarrow & \mathcal{G}' \end{array} .$$

Since $\mathcal{F}$ and $f_*\mathcal{F}$ have same global sections for any $\mathcal{F}$, there are natural identificiations

$$\begin{array}{ccccc} 0 & \longrightarrow & \Gamma(X, \mathcal{F}) & \longrightarrow & \Gamma(X, f_*\mathcal{I}^\bullet) \\ & & \| & & \| \\ 0 & \longrightarrow & \Gamma(Y, f_*\mathcal{F}) & \longrightarrow & \Gamma(Y, f_*\mathcal{I}^\bullet) \end{array} .$$

Taking cohomology gives the desired natural isomorphisms for all $i \geq 0$. □

2. Let $f : X \to Y$ be an affine morphism of schemes (II, Ex. 5.17) with X Noetherian, and let $\mathcal{F}$ be a quasi-coherent sheaf on X. Show that the hypotheses of (Ex. 8.1) are satisfies, and hence that $\mathrm{H}^i(X, \mathcal{F}) \cong \mathrm{H}^i(Y, f_*\mathcal{F})$ for each $i \geq 0$.

Proof. Let V be an open affine subset of Y. By (8.2) and (8.5), we have

$$\mathrm{R}^i f_*(\mathcal{F})\big|_V \cong \mathrm{H}^i(\widetilde{f^{-1}V, \mathcal{F}\big|_{f^{-1}(V)}}),$$

which is zero because $f^{-1}(V)$ is affine (3.7). Hence, $\mathrm{R}^i f_*(\mathcal{F}) = 0$, and the result follows by the previous exercise. □

3. Let $f : X \to Y$ be a morphism of ringed spaces, let $\mathcal{F}$ be an $\mathcal{O}_X$-module, and let $\mathcal{E}$ be a locally free $\mathcal{O}_Y$-module of finite rank. Prove the *projection formula* (cf. (II, Ex. 5.1))

$$\mathrm{R}^i f_*(\mathcal{F} \otimes f^*\mathcal{E}) \cong \mathrm{R}^i f_*(\mathcal{F}) \otimes \mathcal{E}.$$

Proof. The pullback of a locally free sheaf of finite rank is also locally free of finite rank. Localization is an exact monoidal functor from the category of $\mathcal{O}_Y$-modules to the category of Abelian groups, so by checking exactness at stalks, tensoring by $\mathcal{E}$ is an exact functor, and it also has a left adjoint, where for any $\mathcal{O}_Y$-modules $\mathcal{M}$ and $\mathcal{N}$ (II, Ex. 5.1)

$$\begin{aligned}\mathrm{Hom}_{\mathcal{O}_Y}(\mathcal{M}\otimes\mathcal{E}^\vee,\mathcal{N}) &\cong \mathrm{Hom}_{\mathcal{O}_Y}(\mathcal{M},\mathcal{H}om_{\mathcal{O}_Y}(\mathcal{E}^\vee,\mathcal{N}))\\ &\cong \mathrm{Hom}_{\mathcal{O}_Y}(\mathcal{M},\mathcal{N}\otimes(\mathcal{E}^\vee)^\vee)\\ &\cong \mathrm{Hom}_{\mathcal{O}_Y}(\mathcal{M},\mathcal{N}\otimes\mathcal{E}).\end{aligned}$$

Hence, the result follows from the $i=0$ case, (1.3A), and Lemma 1.

It remains to show

$$f_*(\mathcal{F}\otimes f^*\mathcal{E})\cong f_*(\mathcal{F})\otimes\mathcal{E}.$$

We first show there is a natural map

$$\phi : f_*(\mathcal{F})\otimes\mathcal{E}\to f_*(\mathcal{F}\otimes f^*\mathcal{E}).$$

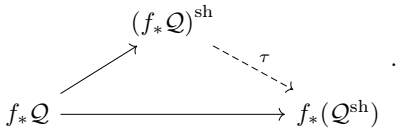
Let $\mathcal{P}$ be the presheaf on Y defined for open subsets $V\subseteq Y$ by

$$V\mapsto f_*(\mathcal{F})(V)\otimes\mathcal{E}(V)$$

so that $\mathcal{P}^{\mathrm{sh}}\cong f_*(\mathcal{F})\otimes\mathcal{E}$, and let $\mathcal{Q}$ be the presheaf on X defined for open subsets $U\subseteq X$ by

$$U\mapsto \mathcal{F}(U)\otimes f^*\mathcal{E}(U)$$

so that $\mathcal{Q}^{\mathrm{sh}}\cong\mathcal{F}\otimes f^*\mathcal{E}$. Sheafification and pushforward do not commute; nonetheless there is a natural morphism $(f_*\mathcal{Q})^{\mathrm{sh}}\to f_*(\mathcal{Q}^{\mathrm{sh}})$. In particular, consider the natural morphism $\mathcal{Q}\to\mathcal{Q}^{\mathrm{sh}}$. Taking the pushforward gives $f_*\mathcal{Q}\to f_*(\mathcal{Q}^{\mathrm{sh}})$. Because $f_*(\mathcal{Q}^{\mathrm{sh}})$ is already a sheaf, by the universal property of sheafification (II, 1.2), there is unique map $\tau:(f_*\mathcal{Q})^{\mathrm{sh}}\to f_*(\mathcal{Q}^{\mathrm{sh}})$ fitting into the commutative diagram



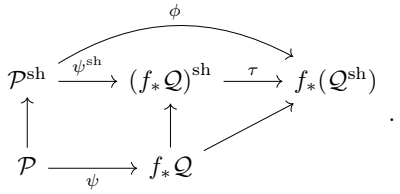
Thus, we define a morphism of presheaves $\psi:\mathcal{P}\to f_*\mathcal{Q}$, which induces a morphism of sheaves $\psi^{\mathrm{sh}}:\mathcal{P}^{\mathrm{sh}}\to(f_*\mathcal{Q})^{\mathrm{sh}}$, and composing with τ gives our desired morphism

$$\phi=\tau\circ\psi^{\mathrm{sh}}:\mathcal{P}^{\mathrm{sh}}\to(f_*\mathcal{Q})^{\mathrm{sh}}\to f_*(\mathcal{Q}^{\mathrm{sh}}).$$

The pushforward for the presheaf $\mathcal{Q}$ distributes over the tensor product, that is

$$\Gamma(V,f_*\mathcal{Q})=f_*(\mathcal{F})(V)\otimes f_*f^*(\mathcal{E})(V).$$

There is a natural morphism of sheaves $\iota:\mathcal{E}\to f_*f^*(\mathcal{E})$, so take $\psi=\mathrm{id}_{f_*\mathcal{F}}\otimes\iota$ to be our morphism of presheaves



Now that we defined a morphism, we remark that the question is local on Y. In particular, let y be a point in Y, and let V be an open subset of Y such that $\mathcal{E}$ is free of finite rank. Then $f^*\mathcal{E}$ is free of finite rank on $U=f^{-1}(V)$. Consider the commutative diagram

$$\begin{array}{ccc} U & \xrightarrow{\bar f} & V \\ {\scriptstyle i}\downarrow & & \downarrow{\scriptstyle j} \\ X & \xrightarrow{f} & Y \end{array}.$$

Because i and j are open immersions, there is an isomorphism of functors

$$j^* f_* \cong \bar{f}_* i^*, \quad i^* f^* \cong \bar{f}^* j^*.$$

Pullback distributes over tensor products, so we have

$$\begin{aligned} j^* f_*(\mathcal{F} \otimes f^*\mathcal{E}) &\cong \bar{f}_* i^*(\mathcal{F} \otimes f^*\mathcal{E}) \\ &\cong \bar{f}_*(i^*\mathcal{F} \otimes i^* f^*\mathcal{E}) \\ &\cong \bar{f}_*(i^*\mathcal{F} \otimes \bar{f}^* j^*\mathcal{E}), \\ j^*(f_*(\mathcal{F}) \otimes \mathcal{E}) &\cong j^* f_*(\mathcal{F}) \otimes j^*\mathcal{E} \\ &\cong \bar{f}_*(i^*\mathcal{F}) \otimes j^*\mathcal{E}. \end{aligned}$$

Open immersions preserve stalks, so we can assume $\mathcal{E}$ to be a free $\mathcal{O}_X$-module of finite rank n. The pullback of a free sheaf of finite rank is also free of finite rank, so we have the isomorphisms

$$\begin{aligned} f_*(\mathcal{F} \otimes f^*\mathcal{E}) &\cong f_*(\mathcal{F} \otimes \mathcal{O}_X^{\oplus n}) \\ &\cong f_*(\mathcal{F}^{\oplus n}) \\ &\cong f_*(\mathcal{F})^{\oplus n} \\ &\cong f_*(\mathcal{F}) \otimes \mathcal{O}_Y^{\oplus n} \\ &\cong f_*(\mathcal{F}) \otimes \mathcal{E}. \end{aligned}$$

The third isomorphism holds because f_* is right adjoint, and right adjoint functors preserve limits, and a direct sum of finite number of terms can be realized as a limit, so the pushforward and finite direct sums commute. □

Lemma 0.1. *Let $\mathcal{A}, \mathcal{B}, \mathcal{C}$ be Abelian categories, and assume $\mathcal{A}$ and $\mathcal{B}$ have enough injectives. Let $F : \mathcal{A} \to \mathcal{B}$ and $G : \mathcal{B} \to \mathcal{C}$ be functors such that F has a left adjoint (or more generally preserves injectives), and either*

(a) F is left exact, and G is exact; or

(b) F is exact, and G is left exact.

In each case, we have the following natural isomorphisms

(a) $\mathrm{R}^i(G \circ F) \cong G \circ \mathrm{R}^i(F)$;

(b) $\mathrm{R}^i(G \circ F) \cong \mathrm{R}^i(G) \circ F$.

Proof. The proof for (b) is similar to the proof of Exercise 1, so we prove (a). Let A be an object in $\mathcal{A}$ and choose an injective resolution

$$0 \to A \to I^\bullet,$$

where one has

$$\mathrm{R}^i F(A) = h^i(FI^\bullet).$$

Applying $G \circ F$ gives a complex in $\mathcal{C}$

$$0 \to (G \circ F)A \to (G \circ F)I^\bullet,$$

and we want to show

$$h^i((G \circ F)I^\bullet) \cong G \circ h^i(FI^\bullet).$$

But this follows from exact functors commuting with cohomology.

□

Chapter 3, Section 9

James Lee

October 11, 2026

3. Some examples of flatness and nonflatness.

(a) If $f : X \to Y$ is a finite surjective morphism of nonsingular varieties over an algebraically closed field k, then f is flat.

(b) Let X be a union of two planes meeting at a point, each of which maps isomorphically to a plane Y. Show that f is not flat. For example, let $Y = \operatorname{Spec} k[x, y]$ and

$$X = \operatorname{Spec} k[x, y, z, w]/(z, w) \cap (x + z, y + w).$$

(c) Again let $Y = \operatorname{Spec} k[x, y]$, but take $X = \operatorname{Spec} k[x, y, z, w]/(z^2, zw, w^2, xz - yw)$. Show that $X_{\text{red}} \cong Y$, X has no embedded points, but that f is not flat.

Proof.

(a) By Lemma 1, we reduce to the case $X = \operatorname{Spec} B$ and $Y = \operatorname{Spec} A$, with A a regular local ring, and B a finite A-algebra containing A such that $B_{\mathfrak{n}}$ is a regular local ring for all maximal ideals $\mathfrak{n} \in \operatorname{Spec} B$. Then f is flat if and only if B is a projective A-module. We use the criterion of (6.12A); that is, we want to show $\operatorname{depth} B = \dim A$. Let $\mathfrak{m}$ be the maximal ideal of A, and let $n = \dim A$. It suffices to show $\operatorname{depth} B \geq n$ by (6.11A), i.e., we show there exists an A-regular sequence for B of length n.

Regular local rings are integrally closed, so A is integrally closed, and B is integral over A, so $\dim B = \dim A$ by (A.M., 11.26). This implies the following: pick any maximal ideal $\mathfrak{n}$ of B, then $B_{\mathfrak{n}}$ is a regular local ring and $\dim B_{\mathfrak{n}} = \dim B$ (A.M., 11.27); in particular, it is a Cohen-Macaulay ring, so a sequence of n elements $x_1, \ldots, x_n$ in $B_{\mathfrak{n}}$ is a $B_{\mathfrak{n}}$-regular sequence for $B_{\mathfrak{n}}$ if and only if $B_{\mathfrak{n}}/(x_1, \ldots, x_n) = \dim B - n = 0$ (II, 8.21A). By the inclusions $A \subseteq B \subseteq B_{\mathfrak{n}}$, a sequence of elements in A is an A-regular sequence for B if and only if it is a $B_{\mathfrak{n}}$-regular sequence for $B_{\mathfrak{n}}$. Thus, we further reduce to showing there exists a $B_{\mathfrak{n}}$-regular sequence contained in A for $B_{\mathfrak{n}}$ of length n.

Since A is a regular local ring, there exist n elements $x_1, \ldots, x_n \in A$ that form a minimal set of generators of $\mathfrak{m}$ and an A-regular sequence for A. Viewing $x_1, \ldots, x_n$ as elements in $B_{\mathfrak{n}}$, the elements $x_1, \ldots, x_n$ generate the extension of $\mathfrak{m}$ in B, so $B_{\mathfrak{n}}/(x_1, \ldots, x_n) \cong B_{\mathfrak{n}}/\mathfrak{m}B_{\mathfrak{n}}$, which is an Artin local ring (A.M., 5.8) and has Krull dimension equal to zero. Thus, $x_1, \ldots, x_n$ is the desired $B_{\mathfrak{n}}$-regular sequence for $B_{\mathfrak{n}}$.

(b) We establish the following notation

$$I = (z, w) \cap (x + z, y + w) \subseteq k[x, y, z, w],$$
$$Q = (x, y) \subseteq k[x, y].$$

Let $K = k(x, y)$ be the function field of Y. Following the proof of (a), we show

$$\dim_k k[x, y, z, w]/(I + Q) \neq \dim_K K[z, w]/I. \tag{1}$$

We start with the left hand side of (1) by computing the image of $I + Q$ under the quotient $k[x, y, z, w] \to k[z, w]$ with kernel Q. It is not hard to see

$$(xz + z^2, yz + zw, xw + zw, yw + w^2) \subseteq I \subseteq (x, y, z, w)^2, \quad (z, w)^2 + Q \subseteq I + Q \subseteq (x, y, z, w)^2 + Q,$$

and the image of $(x, y, z, w)^2 + Q$ in $k[z, w]$ is $(z, w)^2$, so the image of $I + Q$ is also $(z, w)^2$, i.e.,

$$k[x, y, z, w]/(I + Q) \cong k[z, w]/(z, w)^2.$$

Hence, we have

$$\dim_k k[x, y, z, w]/(I + Q) = 3.$$

It remains to compute the right hand side of (1). The $k[x,y]$-module $k[x,y,z,w]/I$ can be generated by $1, z, w$, so it spans $K[z,w]/I$ over K, and $wx - zy$ is in I, which implies z and w are K-linearly dependent in $K[z,w]/I$. Hence, the K-dimension of $K[z,w]/I$ is at most 2.

(c) Let $I = (z^2, zw, w^2, xz - yw)$. It suffices to show

$$\sqrt{I} = (z, w).$$

The generators of I are also in (z,w), a prime ideal, so we have inclusion in one direction, and the left-hand side contains z, w, so we have inclusion in the other direction. Also, I is a (z,w)-primary ideal, so it can only have one associated point (A.M., 4.5). Hence, X has no embedded points. Lastly, by the same reasoning in (b), we have

$$\dim_k k[x,y,z,w]/(I+Q) = 3, \quad \dim_K K[z,w]/I \leq 2.$$

□

4. *Open Nature of Flatness.* Let $f : X \to Y$ be a morphism of finite type of Noetherian schemes. Then $\{x \in X \mid f \text{ is flat at } x\}$ is an open subset of X (possibly empty).

Proof. The question is local on X, so we can reduce to the affine case.

Lemma 0.1. *Let A be a Noetherian ring, B an A-algebra of finite type, and $\mathfrak{q}$ a prime ideal of B. If $B_{\mathfrak{q}}$ is flat over A, then there exists $g \in B - \mathfrak{q}$ such that B_g is flat over A.*

We remark that by Lemma 1, $B_{\mathfrak{q}}$ is flat over A if and only if $B_{\mathfrak{q}}$ is flat over $A_{\mathfrak{p}}$, where $\mathfrak{p}$ is the contraction of $\mathfrak{q}$ in A. Let E be the flat locus of f. Since $X = \operatorname{Spec} B$ is a Noetherian space, we can use the following criterion:

Lemma 0.2 (A.M., Ex. 7.22)**.** *Let X be a Noetherian space, and let E be a subset of X. Then E is open in X if and only if, for each irreducible closed subset $X_0 \subseteq X$, either $E \cap X_0 = \emptyset$, or else $E \cap X_0$ contains a nonempty open subset of X_0.*

In our case, E is open in $X = \operatorname{Spec} B$ if and only if, for each prime ideal $\mathfrak{q}$ of B, either $E \cap V(\mathfrak{q}) = \emptyset$, or else there exists $g \in B - \mathfrak{q}$ such that $V(\mathfrak{q}) \cap X_g \subseteq E \cap V(\mathfrak{q})$. Notice that if $E \cap V(\mathfrak{q})$ is nonempty, then $\mathfrak{q} \in E$, because if there exists a prime ideal $\mathfrak{q}' \supset \mathfrak{q}$ such that $B_{\mathfrak{q}'}$ is flat over A, then $B_{\mathfrak{q}}$ is a localization of $B_{\mathfrak{q}'}$, so $B_{\mathfrak{q}}$ is also flat over A by transitivity of flatness. Thus, we can replace the condition $V(\mathfrak{q}) \cap E$ is nonempty with $\mathfrak{q}$ is in E.

To proceed, we need a couple preliminary algebraic results:

Lemma 0.3 (Matsumura, 22.3)**.** *Let A be a Noetherian ring, I an ideal of A, and M an A-module. If $\bigcap_{n=1}^{\infty} I^n(\mathfrak{a} \otimes_A M) = 0$ for any ideal $\mathfrak{a}$ of A, then the following are equivalent:*

(i) M is flat over A;

(ii) M/IM is flat over A/I, and $\operatorname{Tor}_1^A(M, A/I) = 0$.

(iii) M/I^nM is flat over A/I^n for every $n \geq 1$.

The main application of Lemma 5 is to the following case: suppose B is a Noetherian A-algebra, and I is an ideal in A such that IB is contained in the Jacobson radical of B. Recall that the radical of a ring is the intersection of all maximal ideals. If $\mathfrak{a}$ is any ideal in A, then $\mathfrak{a} \otimes_A B$ is a finitely generated B module, so (A.M., 10.19) gives us

$$\bigcap_{n=1}^{\infty} I^n(\mathfrak{a} \otimes_A B) = \bigcap_{n=1}^{\infty} (IB)^n(\mathfrak{a} \otimes_A B) = 0.$$

Hence, B satisfies the hypothesis of Lemma 5.

We proceed with a partial proof of Lemma 5. The implications $(i) \implies (ii) \implies (iii)$ are clear and holds without any special assumptions on A, I, or M, so we just prove $(iii) \implies (i)$. See (Matsumura, 22.3, 24.3) for more details.

Denote $A_n = A/I^n$ and $M_n = M/I^nM$, and fix an ideal $\mathfrak{a}$ of A. We want to show $\mathfrak{a} \otimes_A M \xrightarrow{j} M$ has zero kernel. By the assumption $\bigcap_n I^n(\mathfrak{a} \otimes_A M) = 0$, it suffices to show $\ker j \subseteq I^n(\mathfrak{a} \otimes_A M)$ for all n. By Krull's Theorem (A.M., 10.11) or (3.1A), for ecah n there exists a sufficiently large integer $k > n$ such that $I^k \cap \mathfrak{a} \subseteq I^n\mathfrak{a}$. Consider the commutative diagram

$$\begin{array}{ccccc} \mathfrak{a} \otimes_A M & \xrightarrow{f} & (\mathfrak{a}/I^k \cap \mathfrak{a}) \otimes_A M & \longrightarrow & (\mathfrak{a}/I^n\mathfrak{a}) \otimes_A M \cong (\mathfrak{a} \otimes_A M)/I^n(\mathfrak{a} \otimes_A M) \\ {\scriptstyle j}\downarrow & & \downarrow & & \\ M & \longrightarrow & M_k & & \end{array}.$$

The A-module $\mathfrak{a}/I^k \cap \mathfrak{a}$ is a natural A_k-module, so we have the isomorphism

$$(\mathfrak{a}/I^k \cap \mathfrak{a}) \otimes_A M \cong (\mathfrak{a}/I^k \cap \mathfrak{a}) \otimes_{A_k} M_k.$$

There is the natural isomorphism $\mathfrak{a}/I^k \cap \mathfrak{a} \cong (\mathfrak{a} + I^k)/I^k$ (A.M., 2.1), which is an ideal in A_k. Since M_k is a flat A_k-module, the map $(\mathfrak{a}/I^k \cap \mathfrak{a}) \otimes_A M \to M_k$ is injective. Hence, we get

$$\ker j \subseteq \ker f \subseteq I^n(\mathfrak{a} \otimes_A M).$$

The last algebraic result we need is the same statement of Lemma 3 but with the additional assumption that A is an integral domain. In fact, a stronger statement holds, which we won't prove:

Lemma 0.4 (Nitsure, 5.11)**.** *Let A be a Noetherian integeral domain, and B an A-algebra of finite type. Then there exists $s \in A - \{0\}$ such that B_s is a free A_s-module.*

We are ready to prove Lemma 3 by using the criterion of Lemma 4. The extended ideal $\mathfrak{p}B_\mathfrak{q}$ is contained in the maximal ideal of $B_{\mathfrak{q}'}$ for any $\mathfrak{q}' \in V(\mathfrak{q})$, $B_{\mathfrak{q}'}$ is flat over A if and only if $B_{\mathfrak{q}'}/\mathfrak{p}B_{\mathfrak{q}'}$ is flat over $A/\mathfrak{p}$ and $\operatorname{Tor}_1^A(B_{\mathfrak{q}'}, A/\mathfrak{p}) = 0$ (Lemma 5). Lemma 2 implies the isomorphism

$$B_\mathfrak{q} \otimes_B \operatorname{Tor}_1^A(B, A/\mathfrak{p}) \cong \operatorname{Tor}_1^A(B_\mathfrak{q}, A/\mathfrak{p}) = 0,$$

and $\operatorname{Tor}_1^A(B, A/\mathfrak{p})$ is a finitely generated B-module because it is a B-submodule of $\mathfrak{p} \otimes_A B$, so there exists $g \in B - \mathfrak{q}$ such that

$$B_g \otimes_B \operatorname{Tor}_1^A(B, A/\mathfrak{p}) \cong \operatorname{Tor}_1^A(B_g, A/\mathfrak{p}) = 0.$$

Thus, $\operatorname{Tor}_1^A(B_{\mathfrak{q}'}, A/\mathfrak{p}) = 0$ for any $\mathfrak{q}' \in V(\mathfrak{q}) \cap X_g$. By Lemma 6, there exists $s \in A - \mathfrak{p}$ such that $B_s/\mathfrak{p}B_s$ is free over $A_s/\mathfrak{p}A_s$, hence flat over $A/\mathfrak{p}$ (Lemma 1). For any $\mathfrak{q}' \in V(\mathfrak{q}) \cap X_s$, $B_{\mathfrak{q}'}/\mathfrak{p}B_{\mathfrak{q}'}$ is a localization of $B_s/\mathfrak{p}B_s$, so it is flat over $A/\mathfrak{p}$ as well. Hence, $V(\mathfrak{q}) \cap X_g \cap X_s$ is a nonempty open subset of $V(\mathfrak{q})$ contained in $E \cap V(\mathfrak{q})$. □

11. Let Y be a nonsingular curve of degree d in $\mathbb{P}^n_k$, over an algebraically closed field k. Show that

$$0 \leq p_a(Y) \leq \frac{1}{2}(d-1)(d-2).$$

Proof. If $n = 2$, the inequalitie becomes an equality even if Y is singular or not irreducible (I, Ex. 7.2b) (Ex. 4.7), so assume $n > 2$. We claim there exists a linear subspace $L \subset \mathbb{P}^n$ of dimension $n-3$ such that $Y \cap L = \emptyset$. In fact, we can find a linear subspace of dimension $n-2$ that does not meet Y, which is what we will prove. A general hyperplane $H \subset \mathbb{P}^n$ intersects Y at d distinct points, and we want to find a hyperplane plane of H, which is a linear subspace of $\mathbb{P}^n$ with dimension $n-2$, that does not meet $Y \cap H$. Thus, the statement reduces to the following: for any set of distinct points $P_1, \ldots, P_d \in \mathbb{P}^n$, there exists a hyperplane $H \subset \mathbb{P}^n$ such that $P_i \notin H$ for every i. Proceeding by induction on n, let $(x_0 : \cdots : x_n)$ be homogenous coordinates for $\mathbb{P}^n$. The base case $n = 1$ is trivial because k is infinite, so assume the statement for $\mathbb{P}^{n-1}$ with $n > 1$. Let U_0 be the affine patch of $\mathbb{P}^n$ defined by $x_0 \neq 0$ so that $\mathbb{P}^n$ is the disjoint union $U_0 \cup \mathbb{P}^{n-1}$. Relabel the points $P_1, \ldots, P_r, Q_1, \ldots, Q_s$ so that $P_i \in U_0$ and $Q_j \in \mathbb{P}^{n-1}$. Applying the inductive step to $\mathbb{P}^{n-1}$ and $\{Q_j\}$ gives a hyperplane H defined by

$$H: \quad f(x_1, \ldots, x_n) = a_1x_1 + \cdots + a_nx_n = 0, \quad a_i \in k,$$

such that $Q_j \notin H$. Taking the projective cone of H in $\mathbb{P}^n$, we want to show there exists $a_0 \in k - \{0\}$ such that the hyperplane defined $a_0x_0 + f(x_1, \ldots, x_n) = 0$ does not meet P_i for every $i, \ldots, r$. Indeed, if

we consider $f(x_1, \ldots, x_n)$ as a function on U_0, then we can pick any $a_0 \in k - \{0\}$ such that $a_0 \neq -f(P_i)$ for every $i = 1, \ldots, r$.

Now, after a suitable automorphism of $\mathbb{P}^n$ we can assume L corresponds to the linear subspace defined by $x_i = 0$ for $i = 0, 1, 2$. Consider the projection map $\pi : \mathbb{P}^n - L \to \mathbb{P}^2$ defined by $(x_0 : \cdots : x_n) \mapsto (x_0 : x_1 : x_2)$. For each $a \in k$, $a \neq 0$, consider the automorphism σ_a of $\mathbb{P}^n$ defined by

$$\begin{aligned} \sigma_a : \mathbb{P}^n &\to \mathbb{P}^n, \\ (x_0 : \cdots : x_n) &\mapsto (x_0 : x_1 : x_2 : ax_3 : \cdots : ax_n). \end{aligned}$$

Further assume π induces a birational from Y to its image (I, Ex. 4.9). For each $a \neq 0$, let $Y_a = \sigma_a(Y)$ with $Y_1 = Y$. Then the Y_a form a flat family parametrized by $\mathbb{A}^1 - \{0\}$, and according to (9.8) this family extends uniquely to a flat family defined over all of $\mathbb{A}^1$, and the fibre Y_0 over 0 agrees set-theoretically with the projection $\pi(Y) \subseteq \mathbb{P}^2$. The degree and arithmetic genus of Y_a is independent of a (9.10), which means

$$p_a(Y_0) = p_a(Y), \quad \deg Y_0 = d.$$

If we take $Y_{(0)}$ to be Y_0 with its reduced induced structure, then we have an isomorphism of schemes $Y_{(0)} \cong \pi(Y)$. Since $\pi(Y)$ is a closed subscheme of $\mathbb{P}^2$, one has

$$p_a(Y_{(0)}) = \frac{1}{2}(d' - 1)(d' - 2),$$

where we take d' to be the degree of $Y_{(0)}$ as a closed subscheme of $\mathbb{P}^2$. The Hilbert polynomial of $Y_{(0)}$ as a closed subscheme of $\mathbb{P}^2$ is same as $Y_{(0)}$ considered as a closed subcheme of $\mathbb{P}^n$. This has the following implication: there is an exact sequence of $\mathcal{O}_{Y_0}$-modules

$$0 \longrightarrow \mathcal{I} \longrightarrow \mathcal{O}_{Y_0} \longrightarrow \mathcal{O}_{Y_{(0)}} \longrightarrow 0,$$

where $\mathcal{I}$ is a nilpotent ideal sheaf of $\mathcal{O}_{Y_0}$. Hilbert polynomials are additive (Ex. 5.1, 5.2a), so one has

$$P_{Y_0}(n) = P_{\mathcal{I}}(n) + P_{Y_{(0)}}(n),$$

where $P_{\mathcal{I}}(n)$, $P_{Y_0}(n)$, and $P_{Y_{(0)}}(n)$ are the Hilbert polynomials of $\mathcal{I}, \mathcal{O}_{Y_0}$, and $\mathcal{O}_{Y_{(0)}}$ with respect to $\mathcal{O}_{\mathbb{P}^n}(1)$, and by our remark, $d' = \deg P_{Y_{(0)}}(n)$. If we can show the following:

(1) $p_a(Y_0) \leq p_a(Y_{(0)})$, and

(2) $\deg Y_{(0)} \leq \deg Y_0$,

then we can combine the inequalities to conclude that

$$0 \leq p_a(Y) = p_a(Y_0) \leq p_a(Y_{(0)}) = \frac{1}{2}(d' - 1)(d' - 2) \leq \frac{1}{2}(d - 1)(d - 2).$$

The degree of the Hilbert polynomial is at most the dimension the dimension of the support, and its leading coefficient is always positive, which proves (2). In addition, we can plug in $n = 0$ to obtain the equality

$$P_{\mathcal{I}}(0) + p_a(Y_0) = p_a(Y_{(0)}),$$

so to prove (1), we want to show $P_{\mathcal{I}}(0) = \chi(\mathcal{I}) \geq 0$. *Incomplete.* □

Lemma 0.5 (Stacks Project, Tag 00HT)**.** *Let $A \to B$ be a ring homomorphism, and let M be a B-module. The following are equivalent:*

(i) M is a flat A-module;

(ii) for every prime ideal $\mathfrak{q}$ of B, $\mathfrak{p}$ its contraction in A, $M_\mathfrak{q}$ is a flat $A_\mathfrak{p}$-module.

Proof. $(i) \implies (ii)$ Suppose M is flat as an A-module. Let $\mathfrak{q}$ be a prime ideal of B, and let $\mathfrak{p}$ be its contraction in A. Then $\mathfrak{a} \otimes_A M \to M$ is injective for any ideal $\mathfrak{a}$ of A. We want to show $\mathfrak{a}A_\mathfrak{p} \otimes_{A_\mathfrak{p}} M_\mathfrak{q} \to M_\mathfrak{q}$ is an $A_\mathfrak{p}$-module monomorphism. By (A.M., 2.15), and because $M_\mathfrak{q}$ is a natural $A_\mathfrak{p}$-module, we have the isomorphisms

$$\begin{aligned}(\mathfrak{a} \otimes_A M) \otimes_B B_\mathfrak{q} &\cong \mathfrak{a} \otimes_A (M \otimes_B B_\mathfrak{q}) \\ &\cong \mathfrak{a} \otimes_A (A_\mathfrak{p} \otimes_{A_\mathfrak{p}} M_\mathfrak{q}) \\ &\cong \mathfrak{a}A_\mathfrak{p} \otimes_{A_\mathfrak{p}} M_\mathfrak{q},\end{aligned}$$

which shows $\mathfrak{a}A_\mathfrak{p} \otimes_{A_\mathfrak{p}} M_\mathfrak{q} \to M_\mathfrak{q}$ is given by localizing the B-module homomorphism $\mathfrak{a} \otimes_A M \to M$ at the prime ideal $\mathfrak{q}$. Localizing preserves injections, so we are done.

$(ii) \implies (i)$ We want to show $\mathfrak{a} \otimes_A M \to M$ is injective for any ideal $\mathfrak{a}$ of A. Because M is a B-module, localizing at $\mathfrak{q}$ gives $(\mathfrak{a} \otimes_A M) \otimes_B B_\mathfrak{q} \to M_\mathfrak{q}$, which is injective by assumption and isomorphism of above. Thus, $\mathfrak{a} \otimes_A M \to M$ is injective as a B-module homomorphism, and therefore injective as an A-module homomorphism. □

Lemma 0.6. *Let $A \to B \to C$ be ring homomorphisms, and suppose C is flat over B. Then for every ideal $\mathfrak{a} \subseteq A$, there is a natural isomorphism*

$$C \otimes_B \operatorname{Tor}_1^A(B, A/\mathfrak{a}) \cong \operatorname{Tor}_1^A(C, A/\mathfrak{a}).$$

Proof. Consider the exact sequence

$$0 \longrightarrow \mathfrak{a} \longrightarrow A \longrightarrow A/\mathfrak{a} \longrightarrow 0.$$

Tensoring by B over A gives an exact sequence with Tor

$$0 \longrightarrow \operatorname{Tor}_1^A(B, A/\mathfrak{a}) \longrightarrow \mathfrak{a} \otimes_A B \longrightarrow B \longrightarrow B/\mathfrak{a}B \longrightarrow 0.$$

Each term in the sequence is a natural B-module, so we can tensor the sequence by C over B

$$0 \longrightarrow C \otimes_B \operatorname{Tor}_1^A(B, A/\mathfrak{a}) \longrightarrow \mathfrak{a} \otimes_A C \longrightarrow C \longrightarrow C/\mathfrak{a}C \longrightarrow 0.$$

Here, we used the fact that

$$(\mathfrak{a} \otimes_A B) \otimes_B C \cong \mathfrak{a} \otimes_A (B \otimes_B C) \cong \mathfrak{a} \otimes_A C.$$

We can also tensor the original sequence by C over A to get

$$0 \longrightarrow \operatorname{Tor}_1^A(C, A/\mathfrak{a}) \longrightarrow \mathfrak{a} \otimes_A C \longrightarrow C \longrightarrow C/\mathfrak{a}C \longrightarrow 0.$$

The A-module homomorphism $B \to C$ induces $\operatorname{Tor}_1^A(B, A/\mathfrak{a}) \to \operatorname{Tor}_1^A(C, A/\mathfrak{a})$, and $\operatorname{Tor}_1^A(C, A)$ is a natural C-module, so there is a natural map

$$\phi : C \otimes_B \operatorname{Tor}_1^A(B, A/\mathfrak{a}) \to \operatorname{Tor}_1^A(C, A/\mathfrak{a})$$

given by multiplication. Consider the commutative diagram

$$\begin{array}{ccccccccc} 0 & \longrightarrow & C \otimes_B \operatorname{Tor}_1^A(B, A/\mathfrak{a}) & \longrightarrow & \mathfrak{a} \otimes_A C & \longrightarrow & C & \longrightarrow & C/\mathfrak{a}C & \longrightarrow & 0 \\ & & \downarrow{\scriptstyle \phi} & & \| & & \| & & \| & & \\ 0 & \longrightarrow & \operatorname{Tor}_1^A(C, A/\mathfrak{a}) & \longrightarrow & \mathfrak{a} \otimes_A C & \longrightarrow & C & \longrightarrow & C/\mathfrak{a}C & \longrightarrow & 0 \end{array}.$$

The five lemma implies ϕ is actually an isomorphism. □

Chapter 3, Section 10

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1. Over a nonperfect field, smooth and regular are not equivalent. For example, let k_0 be a field of characteristic $p > 0$, let $k = k_0(t)$, and let $X \subseteq \mathbb{A}^2_k$ be the curve defined by $y^2 = x^p - t$. Show that every local ring of X is a regular local ring, but X is not smooth over k.

Proof. To show every local ring of X is a regular local ring, it suffices to show the coordinate ring of X defined by $A = k[x, y]/(y^2 - x^p - t)$ is integrally closed. If $p \neq 2$, the result follows from (II, Ex. 6.4), so assume $p = 2$. An immediate consequence is

$$y^2 - x^2 - t = (x + y)^2 - t,$$

which implies the isomorphism

$$A = \frac{k[x, y]}{(x + y)^2 - t} \cong k(\sqrt{t})[z].$$

But $k(\sqrt{t})$ is a field, and polynomial rings over fields are integrally closed, so we conclude that A is integrally closed.

Let $\bar{k}$ be the algebraic closure of k. By (10.2), X is smooth over k if and only if $X \times_k \bar{k}$ is regular, i.e., $A \otimes_k \bar{k}$ defines a nonsingular $\bar{k}$-variety; but $t^{1/p} \in \bar{k}$, so we can factorize

$$y^2 - x^p - t = y^2 - (x + t^{1/p})^p.$$

Thus, $(-t^{1/p}, 0)$ is a singular point of $X \times_k \bar{k}$ (A.M., Ex. 11.1), so X is not smooth over k. □

2. Let $f : X \to Y$ be a proper, flat morphism of varieties over k. Suppose for some point $y \in Y$ that the fiber X_y is smooth over $k(y)$. Then show that there is an open neighborhood V of y in Y such that $f : f^{-1}(V) \to V$ is smooth.

Proof. Flat morphisms of finite type between Noetherian schemes are open (Ex. 9.1), proper morphisms are closed, X and Y are irreducible, so f must be surjective, and the generic point $\xi \in X$ is mapped to the generic point of Y. Some immediate consequence are the following: f trivially satisfies (2) in the definition of smoothness of relative dimension $n = \dim X - \dim Y$, so for every open set $V \subseteq X$, $f : f^{-1}(V) \to V$ satisfies (2). Thus, we want to find an open neighborhood V of y such that $\Omega_{X/Y}$ is free at every point in $f^{-1}(V)$ (10.0.2). For every $x \in X_y$, we have the inequalities

$$\dim X - \dim Y \leq \dim_{k(\xi)} \Omega_{X/Y} \otimes k(\xi) \leq \dim_{k(x)} \Omega_{X/Y} \otimes k(x) = \dim X_y,$$

by (II, Ex. 5.8a) and (II, 8.6A), where the last equality is by smoothness of X_y over $k(y)$ and the fact that relative differentials commute with base change. Since $n = \dim X_y$ (9.6), $\Omega_{X/Y}$ is free of rank n at every $x \in X_y$ (II, 8.9).

Now, let $U = \{x \in X \mid \Omega_{X/Y} \text{ is free at } x\}$, and $Z = X - U$. Being free is an open property for finitely generated modules, so U is an open subset of X. Then f is an open and closed mapping, so $f(U)$ and $X - f(Z)$ are open sets in Y. Also, $f^{-1}(y) = X_y$ is a subset of U, which means y is not in $f(Z)$. Therefore, $f(U) \cap (X - f(Z))$ is a nonempty open set containing y, so we can find an open neighborhood V of y such that $V \subseteq f(U) \cap (X - f(Z))$. Since $f^{-1}(V) \subseteq U$, $\Omega_{X/Y}$ is free at every point in $f^{-1}(V)$, which is what we wanted to show. □

3. A morphism $f : X \to Y$ of schemes of finite type over k is *étale* if it is smooth of relative dimension 0. It is *unramified* if for every $x \in X$, letting $y = f(x)$, we have $\mathfrak{m}_y \cdot \mathcal{O}_{x,X} = \mathfrak{m}_x$, and $k(x)$ is a separable algebraic extension of $k(y)$. Show that the following conditions are equivalent:

(i) f is étale;

(ii) f is flat, and $\Omega_{X/Y} = 0$;

(iii) f is flat and unramified.

Proof. $(i) \iff (ii)$ The forward direction is trivial. On the other hand, $\Omega_{X/Y} = 0$ trivially satisfies condition (3) in the definition of smoothness, and we want to show condition (2) holds as well. According to (9.6), condition (2) is equivalent to saying that every irreducible component of every fibre X_y of f has dimension 0, so let X' be an irreducible component of X_y with its reduced induced structure. Then X' is an integral scheme of finite type over $k(y)$, and $\Omega_{X'/k(y)}$ is a coherent sheaf, so by (II, Ex. 3.20b) and (II, 8.6A), we reduce to showing $\Omega_{X'/k(y)} \otimes k(x) = 0$ for every $x \in X'$. Then

$$\Omega_{X_y/k(y)} \otimes k(x) = \Omega_{X/Y} \otimes k(x) = 0$$

since relative differentials commute with base extension and $\Omega_{X/Y} = 0$. Hence, the result follows from the surjective map of (II, 8.12)

$$\Omega_{X_y/k(y)} \otimes k(x) \longrightarrow \Omega_{X'/k(y)} \otimes k(x) \longrightarrow 0.$$

$(i) \iff (iii)$ Relative differentials commute with base extension, so by the equivalence of (i) and (ii), we reduce to the case when $Y = \operatorname{Spec} k$. Thus, we want to show X is smooth of relative dimension 0 over k if and only if $\mathfrak{m}_x = 0$ and $k(x)$ is a separable algebraic extension of k for every $x \in X$. Consider the exact sequence of (II, 8.4A)

$$\mathfrak{m}_x/\mathfrak{m}_x^2 \longrightarrow \Omega_{X/k} \otimes k(x) \longrightarrow \Omega_{k(x)/k} \longrightarrow 0.$$

In the forward direction, suppose $\Omega_{X/k} = 0$. Then $\Omega_{k(x)/k} = 0$ by the exact sequence, so $k(x)$ is a separable algebraic extension of k (II, 8.6A) Also, X is smooth of relative dimension 0 over k if and only if $\bar{X} = X \times \bar{k}$ is equidimensional of dimension zero and regular, where $\bar{k}$ an algebraic closure of k (10.2). Regularity and dimension of $X \times \bar{k}$ implies $\mathfrak{m}_x \otimes_k \bar{k} = 0$, hence $\mathfrak{m}_x = 0$. On the otherhand, if $\mathfrak{m}_x = 0$ and $k(x)$ is a separable algebraic extension of k. Then $\Omega_{k(x)/k} = 0$ (II, 8.6A), and the term on the left hand of the exact sequence is zero. Hence, $\Omega_{X/k} = 0$. □

5. If x is a point of a scheme X, we define an *étale neighborhood* of x to be an étale morphism $f : U \to X$, together with a point $x' \in U$, such that $f(x') = x$. As an example of the use of étale neighborhoods, prove the following: if $\mathcal{F}$ is a coherent sheaf on X, and if every point of X has an étale neighborhood $f : U \to X$ for which $f^*\mathcal{F}$ is a free $\mathcal{O}_U$-module, then $\mathcal{F}$ is locally free on X.

Proof. Consider the following lemma:

Lemma 0.1. *If B is a faithfully flat A-algebra, then for any A-module M, we have M is A-flat if and only if $M \otimes_A B$ is B-flat.*

Let $\mathfrak{a}$ be an ideal of A. Then

$$\operatorname{Tor}_1^A(M, A/\mathfrak{a}) \otimes_A B \cong \operatorname{Tor}_1^B(M \otimes_A B, B/\mathfrak{a}B) = 0,$$

so by faithfull flatness, we conclude that $\operatorname{Tor}_1^A(M, A/\mathfrak{a}) = 0$ (A.M., Ex. 3.16).

Now, if $f^*\mathcal{F}$ is a free $\mathcal{O}_U$-module, then $(f^*\mathcal{F})_{x'}$ is a free $\mathcal{O}_{x',U}$-module with $(f^*\mathcal{F})_{x'} \cong \mathcal{F}_x \otimes_{\mathcal{O}_{x,X}} \mathcal{O}_{x',U}$, and $\mathcal{O}_{x',U}$ is a faithfully flat $\mathcal{O}_{x,X}$-algebra, so the result follows from Lemma 1 and the fact that for finitely generated modules over Noetherian local rings, being free is equivalent to being flat. □

6. Let Y be the plane nodal cubic curve $y^2 = x^2(x+1)$. Show that Y has a finite étale covering X of degree 2, where X is a union of two irreducible components, each one isomorphic to the normalization of Y.

Proof. For simplicity, assume k is algebraically closed and has characteristic $\neq 2$. Consider two copies of the twisted cubic curve X_1 and X_2 in $\mathbb{A}^3$ with affine coordinates x, y, z given by the parametric equations

$$X_1 : \begin{cases} x = t^2 - 1 \\ y = t^3 - t \\ z = t \end{cases} \quad , \quad X_2 : \begin{cases} x = t^2 - 1 \\ y = t^3 - t \\ z = -t \end{cases} .$$

Each are isomorphic to the normalization of Y, and they intersect at the following three points

$$P_1 = (0, 0, 1), \quad P_2 = (0, 0, -1), \quad Q = (-1, 0, 0).$$

Let X be the closed subscheme of X with two irreducible components X_1 and X_2. There is an obvious map $X \to Y$ given by the projection map onto the xy-plane, and each irreducible component of X maps onto Y. Finally, let $\pi : \tilde{X} \to X$ be obtained from X by blowing up the point Q. Then $\tilde{X} \to Y$ is our desired étale covering of degree 2. $\square$

Chapter 3, Section 11

James Lee

October 11, 2026

1. Show that the result of (11.2) is false without the projective hypothesis. For example, let $X = \mathbb{A}^n_k$, let $P = (0, \dots, 0)$, let $U = X - P$, and let $f : U \to X$ be the inclusion. Then the fibers of f all have dimension 0, but $\mathrm{R}^{n-1} f_* \mathcal{O}_U \neq 0$.

Proof. By (8.2) and (8.5), we have the isomorphism

$$\mathrm{R}^{n-1} f_* \mathcal{O}_U \big|_U \cong \widetilde{\mathrm{H}^{n-1}(U, \mathcal{O}_U)},$$

so it suffices to show $\mathrm{H}^{n-1}(U, \mathcal{O}_U) \neq 0$. This is a generalization of (Ex. 4.3), and we compute $\mathrm{H}^{n-1}(U, \mathcal{O}_U)$ via Čech cohomology. Let $x_1, \dots, x_n$ be affine coordinates of X, i.e., $X = \operatorname{Spec} k[x_1, \dots, x_n]$, and take $\mathfrak{U}$ to be the open covering of U by the open sets $U_i = X - \{x_i \neq 0\}$. Setting $V = \bigcap_{i=1}^n U_i$ and $V_j = \bigcap_{i \neq j} U_i$, the last two terms of the Čech complex are

$$C^{n-2} = \prod_{j=1}^n \Gamma(V_j, \mathcal{O}_U), \quad C^{n-1} = \Gamma(V, \mathcal{O}_U).$$

Now

$$\Gamma(V_j, \Omega) = k\left[x_1, \dots, x_n, \frac{1}{x_1}, \dots, \widehat{\frac{1}{x_j}}, \dots, \frac{1}{x_n}\right],$$
$$\Gamma(V, \Omega) = k\left[x_1, \dots, x_n, \frac{1}{x_1}, \dots, \frac{1}{x_n}\right],$$

and the image of the boundary map $d : C^{n-2} \to C^{n-1}$ consists of all expressions $x_1^{r_1} \cdots x_n^{r_n}$ where at least one of $r_1, \dots, r_n$ is non-negative. Hence, $\mathrm{H}^{n-1}(U, \mathcal{O}_U)$ consists of all expressions $x_1^{r_1} \cdots x_n^{r_n}$ with $r_i < 0$ for all i. □

2. Show that a projective morphism with finite fibers is a finite morphism.

Proof. Let $f : X \to Y$ be a projective morphism of Noetherian schemes, and let $Y' = \mathcal{Spec}\, f_* \mathcal{O}_X$. The natural map $g : Y' \to Y$ is finite, and we get a projective morphism $f' : X \to Y'$ with connected fibres such that f factors into $g \circ f'$ (11.5). The property of quasi-finiteness satisfies the conditions of (II, Ex. 4.8), and finite morphisms are separated, so f' is quasi-finite. The composition of finite morphisms is finite, so it suffices to show f' is a finite morphism. Thus, we reduce to showing projective morphisms with finite and connected fibres is a finite morphism. By (II, Ex. 3.4), we further reduce to the case $Y = \operatorname{Spec} A$ for some Noetherian ring A and show $X = \operatorname{Spec} B$, where B is a finite A-algebra. If f has finite and connected fibres, then f is injective as a map between topological spaces. Since projective morphisms are closed mappings, f is a homeomorphism of X onto a closed subscheme of Y. The result follows from (II, Ex. 3.11). □

3. Let X be a normal, projective variety over an algebraically closed field k. Let $\mathfrak{d}$ be a linear system (of effective Cartier divisors) without base points, and assume that $\mathfrak{d}$ is *not composite with a pencil*, which means that if $f : X \to \mathbb{P}^n_k$ is the morphism determined by $\mathfrak{d}$, then $\dim f(X) \geq 2$. Then show that every divisor in $\mathfrak{d}$ is connected. This improves Bertini's theorem (10.9.1).

4. *Principle of Connectedness.* Let $\{X_t\}$ be a flat family of closed subschemes of $\mathbb{P}^n_k$ parameterized by an irreducible curve T of finite type over k. Suppose there is a nonempty open set $Y \subseteq T$, such that for all closed points $t \in Y$, X_t is connected. Then prove that X_t is connected for all $t \in T$.

5. Let Y be a hypersurface in $X = \mathbb{P}_k^N$ with $N \geq 4$. Let $\hat{X}$ be the formal completion of X along Y (II, §9). Prove that the natural map $\operatorname{Pic} \hat{X} \to \operatorname{Pic} Y$ is an isomorphism.

6. Again let T be a hypersurface in $X = \mathbb{P}_k^N$, this time with $N \geq 2$.

(a) If $\mathcal{F}$ is a locally free sheaf on X, show that the natural map

$$H^0(X, \mathcal{F}) \to H^0(\hat{X}, \hat{\mathcal{F}})$$

is an isomorphism.

(b) Show that the following conditions are equivalent:

(i) for each locally free sheaf $\mathfrak{F}$ on $\hat{X}$, there exists a coherent sheaf $\mathcal{F}$ on X such that $\mathfrak{F} \cong \hat{\mathcal{F}}$ (i.e., $\mathfrak{F}$ is *algebraizable*);

(ii) for each locally free sheaf $\mathfrak{F}$ on $\hat{X}$, there is an integer n_0 such that $\mathfrak{F}(n)$ is generated by global sections for all $n \geq n_0$.

(c) Show that the conditions (i) and (ii) of (b) imply that the natural map $\operatorname{Pic} X \to \operatorname{Pic} \hat{X}$ is an isomorphism.

Note. In fact, (i) and (ii) always hold if $N \geq 3$. This fact, coupled with (Ex. 11.5) leads to Grothendieck's proof of the Lefschetz theorem which says that if Y is a hypersurface in $\mathbb{P}_k^N$ with $N \geq 4$, then $\operatorname{Pic} Y \cong \mathbb{Z}$, and it is generated by $\mathcal{O}_Y(1)$.

8. Let $f : X \to Y$ be a projective morphism, let $\mathcal{F}$ be a coherent sheaf on X which is flat over Y, and assume that $\mathrm{H}^i(X_y, \mathcal{F}_y) = 0$ for some i and some $y \in Y$. Then show that $\mathrm{R}^i f_*(\mathcal{F})$ is 0 in a neighborhood of y.

Proof. Since $\mathrm{R}^i f_*(\mathcal{F})$ is a coherent sheaf on Y (8.8b), it suffices to show $\mathrm{R}^i f_*(\mathcal{F})_y = 0$. The question is local on Y, so we reduce to the case Y is affine, equal to the spectrum of the local Noetherian ring $A = \mathcal{O}_y$, and y is the closed point of Y. Let $\hat{A}$ denote the completion of A with respect to its maximal ideal $\mathfrak{m}$. Consider the natural isomorphism of (11.1)

$$\widehat{\mathrm{R}^i f_*(\mathcal{F})}_y \cong \varprojlim \mathrm{H}^i(X_n, \mathcal{F}_n).$$

We claim it suffices to show the right hand side is zero. Indeed, by (8.2) and (8.5), we have the isomorphism

$$\mathrm{R}^i f_*(\mathcal{F})_y \cong \mathrm{H}^i(X, \mathcal{F}),$$

and according to (A.M., 10.13), we also have the isomorphism

$$\widehat{\mathrm{R}^i f_*(\mathcal{F})}_y \cong \mathrm{H}^i(X, \mathcal{F}) \otimes \hat{A},$$

so the claim follows from the fact that $\hat{A}$ is a fathfully flat A-algebra.

Now, we will show the natural map $\mathrm{H}^i(X_{n+1}) \to \mathrm{H}^i(X_n, \mathcal{F}_n)$ is injective for every $n > 1$. Admitting this for the moment, since $X_1 = X_y$ and $\mathcal{F}_1 = \mathcal{F}_y$, it follows by induction that

$$\mathrm{H}^i(X_n, \mathcal{F}_n) \hookrightarrow \mathrm{H}^i(X_y, \mathcal{F}_y) = 0,$$

which in turn implies the inverse limit is zero, which is what we want.

Consider the exact sequence of A-modules

$$0 \longrightarrow \mathfrak{m}^n/\mathfrak{m}^{n+1} \longrightarrow A/\mathfrak{m}^{n+1} \longrightarrow A/\mathfrak{m}^n \longrightarrow 0.$$

Since $\mathcal{F}$ is flat over Y, tensoring with $\mathcal{F}$ is an exact functor, so we have an exact sequence

$$0 \longrightarrow \mathcal{F} \otimes \mathfrak{m}^n/\mathfrak{m}^{n+1} \longrightarrow \mathcal{F}_{n+1} \longrightarrow \mathcal{F}_n \longrightarrow 0.$$

Taking the long exact sequence in cohomology gives

$$\mathrm{H}^i(X, \mathcal{F} \otimes \mathfrak{m}^n/\mathfrak{m}^{n+1}) \longrightarrow \mathrm{H}^i(X_{n+1}, \mathcal{F}_{n+1}) \longrightarrow \mathrm{H}^i(X_n, \mathcal{F}_n),$$

so it suffices to show the term on the left is zero. Since $\mathfrak{m}^n/\mathfrak{m}^{n+1}$ is a finite-dimensional vector space over $k(y)$, and cohomology commutes with direct sums, we have the isomorphism

$$\mathrm{H}^i(X, \mathcal{F} \otimes \mathfrak{m}^n/\mathfrak{m}^{n+1}) \cong \mathrm{H}^i(X, \mathcal{F} \otimes k(y)) \otimes_{k(y)} \mathfrak{m}^n/\mathfrak{m}^{n+1},$$

where $k(y)$ is the constant sheaf $k(y)$ on the closed point y of Y. Thus, we can apply the isomorphism of (9.4)

$$\mathrm{H}^i(X, \mathcal{F} \otimes k(y)) \cong \mathrm{H}^i(X_y, \mathcal{F}_y) = 0$$

to conclude that $\mathrm{H}^i(X, \mathcal{F} \otimes \mathfrak{m}^n/\mathfrak{m}^{n+1}) = 0$.

□

Chapter 3, Section 12

James Lee

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1. Let Y be a scheme of finite type over an algebraically closed field k. Show that the function

$$\varphi(y) = \dim_k(\mathfrak{m}_y/\mathfrak{m}_y^2)$$

is upper semicontinuous on the set of closed points of Y.

Proof. Apply (12.7.2) to the sheaf of relative differentials $\Omega_{Y/k}$. For each closed point $y \in Y$, we have a natural isomorphism (II, 8.7A)

$$\mathfrak{m}_y/\mathfrak{m}_y^2 \cong \Omega_{Y/k} \otimes k(y).$$

□

2. Let $\{X_t\}$ be a family of hypersurfaces of the same degree in $\mathbb{P}_k^n$. Show that for each i, the function $h^i(X_t, \mathcal{O}_{X_t})$ is a constant function of t.

Proof. Hypersurfaces of same degree in projective space $\mathbb{P}^n$ have the same cohomology. For any hypersurface $X \subset \mathbb{P}^n$ of degree d, we have an exact sequence

$$0 \longrightarrow \mathcal{O}_{\mathbb{P}^n}(-d) \longrightarrow \mathcal{O}_{\mathbb{P}^n} \longrightarrow \mathcal{O}_X \longrightarrow 0,$$

and $h^i(X, \mathcal{O}_X)$ is independent of the choice of X. □

4. Let Y be an integral scheme of finite type over an algebraically closed field k. Let $f : X \to Y$ be a flat projective morphism whose fibers are all integral schemes. Let $\mathcal{L}, \mathcal{M}$ be invertible sheaves on X, and assume for each $y \in Y$ that $\mathcal{L}_y \cong \mathcal{M}_y$ on the fiber X_y. Then show that there is an invertible sheaf $\mathcal{N}$ on Y such that $\mathcal{L} \cong \mathcal{M} \otimes f^*\mathcal{N}$.

Proof. If f is flat, then $\mathcal{L} \otimes \mathcal{M}^{-1}$ is flat over Y. For each $y \in Y$, $(\mathcal{L} \otimes \mathcal{M}^{-1})_y \cong \mathcal{O}_{X_y}$ on the fibre X_y, and each X_y is integral and projective over $k(y)$, so we have

$$h^0(X_y, (\mathcal{L} \otimes \mathcal{M}^{-1})_y) = h^0(X_y, \mathcal{O}_{X_y}) = 1.$$

Thus, $f_*(\mathcal{L} \otimes \mathcal{M}^{-1})$ is locally free of rank one on Y (12.9), say $f_*(\mathcal{L} \otimes \mathcal{M}^{-1}) = \mathcal{N}$ for some $\mathcal{N} \in \operatorname{Pic} Y$. Applying f^* to both sides, and using the fact that the natural morphism

$$f^* f_*(\mathcal{L} \otimes \mathcal{M}^{-1}) \to \mathcal{L} \otimes \mathcal{M}^{-1}$$

is actually an isomorphism because $f_*(\mathcal{L} \otimes \mathcal{M}^{-1})$ is locally free of rank one, we obtain our desired isomorphism. □

5. Let Y be an integral scheme of finite type over an algebraically closed field k. Let $\mathscr{E}$ be a locally free sheaf on Y, and let $X = \mathbb{P}(\mathcal{E})$. Then show that $\operatorname{Pic} X \cong (\operatorname{Pic} Y) \times \mathbb{Z}$. This strengthens (II, Ex. 7.9).

Proof. Let r be the rank of $\mathcal{E}$, $\mathcal{O}_X(1)$ the canonical invertible sheaf of X, and $\pi : X \to Y$ the natural projection morphism. For any point $y \in Y$, let $k(y)$ be the residue field of y, and let $\mathbb{P}_y$ be the fiber of π over y, which is a projective r-space over $k(y)$. There is an obvious map

$$\begin{aligned} \varphi : \operatorname{Pic} Y \times \mathbb{Z} &\to \operatorname{Pic} X \\ (\mathcal{L}, d) &\mapsto \mathcal{O}_X(d) \otimes \pi^*\mathcal{L}. \end{aligned}$$

First, we show φ is injective. If we have an isomorphism

$$\mathcal{O}_X(d) \otimes \pi^*\mathcal{L} \cong \mathcal{O}_X(d') \otimes \pi^*\mathcal{L}',$$

then it is clear by checking on fibres that $d = d'$, so it remains to show $\pi^* : \operatorname{Pic} Y \to \operatorname{Pic} X$ is injective. We have $\pi_*\mathcal{O}_X = \mathcal{O}_Y$ (II, 7.11a), and for any $\mathcal{N} \in \operatorname{Pic} Y$, the projection formula gives

$$\pi_*\pi^*\mathcal{N} \cong \pi_*\mathcal{O}_X \otimes \mathcal{N} \cong \mathcal{N},$$

so if $\pi^*\mathcal{N} \cong \mathcal{O}_X$, then $\mathcal{N} \cong \pi_*\mathcal{O}_X \cong \mathcal{O}_Y$. Hence, φ is injective.

The hard part is to show φ is surjective. Since π is a flat projective morphism, we can apply the results of §12. In particular, if $\mathcal{M}$ is an invertible sheaf on X, then $\mathcal{M}_y$ is an invertible sheaf on $\mathbb{P}_y$ for every $y \in Y$, so we can write $\mathcal{M}_y \cong \mathcal{O}(d_y)$ for some $d_y \in \mathbb{Z}$. If we can show d_y is independent of $y \in Y$, then by (Ex. 12.4) there exists an invertible sheaf $\mathcal{N}$ on Y such that $\mathcal{M} \cong \mathcal{O}_X(d) \otimes \pi^*\mathcal{N}$ for some $d \in \mathbb{Z}$, which proves φ is surjective.

We proceed by showing the minimum value of d_y is obtained when $y = \eta$ is the generic point of Y. Suppose there exists $y' \in Y$ such that $d_{y'} < d_\eta$. After twisting by $\mathcal{O}_X(-d_{y'})$, we can assume $d_{y'} = 0$. By (12.8), the function

$$h^0(y, \mathcal{M}) = \dim_{k(y)} \mathrm{H}^0(\mathbb{P}_y, \mathcal{O}_{\mathbb{P}_y}(d_y))$$

is upper semicontinuous on Y (12.8), and we have the strict inequality

$$\dim_{k(y')} \mathrm{H}^0(\mathbb{P}_{y'}, \mathcal{O}_{\mathbb{P}_{y'}}) < \dim_{k(\eta)} \mathrm{H}^0(\mathbb{P}_\eta, \mathcal{O}_{\mathbb{P}_\eta}(d_\eta))$$

by the explicit calculations of (5.1). But this implies the set $\{y \in Y \mid h^0(y, \mathcal{M}) \geq h^0(\eta, \mathcal{M})\}$ is a closed set containing η and not containing y', a contradiction. Hence, we have $d_\eta \leq d_y$ for all $y \in Y$.

Next, we prove the inequality in the opposite direction. By Serre duality (5.1) (7.7), we have

$$\begin{aligned} h^{r-1}(y, \mathcal{M}) &= \dim_{k(y)} \mathrm{H}^{r-1}(\mathbb{P}_y, \mathcal{O}_{\mathbb{P}_y}(d_y)) \\ &= \dim_{k(y)} \mathrm{H}^{r-1}(\mathbb{P}_y, \mathcal{O}_{\mathbb{P}_y}(-d_y - r)) \\ &= h^0(y, \mathcal{O}_X(-r) \otimes \mathcal{M}^{-1}), \end{aligned}$$

and $h^{r-1}(y, \mathcal{M})$ is also an upper semicontinuous on Y. Suppose there exists $y' \in Y$ such that $d_\eta < d_{y'}$. After twisting by $\mathcal{O}_X(-d_\eta + r)$, we can assume $d_\eta = r$. In particular $d_{y'}$ is a positive integer. Again, we have the strict inequality

$$\dim_{k(y')} \mathrm{H}^0(\mathbb{P}_{y'}, \mathcal{O}_{\mathbb{P}_{y'}}(-d_{y'} - r)) < \dim_{k(\eta)} \mathrm{H}^0(\mathbb{P}_\eta, \mathcal{O}_{\mathbb{P}_\eta}),$$

arriving at the same contradiction as above. Hence, d_y is independent of $y \in Y$. □

6. Let X be an integral projective scheme over an algebraically closed field k, and assume that $\mathrm{H}^1(X, \mathcal{O}_X) = 0$. Let T be a connected scheme of finite type over k.

(a) If $\mathcal{L}$ is an invertible sheaf on $X \times T$, show that the invertible sheaves $\mathcal{L}_t$ on $X = X \times \{t\}$ are isomorphic, for all closed points $t \in T$.

(b) Show that $\operatorname{Pic}(X \times T) = \operatorname{Pic} X \times \operatorname{Pic} T$.

Proof.

(a)

(b)

□

Chapter 4

Curves

Chapter 4, Section 1

James Lee

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1. Let X be a curve, and let $P \in X$ be a point. Then there exists a nonconstant rational function $f \in K(X)$, which is regular everywhere except at P.

Proof. For large $n > 0$, we have $\mathrm{H}^0(\mathcal{O}_X(nP)) \neq 0$ (1.3.2). This implies the existence of an effective divisor D' such that $D' - nP = (f)$, where f is a rational function and (f) is the divisor of zeros of f. Since D' is effective, the only pole of f is at P of order at most n. □

5. For an effective divisor D on a curve X of genus g, show that $\dim |D| \leq \deg D$. Furthermore, equality holds if and only if $D = 0$ or $g = 0$.

Proof. The Riemann-Roch theorem gives

$$\ell(D) - 1 = \ell(K - D) + \deg D - g,$$

so it suffices to show $\ell(K - D) \leq g$ and equality holds if and only if $D = 0$ or $g = 0$.

If D is effective, then we have an exact sequence

$$0 \longrightarrow \mathcal{O}_X(-D) \longrightarrow \mathcal{O}_X \longrightarrow \mathcal{O}_D \longrightarrow 0,$$

Tensoring by $\mathcal{O}_X(K)$ and taking global sections gives an injection of vector spaces $\mathrm{H}^0(\mathcal{O}_X(K - D)) \to \mathrm{H}^0(\mathcal{O}_X(K))$, so we conclude that $\ell(K - D) \leq \ell(K) = g$.

Now, we have $\dim |D| = \deg D$ if and only if $\ell(K - D) = \ell(K)$, and it is clear this equality holds if either $D = 0$ or $g = 0$. Conversely, suppose $D \neq 0$ and $g \neq 0$. We will show $\ell(K - D) < \ell(K)$. Since D is effective, $d = \deg D > 0$, so we proceed by induction on d. If $D = P$ for a point $P \in X$, then $\ell(P) = 1$ because X is not rational. Thus, Riemann-Roch gives $\ell(K - P) = g - 1$.

Assume the statement for $d > 1$, and let D be an effective divisor of degree $d + 1$. Let P be a point in the support of D so that $D' = D - P$ is an effective divisor of degree d. By the inductive hypothesis, we have $\ell(K - D') < \ell(K)$. Consider the exact sequence

$$0 \longrightarrow \mathcal{O}_X(-P) \longrightarrow \mathcal{O}_X \longrightarrow \mathcal{O}_P \longrightarrow 0.$$

Tensoring with $\mathcal{O}_X(K - D')$ and taking global sections gives an inclusion of vector spaces $\mathrm{H}^0(\mathcal{O}_X(K - D)) \to \mathrm{H}^0(\mathcal{O}_X(K - D'))$. Thus, $\ell(K - D) \leq \ell(K - D')$, so combining inequalities, we get

$$\ell(K - D) \leq \ell(K - D') < \ell(K).$$

□

6. Let X be a curve of genus g. Show that there is a finite morphism $f : X \to \mathbb{P}^1$ of degree $\leq g + 1$.

Proof. Fix a point $P \in X$, and consider the divisor $D = (g + 1)P$. By Riemann-Roch, $\ell(D) > 1$, which implies the existence of a nonconstant rational function $f \in K(X)$ with poles no worse than D. This means we can write $f = g/h$ such that the order of vanishing of h at P is at most $g + 1$, and h has no other zeros. The inclusion of fields $k(f) \subseteq K(X)$ induces a finite morphism $f : X \to \mathbb{P}^1$. Now, the degree of $f^{-1}(0)$ is at most $g + 1$ points, so f has degree at most $g + 1$ by (II, 6.9). □

7. A curve X is called *hyperelliptic* if $g \geq 2$ and there exists a finite morphism $f : X \to \mathbb{P}^1$ of degree 2.

(a) If X is a curve of genus $g = 2$, show that the canonical divisor defines a complete linear system $|K|$ of degree 2 and dimension 1, without basepoints. Use (II, 7.8.1) to conclude that X is hyperelliptic.

(b) Show that the curves constructed in (1.1.1) all admit a morphism of degree 2 to $\mathbb{P}^1$. Thus, there exist hyperelliptic curves of any genus $g \geq 2$.

Proof.

(a) By Riemann-Roch, $\deg K = 2g - 2 = 2$ and $\dim |K| = \ell(K) - 1 = 1$. For any $P \in X$, consider the exact sequence

$$0 \longrightarrow \mathcal{O}_X(K - P) \longrightarrow \mathcal{O}_X(K) \longrightarrow k(P) \longrightarrow 0.$$

The linear system $|K|$ is basepoint free if and only if $\mathrm{H}^0(\mathcal{O}_X(K)) \to \mathrm{H}^0(\mathcal{O}_P)$ is surjective for every $P \in X$. Taking the long exact sequence in cohomology gives

$$0 \longrightarrow \mathrm{H}^0(X, \mathcal{O}_X(K - P)) \longrightarrow \mathrm{H}^0(X, \mathcal{O}_X(K)) \longrightarrow \mathrm{H}^0(P, \mathcal{O}_P).$$

Since $\ell(K) = 2$ and $h^0(P, \mathcal{O}_P) = 1$, it is enough to show $h^0(X, \mathcal{O}_X(K - P)) = 1$. Again, by the Riemann-Roch theorem, $\ell(K - P) = \ell(P)$, and $\ell(P) = 1$ because X is not rational. Hence, $\ell(K - P) = 1$.

(b) If D is a divisor of type (a, b) with $a, b > 0$, then D is very ample, so by Bertini's theorem there exists an irreducible nonsingular curve $C \subseteq Q = \mathbb{P}^1 \times \mathbb{P}^1$ that is linearly equivalent to D and has type (a, b) as a divisor in Q. Also, $a = \deg p_1\big|_C$ and $b = \deg p_2\big|_C$. Suppose $(a, b) = (g + 1, 2)$. By (III, Ex. 5.6), C has genus $= (g + 1 - 1)(2 - 1) = g$, and $p_2\big|_C : C \to \mathbb{P}^1$ is a morphism of degree 2. Hence, X is a hyperelliptic curve of genus g.

□

Chapter 4, Section 2

James Lee

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1. Use (2.5.3) to show that $\mathbb{P}^n$ is simply connected.

Proof. Proceed by induction on n. The base case $n = 1$ is (2.5.3). Let $f : X \to \mathbb{P}^n$ be an étale covering. We may assume X is connected. Then X is smooth over k since f is étale (III, 10.1), and X is projective over k since f is finite, so X is a smooth irreducible projective variety of dimension n. It is enough to show the degree of f is 1. By (III, Ex. 11.3), $f^{-1}(H) = X \times_{\mathbb{P}^n} H$ is smooth and irreducible for almost every hyperplane $H \subset \mathbb{P}^n$. Also, the induced morphism $f' : f^{-1}(H) \to H$ is an étale covering of $H \cong \mathbb{P}^{n-1}$, since étale morphisms are stable under base change. By the inductive hypothesis, f' is an isomorphism. Hence, f is an isomorphism. □

2. *Classification of Curves of Genus 2.* Fix an algebraically closed field k of characteristic $\neq 2$.

(a) If X is a curve of genus 2 over k, the canonical linear system $|K|$ determines a finite morphism $f : X \to \mathbb{P}^1$ of degree 2 (Ex. 1.7). Show that it is ramified at exactly 6 points, with ramification index 2 at each one. Note that f is uniquely determined, up to an automorphism of $\mathbb{P}^1$, so X determines an (unordered) set of 6 points of $\mathbb{P}^1$, up to an automorphism of $\mathbb{P}^1$.

(b) Conversely, given six distinct elements $\alpha_1, \ldots, \alpha_6 \in k$, let K be the extension of $k(x)$ determined by the equation $z^2 = (x - \alpha_1) \cdots (x - \alpha_6)$. Let $f : X \to \mathbb{P}^1$ be the corresponding morphism of curves. Show that $g(X) = 2$, the map f is the same as the one determined by the canonical linear system, and f is ramified over the six points $x = \alpha_i$ of $\mathbb{P}^1$, and nowhere else.

(c) Using (I, Ex. 6.6), show that if P_1, P_2, P_3 are three distinct points of $\mathbb{P}^1$, then there exists a unique $\varphi \in \operatorname{Aut} \mathbb{P}^1$ such that $\varphi(P_1) = 0$, $\varphi(P_2) = 1$, $\varphi(P_3) = \infty$. Thus in (a), if we order the six points of $\mathbb{P}^1$, and then normalize by sending the first three to $0, 1, \infty$, respectively, we may assume that X is ramified over $0, 1, \infty, \beta_1, \beta_2, \beta_3$, where $\beta_1, \beta_2, \beta_3$ are three distinct elements of k, $\neq 0, 1$.

(d) Let Σ_6 be the symmetric group on 6 letters. Define an action of Σ_6 on sets of three distinct elements $\beta_1, \beta_2, \beta_3$ of k, $\neq 0, 1$, as follows: reorder the set $0, 1, \infty, \beta_1, \beta_2, \beta_3$ according to a given element $\sigma \in \Sigma_6$, then renormalize as in (c) so that the three become $0, 1, \infty$ again. Then the last three are the new $\beta_1', \beta_2', \beta_3'$.

(e) Summing up, conclude that there is a one-to-one correspondence between the set of isomorphism classes of curves of genus 2 over k, and triples of distinct elements $\beta_1, \beta_2, \beta_3$ of k, $\neq 0, 1$, modulo the action of Σ_6 described in (d). In particular, there are many non-isomorphic curves of genus 2. We say that curves of genus 2 depend on three parameters, since they correspond to the points of an open subset of $\mathbb{A}^3_k$ modulo a finite group.

Proof.

(a) By Hurtwitz's theorem, the degree of the ramification divisor is computed to be

$$\deg R = \deg K_X - 2 \deg K_{\mathbb{P}^1} = 6.$$

The ramification index at a point cannot exceed the degree of the morphism, so f must be ramified at exactly 6 distinct points each having ramification index 2.

(b) Let $B = k[x, z]/(z^2 - g(x))$ be an affine coordinate ring of X with $g(x) = (x - \alpha_1) \cdots (x - \alpha_6)$, and let f be the projection morphism onto the x-axis. By (II, 8.3A), we have

$$\Omega_{B/k} \cong (B\mathrm{d}x \oplus B\mathrm{d}z)/(g'(x)\mathrm{d}x + 2z\mathrm{d}z).$$

Let $P = (x', z')$ be a closed point in X. If $z' \neq 0$, then z is invertible in $\mathcal{O}_{P,X}$, so that

$$\mathrm{d}z = -\frac{g'(x)}{2z}\mathrm{d}x,$$

which means $\mathrm{d}x$ generates $(\Omega_{B/k})_P$. If $z' = 0$, then $x' = \alpha_i$ for some i, and $g'(x)$ is invertible in $\mathcal{O}_{P,X}$, so that

$$\mathrm{d}x = -\frac{2z}{g'(x)}\mathrm{d}z,$$

which means $\mathrm{d}z$ generates $(\Omega_{B/k})_P$. Since $\mathrm{d}x$ is a local parameter of $\mathbb{P}^1$, we see that P is ramified if and only if $z' = 0$ with ramification index $e_P = v_P(\mathrm{d}x/\mathrm{d}z) + 1 = 2$. There are six ramification points corresponding to the six roots of $g(x)$. Hence, $g(X) = 2$ by the Hurwitz formula.

□

3. *Plane Curves.* Let X be a curve of degree d in $\mathbb{P}^2$. For each $P \in X$, let $T_P(X)$ be the tangent line to X at P (I, Ex.7.3). Considering $T_P(X)$ as a point of the dual projective plane $(\mathbb{P}^2)^*$, the map $P \to T_P(X)$ gives a morphism of X to its *dual curve* X^* in $(\mathbb{P}^2)^*$ (I, Ex. 7.3). Note that even though X is nonsingular, X^* in general will have singularities. We assume $\operatorname{char} k = 0$ below.

(a) Fix a line $L \subseteq \mathbb{P}^2$ which is not tangent to X. Define a morphism $\varphi : X \to L$ by $\varphi(P) = L \cap T_P(X)$, for each point $P \in X$. Show that φ is ramified at P if and only if either (1) $P \in L$, or (2) P is an *inflection point* of X, which means that the intersection multiplicity (I, Ex. 5.4) of $T_P(X)$ with X at P is ≥ 3. Conclude that X has only finitely many inflection points.

(b) A line of $\mathbb{P}^2$ is a *multiple tangent* of X if it is tangent to X at more than one point. It is a *bitangent* if it is tangent to X at exactly two points. If L is a multiple tangent of X, tangent to X at the points $P_1, \ldots, P_r$, and if none of the P_i is an inflection point, show that the corresponding point of the dual curve X^* is an *ordinary r-fold point*, which means a point of multiplicity r with distinct tangent directions (I, Ex. 5.3). Conclude that X has only finitely many multiple tangents.

(c) Let $O \in \mathbb{P}^2$ be a point which is not on X, nor on any inflectional or multiple tangent of X. Let L be a line not containing O. Let $\psi : X \to L$ be the morphism defined by projection from O. Show that ψ is ramified at $P \in X$ if and only if the line OP is tangent to X at P, and that the ramification index is 2. Use Hurwitz's theorem and (I, Ex. 7.2) to conclude that there are exactly $d(d-1)$ tangents of X passing through O. Hence, the degree of the dual curve (sometimes called the *class* of X) is $d(d-1)$.

(d) Show that for all but a finite number of points of X, a point O of X lies on exactly $(d+1)(d-2)$ tangents of X, not counting the tangent at O.

(e) Show that the degree of the morphism φ of (a) is $d(d-1)$. Conclude that if $d \geq 2$, then X has $3d(d-2)$ inflection points, properly counted. (If $T_P(X)$ has intersection multiplicity r with X at P, then P should be counted $r-2$ times as an inflection point. If $r = 3$, we called it an *ordinary inflection point.*) Show that an ordinary inflection point of X corresponds to an ordinary cusp of the dual curve X^*.

(f) Now let X be a plane curve of degree $d \geq 2$, and assume that the dual curve X^* has only nodes and ordinary cusps as singularities (which should be true for sufficiently general X.) Then show that X has exactly $\frac{1}{2}d(d-2)(d-3)(d+3)$ bitangents.

(g) For example, a plane cubic curve has exactly 9 inflection points, all ordinary. The line joining any two of them intersects the curve in a third one.

(h) A plane quartic curve has exactly 28 bitangents. (This holds even if the curve has a tangent with four-fold contact, in which case the dual curve X^* has a tacnode.)

Proof.

(a) Find local affine coordinates x, y of $\mathbb{P}^2$ such that

$$X : y - f(x) = 0, \quad L : x = 0.$$

Setting x and y to be local parameters for X at P and L at $\varphi(P)$, respectively, we can write φ in the form

$$\varphi(x) = f(x) - xf'(x).$$

It follows that

$$\varphi^* \mathrm{d}y = \frac{\mathrm{d}y}{\mathrm{d}x}\mathrm{d}x, \quad \frac{\mathrm{d}y}{\mathrm{d}x} = -xf''.$$

By (2.2), φ is ramified at P if and only if $v_P(\mathrm{d}t/\mathrm{d}u) > 0$, where we have

$$v_P\left(\frac{\mathrm{d}y}{\mathrm{d}x}\right) = v_P(x) + v_P(f'').$$

Notice that $v_P(x) > 0$ if and only if $P \in L$. Also, it is not hard to see

$$\mu_P(X, T_P(X)) = v_P(f'') + 2.$$

Thus, $v_P(\mathrm{d}t/\mathrm{d}u) > 0$ if and only if $v_P(x) > 0$ or $v_P(f'') > 0$ if and only if (10)$P \in L$ or (2) P is an inflection point of X. Since $\varphi : X \to L$ ramifies at finitely many points, X has finitely many inflection points.

(b) Find local affine coordinates x, y of $\mathbb{P}^2$ such that

$$X : y - f(x) = 0, \quad L : y = 0.$$

This means x is a local parameter of X, so we can write $\gamma : X \ : X \to X^* \subset (\mathbb{P}^2)^*$ in the form

$$\gamma(x) = (-f'(x), xf'(x) - f(x)).$$

The tangent directions of $Q = \gamma(P_i)$ are precisely the images of the tangent vectors at P_i under the differential of γ. In particular, identifying the tangent vector of X at P_i to the row vector $(1, 0) \in T_{P_i}(X)$ and applying $\mathrm{d}\gamma$ gives

$$v_i = (-f''(x_i), x_i f''(x_i)) \in T_Q(X^*).$$

We have shown in (a) that $f''(x_i) = 0$ if and only if P_i is an inflection point, which implies v_i and v_j define the same direction if and only if $P_i = P_j$. Hence, Q is an ordinary r-fold point with distinct tangent directions. Each multiple tangent of X corresponds to singular point of X^*, but a curve can have only finitely many singular points. Hence, X has finitely many multiple tangents.

(c) Find local affine coordinates x, y of $\mathbb{P}^2$ such that

$$X : y - f(x) = 0, \quad L : x = 0, \quad O = (1 : 0 : 0).$$

We can set $u = x$ and $t = y$ to be local parameters for X at P and L at $Q = \psi(P)$, respectively, and ψ is given by the projection morphism onto the x-axis. It follows that

$$\psi^* \mathrm{d}y = \frac{\mathrm{d}y}{\mathrm{d}x}\mathrm{d}x, \quad \frac{\mathrm{d}y}{\mathrm{d}x} = f'.$$

Hence, ψ is ramified at $P \in X$ if and only if $f'(P) = 0$ if and only if the tangent of X at P passes through O. Suppose ψ is ramified at P. Since O is not on any inflectional or multiple tangents of X, $v_P(f'') = 0$. Hence, $v_P(f') = v_P(f'') + 1 = 1$.

The genus of X is related to d by the degree genus formula. Since ψ has degree d as a morphism of curves, Hurwitz's theorem can be used to compute the degree of the ramification divisor

$$\deg R = \deg K_X - d \deg K_{\mathbb{P}^1} = d(d-1).$$

The ramification index of each ramification point is 2, so the degree of R equals to the number of tangents passing through O.

(d) This is another application of Hurwitz's theorem. The projection morphism $\psi : X - O \to L$ has degree $d - 1$, so the degree of the ramification divisor of ψ is

$$\deg R = \deg K_X - (d-1) \deg K_{\mathbb{P}^1} = (d+1)(d-2).$$

A general point of X does not lie on any multiple tangents of X. Hence, the degree of R equals to the number of tangents passing through O.

(e) Define $\pi : (\mathbb{P}^2)^* - L \to L$ by $L' \mapsto L \cap L'$. Since φ factors into $\varphi = \pi \circ \gamma$, the degree of φ equals to the degree of the image of γ, which is $d(d-1)$. The degree of the ramification divisor of φ is

$$\deg R = \deg K_X - d(d-1) \deg K_{\mathbb{P}^1} = 3d^2 - 5d.$$

The degree of the ramification index also counts the number of points $X \cap L$. In particular, $X \cap L$ is generically d distinct points, and the ramification index of each $P \in X \cap L$ is 2, so we need to subtract d from $\deg R$ to get the total number of inflection points of X, which equals to $3d(d-2)$. If $r = 3$, an ordinary inflection point of X corresponds to a twisted cubic in the dual curve locally, which has an ordinary cusp.

(f) Let β_1 and β_2 be the number of bitangents and inflection points of X, respectively. We have two ways of computing the arithmetic genus of X^*: (1) in terms of $p_a(X)$, and (2) $X^* \subset (\mathbb{P}^2)^*$ as a plane curve of degree $d(d-1)$.

(1) The dual morphism $\gamma : X \to X^*$ defined in (b) is generically one-to-one by, so γ is a birational onto its image. The result of (Ex. 1.8) says

$$p_a(X^*) = p_a(X) + \delta, \quad \delta = \sum_{P \in X^*} \delta_P,$$

where $\delta_P = 1$ if $P \in X^*$ is a node or ordinary cusp, zero otherwise. In particular, $\delta = \beta_1 + \beta_2$ by (c) and (e).

(2) On the otherhand, X^* is a plane curve of degree $d(d-1)$ in $(\mathbb{P}^2)^*$, so the degree-genus formula for plane curves (III, Ex. 4.7) says

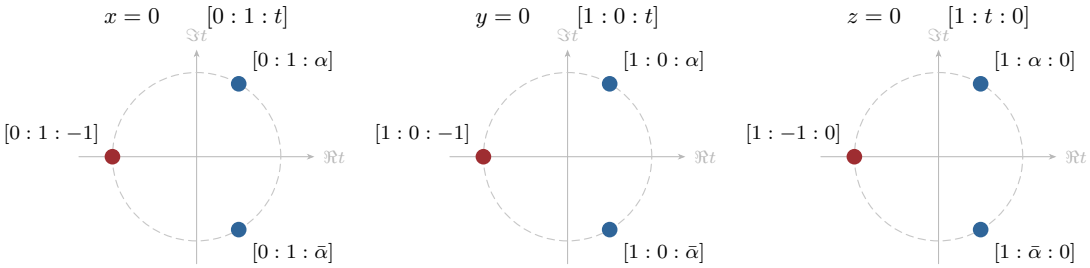
$$p_a(X^*) = \frac{1}{2}(d-2)(d+1)(d^2-d-1)$$

A smooth plane curve of degree d has $d(d-2)$ inflection points. Hence, the number of bitangets of X is

$$\beta_1 = p_a(X^*) - p_a(X) - \beta_2 = \frac{1}{2}d(d-2)(d-3)(d+3).$$

(g) Let $E \subset \mathbb{P}^2$ be a plane cubic curve, let $P, Q \in E$ be two inflection points, and let L the line through P and Q. If $P = Q$, then L is the tangent line through P. In this case, P is an inflection point, so it is counted with a multiplicity of 3, i.e., L meets no other point of X. Otherwise, L cannot be a tangent line for the same reason, and L meets X at a distinct point $R \neq P, Q$. The morphism $\varphi : E \to L$ defined in (a) is ramified at R. By (e), φ has degree $= 3(3-1) = 6$. Since P and Q are are ordinary inflection points, we have the ramification index of φ at P and Q is 2. Hence, R also has ramification index of 2, which is an inflection point.

$$C: \quad x^3 + y^3 + z^3 = 0 \quad \subset \mathbb{P}^2_{\mathbb{C}}$$

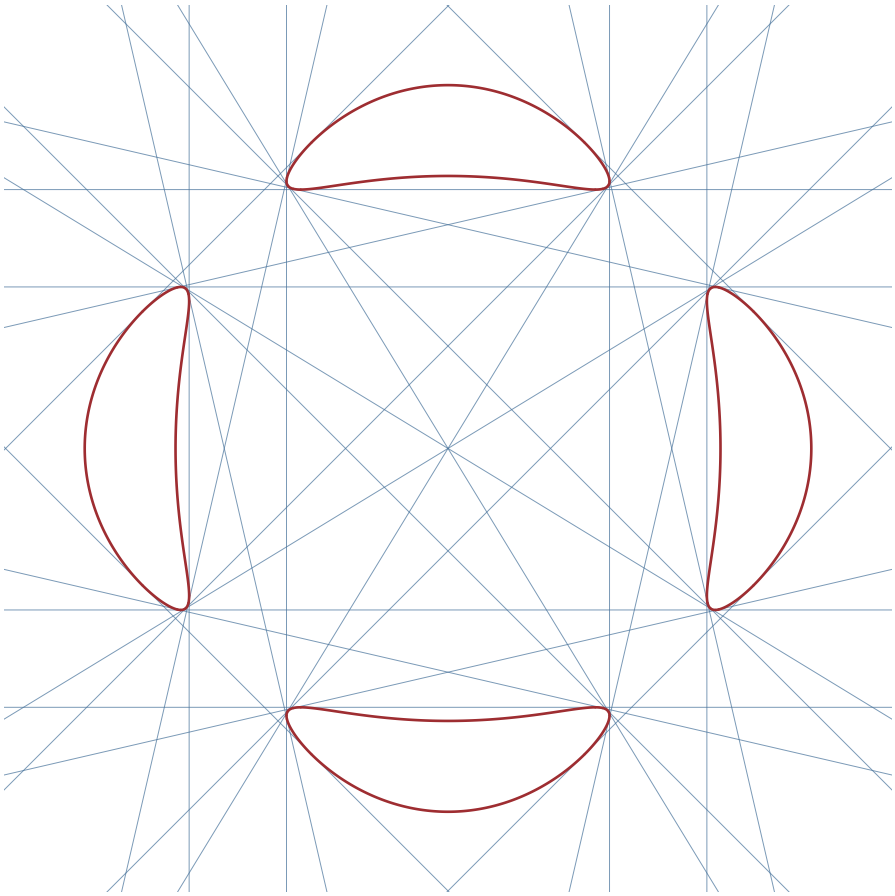


$$\mathrm{Flex}(C) = C \cap \{xyz = 0\}, \qquad t^3 = -1, \qquad \alpha = e^{\pi i/3} = \frac{1 + i\sqrt{3}}{2}.$$

Each panel is a complex parameter plane on one coordinate line.

● 3 real flexes ● 6 nonreal flexes

(h) $144(x^4 + y^4) - 225(x^2 + y^2) + 350x^2y^2 + 81 = 0$



□

4. *A Funny Curve in Characteristic p.* Let X be the plane quartic curve $x^3y + y^3z + z^3x = 0$ over a field of characteristic 3. Show that X is nonsingular, every point of X is an inflection point, the dual curve X^* is isomorphic to X, but the natural map $X \to X^*$ is purely inseparable.

Proof. The partial derivatives of $f = x^3y + y^3z + z^3x$ are

$$\partial_x f = z^3, \quad \partial_y f = x^3, \quad \partial_z f = y^3,$$

so in coordinates, the dual morphism $X \to X^*$ is given by

$$(x : y : z) \mapsto (x^3 : y^3 : z^3),$$

This immediately shows X is nonsingular, every point of X is an inflection point, and the natural map $X \to X^*$ is purely in separable. Also, we can identify $X^* \subset (\mathbb{P}^2)^*$ with the image of X under the k-linear Frobenius morphism $\mathbb{P}^2 \to \mathbb{P}^2$ (2.4.1), and under this identification, $f^3 = 0$ is a defining equation of X^*. Hence, X and X^* are isomorphic as abstract projective k-schemes. □

5. *Automorphisms of a Curve of Genus* ≥ 2. Prove the theorem of Hurtwitz [1] that a curve X of genus $g \geq 2$ over a field of characteristic 0 has at most $84(g-1)$ automorphisms. We will see later (Ex. 5.2) or (V, Ex. 1.11) that the group $G = \operatorname{Aut} X$ is finite. So let G have order n. Then G acts on the function field $K(X)$. Let L be the fixed field. Then the field extension $L \subseteq K(X)$ corresponds to a finite morphism of curves $f : X \to Y$ degree n.

(a) If $P \in X$ is a ramification point, and $e_p = r$, show that $f^{-1}f(P)$ consists of exactly n/r points, each having ramification index r. Let $P_1, \ldots, P_s$ be a maximal set of ramification points of X lying over distinct points of Y, and let $e_{P_i} = r_i$. Then show that Hurtwitz's theorem implies that

$$(2g-2)/n = 2g(Y) - 2 + \sum_{i=1}^{s}(1 - 1/r_i).$$

(b) Since $g \geq 2$, the left hand side of the equation is > 0. Show that if $g(Y) \geq 0$, $s \geq 0$, $r_i \geq 2$, $i = 1, \ldots, s$ are integers such that

$$2g(Y) - 2 + \sum_{i=1}^{s}(1 - 1/r_i) > 0,$$

then the minimum value of this expression is $1/42$. Conclude that $n \leq 84(g-1)$.

Proof.

(a) Each $\sigma \in G$ induces an automorphism of X such that $f \circ \sigma = f$, so there is a group action of G on each fibre $f^{-1}f(P)$. The action is transitive (A.M., Ex. 5.13), and $e_P = e_{\sigma(P)}$ because σ induces an isomorphism of local rings $\mathcal{O}_{\sigma(P_i),X} \to \mathcal{O}_{P_i,X}$, hence sends uniformizer to uniformizer. In particular, if d_i is the size of the orbit of P_i, then $n = d_i r_i$. Hence, we have

$$\sum_{P \in X}(e_P - 1) = \sum_{i=1}^{s} \sum_{P \in f^{-1}f(P_i)} (e_P - 1) = \sum_{i=1}^{s} d_i(r_i - 1) = n\sum_{i=1}^{s}(1 - 1/r_i).$$

(b) The following gives the minimum value

$$g(Y) = 0, \quad s = 3, \quad r_1 = \frac{1}{2}, \quad r_2 = \frac{1}{3}, \quad r_3 = \frac{1}{7}.$$

We conclude that

$$n \leq 42(2g-2) = 84(g-1).$$

□

6. *f_* for Divisors.* Let $f : X \to Y$ be a finite morphism of curves of degree n. We define a homomorphism $f_* : \operatorname{Div} X \to \operatorname{Div} Y$ by $f_*(\sum n_i P_i) = \sum n_i f(P_i)$ for any divisor $D = \sum n_i P_i$ on X.

(a) For any locally free sheaf $\mathcal{E}$ on Y, of rank r, we define $\det \mathcal{E} = \bigwedge^r \mathcal{E} \in \operatorname{Pic} Y$ (II, Ex. 6.11). In particular, for any invertible sheaf $\mathcal{M}$ on X, $f_*\mathcal{M}$ is locally free of rank n on Y, so we can consider $\det f_*\mathcal{M} \in \operatorname{Pic} Y$. Show that for any divisor D on X,

$$\det(f_*\mathcal{O}_X(D)) \cong (\det f_*\mathcal{O}_X) \otimes \mathcal{O}_Y(f_*D).$$

Note in particular that $\det(f_*\mathcal{O}_X(D)) \neq \mathcal{O}_Y(f_*D)$ in general!.

(b) Conclude that f_*D depends only on the linear equivalence class of D, so there is an induced homomorphism $f_* : \operatorname{Pic} X \to \operatorname{Pic} Y$. Show that $f_* \circ f^* : \operatorname{Pic} Y \to \operatorname{Pic} Y$ is just multiplication by n.

(c) Use duality for a finite flat morphisms (III, Ex. 6.10) and (III, Ex. 7.2) to show that

$$\det f_*\Omega_X \cong (\det f_*\mathcal{O}_X)^{-1} \otimes \Omega_Y^{\otimes n}.$$

(d) Now assume that f is separable, so we have the ramification divisor R. We define the *branch divisor* B to be the divisor f_*R on Y. Show that

$$(\det f_*\mathcal{O}_X)^2 \cong \mathcal{O}_Y(-B).$$

Proof.

(a) Let P be any point in X, and let $f_*P = f(P)$ be the image of P. The skyskraper sheaf $k(f_*P)$ has a natural free resolution

$$0 \longrightarrow \mathcal{O}_Y(-f_*P) \longrightarrow \mathcal{O}_Y \longrightarrow k(f_*P) \longrightarrow 0.$$

In particular, $\det k(f_*P) \cong \mathcal{O}_Y(f_*P)$, and any finite free resolution of $k(f_*P)$ induces isomorphic determinant line bundles (II, Ex. 6.11). Also, the sheaves $k(f_*P) \to f_*k(P)$ are isomorphic, so any free resolution of $f_*k(P)$ is also a free resolution of $k(f_*P)$.

Now, let D be any non-zero divisor, and let P be any point. We will show that the formula is true for D if and only if it is true for $D+P$ (1.3). Consider the exact sequence

$$0 \longrightarrow \mathcal{O}_X(-P) \longrightarrow \mathcal{O}_X \longrightarrow k(P) \longrightarrow 0.$$

Tensoring with $\mathcal{O}_X(D+P)$ and applying f_* gives

$$0 \longrightarrow f_*\mathcal{O}_X(D) \longrightarrow f_*\mathcal{O}_X(D+P) \longrightarrow f_*k(P) \longrightarrow 0.$$

By the remarks above, we have an isomorphism

$$\mathcal{O}_Y(f_*P) \cong \det(f_*\mathcal{O}_X(D+P)) \otimes (\det(f_*\mathcal{O}_X(D)))^{-1},$$

which is what we wanted to show.

(b) By (a), since $f_*\mathcal{O}_X(D)$ depends only on the linear equivalence class of D, so does $\mathcal{O}_Y(f_*D)$. For the second part, it suffices to show the statement holds for any point $Q \in Y$. Indeed, it follows by definition $f_* \circ f^*$ is multiplication by some integer. On the other hand, f^* multiplies the degree by n (II, 6.9), and f_* preserves degrees. Hence, $f_* \circ f^*$ must be multiplication by n.

(c) Let $\mathcal{F}$ be a coherent sheaf on Y. By definition, we have

$$f_*f^!\mathcal{F} \cong f_*\mathcal{O}_X^{-1} \otimes \mathcal{F}, \quad f_*\mathcal{O}_X^{-1} = \mathcal{H}om_Y(f_*\mathcal{O}_X, \mathcal{O}_Y).$$

By (III, Ex. 7.3), $f^!\Omega_Y \cong \Omega_X$ is a dualizing sheaf on X. Applying above to $\mathcal{F} = \Omega_Y$ gives

$$\begin{aligned} \det f_*\Omega_X &\cong \det f_*f^!\Omega_Y \\ &\cong \det\big(f_*\mathcal{O}_X^{-1} \otimes \Omega_Y\big) \\ &\cong (\det f_*\mathcal{O}_X)^{-1} \otimes \Omega_Y^{\otimes n}. \end{aligned}$$

(d) Let K_X and K_Y be canonical divisors of X and Y, respectively. They satisfy the following relation (2.3)

$$f^*K_Y - K_X \sim -R.$$

Applying f_* gives

$$nK_Y - f_*K_X \sim -B.$$

On the otherhand, (a) applied to $D = K_X$ gives

$$\mathcal{O}_Y(-f_*K_X) \cong (\det f_*\mathcal{O}_X) \otimes (\det f_*\Omega_X)^{-1}.$$

and by (c), we have the isomorphism

$$\mathcal{O}_Y(nK_Y) \cong (\det f_*\mathcal{O}_X) \otimes (\det f_*\Omega_X),$$

Combining these two isomorphisms gives

$$(\det f_*\mathcal{O}_X)^2 \cong \mathcal{O}_Y(nK_Y - f_*K_X) \cong \mathcal{O}_Y(-B).$$

□

7. *Étale Covers of Degree 2.* Let Y be a curve over a field k of characteristic $\neq 2$. We show there is a one-to-one correspondence between finite étale morphisms $f : X \to Y$ of degree 2, and 2-torsion elements of $\operatorname{Pic} Y$, i.e., invertible sheaves $\mathcal{L}$ on Y with $\mathcal{L}^2 \cong \mathcal{O}_Y$.

(a) Given an étale morphism $f : X \to Y$ of degree 2, there is a natural map $\mathcal{O}_Y \to f_*\mathcal{O}_X$. Let $\mathcal{L}$ be the cokernel. Then $\mathcal{L}$ is an invertible sheaf on Y, $\mathcal{L} \cong \det f_*\mathcal{O}_X$, and so $\mathcal{L}^2 \cong \mathcal{O}_X$ by (Ex. 2.6). Thus an étale cover of degree 2 determines a 2-torsion element in $\operatorname{Pic} Y$.

(b) Conversely, given a 2-torsion element $\mathcal{L}$ in $\operatorname{Pic} Y$, define an $\mathcal{O}_Y$-algebra structure on $\mathcal{O}_Y \oplus \mathcal{L}$ by $\langle a, b\rangle \cdot \langle a', b'\rangle = \langle aa' + \varphi(b \otimes b'), ab' + a'b\rangle$, where φ is an isomorphism of $\mathcal{L} \otimes \mathcal{L} \to \mathcal{O}_Y$. Then take $X = \mathcal{S}pec(\mathcal{O}_Y \oplus \mathcal{L})$. Show that X is an étale cover of Y.

(c) Show that these two processes are inverse to each other.

Proof.

(a) We check that $\mathcal{L}$ is an invertible sheaf on Y. At the level of stalks, the natural map $\mathcal{O}_{Q,Y} \to (f_*\mathcal{O}_X)_Q \cong \mathcal{O}_{Q,Y}^{\oplus 2}$ is the diagonal map, so the cokernel is free of rank one. Hence, $\mathcal{L}$ is locally free of rank one.

(b) By construction, $f : X \to Y$ is a finite morphism, and $f_*\mathcal{O}_X \cong \mathcal{O}_Y \oplus \mathcal{L}$ is locally free of rank 2, so f is a flat morphism. Also, φ is locally given by multiplication, i.e., $\varphi(b \otimes b') = bb'$. If P is a point in Y, then there is an isomorphism

$$(f_*\mathcal{O}_X)_P \cong \mathcal{O}_{P,X}[\varepsilon]/(\varepsilon^2 - 1) \cong \mathcal{O}_{P,X} \times \mathcal{O}_{P,X},$$

so f is unramified at every $P \in Y$. Hence, $f : X \to Y$ is a degree 2 étale cover of Y.

(c) One direction is obvious. Consider the pullback diagram

$$\begin{array}{ccc} X \times_Y X & \longrightarrow & X \\ \downarrow & & \downarrow \\ X & \xrightarrow{f} & Y \end{array}.$$

The projection morphisms $p_1, p_2 : X \times_Y X \to X$ are étale covers of degree 2. Also, the diagonal $\Delta \subset X \times_Y X$ is a closed subscheme of $X \times_Y X$ is isomorphic to X. Thus, the restriction $p_1, p_2 : X \times_Y X - \Delta \to X$ is an étale cover of X of degree 1, which is an isomorphism. Hence, $\tau = p_2 \circ p_1^{-1} : X \to X$ is an automorphism of X over Y which tranposes the fibres of f.

Now, let $f : X \to Y$ be a étale cove of degree 2, and let $X' = \mathcal{S}pec(\mathcal{O}_Y \oplus \mathcal{L})$, where $\mathcal{L}$ is the cokernel of the natural morphism $\mathcal{O}_Y \to f_*\mathcal{O}_X$. We want to show X and X' are isomorphic. It suffices to show the following short exact sequence of $\mathcal{O}_Y$-modules splits

$$0 \longrightarrow \mathcal{O}_Y \longrightarrow f_*\mathcal{O}_X \longrightarrow \mathcal{L} \longrightarrow 0.$$

The automorphism $\tau : X \to X$ of X over Y, which tranposes the fibres of f, induces an $\mathcal{O}_Y$-module automorphism $\tau_* : f_*\mathcal{O}_X \to f_*\mathcal{O}_X$. The kernel of the trace morphism $t = 1 - \tau : f_*\mathcal{O}_X \to f_*\mathcal{O}_X$ is exactly $\mathcal{O}_Y$. Hence, t induces a section $\mathcal{L} \to f_*\mathcal{O}_X$, so the short exact sequence splits. □

Chapter 4, Section 3

James Lee

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1. If X is a curve of genus 2, show that a divisor D is very ample $\iff \deg D \geq 5$. This strengthens (3.3.4).

Proof. Let D be a very ample divisor on X. By the degree-genus formula, a nonsingular plane curve cannot have genus equal to 2, which implies $\dim |D| \geq 3$. Applying the Riemann-Roch theorem to D gives

$$\deg D = \dim |D| - \dim |K - D| + 1 \geq 4 - \dim |K - D|.$$

By (Ex. 1.5), D has degree at least 3 (Ex. 1.5), so $K - D$ has negative degree. Hence, $|K - D|$ has dimension -1. □

3. If X is a curve of genus ≥ 2 which is a complete intersection (II, Ex. 8.4) in some $\mathbb{P}^n$, show that the canonical divisor K is very ample. Conclude that a curve of genus 2 can never be a complete intersection in any $\mathbb{P}^n$.

Proof. Let d be the degree of X in $\mathbb{P}^n$, and assume X is not contained in any hyperplane. Then X is a complete intersection of $n-1$ hypersurfaces of degree d_i, and by (II, Ex. 8.4), we have $\mathcal{O}_X(K) \cong \mathcal{O}_X(e)$, where $e = \sum d_i - n - 1$, so it is enough to show $e \geq 1$. We have two ways of computing $\deg K$, which are $\deg K = ed$ and $\deg K = 2g - 2 \geq 2$. Hence, $e \geq 1$. The linear system given by the canonical sheaf of a curve of genus 2 is the unique g^1_2 (Ex. 1.7), which is never ample. □

6. *Curves of Degree 4.*

(a) If X is a curve of degree 4 in some $\mathbb{P}^n$, show that either
   (1) $g = 0$, in which case X is either the rational normal quartic in $\mathbb{P}^4$ (Ex. 3.4) or the rational quartic curve in $\mathbb{P}^3$ (II, 7.8.6), or
   (2) $X \subseteq \mathbb{P}^2$, in which case $g = 3$, or
   (3) $X \subseteq \mathbb{P}^3$ and $g = 1$.

(b) In the case $g = 1$, show that X is a complete intersection of two irreducible quadric surfaces in $\mathbb{P}^3$ (I, Ex. 5.1).

Proof.

(a) $\boxed{n = 2}$ The degree-genus formula gives

$$g = \frac{(d-1)(d-2)}{2} = 3.$$

$\boxed{n = 3}$ If X is not contained in any hyperplane, then $g < 3$ (III, Ex. 9.11) (Ex. 3.5b). Since $g \neq 2$ by (Ex. 3.1), X is either the rational quartic curve in $\mathbb{P}^3$, or an elliptic curve.

$\boxed{n = 4}$ X is rational by (Ex. 1.5).

(b) Let $\mathcal{I}$ be the ideal sheaf of X in $\mathbb{P}^3$, and consider the exact sequence

$$0 \longrightarrow \mathcal{I} \longrightarrow \mathcal{O}_{\mathbb{P}^3} \longrightarrow \mathcal{O}_X \longrightarrow 0.$$

Twisting by $\mathcal{O}_{\mathbb{P}_3}(2)$ and taking cohomology gives

$$0 \longrightarrow \mathrm{H}^0(\mathbb{P}^3, \mathcal{I}(2)) \longrightarrow \mathrm{H}^0(\mathbb{P}^3, \mathcal{O}_{\mathbb{P}^3}(2)) \longrightarrow \mathrm{H}^0(X, \mathcal{O}_X(2)) \longrightarrow \cdots.$$

Now, X is an elliptic curve, and $\mathcal{O}_X(2)$ corresponds to a divisor D of degree 8. The Riemann Roch theorem gives $\dim \mathrm{H}^0(X, \mathcal{O}_X(2)) = 8$, and the vector space in the middle has dimension 10, so we conclude that $h^0(\mathbb{P}^3, \mathcal{I}(2)) \geq 2$. An element of $\mathrm{H}^0(\mathbb{P}^3, \mathcal{I}(2))$ is a form of degree 2, whose zero-set is a quadric surface containing X. Any such quadric must be irreducible and reduced because X is not contained in any hyperplane. Hence, X is contained in two distinct irreducible quadric surfaces and is a complete intersection because X is a curve of degree 4.

□

9. Prove the following lemma of Bertini: if X is a curve of degree d in $\mathbb{P}^3$, not contained in any plane, then for almost all planes $H \subseteq \mathbb{P}^3$ (meaning a Zariski open subset of the dual projective space $(\mathbb{P}^3)^*$), the intersection $X \cap H$ consists of exactly d distinct points, no three of which are collinear.

Proof. Bertini's theorem says a general plane in $\mathbb{P}^3$ intersects X in d distinct points. To show that no three of these points are collinear generically, it is enough to show a general plane in $\mathbb{P}^3$ does not contain a multisecant of X. The proof of (3.10) shows that the set of multisecants of X has dimension ≤ 1 in $(\mathbb{P}^3)^*$, and a general plane in $\mathbb{P}^3$ corresponds to a line in $(\mathbb{P}^3)^*$. Hence, for almost almost all planes $H \subseteq \mathbb{P}^3$, the intersection $X \cap H$ consists of exactly d distinct points, no three of which are collinear. □

12. For each value of $d = 2, 3, 4, 5$ and r satisfying $0 \leq r \leq \frac{1}{2}(d-1)(d-2)$, show that there exists an irreducible plane curve of degree d with r nodes and no other singularities.

Proof. There is nothing to prove for $d = 2$, so fix any $d \geq 3$, and denote $g = \frac{1}{2}(d-1)(d-2)$.

$$d = 2, 3, 4, 5, \quad g = 0, 1, 3, 6$$

Both extremes $r = 0, g$ are consructed in (3.11.1).

If X is any elliptic curve and P is any point in X, then the linear system $|dP|$ gives an embedding of X in $\mathbb{P}^{d-1}$ as a curve of degree d. Projecting the curve into $\mathbb{P}^2$ by (3.5) and (3.10) gives a plane curve of degree d and $g-1$ nodes. This gives $d = 4$ and $r = 2$.

By (Ex 3.6), the complete intersection of two quadric surfaces in $\mathbb{P}^3$ is an elliptic curve of degree 4. Projecting from a point not in the curve by (3.10) gives a plane curve of degree $d = 4$ and $r = 2$ nodes

Similarly, any divisor of degree ≥ 5 on a genus 2 curve is very ample by (Ex. 1.5), so there exists plane curves of degree $d \geq 5$ with $g - 2$ nodes. This gives $d = 5, r = 4$.

If $Q \subset \mathbb{P}^3$ is the nonsingular quadric suface, then there exists divisors of type (a, b) for any $a, b > 0$ that are linearly equivalent to an irreducible nonsingular curve of degree $a + b$ and genus $(a-1)(b-1)$ as a curve in $\mathbb{P}^3$.

A divisor of type $(2, 3)$ gives a nonsingular curve of degree $2 + 3 = 5$ and genus $(2-1)(3-1) = 2$ in $\mathbb{P}^3$. Projecting into $\mathbb{P}^2$ by (3.10) gives a plane curve of degree $d = 5$ and $r = 4$ nodes. Projecting this curve from one of its own points that does not lie in a plane spanned by coplanar tangent lines gives a plane curve of degree $d = 4$ and $r = 1$ node.

Now, consider a divisor of type $(3, 3)$. It is a nonsingular curve of degree $3+3 = 6$ and genus $(3-1)(3-1) = 4$ in $\mathbb{P}^3$. Projecting this curve from one of its points that does not lie in a plane spanned by coplanar tangent lines gives a plane curve of degree $d = 5$ and $r = 2$ nodes.

Finally, consider the divisor of type $(4, 2)$. It has degree $2 + 4 = 6$ and genus $(4-1)(2-1) = 3$ as a curve in $\mathbb{P}^3$. Projecting this curve from one of its points that does not lie in a plane spanned by coplanar tangent lines gives a plane curve of degree $d = 5$ and $r = 3$ nodes.

More generally, any divisor of degree $d \geq 2g + 1$ on any curve of genus g is very ample by (3.2), so there exists plane curves of degree d and $g - d$ nodes. This gives $d = 5$ and $r = 1$.

□

Chapter 4, Section 4

James Lee

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3. Let the elliptic curve X be embedded in $\mathbb{P}^2$ so as to have the equation $y^2 = x(x-1)(x-\lambda)$. Show that any automorphism of X leaving $P_0 = (0:1:0)$ fixed is induced by an automorphism of $\mathbb{P}^2$ coming from the automorphism of the affine (x,y)-plane given by

$$\begin{cases} x' = ax + b \\ y' = cy. \end{cases}$$

In each of the four cases of (4.7), describe these automorphisms of $\mathbb{P}^2$ explicitly, and hence determine the structure of the group $G = \mathrm{Aut}(X, P_0)$.

Proof. Let : $X \to \mathbb{P}^1$ be the unique g^1_2 with $f(P_0) = \infty$, branched over $0, 1, \lambda, \infty$. If $\sigma \in G$, then by (4.4) there exists an automorphism τ of $\mathbb{P}^1$ which fixes ∞ and satisfies $f \circ \sigma = \tau \circ f$. In particular, τ sends $\{0, 1, \lambda\}$ to $\{0, 1, \lambda\}$ in some order. If $\tau = \mathrm{id}$, then either $\sigma = \mathrm{id}$ or σ is the unique automorphism of order 2 fixing P_0 and interchanging the sheets of f. We call it the *hyperbolic involution* of X, and we denote it by $\iota : X \to X$. It comes from the automorphism of the affine (x,y)-plane corresponding to the reflection across the x-axis:

$$\begin{cases} x' = x \\ y' = -y \end{cases}.$$

The other automorphims are determined using the methods of (4.7). In particular, we look at the orbits of the action of S_3 on $k - \{0, 1\}$ (4.5).

$\boxed{j \neq 0, 1728}$ Since the orbits of λ are all distinct, $\tau = \mathrm{id}$, and $\sigma = \mathrm{id}$ or ι.

$\boxed{j = 1728, \mathrm{char}\, k \neq 3}$ This is the case $\lambda = -1, \frac{1}{2}$, or 2, and λ coincides with one other element of its orbit under the action of S_3. For example, if $\lambda = -1$, then $\lambda = 1/\lambda$, and G is the cyclic group of order 4 generated by

$$\begin{cases} x' = -x \\ y' = iy \end{cases}.$$

$\boxed{j = 0, \mathrm{char}\, k \neq 3}$ This is the case $\lambda = -\omega$ or $-\omega^2$, where ω is a third root of unity, and λ coincides with two other elements of its orbit under S_3. For example, if $\lambda = -\omega$, then

$$\lambda = \frac{1}{1-\lambda} = \frac{\lambda - 1}{\lambda},$$

and G is the cyclic group of order 6 generated by

$$\begin{cases} x' = \omega^2 x + 1 \\ y' = -y \end{cases}.$$

$\boxed{j = 0\ (= 1728), \mathrm{char}\, k = 3}$ If $\mathrm{char}\, k = 3$ and $j = 0$, then $\lambda = -1$, and all six elements of its orbit under S_3 coincide. Note that k has no nontrivial third roots of unity, and the values $\lambda = -1, \frac{1}{2}, 2$ of the previous case coincide too. So G is the cyclic group of order 12 generated by

$$\begin{cases} x' = x + 1 \\ y' = iy \end{cases}.$$

□

4. Let X be an elliptic curve in $\mathbb{P}^2$ given by an equation of the form

$$y^2 + a_1xy + a_3y = x^3 + a_2x^2 + a_4x + a_6.$$

Show that the j-invariant is a rational function of the a_i, with coefficients in $\mathbb{Q}$. In particular, if the a_i are all in some field $k_0 \subseteq k$, then $j \in k_0$ also. Furthermore, for every $\alpha \in k_0$, there exists an elliptic curve defined over k_0 with j-invariant equal to α.

Proof. Complete the square in y to get an equation of the form

$$y^2 = x^3 + Ax^2 + Bx + C,$$

with

$$A = \frac{a_1^2}{4} + a_2, \quad B = \frac{a_1a_3}{2} + a_4, \quad C = \frac{a_3^2}{4} + a_6.$$

Let α, β, γ be the distinct roots of the cubic on the right-hand side. Applying a linear transformation of x, which sends α, β to $0, 1$, transforms the equation of X into

$$y^2 = x(x-1)(x-\lambda), \quad \lambda = \frac{\gamma - \alpha}{\beta - \alpha}.$$

We can plug in λ into the j-invariant formula. Simplifying, we get a rational function of the elementary symmetric polynomials of α, β, γ (which are just A, B, C) with coefficients in $\mathbb{Q}$

$$j = 2^8 \frac{(A^2 - 3B)^3}{A^2B^2 - 4B^3 - 4A^3 + 18ABC - 27C^2}.$$

Since A, B, C are polynomials of the a_i with integer coefficients, the j-invariant is a rational function of the a_i with coefficients in $\mathbb{Q}$. □

5. Let X, P_0 be an elliptic curve having an endormorphism $f : X \to X$ of degree 2.

(a) If we represent X as a 2-1 covering of $\mathbb{P}^1$ by a morphism $\pi : X \to \mathbb{P}^1$ ramified at P_0, then as in (4.4), show that there is another morphism $\pi' : X \to \mathbb{P}^1$ and a morphism $g : \mathbb{P}^1 \to \mathbb{P}^1$, also of degree 2, such that $\pi \circ f = g \circ \pi'$.

(b) For a suitable choice of coordinates in the two copies of $\mathbb{P}^1$, show that g can be taken to be the morphism $x \mapsto x^2$.

(c) Now show that g is branched over two of the branch points of π, and that g^{-1} of the other two branch points of π consists of the four branch points of π'. Deduce a relation involving the invariant λ of X.

(d) Solving the above, show that there are just three values of j corresponding to elliptic curves with an endomorphism of degree 2, and find the corresponding values of λ and j.

Proof.

(a) The morphism $\pi \circ f$ corresponds to the linear system $|f^*(2P_0)|$. Notice that we can write $f^*(2P_0) = 2f^*(P_0)$, which gives a hint as to how to factor $\pi \circ f$. In particular, we can factor $\pi \circ f$ through the morphism $\pi' : X \to \mathbb{P}^1$ given by the linear system $|f^*(P_0)|$ on X, followed by the morphism $g : \mathbb{P}^1 \to \mathbb{P}^1$ given by the linear system $|\pi'_*(f^*P_0)|$ on $\mathbb{P}^1$.

(b) Any degree two morphism $g : \mathbb{P}^1 \to \mathbb{P}^1$ is branched over two points, and we can choose coordinates in the two copies of $\mathbb{P}^1$ so that these two points are $0, \infty$ in the target and source. Then g is given by the morphism $x \mapsto x^2$. Indeed, on some local affine patch, g is given by a degree two polynomial in x, say $g(x) = (x - \alpha)(x - \beta)$. Then g is ramified at $x = \frac{1}{2}(\alpha + \beta)$ and ∞, and we can choose coordinates so that these two points are $0, \infty$. Similarly, on the target, we can choose coordinates so that the two branch points of g are $0, \infty$. Then g is given by a degree two polynomial with roots at $0, \infty$, which is just $x \mapsto x^2$.

(c) The morphism $\pi \circ f$ is generically four-to-one, and it is branched over the branch points of π because any morphism of elliptic curves is unramified by the Riemann-Hurwitz formula. This just says $(\pi \circ f)^{-1}$ of a branch point of π consists of two points. Since g is a morphism of degree two ramified at exactly two points, and π' is ramified at exactly four points, g must be branched over two of the branch points of π,

and g^{-1} of the other two branch points of π consists of the four branch points of π'. Note that one of the points g is branched over is given by

$$g(\pi'(f^{-1}P_0)) = \pi(f(f^{-1}P_0)) = \pi(P_0).$$

Now, choose coordinates in the two copies of $\mathbb{P}^1$ so that the branch points of g are $0, \infty$, the other two branch points of π are $1, \lambda$, and g is given by $x \mapsto x^2$. Then the four branch points of π' are $\pm 1, \pm\sqrt{\lambda}$. Consider the linear transformation of $\mathbb{P}^1$ sending $-1, 1, \sqrt{\lambda}$ to $0, 1, \infty$, respectively, which is given by

$$\phi(x) = \frac{(1-\sqrt{\lambda})(x+1)}{2(x-\sqrt{\lambda})}$$

Then the branch points of π' are sent to $0, 1, \infty, \phi(-\sqrt{\lambda})$, where

$$\lambda' := \phi(-\sqrt{\lambda}) = -\frac{(1-\sqrt{\lambda})^2}{4\sqrt{\lambda}}.$$

Since λ' and λ give the same j-invariant, they must be related by a fractional linear transformation in S_3. In particular, λ' is in the orbit of λ under the action of S_3 on $k - \{0, 1\}$ (4.5).

(d) Solving for λ in the above, we get the following three unique values of λ modulo the action of S_3:

$$\lambda = 2, \quad 3 + 2\sqrt{2}, \quad \frac{1 + 3i\sqrt{7}}{2}.$$

The corresponding values of j are

$$j = 2^6 \cdot 3^3, \quad 2^6 \cdot 5^3, \quad -3^3 \cdot 5^3.$$

□

7. *The Dual of a Morphism.* Let X and X' be elliptic curves over k, with base points P_0, P_0'.

(a) If $f : X \to X'$ is any morphism, use (4.11) to show that $f^* : \operatorname{Pic} X' \to \operatorname{Pic} X$ induces a homomorphism $\hat{f} : (X', P_0') \to (X, P_0)$. We call this the *dual* of f.

(b) If $f : X \to X'$ and $g : X' \to X''$ are two morphisms, then $\widehat{g \circ f} = \hat{f} \circ \hat{g}$.

(c) Assume $f(P_0) = P_0'$, and let $n = \deg f$. Show that if $Q \in X$ is any point, and $f(Q) = Q'$, then $\hat{f}(Q') = n_X(Q)$. Conclude that $f \circ \hat{f} = n_{X'}$ and $\hat{f} \circ f = n_X$.

(d) If $f, g : X \to X'$ are two morphisms preserving the base points P_0, P_0', then $\widehat{f+g} = \hat{f} + \hat{g}$.

(e) Using (d), show that for any $n \in \mathbb{Z}$, $\hat{n}_X = n_X$. Conclude that $\deg n_X = n^2$.

(f) Show for any f that $\deg \hat{f} = \deg f$.

Proof. Let the pair $(X, \mathcal{P})$ (resp. $(X', \mathcal{P}')$) be the Jacobian of (X, P_0) (resp. (X', P_0')). If $\mathcal{C}$ is any category, and T is any object in $\mathcal{C}$, then denote by $h_T : \mathcal{C}^{\mathrm{op}} \to \mathbf{Set}$ the contravariant Hom functor defined by $U \mapsto \operatorname{Hom}_{\mathcal{C}}(U, T)$. Recall that X and $\mathcal{P} \in \operatorname{Pic}^\circ(X/X)$ satisfy the following universal property: if T is any scheme of finite type over k, and for any $\mathcal{M} \in \operatorname{Pic}^\circ(X/T)$, there is a unique morphism $f : T \to X$ such that $f^*\mathcal{P} \cong \mathcal{M}$ in $\operatorname{Pic}^\circ(X/T)$. In other words, $(X, \mathcal{P})$ represents the functor $\operatorname{Pic}^\circ(X/-)$. Lastly, $(X, \mathcal{P})$ is a group object in the category of schemes of finite type over k, so for any T, $h_X(T)$ has a natural group structure induced by $\operatorname{Pic}^\circ(X/T)$.

(a) Any morphism $f : X \to X'$ defines a natural transformation $f^* : \operatorname{Pic}^\circ(X'/-) \to \operatorname{Pic}^\circ(X/-)$ of group functors in the following way (4.10.2). If T is any scheme of finite type over k, then $f' = f \times 1_T : X' \times T \to X \times T$ induces a group homomorphism $f^* : \operatorname{Pic}(X' \times T) \to \operatorname{Pic}(X \times T)$. Let $p : X \times T \to T$ and $p' : X' \times T \to T$ be the projection maps. The image of f^* lies in $\operatorname{Pic}^\circ(X \times T)$, and $p = p' \circ f'$ implies $f^* p'^* \operatorname{Pic}(T) = p^* \operatorname{Pic}(T)$. Hence, we have a well-defined group homomomorphism $f_T^* : \operatorname{Pic}^\circ(X'/T) \to \operatorname{Pic}^\circ(X/T)$ for each scheme T of finite type over k.

It remains to show f^* is functorial. Let $g : S \to T$ be any morphism of schemes of finite type over k. We can apply $\mathrm{Pic}(-)$ to the following commutative diagram

$$\begin{array}{ccc} X \times S & \xrightarrow{f \times 1_S} & X' \times S \\ {\scriptstyle 1_X \times g}\downarrow & & \downarrow{\scriptstyle 1_{X'} \times g} \\ X \times T & \xrightarrow{f \times 1_T} & X' \times T \end{array},$$

which gives a commutative diagram of Picard groups with arrows given by taking inverse images of invertible sheaves.

Now, we identify h_X and $h_{X'}$ with $\mathrm{Pic}^\circ(X/-)$ and $\mathrm{Pic}^\circ(X'/-)$, respectively. Yoneda's lemma says there exists a natural bijection between the set of morphisms $X' \to X$ and the set of natural transformations of functors $h_{X'} \to h_X$. Hence, there exists a unique morphism $\hat{f} \in \mathrm{Hom}(X', X)$ such that $f^* \in \mathrm{Hom}(h_{X'}, h_X)$ is given by composition with $\hat{f}$. The invertible sheaves $\mathcal{O}_X$ and $\mathcal{O}_{X'}$ correspond to $P_0 \in X$ and $P_0' \in X'$, respectively, and structure sheaves pullback, so we conclude that $\hat{f}(P_0') = P_0$.

(b) This is a consequence of the uniqueness property of $\hat{f}$ in $\mathrm{Hom}(X', X)$, and the analogous fact for inverse image of sheaves, i.e., $(g \circ f)^* = f^* \circ g^*$.

(c) The points $Q' \in X'$ and $\hat{f}(Q') \in X$ correspond to $\mathcal{O}_{X'}(Q' - P_0')$ in $\mathrm{Pic}^\circ(X'/k)$ and $f^*\mathcal{O}_{X'}(Q' - P_0')$ in $\mathrm{Pic}^\circ(X/k)$, respectively. Thus, it is enough to show the linear equivalence of the following divisors

$$f^*(Q' - P_0') \sim n_X(Q) - P_0.$$

By the hypothesis, $Q' - P_0' = f_*(Q - P_0)$, which by (Ex. 2.6) implies

$$f^*(Q' - P_0') = n(Q - P_0).$$

The right-hand side is linearly equivalent to $n_X(Q) - P_0$ because the group law on X is induced by that of $\mathrm{Pic}^\circ(X/k)$, so we conclude that $\hat{f} \circ f = n_X$. For the second corollary, observe that f and multiplication by n commutes since f is a morphism of proper group schemes. Hence, we have

$$(f \circ \hat{f})(Q') = f(n_X(Q)) = n_{X'}(f(Q)) = n_{X'}(Q').$$

(d) It is enough to show for any $\mathcal{L}' \in \mathrm{Pic}^\circ(X')$, that

$$(f + g)^*\mathcal{L}' \cong f^*\mathcal{L}' \otimes g^*\mathcal{L}'.$$

Let Q' be the closed point in X' such that $\mathcal{L}' \cong \mathcal{O}_{X'}(Q' - P_0')$. There exists a bijection

$$\begin{aligned} \mathrm{Hom}_{\mathbf{Sch}(k)}(X, X') &\xrightarrow{\sim} \mathrm{Pic}^\circ(X'/X) \\ f : X \to X' &\mapsto \mathcal{L}_f := (f \times 1_{X'})^*\mathcal{P}', \end{aligned}$$

where the group law on the left-hand side is induced by $\mathrm{Pic}^\circ(X'/X)$, i.e., $\mathcal{L}_{f+g} = \mathcal{L}_f \otimes \mathcal{L}_g$ is a representative of $f+g$ in $\mathrm{Pic}(X \times X')$ for any $f, g : X \to X'$. Let $\sigma : X \to X \times X'$ be the section $x \mapsto (x, Q')$. If we assume $\mathcal{P}' = \mathcal{O}_{X' \times X'}(\Delta - X' \times P_0')$ as in (4.11), then $\sigma^*\mathcal{L}_f \cong f^*\mathcal{O}_{X'}(Q')$, so replacing $\mathcal{P}'$ by $\mathcal{P}' \otimes p_2^*\mathcal{O}_{X'}(-P_0)$, we get $\sigma^*\mathcal{L}_f \cong f^*\mathcal{L}'$. Hence,

$$(f + g)^*\mathcal{L}' \cong \sigma^*\mathcal{L}_{f+g} \cong \sigma^*\mathcal{L}_f \otimes \sigma^*\mathcal{L}_g \cong f^*\mathcal{L}' \otimes g^*\mathcal{L}'.$$

(e) Proceeding by induction on n reduces the statement to the case $n = 2$. The dual of the identity is the identity. Hence, $\hat{2}_X = \widehat{1_X + 1_X} = 1_X + 1_X = 2_X$ by (d). According to (c), $\hat{n}_X \circ n_X$ is multiplication by $\deg n_X$. Since $\hat{n}_X \circ n_X = n_X^2$, we conclude that $\deg n_X = n^2$.

(f) By (c) and (e), $\deg \hat{f} \circ f = n^2$. The degree of morphisms of curves is multiplicative with respect to composition. Hence, $\deg \hat{f} = n$.

□

9. We say two elliptic curves X, X' are *isogenous* if there is a finite morphism $f : X \to X'$.

(a) Show that isogeny is an equivalence relation.

(b) For any elliptic curve X, show that the set of elliptic curves X' isogenous to X, up to isomorphism is countable.

Proof.

(a) The identity morphism is trivially finite, so any elliptic curve is isogenous to itself. A composition of finite morphisms is finite, so isogeny satisfies transitivty. Lastly, (Ex. 4.7) implies isogeny is reflexive. Hence, isogeny is an equivalence relation.

(b) Since $\ker f$ is a closed subgroup scheme of X, it is a finite set of points, all of which are torsion under the group law of X. Every torsion point of X lies in the kernel of a multiplication map $n_X : X \to X$ for some $n > 0$. By (Ex. 4.7), n_X is an étale covering of degree n^2, which implies X has exactly n^2 points of order n. Hence, there are countably many torsion points on X. The set of finite subsets of a countable set is countable, so there are only countably many candidates for $\ker f$. Since X' is uniquely determined by X and $\ker f$, the set of elliptic curves isogenous to X is countable.

□

11. Let X be an elliptic curve over $\mathbb{C}$, defined by the elliptic functions with periods $1, \tau$. Let R be the the ring of endormorphisms of X.

(a) If $f \in R$ is a nonzero endormorphism correspondending to complex multiplication by α as in (4.18), show that $\deg f = |\alpha|^2$.

(b) If $f \in R$ corresponds to $\alpha \in \mathbb{C}$ again, show that the dual $\hat{f}$ of (Ex. 4.7) corresponds to the complex conjugate $\bar{\alpha}$ of α.

(c) If $\tau \in \mathbb{Q}(\sqrt{-d})$ happens to be integral over $\mathbb{Z}$, show that $R = \mathbb{Z}[\tau]$.

Proof.

(a) It is not hard to see $\deg f$ is the size of the quotient $|\Lambda/\alpha\Lambda|$. Equivalently, it is the ratio between the areas of the parallelograms with vertices $\alpha, \alpha\tau$, and $1, \tau$. But α just scales the area by $|\alpha|^2$. Hence, $\deg f = |\alpha|^2$.

(b) By (Ex. 7c), $\hat{f} \circ f$ is multiplication by $|\alpha|^2$, but f is multiplication by α, and $\bar{\alpha}$ is the unique element in $\mathbb{C}$ such that $\alpha\bar{\alpha} = |\alpha|^2$. Hence, $\hat{f}$ is multiplication by $\bar{\alpha}$.

(c) The ring of integers of $\mathbb{Q}(\sqrt{-d})$ is given by

$$\begin{cases} \mathbb{Z}[\sqrt{-d}] & d \equiv 1, 2 \mod 4 \\ \mathbb{Z}\left[\dfrac{1+\sqrt{-d}}{2}\right] & d \equiv 3 \mod 4 \end{cases}.$$

By (4.19), if $\tau = r + s\sqrt{-d}$ for some $r, s \in \mathbb{Q}$, the endomorphism ring R is

$$R = \{a + b\tau \mid a, b \in \mathbb{Z}, \quad 2br, b(r^2 + ds^2) \in \mathbb{Z}\}.$$

Suppose τ is integral over $\mathbb{Z}$. If $d \equiv 1, 2 \mod 4$, then $r, s \in \mathbb{Z}$, so $a + b\tau \in R$ for all $a, b \in \mathbb{Z}$. If $d \equiv 3 \mod 4$, then we can write $\tau = (r' + s'\sqrt{-d})/2$ with $r', s' \in \mathbb{Z}$ such that $r' \equiv s' \mod 2$. For any $b \in \mathbb{Z}$, the first condition $2br = br' \in \mathbb{Z}$ is easily satisfied. Also, we have

$$r^2 + ds^2 = \frac{r'^2 + ds'^2}{4} \in \mathbb{Z}$$

for all $b \in \mathbb{Z}$ since $r'^2, s'^2 \equiv 0, 1 \mod 4$. Hence, $R = \mathbb{Z}[\tau]$.

□

12. Again let X be an elliptic curve over $\mathbb{C}$ determined by the elliptic functions with periods $1, \tau$, and assume that τ lies in the region G of (4.15B).

(a) If X has any autmorphisms leaving P_0 fixed other than ± 1, show that either $\tau = i$ or $\tau = \omega$, as in (4.20.1) and (4.20.2). This gives another proof of the fact (4.7) that there are only two curves, up to isomorphism, having automorphisms other than ± 1.

(b) Now show that there are exactly three values of τ for which X admits an endomorphism of degree 2. Can you match these with the three values of j determined in (Ex. 4.5)?

Proof.

(a) By (Ex. 4.11), τ must be a root of unity in G. The only roots of unity in G are i and ω.

(b) By (Ex. 4.11), we need to find the values of τ such that Λ contains an element of norm 2. In particular, if we write $m + n\tau \in \Lambda$, then we need to determine when

$$m^2 + n^2|\tau|^2 + 2mn\,\mathrm{Re}(\tau) = 4$$

admits a solution for some $m, n \in \mathbb{Z}$. Using the bounds on $|\tau|$ and $\mathrm{Re}(\tau)$ as specified in (4.15B), the only possible values of τ are

$$\tau = i, \quad i\sqrt{2}, \quad -\frac{1}{2} + i\frac{\sqrt{7}}{2}.$$

The corresponding values of j are

$$j = 2^6 \cdot 3^3, \quad 2^6 \cdot 5^3, \quad -3^3 \cdot 5^3.$$

□

13. If $p = 13$, there is just one value of j for which the Hasse invariant of the corresponding curve is 0. Find it.

Proof. The polynomial $h_p(\lambda)$ of (4.22) for $p = 13$ is

$$h_{13}(\lambda) = \lambda^6 + 10\lambda^5 + 4\lambda^4 + 10\lambda^3 + 4\lambda^2 + 10\lambda + 1,$$

and an elliptic curve has Hasse invariant 0 if and only if $h_{13}(\lambda) = 0$. On the otherhand, consider the j-invariant

$$j = 2^8 \frac{(\lambda^2 - \lambda + 1)^3}{\lambda^2(\lambda - 1)^2}.$$

Expanding above modulo 13 gives

$$\lambda^6 + 10\lambda^5 + (10j + 6)\lambda^4 + (6j + 6)\lambda^3 + (10j + 6)\lambda^2 + 10\lambda + 1 = 0.$$

Matching coefficients with $h_{13}(\lambda)$ and solving for j gives $j \equiv 5 \mod 13$. □

14. The Fermat curve $X : x^3 + y^3 = z^3$ gives a nonsingular curve in characteristic p for every $p \neq 3$. Determine the set $\mathfrak{P} = \{p \neq 3 \mid X_{(p)} \text{ has Hasse invariant } 0\}$, and observe (modulo Dirichlet's theorem) that it is a set of primes of density 1/2.

Proof. Using (4.21) with $f(x, y, z) = x^3 + y^3 - z^3$, we find that the coefficient of $(xyz)^{p-1}$ in f^{p-1} is zero if and only if $p \equiv 2 \mod 3$. By Dirichlet's theorem, the density of $\mathfrak{P}$ is $1/\phi(3) = 1/2$. □

15. Let X be an elliptic curve over a field k of characteristic p. Let $F' : X_p \to X$ be the k-linear Frobenius morphism (2.4.1). Use (4.10.7) to show that the dual morphism $\hat{F}' : X \to X_p$ is separable if and only if the Hasse invariant of X is 1. Now use (Ex. 4.7) to show that if the Hasse invariant is 1, then the subgroup of points of order p on X is isomorphic to $\mathbb{Z}/p$; if the Hasse invariant is 0, then it is 0.

Proof. By (III, Ex. 8.1) and (4.10.7), if $f : X \to X'$ is any morphism of elliptic curves over k, then the induced map of cohomology $f^* : \mathrm{H}^1(X, \mathcal{O}_{X'}) \to \mathrm{H}^1(X, \mathcal{O}_X)$ is precisely the map of tangent spaces at 0 of the dual map $\hat{f} : X' \to X$. By the Riemann-Hurwitz formula, any separable morphism of elliptic curves is not ramified. Thus, a morphism of elliptic curves $f : X \to X'$ is inseparable if and only if f is ramified at a point, in which case f is ramified everywhere by the group structure of elliptic curves. Also, any morphism of curves is ramified at a point if and only if the map of tangent spaces is not an isomorphism. Hence, $\hat{F}'$ is separable if and only if the map of tangent spaces $F'^* : \mathrm{H}^1(X, \mathcal{O}_X) \to \mathrm{H}^1(X_p, \mathcal{O}_{X_p})$ is not the zero map if and only if the Hasse invariant of X is 1.

The subgroup of points of order p on X is the kernel of the multiplication by p morphism $p_X : X \to X$. Since the k-linear Frobenius morphism F' has degree p, we have $p_X = F' \circ \hat{F}'$ by (Ex. 4.7c). Also, F' is a homeomorphism on the underlying topological spaces, i.e., F' is a one-to-one map. If the Hasse invariant of X is 1, then $\hat{F}'$ is a separable morphism of degree p, so $\hat{F}'$ is a p-to-one map. Hence, p_X is a p-to-one map, and $(\ker p_X)[k] \cong \mathbb{Z}/p$. If the Hasse invariant of X is 0, then $p_X = F' \circ \hat{F}'$ is a purely inseparable morphism of curves. Hence, p_X is a one-to-one map, and $(\ker p_X)[k] = 0$. □

16. Again let X be an elliptic curve over k of characteristic p, and suppose X is defined over the field $\mathbb{F}_q$ of $q = p^r$ elements, i.e., $X \subseteq \mathbb{P}^2$ can be defined by an equation with coefficients in $\mathbb{F}_q$. Assume also that X has a rational point over $\mathbb{F}_q$. Let $F' : X_q \to X$ be the k-linear Frobenius with respect to q.

(a) Show that $X_q \cong X$ as schemes over k, and that under this identification, $F' : X_q \to X$ is the map obtained by the qth-power map on the coordinates of points of X, embedded in $\mathbb{P}^2$.

(b) Show that $1_X - F'$ is a separable morphism and its kernel is just the set $X(\mathbb{F}_q)$ of points of X with coordinates in $\mathbb{F}_q$.

(c) Using (Ex. 4.7), show that $F' + \hat{F}' = a_X$ for some integer a, and that $N = q - a + 1$, where $N = \#X(\mathbb{F}_q)$.

(d) Use the fact that $\deg(m + nF') > 0$ for all $m, n \in \mathbb{Z}$ to show that $|a| \leq 2\sqrt{q}$. This is Hasse's proof of the analogue of the Riemann hypothesis for elliptic curves.

(e) Now assume $q = p$, and show that the Hasse invariant of X is 0 if and only if $a \equiv 0 \mod p$. Conclude that for $p \geq 5$ that X has Hasse invariant 0 if and only if $N = p + 1$.

Proof.

(a) Suppose X is locally isomorphic to $\operatorname{Spec} A$, where $A = k[x, y]/(f)$ and $f(x, y)$ is some degree three polynomial with coefficients in $\mathbb{F}_q \subset k$. Then X_p is locally isomorphic to $\operatorname{Spec}(k \otimes_q A)$, where the copy of k here has the qth-power Frobenius for the structure morphism as a k-algebra. The image of f under the identification

$$\begin{aligned} k \otimes_q k[x, y] &\cong k[x, y] \\ (b \cdot 1)x &\mapsto b^q x, \end{aligned}$$

gives a defining equation $f^{(q)}$ for X_q. Notice that $f^{(q)}$ is obtained from f by raising each coefficient of f to the qth-power. Since each coefficient of f is in $\mathbb{F}_q$, we see that $f^{(q)}$ and f are actually equal in $k[x, y]$. Hence, $X_q \cong X$ as schemes over k.

Under this indentification, $F' : X_q \to X$ is obtained from the restriction of the Frobenius on $\mathbb{P}^2$. The k-linear Frobenius morphism on an affine coordinate ring of $\mathbb{P}^2$ can be written as

$$\begin{aligned} F^* : k[x, y] &\to k[x, y] \\ g(x, y) &\mapsto g(x^q, y^q), \end{aligned}$$

which is exactly the ring homomorphism that gives the qth-power map on the coordinates.

(b) If $t \in \mathcal{O}_{P,X}$ is a local parameter of X, then $(1_X - F')^* t = t - t^p$. Since $\mathrm{d}(t - t^p) = \mathrm{d}t$, the map $1_X - F'$ is ramified nowhere. Hence, $1_X - F'$ is a separable morphism. Under the embedding $X \subset \mathbb{P}^2$, the kernel of $1_X - F'$ consists of the points fixed by the map on X obtained by the qth-power map on the coordinates of points of X. Since $\mathbb{F}_q$ is precisely the fixed field of k under the qth-power Frobenius, the kernel of $1_X - F'$ consists of the points of $X \subset \mathbb{P}^2$ with coordinates in $\mathbb{F}_q$.

(c) Recall from (Ex. 4.7), (Ex. 4.15), and (Ex. 4.16b) that

$$(1_X - F') \circ \widehat{(1_X - F')} = N_X, \quad F' \circ \hat{F}' = q_X.$$

Expanding the left-hand side of the first identity gives

$$(1_X - F') \circ \widehat{(1_X - F')} = 1_X - (F' + \hat{F}') + q_X.$$

Hence, we have

$$F' + \hat{F}' = q_X + 1_X - N_X.$$

(d) For all $m, n \in \mathbb{Z}$, we have the inequality

$$m^2 + n^2 q - mna = \deg(m - nF') > 0.$$

Rearranging gives

$$|a| < \left| \frac{m}{n} + \frac{n}{m} q \right|$$

The real-valued function $f(t) = t + q/t$ has a local extremas at $(\sqrt{q}, \pm 2\sqrt{q})$. The rational numbers are dense in the real numbers, so we conclude that $|a| \leq 2\sqrt{q}$. I wonder if it is every the case $a = 2\sqrt{q}$ when $\sqrt{q}$ is a perfect square.

(e) By (Ex. 4.15), the Hasse invariant of X is 0 if and only if $\hat{F}'$ is inseparable. If $t \in \mathcal{O}_{P,X}$ is a local parameter, then

$$(F' + \hat{F}')^* t = t^p + \hat{F}'^* t,$$

so $\hat{F}'$ is inseparable if and only if $F' + \hat{F}'$ is inseparable. The conclusion follows from the fact that a_X is inseparable if and only if $a \equiv 0 \mod p$. For $p \geq 5$, $a \equiv 0 \mod p$ and $|a| \leq 2\sqrt{p}$ can be both satisfied if and only if $a = 0$. What about the cases for $p < 5$?

□

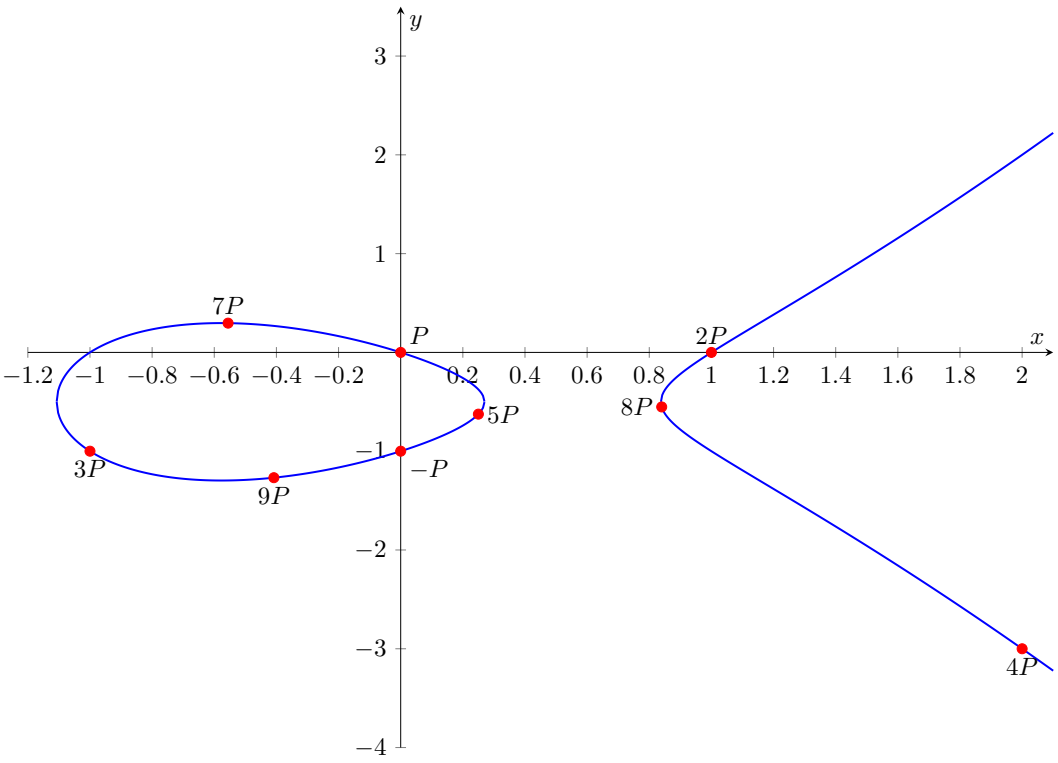
17. Let X be the curve $y^2 + y = x^3 - x$ of (4.23.8).

(a) If $Q = (a, b)$ is a point on the curve, compute the coordinates of the point $P + Q$, where $P = (0, 0)$, as a function of a, b. Use this formula to find the coordinates nP, $n = 1, 2, \ldots, 10$.

(b) This equation defines a nonsingular curve over $\mathbb{F}_q$ for all $p \neq 37$.

Proof.

(a) $P + Q = \left(-\frac{a+b}{a^2}, \frac{b(a+b)}{a^3} - 1 \right)$, $\quad -P = (0, -1)$, $\quad 2P = (1, 0)$



(b) Let X be a cubic curve in $\mathbb{P}^2$ given by an equation of the form

$$y^2 + a_1xy + a_3y = x^3 + a_2x^2 + a_4x + a_6, \quad a_i \in k.$$

Define the following quantities

$$\begin{aligned} b_2 &= a_1^2 + 4a_2 \\ b_4 &= 2a_4 + a_1a_3 \\ b_6 &= a_3^2 + 4a_6 \\ b_8 &= a_1^2a_6 + 4a_2a_6 - a_1a_3a_4 + a_2a_3^2 - a_4^2. \end{aligned}$$

The discriminant of X is given by

$$\Delta = -b_2^2b_8 - 8b_4^3 - 27b_6^2 + 9b_2b_4b_6,$$

and X defines a nonsingular curve over $\mathbb{F}_q$ if and only if $\Delta \neq 0$ in $\mathbb{F}_q$. For our elliptic curve, the quantities b_i are

$$b_2 = 0, \quad b_4 = -2, \quad b_6 = 1, \quad b_8 = -1,$$

so the determinant of X is $\Delta = 37$. Hence, X defines a nonsingular curve over $\mathbb{F}_q$ for all $p \neq 37$.

□

20. Let X be an elliptic curve over a field k of characteristic $p > 0$, and let $R = \mathrm{End}(X, P_0)$ be its ring of endormorphisms.

(a) Let X_p be the curve over k defined by changing the k-structure of X (2.4.1). Show that $j(X_p) = j(X)^{1/p}$. Thus, $X \cong X_p$ over k if and only if $j \in \mathbb{F}_p$.

(b) Show that p_X in R factors into a product of $\pi\hat{\pi}$ of two elements of degree p if and only if $X \cong X_p$. In this case, the Hasse invariant of X is 0 if and only if π and $\hat{\pi}$ are associates in R.

(c) If $\mathrm{Hasse}(X) = 0$ show in any case $j \in \mathbb{F}_{p^2}$.

(d) For any $f \in R$, there is an induced map $f^* : \mathrm{H}^1(\mathcal{O}_X) \to \mathrm{H}^1(\mathcal{O}_X)$. This must be multiplication by an element $\lambda_f \in k$. So we obtain a ring homomorphism $\varphi : R \to k$ by sending f to λ_f. Show that any $f \in R$ commutes with the (nonlinear) Frobenius morphism $F : X \to X$, and conclude that if $\mathrm{Hasse}(X) \neq 0$, then the image of φ is $\mathbb{F}_p$. Therefore, R contains a prime ideal $\mathfrak{p}$ with $R/\mathfrak{p} \cong \mathbb{F}_p$.

Proof.

(a) Some of the statements proved in (Ex. 4.16) hold here as well. In particular, under the idenficiation of the k-linear Frobenius morphism $F^* : k[x, y] \to k[x, y]$ defined by $x, y \mapsto x^p, y^q$, let $f(x, y)$ be a cubic polynomal that locally defines X in $\mathbb{P}^2$. The image of $f(x, y)$ under F^* is $f(x^p, x^p) = 0$, so if we write

$$f(x, y) = y^2 - x(x - 1)(x - \lambda),$$

then

$$F^*f(x, y) = y^{2p} - x^p(x^p - 1)(x^p - \lambda).$$

Since k is algebraically closed, there exists a unique pth root $\lambda^{1/p}$ of λ. Thus, we can factorize

$$y^{2p} - x^p(x^p - 1)(x^p - \lambda) = (y^2 - x(x - 1)(x - \lambda^{1/p}))^p.$$

The radical of the right-hand side gives the defining equation of X_p. Plugging in $\lambda^{1/p}$ gives $j(X_p) = j(X)^{1/p}$.

(b) If X admits a degree p endomorphism $\pi : X \to X$, then $X \cong X_p$ by (2.5). Conversely, if $X_p \cong X$, then p_X factors as $F' \circ \hat{F}'$, where $F' : X_p \to X$ is the k-linear Frobenius morphism (Ex. 4.7). By (Ex. 4.15), the Hasse invariant of X is 0 if and only if $\hat{F}'$ is an inseparable morphism. The Frobenius is the unique purely inseparable morphism of degree p (2.5), so if $X_p \cong X$, then F' and $\hat{F}'$ must differ by an automorphism of X, i.e., they are associates in R. If F' and $\hat{F}'$ are associates, then $\hat{F}'$ is an inseparable morphism since F' is inseparable. Hence, the Hasse invariant of X is 0.

(c) If $\mathrm{Hasse}(X) = 0$, then $\hat{F}' : X \to X_p$ is a purely inseparable morphism of degree p. By (2.5), X and the pth Frobenius twist $(X_p)_p \cong X_{p^2}$ of X_p are isomorphic as abstract k-schemes. Hence, $j(X) = j(X)^{1/p^2}$ by (Ex. 4.20a).

(d) Clearly f and F commute on the underlying topological spaces. The statement holds for any homomorphism between k-algebras, so it is immediate that f and F commute at the level of sheaves as well. If $\mathrm{Hasse}(X) \neq 0$, then the p-linear map of cohomology $F^* : \mathrm{H}^1(X, \mathcal{O}_X) \to \mathrm{H}^1(X, \mathcal{O}_X)$ is non-zero, and commutes with $f^* : \mathrm{H}^1(X, \mathcal{O}_X) \to \mathrm{H}^1(X, \mathcal{O}_X)$. In particular, $F^*(1) \neq 0$, and $(f \circ F)^*(1) = (F \circ f)^*(1)$. Expanding both sides gives

$$(f \circ F)^*(1) = F^*(f^*(1)) = F^*(\lambda_f \cdot 1) = \lambda_f^p \cdot F^*(1)$$
$$(F \circ f)^*(1) = f^*(F^*(1)) = \lambda_f \cdot F^*(1).$$

Since $F^*(1)$ is non-zero, we have $\lambda_f^p = \lambda_f$. Hence, λ_f is in $\mathbb{F}_p$.

□

Chapter 4, Section 5

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1. Show that a hyperelliptic curve can never be a complete intersection in any projective space.

Proof. Combine (Ex. 3.3) and (5.2). □

2. If X is a curve of genus ≥ 2 over a field of characteristic 0, show that the group $\operatorname{Aut} X$ of automorphisms of X is finite.

Proof. $\boxed{X \text{ is hyperelliptic.}}$ Let $\pi : X \to \mathbb{P}^1$ be the unique g^1_2 of X. By the Riemann-Hurwitz formula, π is ramified at $2g+2$ points. If σ is an automorphism of X, then there exists $\tau \in \mathrm{PGL}(1)$ such that the following diagram commutes

$$\begin{array}{ccc} X & \xrightarrow{\sigma} & X \\ \downarrow & & \downarrow \\ \mathbb{P}^1 & \xrightarrow{\tau} & \mathbb{P}^1 \end{array}.$$

This means $\mathrm{Aut}(X)$ permutes the branch points of π, so there is a group homorphism $\mathrm{Aut}(X) \to S_{2g+2}$. The latter is a finite group, so it remains to show it has finite kernel. If σ fixes every branch point, then τ is the identity, so σ is either the identity map, or the unique hyperelliptic involution (think in terms of function fields and the fact that the hyperellitpic involution is induced by the unique non-trivial element in $\mathrm{Gal}(K(X)/K(\mathbb{P}^1))$).

$\boxed{X \text{ is nonhyperelliptic.}}$ Consider the canonical embedding $X \hookrightarrow |K| \cong \mathbb{P}^{g-1}$. If σ is an automorphism of X, then there exists $\tau \in \mathrm{PGL}(g-1)$ such that we have the commutative diagram

$$\begin{array}{ccc} X & \xrightarrow{\sigma} & X \\ \downarrow & & \downarrow \\ \mathbb{P}^{g-1} & \xrightarrow{\tau} & \mathbb{P}^{g-1} \end{array}.$$

By (Ex. 4.6), X has $g(g^2-1)$ hyperoscullation points in this embedding, and τ must permute them because linear transformations preserve incidence structures. So there is a group homomorphism $\mathrm{Aut}(X) \to S_{g(g^2-1)}$, and since $S_{g(g^2-1)}$ is finite, it remains to show the kernel is finite. For every $g \geq 3$, $X \cap H$ can contain at most point two hyperoscullation points for any hyperplane $H \subset \mathbb{P}^{g-1}$, i.e., no three hyperoscullation points are collinear. It is a general fact that any automorphism in $\mathrm{PGL}(n)$ is the identity map if it fixes at least $n+2$ points in general position. Since $g(g^2-1) \geq g+1$ for every $g \geq 3$, if τ fixes the hyperoscullation points, it must be the identity. Hence, σ is the identity. As a corollary, any automorphism of a nonhyperelliptic is induced by an automorphism of the canonical embedding, i.e., there exists an injective map $\mathrm{Aut}(X) \to \mathrm{PGL}(g-1)$. □

3. *Moduli of Curves of Genus 4.* The hyperelliptic curves of genus 4 form an irreducible family of dimension 7. The nonhyperelliptic ones form an irreducible family of dimension 9. The subset of those having only one g^1_3 is an irreducible family of dimension 8.

Proof. Hyperelliptic elliptic curves of genus 4 are determined as two-fold coverings of $\mathbb{P}^1$, ramified at $0, 1, \infty$, and $2g-1 = 7$ additional points, up to the action of a certain finite group. So the hyperelliptic curves form an irreducible subvariety of dimension 7 of $\mathfrak{M}_4$ (5.5.5).

The canonical model of a nonhyperelliptic curve of genus 4 is the complete intersection of a unique quadric surface and a cubic surface in $\mathbb{P}^3$ (5.2.2). The family of all these curves is parametrized by the space $\mathfrak{S}$

of all complete intersections of a quadric surface and a cubic surface. We can think of $\mathfrak{S}$ as the set of all homogenous ideals generated by a degree two and three homogenous polynomial in $k[x_1, \ldots, x_4]$. So there is a morphism $\mathfrak{S} \to \mathfrak{M}_4$, whose fibres are images of the group PGL(3), which has dimension 15. It remains to compute the dimension of $\mathfrak{S}$. There is a projection map $\mathfrak{S} \to \mathfrak{Q}$, where $\mathfrak{Q}$ is the space of all irreducible quadric surfaces in $\mathbb{P}^3$. It is a locally closed subset of $|\,\mathrm{H}^0(\mathbb{P}^3, \mathcal{O}_{\mathbb{P}^3}(2))|$, which has dimension $\binom{3+2}{2} - 1 = 9$. The fibres of $\mathfrak{S} \to \mathfrak{Q}$ is the image of pullback of the linear system of cubic surfaces in $\mathbb{P}^3$, which is just $|\,\mathrm{H}^0(Q, \mathcal{O}_Q(3))|$ for some quadric surface $Q \subset \mathbb{P}^2$. Note that the dimension of this space is independent of the choice of Q. Consider the short exact sequence

$$0 \longrightarrow \mathcal{O}_{\mathbb{P}^3}(-2) \longrightarrow \mathcal{O}_{\mathbb{P}^3} \longrightarrow \mathcal{O}_Q \longrightarrow 0.$$

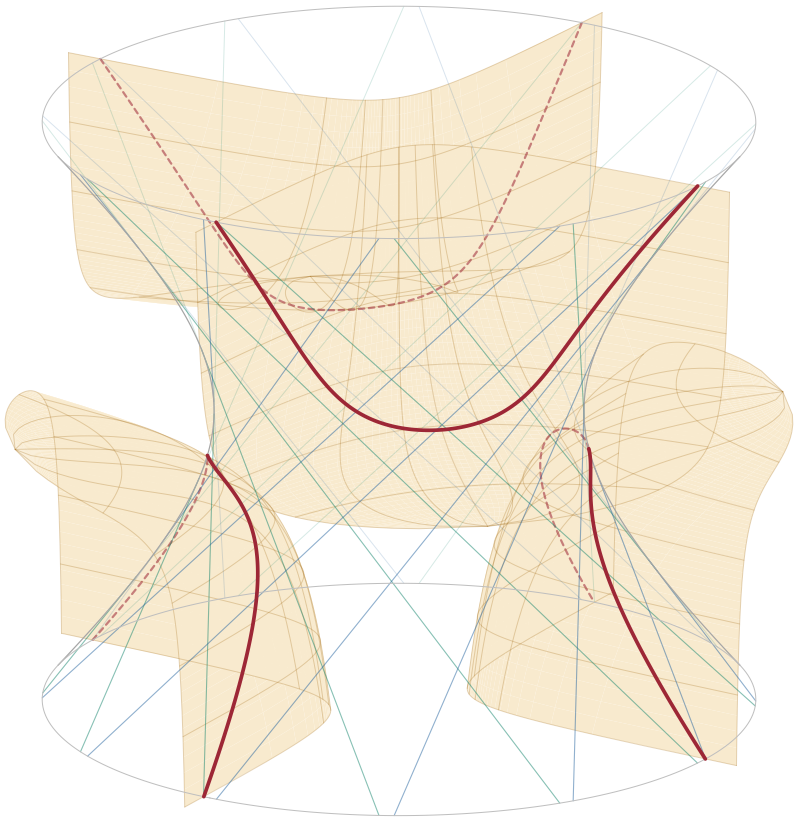
Tensoring by $\mathcal{O}_{\mathbb{P}^3}(3)$ and taking cohomology gives

$$0 \longrightarrow \mathrm{H}^0(\mathbb{P}^3, \mathcal{O}_{\mathbb{P}^3}(1)) \longrightarrow \mathrm{H}^0(\mathbb{P}^3, \mathcal{O}_{\mathbb{P}^3}(3)) \longrightarrow \mathrm{H}^0(Q, \mathcal{O}_Q(3)) \longrightarrow 0.$$

By exactness, we have

$$\dim |\,\mathrm{H}^0(Q, \mathcal{O}_Q(3))| = \binom{3+3}{3} - \binom{3+1}{1} - 1 = 15,$$

which implies the fibres of $\mathfrak{S} \to \mathfrak{Q}$ have dimension 15, so $\mathfrak{S}$ has dimension $9 + 15 = 24$. Hence, $\mathfrak{M}_4$ has dimension $24 - 15 = 9$. The subspace of $\mathfrak{Q}$ consisting of quadric cones have dimension 8, so the space of these curves have codimension one in $\mathfrak{M}_4$. Hence, the subset of of nonhyperelliptic curves having only one g^1_3 is an irreducible family of dimension 8. □



$$Q:\ x^2 + y^2 - z^2 = w^2, \qquad S:\ 8xyz = w^3.$$

$$C \sim (3,3), \qquad g(C) = 4, \qquad \omega_C \cong \mathcal{O}_C(1).$$

5. *Curves of Genus 5.* Assume X is not hyperelliptic.

(a) The curves of genus 5 whose canonical model in $\mathbb{P}^4$ is a complete intersection $F_2.F_2.F_2$ form a family of dimension 12.

(b) X has a g^1_3 if and only if it can be represented as a plane quintic with one node. These form an irreducible family of dimension 11.

(c) In that case, the conics through the node cut out the canonical system (not counting the fixed points at the node). Mapping $\mathbb{P}^2 \to \mathbb{P}^4$ by this linear system of conics, show that the canonical curve X is contained in a cubic surface $V \subseteq \mathbb{P}^4$, with V isomorphic to $\mathbb{P}^2$ with one point blown up (II, Ex. 7.7). Furthermore, V is the union of all the trisecants of X corresponding to the g^1_3 (5.5.3), so V is contained in the intersection of all the quadric hypersurfaces containing X. Thus V and g^1_3 are unique.

Proof.

(a) A curve of genus 5 whose canonical model in $\mathbb{P}^4$ is a complete intersection of three quadrics is given by a three-dimensional subspace of $\mathrm{H}^0(\mathbb{P}^4, \mathcal{O}_{\mathbb{P}^4}(2))$, which has dimension $\binom{4+2}{2} = 15$. Thus, the family of all these curves is parametrized by an open subset of the Grassmannian $\mathrm{Gr}(3, 15)$, which has dimension $3(15-3) = 36$. There is a natural action of $\mathrm{PGL}(4)$ on $\mathrm{Gr}(3, 15)$, and since the embedding is canonical, two three-dimensional subspaces in $\mathrm{Gr}(3, 15)$ give isomorphic curves if and only if they are in the same orbit of $\mathrm{PGL}(4)$. So there is a morphism $\mathrm{Gr}(3, 15) \dashrightarrow \mathfrak{M}_5$, whose fibres are the images of the group $\mathrm{PGL}(4)$ which has dimension $(4+1)^2 - 1 = 24$. Since any individual curve has only finitely many automorphisms (Ex. 5.2), the fibres have dimension $= 24$, and so the image of $\mathrm{Gr}(3, 15)$ has dimension $36 - 24 = 12$.

(b) A canonical curve X of genus 5 in $\mathbb{P}^4$ has a g^1_3 if and only if it has a trisecant. This means there exists distinct points there exists a divisor D of degree 3 such that $\dim |K - D| = 2$ (5.5.3). The complete linear system $|K - D|$ is base-point free, since otherwise if P is a base-point, then

$$\dim |D + P| = \dim |K - D - P| = 2,$$

which means $|D + P|$ is a g^2_4 on X. This is impossible for a nonhyperelliptic curve of genus 5, since the image is either a conic or a quartic curve. The former corresponds to a g^1_2 on X, and the latter implies X has genus at most $\binom{4-1}{2} = 3$. Therefore, $X \to |K - D| \cong \mathbb{P}^2$ must be birational by reasons of degree, and by (3.11.1) the image is a plane quintic with a singular point of multiplicity 2, which is either a cusp or a node.

Conversely, suppose X can be represented as a plane quintic $X' \subseteq \mathbb{P}^2$ with one node at $P \in X'$. Let $\pi : V \to \mathbb{P}^2$ be the blowing up of $\mathbb{P}^2$ with center at P, and let $E \subset V$ be the exceptional divisor of π. Let H be any hyperplane in $\mathbb{P}^2$ so that $|2H|$ is the linear system of conics in $\mathbb{P}^2$. Using the results of (V, §3) and (V, §4), the linear system $|2H - P|$ of conics passing through P has dimension $5 - 1 = 4$ with an assigned basepoint at P. By (II, 7.17.3), the rational map $\mathbb{P}^2 \dashrightarrow |2H - P| \cong \mathbb{P}^4$ extends to a morphism $V \to \mathbb{P}^4$. It is an embedding of V into $\mathbb{P}^4$ because it corresponds to the linear system $|D|$ with $D = 2\pi^*H - E$ on V, which is very ample by (V, 4.2.2). The strict transform $\tilde{X}'$ of X' is nonsingular, i.e., $\tilde{X}' \cong X$, and by (V, 3.6), it relates to X' as divisors in V in the following way

$$\pi^* X' = \tilde{X}' + 2E.$$

Using the fact that $X' \sim 5H$ in $\mathbb{P}^2$, by (V, Ex. 1) the degree of $\tilde{X}'$ as a curve in $\mathbb{P}^4$ equals to

$$\tilde{X}'.D = (5\pi^*H - 2E).(2\pi^*H - E) = 10(\pi^*H)^2 + 2E^2 = 10 - 2 = 8.$$

By the Riemann-Roch theorem, the only degree $2g-2$ invertible sheaf with at least g sections on a curve of genus g is the canonical sheaf, so this embedding of X in $\mathbb{P}^4$ is canonical. Now, the identification of V with a rational scroll $V \to \mathbb{P}^1$ of degree 3 in $\mathbb{P}^4$ (V, 2.19.1) induces the following correspondence of divisors on V

$$a\pi^*H - bE = (a-b)C_0 + af,$$

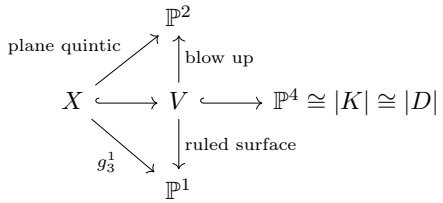
where C_0 is a section and f is any fibre of the ruling $V \to \mathbb{P}^1$. Rewriting $\tilde{X}'$ in this way gives

$$\tilde{X}' = 3C_0 + 5f.$$

Since $\tilde{X}'.f = 3$, a fibre of $V \to \mathbb{P}^1$ and $\tilde{X}'$ intersect at three points. Also, V is a rational scroll in $\mathbb{P}^4$, so any fibre of V is a line in $\mathbb{P}^4$. Hence, any fibre of $V \to \mathbb{P}^1$ is a trisecant of X, so X has a 1-parameter family of trisecants in $\mathbb{P}^4$.

By (3.11.1), the space $V_{d,r}$ of irreducible plane curves of degree d and r nodes is irreducible and has dimension $\frac{1}{2}d(d+3) - r$. In our case, $V_{5,1}$ has dimension 19, and we have just shown $V_{5,1}$ parametrizes the space of all genus 5 curves with a g^1_3. So there is a morphism $V_{5,1} \to \mathfrak{M}_5$, whose fibres are images of the group PGL(2), which has dimension 8. Hence, the family of genus 5 curves with a g^1_3 form an irreducible family of dimension $19 - 8 = 11$.

(c) The results of this exercise combined with (5.5.3) can be summarized by the commutative diagram



where K is a canonical divisor of X, and D is the divisor of V from (Ex. 5.5b). Thus, we have three different representations of nonhyperelliptic curves of genus 5 with a g^1_3.

□

6. Show that a nonsingular plane curve of degree 5 has no g^1_3. Show that there are nonhyperelliptic curves of genus 6 which cannot be represented as a nonsingular plane quintic curve.

Proof. Let $X \subset \mathbb{P}^2$ be a nonsingular plane curve of degree 5. The Veronese embedding $\mathbb{P}^2 \xrightarrow{\sim} V \subset \mathbb{P}^5$ realizes $\mathbb{P}^2$ as a surface of degree 4, and realizes X as a curve of degree 10 in $\mathbb{P}^5$. Since $g = 6$, this embedding is the canonical embedding. In particular, the only degree $2g-2$ invertible sheaf with at least g sections on a curve of genus g is the canonical sheaf. The proof of (5.5.3) applies to any nonhyperelliptic curve. In particular, X has a g^1_3 if and only if its canonical embedding in $\mathbb{P}^5$ has a trisecant line. Suppose that $L \subset \mathbb{P}^5$ is a trisecant line to the canonical image of X, meeting X at three points P, Q, R. Since $X \subset V$, the line L meets the Veronese surface in P, Q, R. These three points determine the g^1_3 on X, and hence $\dim |P + Q + R| = 1$. On the other hand, three points of the Veronese surface that are collinear in $\mathbb{P}^5$ must be collinear in $\mathbb{P}^2$, because the line through P, Q, R intersects V at three points, and V has degree 2 in $\mathbb{P}^5$, so the line must be contained in V. Thus P, Q, R are collinear in $\mathbb{P}^2$. But there is only one line in $\mathbb{P}^2$ passing through three distinct collinear points, contradicting the dimension of $|P + Q + R|$. Therefore, X has no trisecant line in its canonical embedding, and consequently X has no g^1_3.

For the second part, it is enough to show there exists nonhyperelliptic curves of genus 6 with a g^1_3. Consider any irreducible nonsingular curve X associated to a divisor of type $(3, 4)$ on the nonsingular quadric surface Q. It has genus $(3-1)(4-1) = 6$, and one of the families of lines on Q cuts out a g^1_3 on X. Hence, X cannot be represented as a nonsingular plane quintic curve. The canonical sheaf of Q is $\omega_Q = \mathcal{O}_Q(-2, -2)$, so by (II, 8.20), the canonical sheaf of X is $\omega_X \cong \mathcal{O}_Q(1, 2) \otimes \mathcal{O}_X$. Since $\mathcal{O}_Q(1, 2)$ is a very ample invertible sheaf on Q, ω_C is very ample as well. Hence, X is not hyperelliptic by (5.2). □

Chapter 4, Section 6

James Lee

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1. A rational curve of degree 4 in $\mathbb{P}^3$ is contained in a unique quadric surface Q, and Q is necessarily nonsingular.

Proof. Let X be a rational curve of degree 4 in $\mathbb{P}^3$, and let $\mathcal{I}$ be its ideal sheaf. Then we have an exact sequence

$$0 \longrightarrow \mathcal{I} \longrightarrow \mathcal{O}_{\mathbb{P}^3} \longrightarrow \mathcal{O}_X \longrightarrow 0.$$

Twisting by $\mathcal{O}_{\mathbb{P}^3}(2)$ and taking cohomology gives

$$0 \longrightarrow \mathrm{H}^0(\mathbb{P}^3, \mathcal{I}(2)) \longrightarrow \mathrm{H}^0(\mathbb{P}^3, \mathcal{O}_{\mathbb{P}^3}(2)) \longrightarrow \mathrm{H}^0(X, \mathcal{O}_X(2)) \longrightarrow \cdots.$$

Now the middle vector space has dimension 10, and $\mathcal{O}_X(2)$ corresponds to a linear system of degree 8, so it has dimension at most 9 by Riemann-Roch. Then, $\mathrm{H}^0(\mathbb{P}^3, \mathcal{I}(2))$ has dimension at least 1, so X is contained in a quadric surface Q. If X is contained in another quadric surface, then it must be a complete intersection by reason of its degree, which implies $\omega_X = \mathcal{O}_X$ (II, Ex. 8.4), a contradiction. If Q is singular, then X must have genus 1 (6.4.1), which is a contradiction. Hence, Q must be a nonsingular quadric surface. □

2. A rational curve of degree 5 in $\mathbb{P}^3$ is always contained in a cubic surface, but there are such curves which are not contained in any quadric surface.

Proof. Let X be a rational curve of degree 5 in $\mathbb{P}^3$, and let $\mathcal{I}$ be its ideal sheaf. Then we have an exact sequence

$$0 \longrightarrow \mathcal{I} \longrightarrow \mathcal{O}_{\mathbb{P}^3} \longrightarrow \mathcal{O}_X \longrightarrow 0.$$

Twisting by $\mathcal{O}_{\mathbb{P}^3}(3)$ and taking cohomology gives

$$0 \longrightarrow \mathrm{H}^0(\mathbb{P}^3, \mathcal{I}(3)) \longrightarrow \mathrm{H}^0(\mathbb{P}^3, \mathcal{O}_{\mathbb{P}^3}(3)) \longrightarrow \mathrm{H}^0(X, \mathcal{O}_X(3)) \longrightarrow \cdots.$$

Now, the middle vector space has dimension 20, and the vector space on the right has dimension 16 by Riemann-Roch on X (note that $\mathcal{O}_X(3)$ corresponds to a divisor of degree 15). So we conclude that $h^0(\mathbb{P}^3, \mathcal{I}(3)) \geq 4$.

A degree 5 curve in $\mathbb{P}^3$ contained in a quadric cone must have genus 2 (6.4.1), so if X is contained in a quadric surface, then it must be a nonsingular quadric surface. Choosing a degree 5 embedding of $\mathbb{P}^1$ in $\mathbb{P}^3$ is the same as picking a 4-dimensional subspace of $\mathrm{H}^0(\mathbb{P}^1, \mathcal{O}_{\mathbb{P}^1}(5))$, which has dimension 6. Thus, the family of embeddings $\mathbb{P}^1 \to \mathbb{P}^3$ of degree 5 is parametrized by the Grassmannian $\mathrm{Gr}(4, 6)$, which has dimension $4(6-4) = 8$. On the other hand, if X is contained in a nonsingular quadric surface $Q \subset \mathbb{P}^3$, then it corresponds to a divisor of type $(1, 4)$ on Q. So choosing an embedding of $\mathbb{P}^1$ as a curve of type $(1, 4)$ in Q is the same as picking a 2-dimensional subspace of $\mathrm{H}^0(\mathbb{P}^1, \mathcal{O}_{\mathbb{P}^1}(1)$ and a 2-dimensional subspace of $\mathrm{H}^0(\mathbb{P}^1, \mathcal{O}_{\mathbb{P}^1}(4))$. Thus, the family of embeddings $\mathbb{P}^1 \to Q$ is parametrized by $\mathrm{Gr}(2, 2) \times \mathrm{Gr}(2, 5)$, which has dimension $1(2-1) + 2(5-2) = 7$. Since the family of embeddings $Q \to \mathbb{P}^3$ has dimension one, there are plenty of embeddings of $\mathbb{P}^1$ as a degree 5 curve in $\mathbb{P}^3$ which are not contained in any quadric surface. □

3. A curve of degree 5 and genus 2 in $\mathbb{P}^3$ is contained in a unique quadric surface Q. Show that for any abstract curve X of genus 2, there exist embeddings of degree 5 in $\mathbb{P}^3$ for which Q is nonsingular, and there exist other embeddings of degree 5 for which Q is singular.

Proof. Let X be a curve of degree 5 and genus 2 in $\mathbb{P}^3$, and let $\mathcal{I}$ be its ideal sheaf. Then we have an exact sequence

$$0 \longrightarrow \mathcal{I} \longrightarrow \mathcal{O}_{\mathbb{P}^3} \longrightarrow \mathcal{O}_X \longrightarrow 0.$$

Twisting by $\mathcal{O}_{\mathbb{P}^3}(2)$ and taking cohomology gives

$$0 \longrightarrow \mathrm{H}^0(\mathbb{P}^3, \mathcal{I}(2)) \longrightarrow \mathrm{H}^0(\mathbb{P}^3, \mathcal{O}_{\mathbb{P}^3}(2)) \longrightarrow \mathrm{H}^0(X, \mathcal{O}_X(2)) \longrightarrow \cdots.$$

The middle vector space has dimension 10, and the vector space on the right has dimension 9 by Riemann-Roch on X (note that $\mathcal{O}_X(2)$ corresponds to a nonspecial divisor of degree 10). So we conclude that $h^0(\mathbb{P}^3, \mathcal{I}(2)) \geq 1$. By reason of degree, X cannot be contained in two distinct quadric surfaces, so there is a unique irreducible quadric surface Q containing X.

Similar to (Ex. 6.2), if X is a genus 2 curve of degree 5 in a nonsingular quadric surface $Q \subset \mathbb{P}^3$, then it corresponds to a divisor of type $(2, 3)$ on Q. Thus, choosing an embedding of X as a curve of type $(2, 3)$ in $Q \subset \mathbb{P}^3$ is the same as picking a g^1_2 and g^1_3 on X. Since X is hyperelliptic, it always has a g^1_2 and g^1_3, so there are plenty of embeddings of X into a nonsingular quadric surface. In fact, any g^1_3 on X is obtained by adding a point to the unique g^1_2, so the family of g^1_3's on X has dimension one, implying that the family of embeddings of X into a nonsingular quadric has dimension one. On the other hand, any effective divisor of degree 5 gives an embedding of X in $\mathbb{P}^3$ (Ex. 3.1), which is the same as picking 5 points in X, so the family of embeddings $X \to \mathbb{P}^3$ has dimension at least 5.

□

4. There is no curve of degree 9 and genus 11 in $\mathbb{P}^3$.

Proof. Suppose there exists a curve $X \subseteq \mathbb{P}^3$ of degree 9 and genus 11, and let $\mathcal{I}$ be its ideal sheaf. We have an exact sequence

$$0 \longrightarrow \mathrm{H}^0(\mathbb{P}^3, \mathcal{I}(2)) \longrightarrow \mathrm{H}^0(\mathbb{P}^3, \mathcal{O}_{\mathbb{P}^3}(2)) \longrightarrow \mathrm{H}^0(X, \mathcal{O}_X(2)) \longrightarrow \cdots.$$

Then $\mathcal{O}_X(2)$ is a very ample line bundle corresponding to a linear system of degree 18, say $|\mathcal{O}_X(2)| = |D|$ for some very ample divisor D of degree 18. Riemann-Roch gives

$$\dim |D| - \dim |K - D| = 8.$$

If D is nonspecial, then $\dim |D| = 8$. If D is special, Clifford's theorem (5.4) says

$$\dim |D| \leq \frac{1}{2} \deg D = 9,$$

and we have an equality if and only if X is hyperelliptic and D is a multiple of the unique g^1_2 on X. But D is very ample, so it cannot be a multiple of the g^1_2, and the inequality must be strict. We conclude that

$$h^0(X, \mathcal{O}_X(2)) = 1 + \dim |D| \leq 9,$$

and therefore, $\mathrm{H}^0(\mathbb{P}^3, \mathcal{I}(2)) \neq 0$. Hence, X must be contained in a quadric surface.

Following the discussion in (6.4.1), if X is contained in nonsingular quadric surface, then it corresponds to a divisor of type (a, b) such that

$$a + b = 9, \quad ab - a - b + 1 = 11,$$

which is not possible. If X is contained in a singular quadric surface, then X must have genus 12. Hence, there is no curve of degree 9 and genus 11 in $\mathbb{P}^3$. □

5. If X is a complete intersection of surfaces of degree a, b in $\mathbb{P}^3$, then X does not lie on any surface of degree $< \min(a, b)$.

Proof. If X is a complete intersection of surfaces H, H' of degree a, b respectively, in $\mathbb{P}^3$, then X has degree ab. Without loss of generality, suppose $a \leq b$. If H'' is surface of degree $c < a$ containing X, then the irreducible components of $H \cap H''$ can have degree at most ac (I, 7.7), which is strictly less than ab, so X cannot be an irreducible component of $H \cap H''$. Hence, X cannot lie on any surface of degree $< a$. □

Chapter 5

Surfaces

Chapter 5, Section 1

James Lee

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1. Let C, D be any two divisors on a surface X, and let the corresponding invertible sheaves be $\mathcal{L}$ and $\mathcal{M}$. Show that

$$C.D = \chi(\mathcal{O}_X) - \chi(\mathcal{L}^{-1}) - \chi(\mathcal{M}^{-1}) + \chi(\mathcal{L}^{-1} \otimes \mathcal{M}^{-1}).$$

Proof. Applying the Riemann-Roch theorem to the divisor $-C - D$ gives

$$\chi(\mathcal{O}_X(-C-D)) = \frac{1}{2}(-C-D).(-C-D-K) + \chi(\mathcal{O}_X)$$

The first term on the right hand side simplies to

$$(-C-D).(-C-D-K) = C^2 + 2C.D. + D^2 + C.K + D.K.$$

Substitutting back to above and isolating $C.D$ gives

$$\begin{aligned} C.D &= -\chi(\mathcal{O}_X) + \chi(\mathcal{O}_X(-C-D)) - \frac{1}{2}C.(C+K) - \frac{1}{2}C.(C+K) \\ &= \chi(\mathcal{O}_X) + \chi(\mathcal{O}_X(-C-D)) - \left(\chi(\mathcal{O}_X) - \frac{1}{2}C.(-C-K)\right) - \left(\chi(\mathcal{O}_X) - \frac{1}{2}D.(-D-K)\right) \\ &= \chi(\mathcal{O}_X) + \chi(\mathcal{O}_X(-C-D)) - \chi(\mathcal{O}_X(-C)) - \chi(\mathcal{O}_X(-D)). \end{aligned}$$

□

2. Let H be a very ample divisor on the surface X, corresponding to a projective embedding $X \subseteq \mathbb{P}^N$. If we write the Hilbert polynomial of X (III, Ex. 5.2) as

$$P(z) = \frac{1}{2}az^2 + bz + c,$$

show that $a = H^2$, $b = \frac{1}{2}H^2 + 1 - \pi$, where π is the genus of a nonsingular curve representing H, and $c = 1 + p_a$. Thus the *degree* of X in $\mathbb{P}^N$, as defined in (I, §7), is just H^2. Show also that if C is any curve in X, then the degree of C in $\mathbb{P}^N$ is just $C.H$.

Proof. By the Riemann-Roch theorem, we have the equality

$$P(z) = \frac{1}{2}H^2z^2 - \frac{1}{2}(H.K)z + 1 + p_a.$$

Matching coefficients readily gives a, b, and c. We can rewrite the coefficient of z as

$$-\frac{1}{2}(H.K) = \frac{1}{2}H^2 - \frac{1}{2}H.(H+K),$$

and the adjunction formula (1.5) gives

$$2\pi - 2 = H.(H+K).$$

We conclude that

$$b = \frac{1}{2}H^2 + 1 - \pi.$$

If C is any curve in X, then the degree of C in $\mathbb{P}^N$ is just the degree of $\mathcal{O}_X(H) \otimes \mathcal{O}_C$ as a line bundle on C, which is equal to $C.H$ by (1.3). □

3. Recall that the *arithmetic genus* of a projective scheme D of dimension 1 is defined as $p_a(D) = 1 - \chi(\mathcal{O}_D)$ (III, Ex. 5.3).

(a) If D is an effective divisor on a surface X, use (1.6) to show that $2p_a(D) - 2 = D.(D + K)$.

(b) $p_a(D)$ depends only on the linear equivalence class of D on X.

(c) More generally, for *any* divisor D on X, we define the *virtual arithmetic genus* (which is equal to the ordinary arithmetic genus if D is effective) by the same formula: $2p_a - 2 = D.(D + K)$. Show that for any two divisors C, D, we have

$$p_a(-D) = D^2 - p_a(D) + 2,$$

and

$$p_a(C + D) = p_a(C) + p_a(D) + C.D - 1.$$

Proof.

(a) We have the exact sequence

$$0 \longrightarrow \mathcal{O}_X(-D) \longrightarrow \mathcal{O}_X \longrightarrow \mathcal{O}_D \longrightarrow 0,$$

and taking Euler characteristics gives

$$\chi(\mathcal{O}_D) = \chi(\mathcal{O}_X) - \chi(\mathcal{O}_X(-D)).$$

On the other hand, the Riemann-Roch theorem gives

$$\chi(\mathcal{O}_X(-D)) = \frac{1}{2}D.(D + K) + \chi(\mathcal{O}_X).$$

Substituting gives

$$1 - \chi(\mathcal{O}_D) = 1 + \frac{1}{2}D.(D + K).$$

(b) Follows from (a) and (1.1(4)), which states $D.(D + K)$ depends only on the linear equivalence class of D on X.

(c)

$$\begin{aligned} 2p_a(-D) - 2 &= -D.(-D + K) \\ &= D.(D - K) \\ &= 2D^2 - D.(D + K) \\ &= 2D^2 - 2p_a(D) + 2 \\ 2p_a(C + D) - 2 &= (C + D).(C + D + K) \\ &= C.(C + K) + D.(D + K) + 2(C.D) \\ &= 2p_a(C) - 2 + 2p_a(D) - 2 + 2(C.D) \end{aligned}$$

□

4. (a) If a surface X of degree d in $\mathbb{P}^3$ contains a straight line $C = \mathbb{P}^1$, show that $C^2 = 2 - d$.

(b) Assume $\operatorname{char} k = 0$, and show for every $d \geq 1$, there exists a nonsingular surface X of degree d in $\mathbb{P}^3$ containing the line $x = y = 0$.

Proof.

(a) By (Ex. 1.5), $C^2 = -C.K - 2$. By (1.3), $C.K = d - 4$, because C is a straight line and $\mathcal{O}_X(K) = \mathcal{O}_X(d-4)$. Hence, $C^2 = 2 - d$.

(b) We can prove the following more general statement:

Proposition 0.1. *For every $n \geq 3$ and $d > 0$, and for every linear subspace $H \subset \mathbb{P}^n$ of codimension at least 2, there exists a nonsingular hypersurface of degree d in $\mathbb{P}^n$ containing H.*

We can assume H has codimension two and is defined by the vanishing of $x = y = 0$. Bertini's theorem (III, 10.9.2) says a general hypersurface X of degree d in $\mathbb{P}^3$ containing H is nonsingular away from H. Also, any such X is defined by the vanishing of $xF + yG = 0$, where F, G define surfaces of degree $d-1$. If P is a point in H, then X is singular at P if and only if both F, G vanish at P. This is a closed condition, so we can find plenty of choices of F, G which give a nonsingular hypersurface containing H.

Example 0.2. For $n = 3$ and $d \geq 2$, the surface defined by

$$x(z^{d-1} + x^{d-1}) + y(w^{d-1} + y^{d-1}) = 0$$

is nonsingular and contains L.

□

5. (a) If X is a surface of degree d in $\mathbb{P}^3$, then $K^2 = d(d-4)^2$.

(b) If X is a product of two nonsingular curves C, C', of genus g, g' respectively, then $K^2 = 8(g-1)(g'-1)$.

Proof.

(a) The adjunction formula gives $\mathcal{O}_X(K) = \mathcal{O}_X(d-4)$. If H is a general hyperplane section of X in $\mathbb{P}^3$, then $K = (d-4)H$, and $H^2 = d$ by (Ex. 1.2). Hence,

$$K^2 = (d-4)^2 H^2 = d(d-4)^2.$$

(b) By (II, Ex. 8.3), there is an isomorphism $\omega_X \cong p_1^* \omega_C \otimes p_2^* \omega_{C'}$. In particular, if $\mathfrak{k}$ and $\mathfrak{k}'$ are canonical divisors of C and C', respectively, then $p_1^*\mathfrak{k} + p_2^*\mathfrak{k}'$ is a canonical divisor of X. Denote $D = p_1^*\mathfrak{k}$ and $D' = p_2^*\mathfrak{k}'$. The support of D consists of $2g-2$ copies of C' which are the fibres of the points in $\mathfrak{k}$ with respect to $p_1 : X \to C$, similarly for D'. It follows that

$$D.D' = (\deg K_C) \cdot (\deg K_{C'}) = 4(g-1)(g'-1).$$

Also, we can always find two canonical divisors with disjoint support, and since the intersection pairing only depends on the linear equivalence class, we have $D^2 = D'^2 = 0$. We conclude that

$$K^2 = 2D.D' = 8(g-1)(g'-1).$$

□

6. (a) If C is a curve of genus g, show that the diagonal $\Delta \subseteq C \times C$ has self-intersection $\Delta^2 = 2 - 2g$. (Use the definition of ω_C in (II, §8).)

(b) Let $l = C \times \text{pt}$ and $m = \text{pt} \times C$. If $g \geq 1$, show that l, m, and Δ are linearly independent in $\text{Num}(C \times C)$. Thus $\text{Num}(C \times C)$ has rank ≥ 3, and in particular, $\text{Pic}(C \times C) \neq p_1^* \operatorname{Pic} C \oplus p_2^* \operatorname{Pic} C$.

Proof.

(a) By definition, $\omega_C \cong \mathcal{O}_X(-\Delta) \otimes \mathcal{O}_\Delta$, and by (1.3), $\deg_\Delta(\mathcal{O}_X(-\Delta) \otimes \mathcal{O}_\Delta) = -\Delta^2$. Hence, we have

$$\Delta^2 = -\deg_\Delta(\mathcal{O}_X(-\Delta) \otimes \mathcal{O}_\Delta) = -\deg(\omega_C) = 2 - 2g.$$

(b) Suppose $D = al + bm + c\Delta = 0$ in $\text{Num}(C \times C)$ for some integers $a, b, c \in \mathbb{Z}$. It is not hard to see

$$l^2 = m^2 = 0, \quad l.m = 1, \quad \Delta.l = \Delta.m = 1,$$

which implies

$$D.m = b + c = 0, \quad D.l = a + c = 0, \quad D.\Delta = a + b + c(2-2g) = 0.$$

Combining the first two gives $a + b + 2c = 0$. The condition $g \geq 1$ forces $a, b, c = 0$, which is what we wanted to show.

□

7. *Algebraic Equivalence of Divisors.* Let X be a surface. Recall that we have defined an algebraic family of effective divisors on X, parametrized by a nonsingular curve T, to be an effective Cartier divisor D on $X \times T$, flat over T (III, 9.8.5). In this case, for any two closed points $0, 1 \in T$, we say that the corresponding divisors D_0, D_1 on X are prealgebraically equivalent. Two arbitrary divisors are prealgebraically equivalent if they are differences of prealgebraically equivalent divisors. Two divisors D, D' are *algebraically equivalent* if there is a finite sequence $D = D_0, D_1, \ldots, D_n = D'$ with D_i and D_{i+1} prealgebraically equivalent for each i.

(a) Show that the divisors algebraically equivalent to 0 form a subgroup of $\operatorname{Div} X$.

(b) Show that linearly equivalent divisors are algebraically equivalent.

(c) Show that algebraically equivalent divisors are numerically equivalent.

Proof.

(a) A divisor D is prealgebraically equivalent to 0 if and only if $D = E - E'$, where E and E' are prealgebraically equivalent effective divisors. Thus, D is algebraically equivalent to 0 if and only if we can write $D = \sum(E_i - E_i')$, where E_i and E_i' are prealgebraically equivalent effective divisors for each i. If $D' = \sum(F_j - F_j')$ is another divisor algebraically equivalent to 0, then

$$D - D' = \sum(E_i - E_i') + \sum(F_j' - F_j),$$

so $D - D'$ is also algebraically equivalent to 0. Hence, the divisors algebraically equivalent to 0 form a subgroup of $\operatorname{Div} X$.

(b) We want to show any principal divisor $D = (f)$ is algebraically equivalent to 0, where f is some rational function on X. Then we can write $D = E - E'$, where E and E' are effective divisors, and f can locally be expressed as a quotient g/h, where g and h are locally defining equations for E and E' as Cartier divisors. We will show E and E' are prealgebraically equivalent. Consider the effective Cartier divisor $\tilde{D}$ on $X \times \mathbb{A}^1$ defined by $F = (1-t)g + th$. If $f = g'/h'$ is any another representative so that $g'h = gh'$, define $F' = (1-t)g' + th'$. To show H is well-defined, it is enough to show $F/F' \in \mathcal{O}^*_{X\times\mathbb{A}^1}$. It is easily check that

$$\frac{(1-t)g + th}{(1-t)g' + th'} \cdot \frac{h}{h} = \frac{(1-t)gh + th^2}{(1-t)gh' + thh'} = \frac{h}{h'} \in \mathcal{O}_X^*,$$

hence $F/F' \in \mathcal{O}^*_{X\times\mathbb{A}^1}$. Notice that $\tilde{D}$ is flat over $\mathbb{A}^1$ since it is surjective onto $\mathbb{A}^1$ (III, 9.7), and the divisors on X corresponding to the fibres over $t = 0, 1$ are E, E', respectively. Hence, E and E' are algebraically equivalent.

(c) We want to show if any divisor D on X algeberaically equivalent to 0 is numerically equivalent to 0. Since divisors algebraically equivalent to zero can be written as a sum of differences of prealgebraically equivalent to 0, it is enough to assume $D = E_0 - E_1$, where E_0 and E_1 are prealgebraically equivalent effective divisors. Let T be a nonsingular irreducible curve, and let E be an effective Cartier divisor on $X \times T$, flat over T, such that the divisors corresponding to the fibres t are $E_t \subset X$. It is enough to show for any irreducible curve C in X, such that C is transveral to E_0 and E_1, we have $C.E_0 = C.E_1$. Since X is a projective scheme, we can view $X \times T$ as a closed subscheme of $\mathbb{P}^n_T$ for some $n > 0$, as well as $C \times T \subset X \times T$. The hypothesis of (9.9) is satisfied for $C \times T$ since $C \times T \to T$ is a flat morphism, so applying (9.9) to the invertible sheaf $\mathcal{L} = \mathcal{O}_{X\times T}(E) \otimes \mathcal{O}_{C\times T}$ on $C \times T$ shows the Hilbert polynomial $P_t(z)$ of $\mathcal{L}_t = \mathcal{O}_{X\times T}(E) \otimes \mathcal{O}_{C\times t}$ is independent of t. Notice that $C.E_t = \deg \mathcal{L}_t$ for $t = 0, 1$ by (1.3), where $\deg \mathcal{L}_t$ is the leading coefficient of $P_t(z)$. Hence, $C.E_t$ is independent of t for $t = 0, 1$, so D is numerically equivalent to zero.

□

8. *Cohomology Class of a Divisor.* For any divisor D on the surface X, we define its cohomology class $c(D) \in \mathrm{H}^1(X, \Omega_X)$ by using the isomorphism $\operatorname{Pic} X \cong \mathrm{H}^1(X, \mathcal{O}_X^*)$ of (III, Ex. 4.5) and the sheaf homomorphism $\mathrm{d}\log : \mathcal{O}_X^* \to \Omega_X$ (III, Ex. 7.4c). Thus we obtain a group homomorphism $c : \operatorname{Pic} X \to \mathrm{H}^1(X, \Omega_X)$. On the other hand, $\mathrm{H}^1(X, \Omega_X)$ is dual to itself by Serre duality (III, 7.13), so we have a nondegenerate bilinear map

$$\langle \quad , \quad \rangle : \mathrm{H}^1(X, \Omega_X) \times \mathrm{H}^1(X, \Omega_X) \to k.$$

(a) Prove that this is compatible with the intersection pairing, in the following sense: for any two divisors D, E on X, we have
$$\langle c(D), c(E)\rangle = (D.E) \cdot 1$$
in k.

(b) If $\operatorname{char} k = 0$, use the fact that $\mathrm{H}^1(X, \Omega_X)$ is a finite-dimensional vector space to show that $\operatorname{Num} X$ is a finitely generated free abelian group.

Proof.

(a) Consider the case when D and E are nonsingular irreducible curves meeting transversally. The pairing of cohomology is defined as the following: the natural map $\Omega_X \otimes \mathcal{O}_D \to \Omega_D$ induces a linear map in cohomology $\mathrm{H}^1(X, \Omega_X) \to \mathrm{H}^1(D, \Omega_D)$. Since Ω_D is a dualizing sheaf for D, we have a trace map $t_D : \mathrm{H}^1(D, \Omega_D) \to k$. Composing, we obtain a linear map $\theta_D : \mathrm{H}^1(X, \Omega_X) \to k$. If E is another irreducible divisor of X, then we can consider its image in $\mathrm{H}^1(X, \Omega_X)$ under c. Hence, $\langle c(D), c(E)\rangle = \theta_D(c(E)) = \theta_E(c(D))$.

Generalizing, we want to show the following diagram commutes

$$\begin{array}{ccc} \operatorname{Pic}(X) & \xrightarrow{(D.\cdot)\cdot 1} & \mathbb{Z} \\ {\scriptstyle c}\downarrow & & \downarrow \\ \mathrm{H}^1(X, \Omega_X) & \xrightarrow{\theta_D} & k \end{array}.$$

The top row factos into

$$\begin{array}{ccc} & & \operatorname{Pic} D \\ & \nearrow & \downarrow{\scriptstyle \deg} \\ \operatorname{Pic} X & \xrightarrow{(D.\cdot)\cdot 1} & \mathbb{Z} \end{array}.$$

The bottom row factors into

$$\begin{array}{ccc} & & \mathrm{H}^1(D, \Omega_D) \\ & \nearrow & \downarrow{\scriptstyle t_D} \\ \mathrm{H}^1(X, \Omega_X) & \xrightarrow{\theta_D} & k \end{array}.$$

The left column factors into

$$\begin{array}{ccc} \operatorname{Pic} X & \longrightarrow & \operatorname{Pic} D \\ \downarrow{\scriptstyle c} & & \downarrow{\scriptstyle c} \\ \mathrm{H}^1(X, \Omega_X) & \longrightarrow & \mathrm{H}^1(D, \Omega_D) \end{array}.$$

The result of (III, Ex. 7.4) gives the commutative diagram

$$\begin{array}{ccc} \operatorname{Pic} D & \xrightarrow{\deg} & \mathbb{Z} \\ {\scriptstyle c}\downarrow & & \downarrow \\ \mathrm{H}^1(D, \Omega_D) & \xrightarrow{t_D} & k \end{array}.$$

So the original diagram commutes.

(b) We have an injection $c : \operatorname{Num} X \hookrightarrow \mathrm{H}^1(X, \Omega_X)$, and in characteristic 0, this means $k \otimes_{\mathbb{Z}} \operatorname{Num} X \hookrightarrow \mathrm{H}^1(X, \Omega_X)$ is also an injection. Since $\mathrm{H}^1(X, \Omega_X)$ is a finite-dimensional vector space, this means $\operatorname{Num} X$ is finitely generated. Also, $\operatorname{Num} X$ has no torsion, so $\operatorname{Num} X$ is actually a free finitely generated $\mathbb{Z}$-module.

□

9. (a) If H is an ample divisor on the surface X, and if D is any divisor, show that
$$(D^2)(H^2) \leq (D.H)^2.$$

(b) Now let X be a product of two curves $X = C \times C'$. Let $l = C \times \text{pt}$, and $m = \text{pt} \times C'$. For any divisor D on X, let $a = D.l$, $b = D.m$. Then we say D has *type* (a, b). If D has type (a, b), with $a, b \in \mathbb{Z}$, show that

$$D^2 \leq 2ab,$$

and equality holds if and only if $D = bl + am$.

Proof.

(a) By (1.9), (1.9.1), and (Ex. 1.8), the statement reduces to the Cauchy-Schwartz inequality for Lorenztian inner products.

(b) By Nakai-Moishezon (1.10), $H = l + m$ is ample, with $H^2 = 2$. Apply the previous exercise to $D' = D + al + bm$ and H, which simplifies to

$$D'^2 \leq 2(a + b)^2.$$

Expanding the left-hand side gives

$$D'^2 = D^2 + 2a^2 + 2b^2 + 2ab.$$

Substitutting and simplifying gives

$$D^2 \leq 2ab.$$

□

Example 0.3. Consider the quadric surface $Q = \mathbb{P}^1 \times \mathbb{P}^1$ and the divisor H of type $(1, 1)$. If D be a divisor of type (a, b), then by (1.4.3), the left-hand side is equal to $2ab$, while the right-hand side is equal to $(a + b)^2$ (IV, 6.4.1) (Ex. 1.1).

10. *Weil's Proof [2] of the Analogue of the Riemann Hypothesis for Curves.* Let C be a curve of genus g defined over the finite field $\mathbb{F}_q$, and let N be the number of points of C rational over $\mathbb{F}_q$. Then $N = 1 - a + q$, with $|a| \leq 2g\sqrt{q}$. To prove this, we consider C as a curve over the algebraic closure k of $\mathbb{F}_q$. Let $f : C \to C$ be the k-linear Frobenius morphism obtained by taking qth powers, which makes sense since C is defined over $\mathbb{F}_q$, so $X_q \cong X$ (IV, 2.4.1). Let $\Gamma \subseteq C \times C$ be the graph of f, and let $\Delta \subseteq C \times C$ be the diagonal. Show that $\Gamma^2 = q(2 - 2g)$, and $\Gamma.\Delta = N$. Then apply (Ex. 1.9) to $D = r\Gamma + s\Delta$ for all r and s to obtain the result.

Proof. Applying the adjunction formula (1.5) to Γ gives

$$\Gamma^2 = 2g - 2 - \Gamma.K,$$

where K is a canonical divisor of $C \times C$. We can choose K to be $K = p_1^* K_C + p_2^* K_C$, where K_C is a canonical divisor of C. Each $p_i^* K_C$ is a sum of $2g - 2$ lines, either $l = C \times \text{pt}$ or $m = \text{pt} \times C$ depending on $i = 1, 2$. Thus, $K = (2g - 2)(l + m)$. Since Γ is the graph of f, $\Gamma.l = \deg f = q$ and $\Gamma.m = 1$.

Now, we can compute $\Gamma.K$

$$\Gamma.K = (2g - 2)(\Gamma.(l + m)) = (2g - 2)(q + 1).$$

Plugging into (1.5) gives

$$\Gamma^2 = 2g - 2 - \Gamma.K = q(2 - 2g).$$

Notice that $\Gamma \cap \Delta$ set-theoretically consists of all the closed points of (P, P) such that $P \in C$ is a fixed point of f. By (1.4), we conclude that $\Gamma.\Delta = N$.

We have

$$D.l = r\Gamma.l + s\Delta.l = rq + s, \quad D.m = r\Gamma.m + s\Delta.m = r + s.$$

By (Ex. 1.8), we have

$$D^2 \leq 2(rq + s)(r + s).$$

Expanding the left-hand side gives

$$D^2 = r^2\Gamma^2 + 2rs\Gamma.\Delta + s^2\Delta^2 = r^2 q(2 - 2g) + 2rsN + s^2(2 - 2g).$$

Hence, we have

$$r^2q(2-2g) + 2rsN + s^2(2-2g) \leq 2(rq+s)(r+s).$$

Plugging in $N = 1 - a + q$ and simplifying in a similar manner to (IV, Ex. 4.16) gives the inequality

$$|a| \leq \left| tqg + \frac{g}{t} \right|$$

for all $t \in \mathbb{R}$. The local minimum and maximum of the right-hand side is $\pm 2g\sqrt{q}$. Hence, $|a| \leq 2g\sqrt{q}$. $\square$

Chapter 5, Section 2

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1. If X is a birationally ruled surface, show that the curve C, such that X is birationally equivalent to $C \times \mathbb{P}^1$, is unique up to isomorphism.

Proof. Let C and C' be two nonsingular curves such that X is birational to $C \times \mathbb{P}^1$ and $C' \times \mathbb{P}^1$, and suppose $\varphi : C \times \mathbb{P}^1 \to C' \times \mathbb{P}^1$ is a birational map with inverse ψ. The case is trivial if either C or C' has genus 0, so assume otherwise. For any fiber f of $C \times \mathbb{P}^1$, consider the composition map $p_1 \circ \varphi : f \to C'$. Since both f and C' are complete curves, $p_1 \circ \varphi$ extends to a morphism, and if C' has genus > 0, then it must be constant. Therefore, $p_1 \circ \varphi$ factors through $p_1 : C \times \mathbb{P}^1 \to C$, which extends to a morphism $C \to C'$. Doing the same thing for ψ gives an isomorphism of C with C'. □

2. Let X be the ruled surface $\mathbb{P}(\mathcal{E})$ over a curve C. Show that $\mathcal{E}$ is decomposable if and only if there exist two sections C', C'' of X such that $C' \cap C'' = \emptyset$.

Proof. If $\mathcal{E}$ is decomposable, then $\mathcal{E} = \mathcal{O} \oplus \mathcal{L}$, where $\mathcal{L} = \mathcal{O}_C(\mathfrak{e})$ for some divisor $\mathfrak{e} \in \operatorname{Div} C$. By (2.8.1), (2.11.1), and (2.11.2), the two natural projections $\mathcal{E} \to \mathcal{L}$ and $\mathcal{E} \to \mathcal{O}_C$ correspond to sections $C' \sim C_0$ and $C'' \sim C_0 - \mathfrak{e}f$, respectively, such that $C'.C'' = 0$. Hence, $C' \cap C'' = \emptyset$. Conversely, suppose there exist two sections C', C'' of X such that $C' \cap C'' = \emptyset$. The corresponding surjective morphisms $\mathcal{E} \to \mathcal{L}'$ and $\mathcal{E} \to \mathcal{L}''$ induce an $\mathcal{O}_C$-module homomorphism $\mathcal{E} \to \mathcal{L}' \oplus \mathcal{L}''$. For every $P \in C$, the morphism of stalks $\mathcal{E}_P \to \mathcal{L}'_P \oplus \mathcal{L}''_P$ is an isomorphism because C' and C'' are disjoint, so $\mathcal{L}'_P \oplus \mathcal{L}''_P$ is generated by $\mathcal{E}_P$. Hence, $\mathcal{E} \to \mathcal{L}' \oplus \mathcal{L}''$ is an isomorphism. □

3. (a) If $\mathcal{E}$ is a locally free sheaf of rank r on a nonsingular curve C, then there is a sequence

$$0 = \mathcal{E}_0 \subseteq \mathcal{E}_1 \subseteq \cdots \subseteq \mathcal{E}_r = \mathcal{E}$$

of subsheaves such that $\mathcal{E}_i/\mathcal{E}_{i-1}$ is an invertible sheaf for each $i = 1, \ldots, r$. We say that $\mathcal{E}$ is a *successive extension* of invertible sheaves.

(b) Show that this is false for varieties of dimension ≥ 2. In particular, the sheaf of differentials $\Omega_{\mathbb{P}^2}$ on $\mathbb{P}^2$ is not an extension of invertible sheaves.

Proof.

(a) We proceed by induction on r. The base case $r = 1$ is trivial, so assume the statement for any $r > 1$. We can twist $\mathcal{E}$ by a very ample sheaf $\mathcal{L}$ so that $\mathcal{E}' = \mathcal{E} \otimes \mathcal{E}$ is generated by global sections. If we prove the statement for $\mathcal{E}'$ so we have a filtration $(\mathcal{E}'_i)$, then tensoring by $\mathcal{L}^{-1}$ gives a filtration $(\mathcal{E}_i = \mathcal{E}'_i \otimes \mathcal{L}^{-1})$ of $\mathcal{E}$ such that

$$\mathcal{E}_i/\mathcal{E}_{i-1} \cong (\mathcal{E}'_i/\mathcal{E}'_{i-1}) \otimes \mathcal{L}^{-1},$$

which is an invertible sheaf for each $i = 1, \ldots, r$. Hence, we can assume $\mathcal{E}$ is generated by global sections. The hypothesis of (II, Ex. 8.2) is satisfied for $\mathcal{E}$, so there exists a short exact

$$0 \longrightarrow \mathcal{O}_C \longrightarrow \mathcal{E} \longrightarrow \mathcal{F} \longrightarrow 0,$$

where $\mathcal{F}$ is also globally generated locally free of rank $r - 1$. By the inductive hypothesis, this means we can find a filtration $(\mathcal{F}_i)$ of $\mathcal{F}$ such that each successive quotient is an invertible sheaf. Consider the preimage $(\mathcal{E}_j)$ of $(\mathcal{F}_i)$, where we relabel $j = i + 1$. Each $\mathcal{E}_j$ is also locally free, $\mathcal{E}_1 = \operatorname{im}(\mathcal{O}_C)$ by exactness, and there is a canonical isomorphism $\mathcal{E}_j/\mathcal{E}_{j-1} \cong \mathcal{F}_{j-1}/\mathcal{F}_{j-2}$. Extending $(\mathcal{E}_j)$ with $\mathcal{E}_0 = 0$ gives a successive extension of sheaves.

(b) Suppose we have an exact sequence

$$0 \longrightarrow \mathcal{L} \longrightarrow \Omega \longrightarrow \mathcal{M} \longrightarrow 0,$$

where $\mathcal{L}$ and $\mathcal{M}$ are invertible sheaves on $\mathbb{P}^2$. Then $\mathrm{H}^1(\mathcal{L}), \mathrm{H}^1(\mathcal{M}) = 0$, so by the long exact sequence in cohomology, we have $\mathrm{H}^1(\Omega) = 0$. But by (III, Ex. 7.2), $\mathrm{H}^1(\Omega) = k$, a contradiction.

□

4. Let C be a curve of genus g, and let X be the ruled surface $C \times \mathbb{P}^1$. We consider the question, for what integers $s \in \mathbb{Z}$ does there exist a section D of X with $D^2 = s$? First show that s is always an even integer, say $s = 2r$.

(a) Show that $r = 0$ and $r \geq g + 1$ are always possible.

(b) If $g = 3$, show that $r = 1$ is not possible, and just one of the two values $r = 2, 3$ is possible, depending on whether C is hyperelliptic or not.

Proof. Fix a section $C_0 = C \times P_0 \subset X$, where P_0 is a point of $\mathbb{P}^1$. The ruled surface $X = C \times \mathbb{P}^1$ corresponds to the projective bundle of $\mathcal{E} = \mathcal{O}_C \oplus \mathcal{O}_C$. The invariant of X is $e = 0$, which means $C_0^2 = -e = 0$. By (2.9), if D is any section of X corresponding to a surjection $\mathcal{E} \to \mathcal{O}_C(\mathfrak{d})$ for some divisor $\mathfrak{d}$ of C with $r = \deg \mathfrak{d}$, then $r = C_0.D$ and $D \sim C_0 + \mathfrak{d}f$. Hence, $s = 2r$.

(a) There is a one-to-one correspondence between sections D of X and finite separable morphisms $f : C \to \mathbb{P}^1$. In particular, a section D is the graph of a morphism $f : C \to \mathbb{P}^1$, and its self-intersection is $D^2 = 2 \cdot \deg f$. By (IV, Ex. 1.6), $g + 1$ is an upperbound on the minimum of $\deg f$. Hence, $r = 0$ and $r \geq g + 1$ is always possible.

(b) If $g > 0$, then there is no morphism $C \to \mathbb{P}^1$ of degree 1. If C is a hyperelliptic curve, then it has a g^1_2, so $r = 2, 3$ is possible. Otherwise, nonhyperelliptic curves of genus 3 always have a g^1_3 (IV, 5.5.2), so $r = 3$ is always possible.

□

5. *Values of e.* Let C be a curve of genus $g \geq 1$.

(a) Show that for each $0 \leq e \leq 2g - 2$ there is a ruled suface X over C with invariant e, corresponding to an indecomposable $\mathcal{E}$.

(b) Let $e < 0$, let D be any divisor of degree $d = -e$, and let $\xi \in \mathrm{H}^1(X, \mathcal{O}(-D))$ be a nonzero element defining an extension

$$0 \longrightarrow \mathcal{O} \longrightarrow \mathcal{E} \longrightarrow \mathcal{O}(D) \longrightarrow 0.$$

Let $H \subseteq |D + K|$ be the sublinear system of codimension 1 defined by $\ker \xi$, where ξ is considered as a linear functional on $\mathrm{H}^0(\mathcal{O}(D + K))$. For any effective divisor E of degree $d - 1$, let $L_E \subseteq |D + K|$ be the sublinear system $|D + K - E| + E$. Show that $\mathcal{E}$ is normalized if and only if for each E as above, $L_E \nsubseteq H$.

(c) Now show that if $-g \leq e < 0$, there exists a ruled surface X over C with invariant e.

(d) For $g = 2$, show that $e \geq -2$ is also necessary for the existence of X.

Proof.

(a) Let $\mathfrak{e}$ be a divisor of degree $-e$ on C, and let $\mathfrak{k}$ is a canonical divisor of C. We need to first construct an indecomposable extension $\mathcal{E}$ of $\mathcal{O}(\mathfrak{e})$ by $\mathcal{O}$. By (III, Ex. 6.1), an extension of vector bundles

$$0 \longrightarrow \mathcal{O} \longrightarrow \mathcal{E} \longrightarrow \mathcal{O}(\mathfrak{e}) \longrightarrow 0$$

corresponds to an element in $\mathrm{Ext}^1(\mathcal{O}(\mathfrak{e}), \mathcal{O})$. In this case, $\mathrm{Ext}^1(\mathcal{O}(\mathfrak{e}), \mathcal{O}) \cong \mathrm{H}^1(\mathcal{O}(-\mathfrak{e}))$ (III, 6.3) and (III, 6.7). This is dual to $\mathrm{H}^0(\mathcal{O}(\mathfrak{k} + \mathfrak{e}))$, so it is enough to find $\mathfrak{e}$ such that $\mathrm{H}^0(\mathcal{O}(\mathfrak{k} + \mathfrak{e})) \neq 0$. The image of the canonical morphism $C \to |\mathfrak{k}| \cong \mathbb{P}^{g-1}$ is either a rational curve of degree $g - 1$, or an embedding of degree $2g - 2$, depending on whether or not C is hyperelliptic. If $-\mathfrak{e}$ is effective, then $\mathrm{H}^0(\mathcal{O}(\mathfrak{k} + \mathfrak{e}))$ is the space

of all hyperplane sections in $\mathbb{P}^{g-1}$ with support containing $\mathfrak{e}$. In any case, a hyperplane section of C in $\mathbb{P}^{g-1}$ consists of $2g-2$ points counted with multiplicity. Hence, for each $0 \le e \le 2g-2$, there is a divisor $\mathfrak{e}$ of degree $-e$ such that $\mathrm{H}^0(\mathcal{O}(\mathfrak{k}+\mathfrak{e})) \neq 0$, so there exists indecomposable extensions of $\mathcal{O}(\mathfrak{e})$ by $\mathcal{O}$.

Now, let $\mathcal{E}$ be an extension of $\mathcal{O}(\mathfrak{e})$ defined by a nonzero element in $\mathrm{Ext}^1(\mathcal{O}(\mathfrak{e}), \mathcal{O})$. It remains to show $\mathcal{E}$ is normalized and indecomposable. Clearly $\mathrm{H}^0(\mathcal{E}) \neq 0$ by construction. If $\mathcal{M}$ is any invertible sheaf, then we have an exact sequence

$$0 \longrightarrow \mathcal{M} \longrightarrow \mathcal{E} \otimes \mathcal{M} \longrightarrow \mathcal{O}(\mathfrak{e}) \otimes \mathcal{M} \longrightarrow 0.$$

If $\deg \mathcal{M} < 0$, then we have $\mathrm{H}^0(\mathcal{M}) = 0$ and $\mathrm{H}^0(\mathcal{O}(\mathfrak{e}) \otimes \mathcal{M}) = 0$, and therefore also $\mathrm{H}^0(\mathcal{E} \otimes \mathcal{M}) = 0$. Furthermore, if $\mathcal{E}$ is decomposable, then it would be $\mathcal{O} \oplus \mathcal{L}$ for some $\mathcal{L}$, by (2.12). But $\bigwedge^2 \mathcal{E} \cong \mathcal{O}_C(\mathfrak{e})$, so $\mathcal{L} \cong \mathcal{O}(\mathfrak{e})$, so in fact this extension would have to split, which it doesn't. Hence, for each $0 \le e \le 2g-2$, there is a ruled surface X over C with invariant e, corresponding to an indecomposable $\mathcal{E}$.

(b) Let $\xi \in \mathrm{H}^1(\mathcal{O}(-D))$ be the (non-zero) element defining the extension $\mathcal{E}$. If E is any effective divisor of degree $d-1$ on C, then we have a commutative diagram

$$\begin{array}{ccc} \mathrm{H}^0(\mathcal{O}) & \xrightarrow{\delta} & \mathrm{H}^1(\mathcal{O}(-D))) \\ \downarrow{\scriptstyle \alpha} & & \downarrow{\scriptstyle \beta} \\ \mathrm{H}^0(\mathcal{O}(E)) & \xrightarrow{\gamma} & \mathrm{H}^1(\mathcal{O}(E-D)) \end{array}$$

where $\delta(1) = \xi$, $\alpha(1) = t$, a non-zero section defining E, and β is induced from the map $\mathcal{O} \to \mathcal{O}(E)$ corresponding to t. On the other hand, β is dual to the map

$$\beta^\vee : \mathrm{H}^0(\mathcal{O}(D+K-E)) \to \mathrm{H}^0(\mathcal{O}(D+K))$$

also induced by t. The subspace of $\mathrm{H}^0(\mathcal{O}(D+K))$ corresponding to L_E is precisely the image of $\beta^\vee$, so $\beta(\xi) = 0$ if and only if $L_E \subseteq H$.

If $\mathcal{E}$ is normalized, then γ is injective for any effective divisor E of degree $d-1$. By the commutative diagram, we must have $\beta(\xi) \neq 0$. Hence, $L_E \nsubseteq H$.

Conversely, if $\mathcal{E}$ is not normalized, then $\mathrm{H}^0(\mathcal{E} \otimes \mathcal{M}) \neq 0$ for some invertible sheaf $\mathcal{M}$ with $\deg \mathcal{M} < 0$, and let $m = -\deg \mathcal{M}$. We have the exact sequence

$$0 \longrightarrow \mathcal{M} \longrightarrow \mathcal{E} \otimes \mathcal{M} \longrightarrow \mathcal{O}(D) \otimes \mathcal{M} \longrightarrow 0 \ .$$

If $m \ge d$, then we have $\mathrm{H}^0(\mathcal{E} \otimes \mathcal{M}) = 0$ since $\deg \mathcal{O}(D) \otimes \mathcal{M} < 0$, and if $m = d$, then either $\mathrm{H}^0(\mathcal{O}(D) \otimes \mathcal{M}) = 0$, or $\mathcal{M} = \mathcal{O}(-D)$, whence the coboundary map δ is injective, hence $\mathrm{H}^0(\mathcal{E} \otimes \mathcal{M}) = 0$. So we can assume $0 < m < d$. The long exact sequence of cohomology gives

$$0 \longrightarrow \mathrm{H}^0(\mathcal{E} \otimes \mathcal{M}) \longrightarrow \mathrm{H}^0(\mathcal{O}(D) \otimes \mathcal{M}) \longrightarrow \mathrm{H}^1(\mathcal{M}) \ ,$$

where $\mathrm{H}^0(\mathcal{M}) = 0$. The middle term is non-zero, so if $\mathcal{M} \cong \mathcal{O}(-M)$ with $m = \deg M$, then $D - M \sim E$ for some effective divisor E of degree $d-m$. For any effective divisor $\tilde{E}$ of degree $d-1$ with $\tilde{E} - E \ge 0$, we have a commutative diagram,

$$\begin{array}{ccc} \mathrm{H}^0(\mathcal{O})) & \xrightarrow{\delta} & \mathrm{H}^1(\mathcal{O}(-D)) \\ \downarrow{\scriptstyle \alpha} & & \downarrow{\scriptstyle \beta} \\ \mathrm{H}^0(\mathcal{O}(E)) & \xrightarrow{\gamma} & \mathrm{H}^1(\mathcal{O}(E-D)) \\ \downarrow{\scriptstyle \phi} & & \downarrow{\scriptstyle \psi} \\ \mathrm{H}^0(\mathcal{O}(\tilde{E})) & \xrightarrow{\eta} & \mathrm{H}^1(\mathcal{O}(\tilde{E}-D)) \end{array} \quad \tilde{\alpha} \circlearrowright \ , \ \tilde{\beta} \circlearrowright$$

where $\delta(1) = \xi$, α, β is induced by a section $t \in \mathrm{H}^0(\mathcal{O}(E))$ defining E, and ϕ, ψ is induced by a section s defining $\tilde{E} - E$.

Now, if $\beta(\xi) = 0$, then we are done. Otherwise, dualizing the column on the right of the commutative diagram gives

$$\begin{array}{ccc} \mathrm{H}^0(\mathcal{O}(D+K-\tilde{E})) & & \\ \downarrow{\scriptstyle \psi^\vee} & \searrow^{\tilde{\beta}^\vee} & \\ \mathrm{H}^0(\mathcal{O}(D+K-E)) & \xrightarrow{\beta^\vee} & \mathrm{H}^0(\mathcal{O}(D+K)) \end{array} .$$

Thus, for every $\tilde{E}$, we have $L_{\tilde{E}} \subseteq L_E$. By the assumption $\beta(\xi) \neq 0$, we have $L_E \nsubseteq H$, so $\mathrm{im}(\beta^\vee) \cap \ker(\xi)$ is a nonempty proper subspace of $\mathrm{im}(\beta^\vee)$. By the Riemann-Roch theorem, we have

$$\dim \mathrm{H}^0(\mathcal{O}(D+K-\tilde{E})) = g, \quad \dim \mathrm{H}^0(\mathcal{O}(D+K-E)) = g+m-1.$$

The space of possible $\tilde{E}$ has dimension $m-1$, so the image of $\psi^\vee$ as $\tilde{E}$ varies is dense inside $\mathrm{H}^0(\mathcal{O}(D+K-E))$. Consequently, we can find at least one $\tilde{E}$ such that $\mathrm{im}(\tilde{\beta}^\vee) \nsubseteq \ker(\xi)$, which is what we wanted to show.

(c) Let $\mathbb{P}^N = |K+D|$ with $N = g+d-2$. For any effective divisor E of degree $d-1$ on C, let us consider the set

$$B_E = \{\text{hyperplanes } H \subset \mathbb{P}^N \mid L_E \subseteq H\} \subset |H|,$$

where $|H|$ is the complete linear system of hyperplanes in $\mathbb{P}^N$, i.e., $|H| = |K+D|$. A hyperplane in $\mathbb{P}^N$ is determined by a nonzero global section $f \in V = \mathrm{H}^0(\mathbb{P}^N, \mathcal{O}_{\mathbb{P}^N}(1))$. Viewing L_E as a linear subspace of $\mathbb{P}^N$, let $W_E = \mathrm{H}^0(L_E, \mathcal{O}_{L_E}(1))$. The space of effective divisors of degree $d-1$ is parameterized by the closed points of the symmetric product $C^{(d-1)}$, which has dimension $d-1$. Consider the restriction map

$$\varphi_E : V \to W_E.$$

It is clearly surjective, and the hyperplanes $H \in B_E$ correspond exactly to those $f \in \ker \varphi_E$. We have $\dim V = g+d-1$ and $\dim W_E = g$, so $\dim B_E = \dim(\ker \varphi_E) - 1 = d-2$.

Now, consider the subset $B \subset C^{(d-1)} \times |H|$ consisting of all pairs $\langle E, H\rangle$ such that $E \in C^{(d-1)}$ is a closed point and $H \in B_E$. Clearly B is the set of closed points of a $C^{(d-1)} \times |H|$, which we denote also by B, and which we give a reduced induced scheme structure. The first projection $p_1 : B \to C^{(d-1)}$ is surjective, with fibre a projective space of dimension $d-2$. So B has dimension $(d-1)+(d-2) = 2d-3$, which is strictly smaller than $\dim |H| = g+d-2$. If $H \in |H| - p_2(B)$, then $H \not\supseteq L_E$ for every $E \in C^{(d-1)}$, so the extension of $\mathcal{O}(D)$ by $\mathcal{O}$ given by the element in $\mathrm{H}^0(\mathcal{O}(K+D)) \cong \mathrm{H}^1(\mathcal{O}(-D))$ corresponding to H satisfies the requirement of (b). Hence, if $-g \leq e < 0$, then there exists a ruled surface X over C with invariant e.

(d) Let C be a curve of genus $g = 2$, let $\mathcal{E}$ be a normalized locally free sheaf of rank 2 on C with invariant e, let $d = -e$, and let $X = \mathbb{P}(\mathcal{E})$ be the ruled surface associated to $\mathcal{E}$. As in the proof of (2.15), we want to show $e = -3, -4$ does not occur.

If it did, we would have a normalized bundle $\mathcal{E}$ with an exact sequence

$$0 \longrightarrow \mathcal{O} \longrightarrow \mathcal{E} \longrightarrow \mathcal{O}(D) \longrightarrow 0 \ ,$$

where D is an effective divisor of degree $d = -e$. Let $\xi \in \mathrm{H}^1(\mathcal{O}(-D))$ be the element defining the extension $\mathcal{E}$. If E is any effective divisor, then applying $\mathrm{Hom}(\mathcal{O}(D-E), -)$ and taking the long exact sequence in cohomology gives

$$\mathrm{Hom}(\mathcal{O}(D-E), \mathcal{E}) \longrightarrow \mathrm{Hom}(\mathcal{O}(D-E), \mathcal{O}(D)) \xrightarrow{\delta_E} \mathrm{Ext}^1(\mathcal{O}(D-E), \mathcal{O}) \ ,$$

so by exactness, the inclusion $i : \mathcal{O}(D-E) \to \mathcal{O}(D)$ factors through $\mathcal{E} \to \mathcal{O}(D)$ if and only if $\delta_E(i) = 0$. By naturality, we have a commutative diagram

$$\begin{array}{ccc} \mathrm{Hom}(\mathcal{O}(D), \mathcal{O}(D)) & \xrightarrow{\delta} & \mathrm{Ext}^1(\mathcal{O}(D), \mathcal{O}) \\ \downarrow{\scriptstyle i^*} & & \downarrow{\scriptstyle i^*} \\ \mathrm{Hom}(\mathcal{O}(D-E), \mathcal{O}(D)) & \xrightarrow{\delta_E} & \mathrm{Ext}^1(\mathcal{O}(D-E), \mathcal{O})) \end{array} ,$$

where $\delta(1) = \xi$ and $i^*(1) = i$, so $\delta_E(i) = 0$ if and only if $i^*(\xi) = 0$. Now, if there exists E of degree $< d$ such that $i : \mathcal{O}(D-E) \to \mathcal{E}$ factors through $\mathcal{E} \to \mathcal{O}(D)$, then $\mathcal{O}(D-E) \to \mathcal{E}$ must also be injective, which is a contradiction of $\mathcal{E}$ being normalized, so it is enough to show there exists E such that $i^*(\xi) = 0$.

The discussion above has the following geometric interpretation. By Serre duality, there is a natural identification

$$|D+K| \cong \mathbb{P}(\mathrm{Ext}^1(\mathcal{O}(D),\mathcal{O})) \cong \mathbb{P}(\mathrm{H}^0(\mathcal{O}(D+K))^\vee),$$

so we can view the linear span of ξ in $\mathrm{Ext}^1(\mathcal{O}(D),\mathcal{O})$ as a point in $|D+K|$, which we will also denote by ξ. It follows that $i^*(\xi)=0$ if and only if ξ is contained in every hyperplane in $|D+K|$ which also contains E.

$\boxed{d=3}$ In either case, $D+K$ is a very ample divisor on C (IV, Ex. 3.1), so consider the embedding of C in $|D+K| \cong \mathbb{P}^3$. The secant variety of C fills up the entire space (IV, §3), so there is a secant line $L \subseteq |D+K|$ which contains ξ. A line is uniquely determined by two points, so we can take E to be any effective divisor of degree 2 with support contained in $C \cap L$. Hence, any hyperplane $H \subseteq |D+K|$ containing E also contains L, so ξ must be a point in H as well.

$\boxed{d=4}$ Again, consider C in its embedding in $|D+K| \cong \mathbb{P}^4$. A plane is uniquely determined by three points, and the union of all planes going through at least three points of C fills up all of $|D+K|$. This means we can find a plane P in $|D+K|$ which contains ξ and at least three points of C. So we can take E to be any effective divisor of degree 3 with support contained in $C \cap P$. Hence, any hyperplane $H \subseteq |D+K|$ containing E also contains P, so ξ must be in H as well.

□

6. Show that every locally free sheaf of finite rank on $\mathbb{P}^1$ is isomorphic to a direct sum of invertible sheaves.

Proof. We call any locally free sheaf $\mathcal{E}$ on a curve C *normalized* if $\mathrm{H}^0(C,\mathcal{E}) \neq 0$ and $\mathrm{H}^0(C,\mathcal{E}\otimes\mathcal{L})=0$ for all invertible sheaves $\mathcal{L}$ on C with $\deg \mathcal{L} < 0$. With this definition, the statement of (2.8) can be made sense for $\mathcal{E}$ of arbitrary rank:

Lemma 0.1. *For every locally free sheaf $\mathcal{E}$ of finite rank on a curve C, there exists an invertible sheaf $\mathcal{L}$ on C such that $\mathcal{E}\otimes\mathcal{L}$ is normalized.*

Going back to the original statement, we can assume $\mathcal{E}$ is normalized. We proceed by induction on the rank r of $\mathcal{E}$. The base case $r=1$ is trivial, so assume the statement holds for any $< r$. Pick a subinvertible sheaf $0 \to \mathcal{O} \to \mathcal{E}$ of maximal degree, and consider the short exact sequence

$$0 \longrightarrow \mathcal{O} \longrightarrow \mathcal{E} \longrightarrow \mathcal{F} \longrightarrow 0\ .$$

The quotient sheaf $\mathcal{F}=\mathcal{E}/\mathcal{O}$ is localy free of rank $r-1$ because of the choice of $0 \to \mathcal{O} \to \mathcal{E}$ (2.8), so by the inductive hypothesis, $\mathcal{F}$ is decomposable, say $\mathcal{F}=\bigoplus \mathcal{O}(b_i)$ for some $b_i \in \mathbb{Z}$. Tensoring by $\mathcal{O}(-1)$ and taking cohomology gives

$$0 \longrightarrow \mathrm{H}^0(\mathcal{O}(-1)) \longrightarrow \mathrm{H}^0(\mathcal{E}(-1)) \longrightarrow \mathrm{H}^0(\textstyle\bigoplus_i \mathcal{O}(b_i-1)) \longrightarrow 0\ .$$

Since $\mathcal{E}$ is normalized, the middle vector space is zero, so by exactness we have $b_i \leq 0$. On the other hand, the space of extensions of $\mathcal{F}$ by $\mathcal{O}$ can be computed by Serre duality, which gives

$$\mathrm{Ext}^1(\mathcal{F},\mathcal{O}) \cong \mathrm{H}^0(\textstyle\bigoplus \mathcal{O}(b_i-1)) = 0,$$

so any extension of $\mathcal{F}$ by $\mathcal{O}$ is trivial. Hence, the short exact sequence splits, and there is an isomorphism $\mathcal{E} \cong \mathcal{O}\oplus\mathcal{F}$. □

7. On the elliptic ruled surface X of (2.11.6), show that the sections C_0 with $C_0^2=1$ form a one-dimensional algebraic family, parametrized by the points of the base curve C, and that no two are linearly equivalent.

Proof. Combine (III, Ex. 11.5), (2.9), (2.16) and the fact that any divisor of positive degree on an elliptic curve is linearly equivalent to an effective divisor. The correspondence is given by for each closed point $Q \in C$, there is a section linearly inequivalent to $C_0+(Q-P)f$, i.e., there is a surjection $\mathcal{E} \to \mathcal{O}_C(Q-P)$ of $\mathcal{O}_C$-modules. By (2.1), $\pi_*(\mathcal{O}_X(C_0+(Q-P)f))$ is isomorphic to $\mathcal{E}\otimes\mathcal{O}_C(Q-P)$, and by (2.16) these locally free sheaves of rank 2 are in one-to-one corresponding with the closed points of C. □

8. A locally free sheaf $\mathcal{E}$ on a curve C is said to be *stable* if for every quotient locally free sheaf $\mathcal{E} \to \mathcal{F} \to 0$, $\mathcal{F} \neq \mathcal{E}$, $\mathcal{F} \neq 0$, we have

$$(\deg \mathcal{F})/\operatorname{rank} \mathcal{F} > (\deg \mathcal{E})/\operatorname{rank} \mathcal{E}$$

Replacing $>$ by $\geq$ defines *semistable*.

(a) A decomposable $\mathcal{E}$ is never stable.

(b) If $\mathcal{E}$ has rank 2 and is normalized, then $\mathcal{E}$ is stable (respectively, semistable) if and only if $\deg \mathcal{E} > 0$ (respectively, ≥ 0).

(c) Show that the indecomposable locally free sheaves $\mathcal{E}$ of rank 2 that are not semistable are classified, up to isomorphism, by giving (1) an integer $0 < e \leq 2g-2$, (2) an element $\mathcal{L} \in \operatorname{Pic} C$ of degree $-e$, and (3) a nonzero $\xi \in \mathrm{H}^1(C, \mathcal{L}^{-1})$, determined up to a nonzero scalar multiple.

Proof.

(a) Denote $\mu(\mathcal{E}) = (\deg \mathcal{E})/\operatorname{rank} \mathcal{E}$ the slope of $\mathcal{E}$, and let $\mathcal{F} \subseteq \mathcal{E}$ consist of the summands of $\mathcal{E}$ that has degree $\leq \mu(\mathcal{E})$, which there must be at least one. There is a surjective morphism of locally free sheaves $\mathcal{E} \to \mathcal{F} \to 0$, and we have $\mu(\mathcal{F}) \leq \mu(\mathcal{E})$. Hence, $\mathcal{E}$ is not stable.

(b) If $\mathcal{E}$ is normal with invariant $e = -\deg \mathcal{E}$, then we have a quotient locally sheaf $\mathcal{E} \to \mathcal{L} \to 0$, where $\mathcal{L}$ is an invertible sheaf of degree $-e$. If $\mathcal{E}$ is stable (respectively, semistable), then

$$-e = \mu(\mathcal{L}) > \mu(\mathcal{E}) = -e/2,$$

(respectively, $\mu(\mathcal{L}) \geq \mu(\mathcal{E})$), which implies $\deg \mathcal{E} = -e > 0$ (respectively, ≥ 0). Conversely, suppose $\deg \mathcal{E} > 0$ (respectively, ≥ 0), and let $\mathcal{E} \to \mathcal{F} \to 0$ be a quotient locally free sheaf. If $\mathcal{F}$ has rank 2, then $\mathcal{E} \to \mathcal{F} \to 0$ is an isomorphism (II, Ex. 7.1). Otherwise, $\mathcal{F}$ is an invertible sheaf, and because of the normalization of $\mathcal{E}$, we must have $\deg \mathcal{F} \geq \deg \mathcal{E}$. Hence, $\mu(\mathcal{F}) = \deg \mathcal{F} \geq \deg \mathcal{E} > \mu(\mathcal{E})$ (respectively, $\geq \mu(\mathcal{E})$) if $\deg \mathcal{E} > 0$ (respectively, $\deg \mathcal{E} \geq 0$).

(c) There is a small typo in the problem statement. The classification should concern *normalized* indecomposable locally free sheaves of rank 2 that are not semistable. Combine (III, Ex. 6.1), (III, Ex. 12.5), (b), and the fact that the extension $0 \to \mathcal{O} \to \mathcal{E} \to \mathcal{L} \to 0$ defined by any $\xi \in \mathrm{H}^1(C, \mathcal{L}^{-1})$ is normalized if $e > 0$.

□

9. Let Y be a nonsingular curve on a quadric cone X_0 in $\mathbb{P}^3$. Show that either Y is a complete intersection of X_0 with a surface of degree $a \geq 1$, in which case $\deg Y = 2a$, $g(Y) = (a-1)^2$, or $\deg Y$ is odd, say $2a+1$, and $g(Y) = a^2 - a$.

Proof. A general hyperplane section of X_0 in $\mathbb{P}^3$ is a conic curve C, and X_0 is a cone over C in $\mathbb{P}^3$ with vertex P_0. If we blow up the point P_0 as in (2.11.4), then we obtain the rational ruled surface $\mathbb{F}$ with invariant $e = 2$, such that if $\pi : \mathbb{F} \to X_0$ is the projection map, then $\pi^{-1}(P_0)$ is a section C_0 with $C_0^2 = -2$. In other words, $\mathbb{F}$ is isomorphic to the projective bundle of $\mathcal{E} = \mathcal{O}_{\mathbb{P}^1} \oplus \mathcal{O}_{\mathbb{P}^1}(-2)$ over $\mathbb{P}^1$. The hyperplane sections of $\pi : \mathbb{F} \to X_0 \hookrightarrow \mathbb{P}^3$ correspond to the linear system $|\mathcal{O}_{\mathbb{F}}(1)| = |C_0 + 2f|$, and by (2.11), $K = -2C_0 - 4f$ is a canonical divisor of $\mathbb{F}$.

If Y is a complete intersection of X_0 with a surface of degree $a \geq 1$, then $Y \sim a(C_0 + 2f)$, so $\deg Y = Y.(C_0 + 2f) = 2a$, and the adjunction formula gives

$$g = \frac{1}{2} Y.(Y + K) + 1 = (a-1)^2.$$

Conversely, if Y has even degree, say $\deg Y = 2a$, then $Y \sim a(C_0 + 2f) - bC_0$ with $0 \leq b < a$ (1.3) (2.20). Let $\mathcal{I} = \mathcal{O}_{\mathbb{F}}(-Y)$ be the ideal sheaf of Y as a closed subscheme of $\mathbb{F}$. We can tensor by $\mathcal{O}_{\mathbb{F}}(a)$, take global sections, and apply (2.4), which gives

$$\mathrm{H}^0(\mathbb{F}, \mathcal{I}(a)) \cong \mathrm{H}^0(\mathbb{P}^1, S^b(\mathcal{E})),$$

where we have

$$S^b(\mathcal{E}) \cong \bigoplus_{i=0}^{b} \mathcal{O}_{\mathbb{P}^1}(-2i).$$

It follows that there exists a section in $|\mathcal{O}_{\mathbb{F}}(a)|$ containing Y, and since the restriction map

$$\mathrm{H}^0(\mathbb{P}^3, \mathcal{O}_{\mathbb{P}^3}(a)) \to \mathrm{H}^0(\mathbb{F}, \mathcal{O}_{\mathbb{F}}(a)) \to 0$$

is surjective, we can find a homogenous polynomial of degree a in $\mathbb{P}^3$ containing Y. So by reasons of degree, Y is the complete intersection of X_0 with a surface of degree a.

If Y has odd degree, say $\deg Y = 2a+1$, then Y is contained in a surface of degree $a+1$. If Z is any such surface in $\mathbb{P}^3$, by (I, 7.7) $X_0 \cap Z$ has two irreducible components each of multiplicity one, consisting of Y and a line L in $\mathbb{P}^3$. The latter must be a member of the one-dimensional family of lines in X_0 which go through the vertex P_0, and the inverse image $\pi^* L = L \times_{X_0} X$ of any of these lines is linearly eqiuvalent to $C_0 + f$, which means

$$Y \sim (a+1)(C_0 + 2f) - (C_0 + f) = aC_0 + (2a+1)f.$$

To conclude, we can compute the genus of Y using the adjunction formula of (1.5)

$$g = \frac{1}{2} Y.(Y + K) + 1 = a^2 - a.$$

□

10. For any $n > e \geq 0$, let X be the rational scroll of degree $d = 2n - e$ in $\mathbb{P}^{d+1}$ given by (2.19). If $n \geq 2e - 2$, show that X contains a nonsingular curve Y of genus $g = d + 2$ which is a canonical curve in this embedding. Conclude that for every $g \geq 4$, there exists a nonhyperelliptic curve which has a g^1_3.

Proof. Let D be the divisor $C_0 + nf$ which gives the emebdding of X as a rational scrolled of degree d in $\mathbb{P}^{d+1}$. If Y is a nonsingular curve in X of genus $g = d + 2$ and of type (a, b) satisfying (2.18), then $Y.D$ is the degree of Y under this embedding, and the genus of Y can computed using the adjunction formula (1.5). For Y to be a canonical curve in $\mathbb{P}^{d+1}$, it is necessary and sufficient that

$$\begin{aligned} Y.D &= 2g - 2, \\ Y.(Y + K) &= 2g - 2. \end{aligned}$$

If $n \geq 2e - 2$, then it can be easily checked that $Y = 3C_0 + (n + e + 2)f$ is very ample and satsifies above. So for any $g \geq 4$, the linear system $|3C_0 + (n + e + 2)f|$ with $n = 2g - 2$ and $e = g$ contains a nonsingular curve Y of genus g which is a canonical curve in $\mathbb{P}^{d+1}$. For any such Y, the restriction of the projection map $Y \hookrightarrow X \to \mathbb{P}^1$ is a g^1_3 of Y. Hence, for every $g \geq 4$, there exists a nonhyperelliptic curve which has a g^1_3. □

11. Let X be a ruled surface over the curve C, defined by a normalized bundle $\mathcal{E}$, and let $\mathfrak{e}$ be the divisor on C for which $\mathcal{O}_C(\mathfrak{e}) \cong \bigwedge^2 \mathcal{E}$ (2.8.1). Let $\mathfrak{b}$ be any divisor on C.

(a) If $|\mathfrak{b}|$ and $|\mathfrak{b} + \mathfrak{e}|$ have no base points, and if $\mathfrak{b}$ is nonspecial, then there is a section $D \sim C_0 + \mathfrak{b}f$, and $|D|$ has no base points.

(b) If $\mathfrak{b}$ and $\mathfrak{b} + \mathfrak{e}$ are very ample on C, and for every point $P \in C$, we have $\mathfrak{b} - P$ and $\mathfrak{b} + \mathfrak{e} - P$ nonspecial, then $C_0 + \mathfrak{b}f$ is very ample.

Proof.

(a) Consider the short exact sequence

$$0 \longrightarrow \mathcal{O}_C \longrightarrow \mathcal{E} \longrightarrow \mathcal{O}_C(\mathfrak{e}) \longrightarrow 0 \ .$$

By the assumption that $\mathfrak{b}$ is nonspecial, we have

$$\mathrm{Ext}^1(\mathcal{O}_C(\mathfrak{e}), \mathcal{O}_C(\mathfrak{b} + \mathfrak{e})) \cong \mathrm{H}^1(\mathcal{O}_C(\mathfrak{b})) = 0,$$

which implies we have a short exact sequence

$$0 \longrightarrow \mathrm{Hom}(\mathcal{O}_C(\mathfrak{e}), \mathcal{O}_C(\mathfrak{b} + \mathfrak{e})) \longrightarrow \mathrm{Hom}(\mathcal{E}, \mathcal{O}_C(\mathfrak{b} + \mathfrak{e})) \longrightarrow \mathrm{Hom}(\mathcal{O}_C, \mathcal{O}_C(\mathfrak{b} + \mathfrak{e})) \longrightarrow 0 \ .$$

Thus, giving a morphism $\mathcal{E} \to \mathcal{O}_C(\mathfrak{b}+\mathfrak{e})$ is equivalent to giving nonzero sections $s \in \mathrm{H}^0(C, \mathcal{O}_C(\mathfrak{b}))$ and $t \in \mathrm{H}^0(C, \mathcal{O}_C(\mathfrak{b}+\mathfrak{e}))$, and the corresponding morphism $\mathcal{E} \to \mathcal{O}_C(\mathfrak{b}+\mathfrak{e})$ is locally defined by

$$(f, g) \mapsto fs + gt.$$

Similar to the proof of (2.17), for the corresponding map $\mathcal{E} \to \mathcal{O}_C(\mathfrak{b}+\mathfrak{e})$ to be surjective, it is enough that the divisors corresponding to s and t do not meet. Consider the subset $B \subset X \times |\mathfrak{b}| \times |\mathfrak{b}+\mathfrak{e}|$ consisting of all triples $\langle P, D, E\rangle$ such that P is in the support of both D and E. If $|\mathfrak{b}|$ and $|\mathfrak{b}+\mathfrak{e}|$ are base point free, then the fibers of $p_1 : B \to X$ have codimension 2 in $|\mathfrak{b}| \times |\mathfrak{b}+\mathfrak{e}|$ by (IV, 3.1), so B has dimension one less than that of $|\mathfrak{b}| \times |\mathfrak{b}+\mathfrak{e}|$. It follows that the image of B under the second projection map $p_2 : B \to |\mathfrak{b}| \times |\mathfrak{b}+\mathfrak{e}|$ is proper, so there exists sections $s \in \mathrm{H}^0(C, \mathcal{O}_C(\mathfrak{b}))$ and $t \in \mathrm{H}^0(C, \mathcal{O}_C(\mathfrak{b}+\mathfrak{e}))$ that do not both vanish at any point in C. Hence, we have a short exact sequence of locally free sheaves on C

$$0 \longrightarrow \mathcal{O}_C(-\mathfrak{b}) \longrightarrow \mathcal{E} \longrightarrow \mathcal{O}_C(\mathfrak{b}+\mathfrak{e}) \longrightarrow 0\ ,$$

so by (2.9), there is a section $D \sim C_0 + \mathfrak{b}f$, and it remains to show $|D|$ has no base points. We will show for every $P \in X$, there exists $D' \in |D|$ such that $P \notin \operatorname{Supp} D'$. Let $\pi : X \to C$ be the projection map, and let $\sigma : C \to X$ be the map corresponding to the section C_0.

$\boxed{P \notin C_0}$ A divisor of the form $D' = C_0 + \mathfrak{b}'f$ where $\mathfrak{b}' \in |\mathfrak{b}|$ misses $\pi(P)$ will do.

$\boxed{P \in C_0}$ Consider the short exact sequence

$$0 \longrightarrow \mathcal{O}_X(-C_0) \longrightarrow \mathcal{O}_X \longrightarrow \mathcal{O}_{C_0} \longrightarrow 0\ .$$

Tensoring by $\mathcal{O}_X(D)$ and taking cohomology gives

$$\mathrm{H}^0(X, \mathcal{O}_X(D)) \xrightarrow{\alpha} \mathrm{H}^0(C_0, \mathcal{O}_{C_0}(\mathfrak{b}+\mathfrak{e})) \longrightarrow \mathrm{H}^1(X, \mathcal{O}_X(\mathfrak{b}f))\ .$$

Since $\mathfrak{b}$ is nonspecial, by (2.4) we have

$$\mathrm{H}^1(X, \mathcal{O}_X(\mathfrak{b}f)) \cong \mathrm{H}^1(C, \mathcal{O}_C(\mathfrak{b})) = 0,$$

so α is a surjection by exactness. This means for any divisor $\mathfrak{b}' \in |\mathfrak{b}+\mathfrak{e}|$ on $C_0 \cong C$, we can find $D' \in |D|$ such that $\sigma^* D' = \mathfrak{b}'$. If $|\mathfrak{b}+\mathfrak{e}|$ has no base points, then we can find a divisor in $|\mathfrak{b}+\mathfrak{e}|$ that misses $P \in C_0$, and we can lift it to a divisor in $|D|$ which also misses P by construction.

Hence, $|D|$ has no base points.

(b) We will show D is very ample by showing that the linear system $|D|$ separates points $P \neq Q$ and tangent vectors $t \in T_P X$. Note that if $\mathfrak{d}$ is any divisor on C, and if there exists a point $P \in C$ such that $\mathfrak{d} - P$ is nonspecial, then $\mathfrak{d}$ is a nonspecial divisor on C. In particular, $\mathfrak{b}$ and $\mathfrak{b}+\mathfrak{e}$ are nonspecial, so we can resume from (a).

$\boxed{\text{Case 1: } P, Q \text{ not both in } C_0, \text{ and not both in any fiber.}}$ Since $\mathfrak{b}$ is very ample, there exists a divisor $\mathfrak{b}' \in |\mathfrak{b}|$ which separates $\pi(P), \pi(Q)$ in C, so $C_0 + \mathfrak{b}'f \in |D|$ separates P, Q in X.

$\boxed{\text{Case 2: } P, t \text{ not both in } C_0, \text{ and not both in any fiber.}}$ Again, $|\mathfrak{b}|$ is very ample, so if t is not in any fiber, then its pushforward $\pi_* t \in T_{\pi(P)} C$ is nonzero. Thus, we can find a divisor of the form $C_0 + \mathfrak{b}'f$ such that $\mathfrak{b}' \in |\mathfrak{b}|$ contains $\pi(P)$ in its support and separates $\pi_* t$ so that $C_0 + \mathfrak{b}'f$ contains P and separates t in X.

$\boxed{\text{Case 3: } P, Q \text{ or } P, t \text{ are both in } C_0.}$ Consider the short exact sequence

$$0 \longrightarrow \mathcal{O}_X(-C_0) \longrightarrow \mathcal{O}_X \longrightarrow \mathcal{O}_{C_0} \longrightarrow 0\ .$$

Tensoring by $\mathcal{O}_X(D)$ and taking cohomology gives

$$\mathrm{H}^0(X, \mathcal{O}_X(D)) \xrightarrow{\beta} \mathrm{H}^0(C_0, \mathcal{O}_X(D) \otimes \mathcal{O}_{C_0}) \longrightarrow \mathrm{H}^1(X, \mathcal{O}_X(\mathfrak{b}f))\ .$$

Since $\mathcal{O}_X(D) \otimes \mathcal{O}_{C_0} \cong \mathcal{O}_{C_0}(\mathfrak{b}+\mathfrak{e})$ is very ample on $C_0 \cong C$, it suffices to show β is a surjection. By (2.4) and the assumption that $|\mathfrak{b}|$ is nonspecial, we have

$$\mathrm{H}^1(X, \mathcal{O}_X(\mathfrak{b}f)) \cong \mathrm{H}^1(C, \pi_* \mathcal{O}_X(\mathfrak{b}f)) \cong \mathrm{H}^1(C, \mathcal{O}_C(\mathfrak{b})) = 0.$$

Hence, β is a surjection by exactness.

Case 4: P, Q or P, t are both in the same fiber. Let f be the fiber containing P, Q or P, t. We have the identification $f \sim \pi^*(\pi(P)) = \pi(P)f$. Consider the short exact sequence

$$0 \longrightarrow \mathcal{O}_X(-f) \longrightarrow \mathcal{O}_X \longrightarrow \mathcal{O}_f \longrightarrow 0 \ .$$

Tensoring by $\mathcal{O}_X(D)$ and taking cohomology gives

$$\mathrm{H}^0(X, \mathcal{O}_X(D)) \xrightarrow{\gamma} \mathrm{H}^0(f, \mathcal{O}_X(D) \otimes \mathcal{O}_f) \longrightarrow \mathrm{H}^1(X, \mathcal{O}_X(D-f)) \ ,$$

Since $D.f = 1$, the invertible sheaf $\mathcal{O}_X(D) \otimes \mathcal{O}_f$ is very ample on $f \cong \mathbb{P}^1$. Thus, to separate P, Q or P, t, it will be sufficient to show that γ is a surjection. The cokernel of this map lands in $\mathrm{H}^1(X, \mathcal{O}_X(D-f))$, which by (2.4) is isomorphic to

$$\mathrm{H}^1(C, \pi_*\mathcal{O}_X(D-f)).$$

On the other hand, we have

$$\pi_*(\mathcal{O}_X(D-f)) \cong \pi_*(\mathcal{O}_X(C_0)) \otimes \mathcal{O}_C(\mathfrak{b} - \pi(P)) \cong \mathcal{E} \otimes \mathcal{O}_C(\mathfrak{b} - \pi(P)).$$

Consider the following exact sequence

$$0 \longrightarrow \mathcal{O}_C \longrightarrow \mathcal{E} \longrightarrow \mathcal{O}_C(\mathfrak{e}) \longrightarrow 0.$$

Tensoring by $\mathcal{O}_C(\mathfrak{b} - \pi(P))$ gives

$$0 \longrightarrow \mathcal{O}_C(\mathfrak{b} - \pi(P)) \longrightarrow \mathcal{E} \otimes \mathcal{O}_C(\mathfrak{b} - \pi(P)) \longrightarrow \mathcal{O}_C(\mathfrak{b} + \mathfrak{e} - \pi(P)) \longrightarrow 0 \ ,$$

and taking cohomology gives

$$\mathrm{H}^1(C, \mathcal{O}_C(\mathfrak{b} - \pi(P))) \longrightarrow \mathrm{H}^1(C, \mathcal{E} \otimes \mathcal{O}_C(\mathfrak{b} - \pi(P))) \longrightarrow \mathrm{H}^1(C, \mathcal{O}_C(\mathfrak{b} + \mathfrak{e} - \pi(P))).$$

The terms on the left and right are zero since we assumed $\mathfrak{b} - Q$ and $\mathfrak{b} + \mathfrak{e} - Q$ are nonspecial for any $Q \in C$, so the term in the middle is zero by exactness. We conclude that γ is a surjection.
Hence, D is very ample.

□

12. Let X be a ruled surface with invariant e over an elliptic curve C, and let $\mathfrak{b}$ be a divisor on C.

(a) If $\deg \mathfrak{b} \geq e + 2$, then there is a section $D \sim C_0 + \mathfrak{b}f$ such that $|D|$ has no base points.

(b) The linear system $|C_0 + \mathfrak{b}f|$ is very ample if and only if $\deg \mathfrak{b} \geq e + 3$.

Proof. $e \geq 0$ If $\mathfrak{d}$ is a divisor of degree d on C, then $d \geq 1$ if and only if $\mathfrak{d}$ is ample and nonspecial, $d \geq 2$ if and only if $\mathfrak{d}$ is base point free, and $d \geq 3$ if and only if $\mathfrak{d}$ is very ample. If $\deg \mathfrak{b} \geq e + 2 \geq 2$, then $|\mathfrak{b}|$ and $|\mathfrak{b} + \mathfrak{e}|$ have no base points, and $\mathfrak{b}$ is nonspecial, so by (Ex. 2.11) there is a section $D \sim C_0 + \mathfrak{b}f$ such that $|D|$ has no base points.

If $C_0 + \mathfrak{b}f$ is very ample and $e \geq -1$, then $\mathcal{O}_X(C_0 + \mathfrak{b}f) \otimes \mathcal{O}_{C_0}$ is very ample on C_0, which is so if and only if

$$(\deg \mathfrak{b}) - e = C_0.(C_0 + \mathfrak{b}f) = \deg \mathcal{O}_X(C_0 + \mathfrak{b}f) \otimes \mathcal{O}_{C_0} \geq 3.$$

Conversely, if $\deg \mathfrak{b} \geq e + 3$, then $\mathfrak{b}$ and $\mathfrak{b} + \mathfrak{e}$ are very ample on C, and $\mathfrak{b} - P, \mathfrak{b} + \mathfrak{e} - P$ have positive degrees, so they are nonspecial. Hence, by (Ex. 2.11) $C_0 + \mathfrak{b}f$ is very ample.

$e = -1$ Pick a base point $P \in C$ such that $\mathfrak{e} = P$, $X = \mathbb{P}(\mathcal{E})$, and $\mathcal{E}$ fits into the short exact sequence

$$0 \longrightarrow \mathcal{O}_C \longrightarrow \mathcal{E} \longrightarrow \mathcal{O}_C(P) \longrightarrow 0 \ .$$

For the first statement, if $\deg \mathfrak{b} \geq 2$, then we return to the situation $e \geq 0$, so assume $\mathfrak{b} = Q$ for some $Q \in C$. We want to show there is a section $D \sim C_0 + Qf$ such that $|D|$ has no base points. Take a point $R \in C$

distinct from P and Q such that $2R \sim P+Q$. This is possible, because the linear system $|P+Q|$ defines a two-to-one map of C to $\mathbb{P}^1$, ramified at four points, and we can take R to be one of them. Imitating the proof of (Ex. 2.11), we can find sections in $\mathrm{H}^0(C, \mathcal{O}_C(Q))$ and $\mathrm{H}^0(C, \mathcal{O}_C(P+Q)) \cong \mathrm{H}^0(C, \mathcal{O}_C(2R))$ that do not both vanish at every point in C since $Q \neq R$. Hence, we have a short sequence of locally free sheaves

$$0 \longrightarrow \mathcal{O}_C(-Q) \longrightarrow \mathcal{E} \longrightarrow \mathcal{O}_C(P+Q) \longrightarrow 0 \ ,$$

so there is a section $D \sim C_0 + Qf$, and it remains to show $|D|$ has no base points. From now, if x is any point in X, then denote $P_x = \pi(x) \in C$, and if Q is any point in C, then denote $f_Q = \pi^{-1}(Q)$. In particular, $x \in X$ lies on the fiber $f_{\pi(x)}$. We only need to check the points in the support of $C_0 + Qf$, and since $D.f_Q = 1$ and $D.C_0 = 2$, $\mathcal{O}_X(D)$ restricts to a base-point free line bundle on f_Q and C_0, so it is enough to show the natural maps

$$\mathrm{H}^0(X, \mathcal{O}_X(D)) \to \mathrm{H}^0(f_Q, \mathcal{O}_X(D) \otimes \mathcal{O}_{f_Q}), \tag{1}$$

$$\mathrm{H}^0(X, \mathcal{O}_X(D)) \to \mathrm{H}^0(C_0, \mathcal{O}_X(D) \otimes \mathcal{O}_{C_0}), \tag{2}$$

are surjective. We begin with f_Q. The cokernel of (1) lands in $\mathrm{H}^1(X, \mathcal{O}_X(D - f_Q))$, which by (2.4) is isomorphic to $\mathrm{H}^1(C, \pi_*\mathcal{O}_X(D - f_Q))$. On the other hand, $D - f_Q \sim C_0$, so

$$\pi_*\mathcal{O}_X(D - f_Q) \cong \mathcal{E},$$

and $\mathrm{H}^1(C, \mathcal{E}) = 0$ since $\mathcal{E}$ is the unique nontrivial extension of $\mathcal{O}_C(P)$ by $\mathcal{O}_C$. Therefore, (1) is surjective. Moving on to C_0, the cokernel of (2) lands in $\mathrm{H}^1(X, \mathcal{O}_X(D - C_0))$, which by (2.4) is isomorphic to $\mathrm{H}^1(C, \pi_*\mathcal{O}_X(D - C_0))$. On the other hand, $D - C_0 \sim f_Q$, so

$$\pi_*\mathcal{O}_X(D - C_0) \cong \mathcal{O}_C(Q),$$

which is nonspecial. Therefore, $\mathrm{H}^1(X, \mathcal{O}_X(D - C_0)) = 0$, and (2) is surjective.

Now suppose $\mathfrak{b}$ is a divisor on C with $\deg \mathfrak{b} \geq e + 3 = 2$. If $\deg \mathfrak{b} \geq 3$, then we can apply the same argument as in $e \geq 0$, so assume $\deg \mathfrak{b} = 2$, which means $\mathfrak{b}$ is a g^1_2 on C. Let $x \neq y$ be points in X, and let t be a tangent vector at x. We will show $D = C_0 + \mathfrak{b}f$ is very ample by showing that the linear system $|D|$ separates points and tangent vectors.

Case 1: x, y not both in C_0, and not both in any fiber. Without loss of generality, assume $y \notin C_0$, and let Q_x be a point in C such that $P_x + Q_x \in |\mathfrak{b}|$. If $P_y \neq Q_x$, then the divisor $C_0 + (P_x + Q_x)f$ separates x and y. Otherwise, we have $P_x + P_y \in |\mathfrak{b}|$, and for any point $z \in X$, consider the section $C_z = C_0 + (P_z - P)f$ (Ex. 2.7). Note that if $P_z \neq P$, then $C_0 \cap C_z = P_z$. If $P = P_x$, then $C_y + 2f_P \in |D|$ goes through x but not y. The hard case is $P = P_y$, and we further divide to two cases.

Case 1a. $x \in C_0$ Since $y \notin C_0$, let $S + T \in |\mathfrak{b}|$ such that $S, T \neq P_y$, then then divisor $C_0 + (S+T)f \in |D|$ goes through x but not y.

Case 1b. $x \notin C_0$ The section $C_x = C_0 + (P_x - P)f$ does not pass through x, and since $2P = 2P_y \in |\mathfrak{b} + P - P_x|$, the section $C_x + 2f_P \in |D|$ goes through y but not x.

Case 2: x, t not both in C_0, and not both in any fiber. A divisor of the form $C_0 + (P_x + Q_x)f$ will do.

Case 3: x, y or x, t are both in C_0. Since $C_0.D = 3$, the invertible sheaf $\mathcal{O}_X(D) \otimes \mathcal{O}_{C_0}$ is very ample on $C_0 \cong C$, so it is enough to show the natural map

$$\mathrm{H}^0(X, \mathcal{O}_X(D)) \to \mathrm{H}^0(C_0, \mathcal{O}_X(D) \otimes \mathcal{O}_{C_0})$$

is surjective. The cokernel of this map lands in $\mathrm{H}^1(X, \mathcal{O}_X(D - C_0))$, which by (2.4) is isomorphic to $\mathrm{H}^1(C, \pi_*\mathcal{O}_X(D - C_0))$. On the other hand, $D - C_0 \sim \mathfrak{b}f$, so

$$\pi_*\mathcal{O}_X(D - C_0) \cong \mathcal{O}_C(\mathfrak{b}),$$

which is nonspecial on C. Hence, $\mathrm{H}^1(X, \mathcal{O}_X(D - C_0)) = 0$, and the map above is surjective.

Case 4: x, y or x, t are both in the same fiber $f = f_Q$ for some $Q \in C$. Since $D.f = 1$, the invertible sheaf $\mathcal{O}_X(D) \otimes \mathcal{O}_f$ is very ample on $f \cong \mathbb{P}^1$, so it is enough to show the natural map

$$\mathrm{H}^0(X, \mathcal{O}_X(D)) \to \mathrm{H}^0(f, \mathcal{O}_X(D) \otimes \mathcal{O}_f)$$

is surjective. The cokernel of this map lands in $\mathrm{H}^1(X, \mathcal{O}_X(D-f))$, which by (2.4) is isomorphic to $\mathrm{H}^1(C, \pi_*\mathcal{O}_X(D-f))$. On the other hand, $D - f \sim C_0 + (\mathfrak{b} - Q)f$, so if we write $R \sim \mathfrak{b} - Q$ for some $R \in C$, then

$$\pi_*\mathcal{O}_X(D-f) \cong \mathcal{E} \otimes \mathcal{O}_C(R)$$

by the projection formula. We have a short exact sequence

$$\mathrm{H}^1(C, \mathcal{O}_C(R)) \longrightarrow \mathrm{H}^1(C, \mathcal{E} \otimes \mathcal{O}_C(R)) \longrightarrow \mathrm{H}^1(C, \mathcal{O}_C(P+R)) \ ,$$

and since the terms on the left and right are zero, the term in the middle is zero. Hence, $\mathrm{H}^1(X, \mathcal{O}_X(D-f)) = 0$, and the map above is surjective. □

13. For every $e \geq -1$ and $n \geq e+3$, there is an elliptic scroll of degree $d = 2n - e$ in $\mathbb{P}^{d-1}$. In particular, there is an elliptic scroll of degree 5 in $\mathbb{P}^4$.

Proof. For any divisor $\mathfrak{b}$ on C of degree n, we use the divisor $D = C_0 + \mathfrak{b}f$. By (Ex. 2.12), $C_0 + \mathfrak{b}f$ is very ample, and since $D.f = 1$ for any fiber f, the image of X in $|D| \cong \mathbb{P}^N$ is a scroll. The degree of the image is

$$D^2 = 2(\deg \mathfrak{b}) - e = 2n - e,$$

and it remains to compute N. By (2.4), we find that

$$\mathrm{H}^0(X, \mathcal{O}_X(D)) = \mathrm{H}^0(C, \pi_*\mathcal{O}_X(D)) = \mathrm{H}^0(C, \mathcal{E} \otimes \mathcal{O}_C(\mathfrak{b})),$$

and as in the proof of (Ex. 2.12), we have a short exact sequence

$$0 \longrightarrow \mathrm{H}^0(C, \mathcal{O}_C(\mathfrak{b})) \longrightarrow \mathrm{H}^0(C, \mathcal{E} \otimes \mathcal{O}_C(\mathfrak{b})) \longrightarrow \mathrm{H}^0(C, \mathcal{O}_C(\mathfrak{b}+P)) \longrightarrow 0.$$

Hence, we can compute N using exactness of above

$$\begin{aligned} N &= h^0(C, \mathcal{E} \otimes \mathcal{O}_C(\mathfrak{b})) - 1 \\ &= h^0(C, \mathcal{O}_C(\mathfrak{b})) + h^0(C, \mathcal{O}_C(\mathfrak{b}+\mathfrak{e})) - 1 \\ &= 2(\deg \mathfrak{b}) + \deg \mathfrak{e} - 1 \\ &= 2n - e - 1. \end{aligned}$$

To obtain an elliptic scroll of degree 5 in $\mathbb{P}^4$, take $n = 2$ and $e = -1$. □

14. Let X be a ruled surface over a curve of genus g, with invariant $e < 0$, and assume that $\operatorname{char} k = p > 0$ and $g \geq 2$.

(a) If $Y \equiv aC_0 + bf$ is an irreducible curve $\neq C_0, f$, then either $a = 1$, $b \geq 0$, or $2 \leq a \leq p-1$, $b \geq \frac{1}{2}ae$, or $a \geq p$, $b \geq \frac{1}{2}ae + 1 - g$.

(b) If $a > 0$ and $b > a(\frac{1}{2}e + (1/p)(g-1))$, then any divisor $D \equiv aC_0 + bf$ is ample. On the other hand, if D is ample, then $a > 0$ and $b > \frac{1}{2}ae$.

Proof.

(a) Let $\tilde{Y}$ be the normalization of Y, and consider the composition of the natural map $\tilde{Y} \to Y$ with the projection $\pi : Y \to C$. If $a \leq p-1$, then π is separable, so we can apply the same argument as in (2.21). If $a \geq p$, then $\pi : \tilde{Y} \to C$ may not be separable. By (IV, 2.5.4), $\pi : \tilde{Y} \to C$ factors as the composite $\tilde{Y} \to Z \to C$, where Z is a smooth curve over k, $\tilde{Y} \to Z$ is purely inseparable of degree p^n for some $n \geq 0$, and $Z \to C$ is finite, separable of degree d such that $a = dp^n$. By Hurwitz's theorem and the fact that $p_a(Y) \geq g(\tilde{Y})$, we have

$$2p_a(Y) - 2 \geq 2g(\tilde{Y}) - 2 = 2g(Z) - 2 \geq d(2g-2).$$

and the adjunction formula (1.5) gives

$$2p_a(Y) - 2 = Y.(Y + K),$$

where K is a canonical divisor of X. Substituting $Y \equiv aC_0 + bf$ and $K \equiv -2C_0 + (2g-2-e)f$ from (2.11), and combining with the inequality above, we find that

$$b(a-1) \geq \frac{1}{2}ae(a-1) + (d-a)(g-1).$$

If $a \geq p$, then $\frac{d-a}{a-1} \geq -1$, so

$$b \geq \frac{1}{2}ae + 1 - g.$$

(b) If D is ample, then by (1.10) $D.f = a > 0$, and $D^2 = 2ab - a^2e > 0$, so $b > \frac{1}{2}ae$. If $a > 0$ and $b > a(\frac{1}{2}e + (1/p)(g-1))$, then $D.f = a > 0$, $D.C_0 = a((1/p)(g-1) - \frac{1}{2}e) > 0$, and

$$D^2 = a(2b - ae) > \frac{2a}{p}(g-1) > 0.$$

If $Y \equiv a'C_0 + b'f$ is any irreducible curve $\neq C_0, f$, then

$$D.Y = ab' + a'b - aa'e.$$

Note that $b > \frac{1}{2}ae$, so the same argument as in (2.21) for when $1 \leq a' \leq p-1$ applies here as well. Now if $a' = 1$, then $b' \geq 0$, so

$$D.Y > \frac{1}{2}ae - ae = -\frac{1}{2}ae > 0.$$

If $2 \leq a' \leq p-1$, then $b' \geq \frac{1}{2}a'e$, so

$$D.Y > \frac{1}{2}aa'e + \frac{1}{2}aa'e - aa'e = 0.$$

If $a' \geq p$, then $b' \geq \frac{1}{2}a'e + 1 - g$, so

$$D.Y > a(\frac{1}{2}a'e + 1 - g) + a'a(\frac{1}{2}e + (1/p)(g-1)) - aa'e = \left(\frac{a'}{p} - 1\right)a(g-1) > 0.$$

Therefore by (1.10), D is ample.

□

16. Let C be a nonsingular affine curve. Show that two locally free sheaves $\mathcal{E}, \mathcal{E}'$ of the same rank are isomorphic if and only if their classes in the Grothendieck group $K(X)$ are the same. This is false for a projective curve.

Proof. Let A be a be an affine coordinate ring of a nonsingular affine curve, and let K be its function field. The category of coherent $\mathcal{O}_C$-modules is equivalent to the category of finitely generated A-modules. Under this identification, $K(X)$ is naturally isomorphic to the Grothendieck group of flat finitely generated A-modules $K_0(A)$ (A.M., Ex. 7.26). The structure theorem of modules over Dedekind domains states $K_0(A) \cong \mathbb{Z} \oplus \mathrm{Cl}(A)$, where if M is any flat finitely generated A-module, then $M \cong A^r \oplus I$, with $r+1 = \dim_K M \otimes_A K$, and I is some fractional ideal in $\mathrm{Cl}(A)$. If $N \cong A^s \oplus J$ is another A-module, then $M \cong N$ if and only if $r = s$ and $I \cong J$ in $\mathrm{Cl}(A)$. □

17. (a) Let $\varphi : \mathbb{P}^1_k \to \mathbb{P}^3_k$ be the 3-uple embedding. Let $\mathcal{I}$ be the sheaf of ideals of the twisted cubic curve C which is the image of φ. Then $\mathcal{I}/\mathcal{I}^2$ is a locally free sheaf of rank 2 on C, so $\varphi^*(\mathcal{I}/\mathcal{I}^2)$ is a locally free sheaf of rank 2 on $\mathbb{P}^1$. By (2.14), therefore, $\varphi^*(\mathcal{I}/\mathcal{I}^2) \cong \mathcal{O}(l) \oplus \mathcal{O}(m)$ for some $l, m \in \mathbb{Z}$. Determine l and m.

(b) Repeat part (a) for the embedding $\varphi : \mathbb{P}^1 \to \mathbb{P}^3$ given by $x_0 = t^4, x_1 = t^3u, x_2 = tu^3, x_3 = u^4$, whose image is a nonsingular rational quartic curve.

Proof.

(a) Let t, u be homogenous coordinates for C, and let x_0, x_1, x_2, x_3 be homogenous coordinates for $\mathbb{P}^3$, so that φ is expicitly given by

$$(t : u) \mapsto (t^3 : t^2u : tu^2 : u^3).$$

It is enough to determine $\mathcal{N}_{C/\mathbb{P}^3} = \varphi^*(\mathcal{I}/\mathcal{I}^2)^\vee$. If Q is any irreducible quadric surface containing C in $\mathbb{P}^3$, then C has type $(1, 2)$ as a divisor on Q, and we can use the short exact sequence of normal bundles

$$0 \longrightarrow \mathcal{N}_{C/Q} \longrightarrow \mathcal{N}_{C/\mathbb{P}^3} \longrightarrow \mathcal{N}_{Q/\mathbb{P}^3}\big|_C \longrightarrow 0,$$

where we have

$$\mathcal{N}_{C/Q} \cong \mathcal{O}_C(4), \quad \mathcal{N}_{Q/\mathbb{P}^3}\big|_C \cong \mathcal{O}_C(6).$$

By exactness, we have two possibilities for the isomorphism class of $\mathcal{N}_{C/\mathbb{P}^3}$, which are distinguished by whether $\mathcal{N}_{C/\mathbb{P}^3}(-6)$ has global sections or not. In particular, we have

$$\mathcal{N}_{C/\mathbb{P}^3} \cong \begin{cases} \mathcal{O}_C(5) \oplus \mathcal{O}_C(5), & \text{if } h^0(\mathcal{N}_{C/\mathbb{P}^3}(-6)) = 0, \\ \mathcal{O}_C(4) \oplus \mathcal{O}_C(6), & \text{otherwise.} \end{cases}$$

The two relevant short exact sequences for this computation are

$$0 \longrightarrow \mathcal{O}_{\mathbb{P}^3} \longrightarrow \mathcal{O}_{\mathbb{P}^3}(1)^{\oplus 4} \longrightarrow \mathcal{T}_{\mathbb{P}^3} \longrightarrow 0 \quad (*)$$

$$0 \longrightarrow \mathcal{T}_C \longrightarrow \mathcal{T}_{\mathbb{P}^3}\big|_C \longrightarrow \mathcal{N}_{C/\mathbb{P}^3} \longrightarrow 0 \quad (**)$$

Restricting to C, twisting by $\mathcal{O}_C(-6)$, and taking the long exact sequence in cohomology of (*) gives

$$0 \longrightarrow \mathrm{H}^0(\mathcal{O}_C(-6)) \longrightarrow \mathrm{H}^0(\mathcal{O}_C(-3)^{\oplus 4}) \longrightarrow \mathrm{H}^0(\mathcal{T}_{\mathbb{P}^3}\big|_C(-6)) \longrightarrow$$

$$\longrightarrow \mathrm{H}^1(\mathcal{O}_C(-6)) \xrightarrow{\beta} \mathrm{H}^1(\mathcal{O}_C(-3)^{\oplus 4}) \longrightarrow \mathrm{H}^1(\mathcal{T}_{\mathbb{P}^3}\big|_C(-6)) \longrightarrow 0.$$

The first two terms on the top row vanish since $\mathrm{H}^0(\mathcal{O}_C(-n)) = 0$ for any $n > 0$, the first two terms on the bottom row can be computed using Serre duality

$$h^1(\mathcal{O}_C(-6)) = h^0(\mathcal{O}_C(4)) = 5, \quad h^1(\mathcal{O}_C(-3)^{\oplus 4}) = h^0(\mathcal{O}_C(1)^{\oplus 4}) = 8,$$

and β is dual to

$$\beta' : \mathrm{H}^0(\mathcal{O}_C(1)^{\oplus 4}) \to \mathrm{H}^0(\mathcal{O}_C(4)),$$

which is explicitly given by

$$(g_1, g_2, g_3, g_4) \mapsto g_1t^3 + g_2t^2u + g_3tu^2 + g_4u^3.$$

It is not hard to see β' is surjective, which means β is injective. So by exactness,

$$h^0(\mathcal{T}_{\mathbb{P}^3}\big|_C(-6)) = 0, \quad h^1(\mathcal{T}_{\mathbb{P}^3}\big|_C(-6)) = 3.$$

Doing the same calculation for (**) gives

$$0 \longrightarrow \mathrm{H}^0(\mathcal{T}_C(-6)) \longrightarrow \mathrm{H}^0(\mathcal{T}_{\mathbb{P}^3}\big|_C(-6)) \longrightarrow \mathrm{H}^0(\mathcal{N}_{C/\mathbb{P}^3}(-6)) \longrightarrow$$

$$\longrightarrow \mathrm{H}^1(\mathcal{T}_C(-6)) \xrightarrow{\alpha} \mathrm{H}^1(\mathcal{T}_{\mathbb{P}^3}\big|_C(-6)) \longrightarrow \mathrm{H}^1(\mathcal{N}_{C/\mathbb{P}^3}(-6)) \longrightarrow 0.$$

The first two terms on the top row vanish, and by Serre duality,

$$h^1(\mathcal{T}_C(-6)) = h^0(\Omega_C(4)) = 3,$$

and α is dual to the map

$$\alpha' : \mathrm{H}^0(\Omega_{\mathbb{P}^3}\big|_C(4)) \to \mathrm{H}^0(\Omega_C(4)).$$

The vector space $\mathrm{H}^0(\Omega_{\mathbb{P}^3}\big|_C(4))$ can be identified with $\ker \beta'$, and $\mathrm{H}^0(\Omega_C(4)) \cong \mathrm{H}^0(\mathcal{O}_C(2))$. This means $\mathrm{H}^0(\Omega_{\mathbb{P}^3}\big|_C(4))$ is generated by the syzygies of β', so we can write a basis

$$s_0 = u\mathrm{d}x_0 - t\mathrm{d}x_1, \quad s_1 = u\mathrm{d}x_1 - t\mathrm{d}x_2, \quad s_2 = u\mathrm{d}x_2 - t\mathrm{d}x_3.$$

Now, we can compute the effect of α' on each s_i,

$$\begin{aligned}
\alpha'(s_0) &= u\,\mathrm{d}\big(t^3\big) - t\,\mathrm{d}\big(t^2u\big) = t^2(u\mathrm{d}t - t\mathrm{d}u),\\
\alpha'(s_1) &= u\,\mathrm{d}\big(t^2u\big) - t\,\mathrm{d}\big(tu^2\big) = tu(u\mathrm{d}t - t\mathrm{d}u),\\
\alpha'(s_2) &= u\,\mathrm{d}\big(tu^2\big) - t\,\mathrm{d}\big(u^3\big) = u^2(u\mathrm{d}t - t\mathrm{d}u),
\end{aligned}$$

so α is an isomorphism. Hence, we have

$$\mathcal{N}_{C/\mathbb{P}^3} \cong \mathcal{O}_C(5) \oplus \mathcal{O}_C(5).$$

(b) We can do the same calculation as above, except in characteristic 2, the corresponding α' has a nontrivial kernel of rank 1. Explicitly, we have

$$\alpha' : \mathrm{H}^0(\Omega_{\mathbb{P}^3}\big|_C(6)) \to \mathrm{H}^0(\Omega_C(6)),$$

where we can identify $\Omega_C(6) \cong \mathcal{O}_C(4)$, and

$$\mathrm{H}^0(\Omega_{\mathbb{P}^3}\big|_C(6)) = \ker(\beta' : \mathrm{H}^0(\mathcal{O}_C(2)^{\oplus 4}) \to \mathrm{H}^0(\mathcal{O}_C(6))).$$

Consider the section of $\Omega_{\mathbb{P}^3}\big|_C(6)$ defined by

$$s = u^2\mathrm{d}x_1 - t^2\mathrm{d}x_2.$$

Its image under β can be computed by plugging in $x_2 = t^3u$ and $x_3 = tu^3$,

$$\begin{aligned}
\beta'(s) &= u^2\,\mathrm{d}\big(t^3u\big) - t^2\,\mathrm{d}\big(tu^3\big)\\
&= u^2(3t^2u\mathrm{d}t + t^3\mathrm{d}u) - t^2(u^3\mathrm{d}t + 3tu^2\mathrm{d}u)\\
&= 2t^2u^2(u\mathrm{d}t - t\mathrm{d}u),
\end{aligned}$$

which is zero if $\operatorname{char} k = 2$. To summarize,

$$\mathcal{N}_{C/\mathbb{P}^3} = \begin{cases} \mathcal{O}_C(6) \oplus \mathcal{O}_C(8) & \operatorname{char} k = 2,\\ \mathcal{O}_C(7) \oplus \mathcal{O}_C(7) & \text{otherwise.} \end{cases}$$

□

Chapter 5, Section 3

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1. Let X be a nonsingular projective variety of any dimension, let Y be a nonsingular subvariety, and let $\pi : \tilde{X} \to X$ be obtained by blowing up Y. Show that $p_a(\tilde{X}) = p_a(X)$.

Proof. The proof of (3.4) follows through for the more general case by (II, 8.24).

Proposition 0.1. *In the above setting, we have $\pi_*\mathcal{O}_{\tilde{X}} = \mathcal{O}_X$, and $\mathrm{R}^i\,\pi_*\mathcal{O}_{\tilde{X}} = 0$ for $i > 0$. Therefore, $\mathrm{H}^i(X, \mathcal{O}_X) \cong \mathrm{H}^i(\tilde{X}, \mathcal{O}_{\tilde{X}})$ for all $i \geq 0$.*

□

5. Let $a_1, \ldots, a_r$, $r \geq 5$, be distinct elements of k, and let C be the curve in $\mathbb{P}^2$ given by the (affine) equation $y^2 = \prod_{i=1}^r (x - a_i)$. Show that the point P at infinity on the y-axis is a singular point. Compute δ_P and $g(\tilde{Y})$, where $\tilde{Y}$ is the normalization of Y. Show in this way that one obtains hyperelliptic curves of genus $g \geq 2$.

Proof. The point P at infinity on the y-axis has homogenous coordinates $(0 : 1 : 0)$, and the local equation of Y at P at the affine patch defined by $y \neq 0$ is given by $z^{r-2} = \prod_{i=1}^r (x - a_i z)$, so Y has multiplicity $\mu_P(Y) = r - 2$. Let us see what happens when we blow up $\mathbb{P}^2$ at P. The substitution $z = ux$ gives

$$x^{r-2}\left(u^{r-2} - x^2 \prod_{i=1}^r (1 - a_i u)\right) = 0,$$

so the strict transform Y_1 of Y is

$$u^{r-2} - x^2 \prod_{i=1}^r (1 - a_i u) = 0.$$

Since Y_1 is still singular at $(0, 0)$, we blow up Y_1 by making the substitution $x = vu$, which gives

$$u^2\left(u^{r-4} - v^2 \prod_{i=1}^r (1 - a_i u)\right) = 0,$$

so the strict transform Y_2 of Y_1 is

$$u^{r-4} - v^2 \prod_{i=1}^r (1 - a_i u) = 0.$$

We can continue this process until the exponent of u is either 0 or 1, which depends on whether r is even or odd. In the former case, r is even, and the equation of the strict transform of Y after $r/2$ blow-ups is given by

$$1 - w^2 \prod_{i=1}^r (1 - a_i u) = 0,$$

which is nonsingular. The multiplicities at the points that were blown up were $r - 2, 2, \ldots, 2$, so the genus of $\tilde{Y}$ can be computed using (3.9.2)

$$g(\tilde{Y}) = \frac{(r-1)(r-2)}{2} - \frac{(r-2)(r-3)}{2} - \frac{r-4}{2} = \boxed{\frac{r-2}{2}}.$$

If r is odd, then the strict transform of Y after $(r-1)/2$ blow-ups is given by

$$u - w^2 \prod_{i=1}^{r} (1 - a_i u) = 0,$$

which is also nonsingular. Similar to the case r is even, the genus of $\tilde{Y}$ can be computed using (3.9.2)

$$g(\widetilde{Y}) = \frac{(r-1)(r-2)}{2} - \frac{(r-2)(r-3)}{2} - \frac{r-3}{2} = \boxed{\frac{r-1}{2}}.$$

Composing the projection of $Y \to \mathbb{P}^1$ onto the x-axis with the normalization $\tilde{Y} \to Y$ is a g^1_2 on $\tilde{Y}$, so $\tilde{Y}$ is a hyperelliptic curve. □

7. For each of the following singularities at $(0,0)$ in the plane, give an embedded resolution, compute δ_P, and decide which ones are equivalent.

(a) $x^3 + y^5 = 0$
(b) $x^3 + x^4 + y^5 = 0$
(c) $x^3 + y^4 + y^5 = 0$
(d) $x^3 + y^5 + y^6 = 0$
(e) $x^3 + xy^3 + y^5 = 0$

Proof.

(a) The substitution $x = uy$ gives

$$y^3(u^3 + y^2) = 0,$$

so after the first blow-up, we obtain a cuspidal curve. Thus,

$$E_1 : y = 0, \quad C_1 : u^3 + y^2 = 0,$$

where E_1 is the exceptional locus and C_1 is the strict transform of C. Next, the substitution $y = uv$ gives

$$u^2(u + v^2) = 0,$$

so the second blow-up gives

$$E_2 : u = 0, \quad C_2 : u + v^2 = 0.$$

The substitution $u = vw$ then gives

$$v(w + v) = 0,$$

so the third blow-up gives

$$E_3 : v = 0, \quad C_3 : w + v = 0.$$

Finally, the substitution $v = wt$ gives

$$w(t + 1) = 0,$$

so the fourth blow-up gives

$$E_4 : w = 0, \quad C_4 : t + 1 = 0.$$

The total transform of the original curve is

$$w^{15} t^9 (t + 1) = 0,$$

and we compute δ_P using (3.9.3):

$$\delta_P = \frac{1}{2} 3(3-1) + \frac{1}{2} 2(2-1) + \frac{1}{2} 1(1-1) + \frac{1}{2} 1(1-1) = 4.$$

(b) The substitution $x = uy$ gives

$$y^3(u^3 + u^4y + y^2) = 0,$$

so the first blow-up gives

$$E_1 : y = 0, \quad C_1 : u^3 + u^4y + y^2 = 0.$$

Next, the substitution $y = uv$ gives

$$u^2(u + u^3v + v^2) = 0,$$

so the second blow-up gives

$$E_2 : u = 0, \quad C_2 = u + u^3v + v^2 = 0.$$

The substitution $u = vw$ then gives

$$v(w + v^3w^3 + v) = 0,$$

so the third blow-up gives

$$E_3 : v = 0, \quad C_3 : w + v^3w^3 + v = 0.$$

Finally, the substitution $v = wt$ gives

$$w(1 + w^5t^3 + t) = 0,$$

so the fourth blow-up gives

$$E_4 : w = 0, \quad C_4 : 1 + w^5t^3 + t = 0.$$

The total transform of the original curve is

$$w^{15}t^9(1 + w^5t^3 + t) = 0,$$

and we compute δ_P using (3.9.3):

$$\delta_P = \frac{1}{2}3(3-1) + \frac{1}{2}2(2-1) + \frac{1}{2}1(1-1) + \frac{1}{2}1(1-1) = 4.$$

(c) The substitution $x = uy$ gives

$$y^3(u^3 + y + y^2) = 0,$$

so the first blow-up gives

$$E_1 : y = 0, \quad C_1 : u^3 + y + y^2 = 0.$$

Next, the substitution $y = uv$ gives

$$u(u^2 + v + uv^2) = 0,$$

so the second blow-up gives

$$E_2 : u = 0, \quad C_2 : u^2 + v + uv^2 = 0.$$

The substitution $v = uw$ then gives

$$u(u + w + u^2w^2) = 0,$$

so the third blow-up gives

$$E_3 : u = 0, \quad C_3 : u + w + u^2w^2 = 0.$$

Finally, the substitution $u = wt$ gives

$$w(t + 1 + w^3t^2) = 0,$$

so the fourth blow-up gives

$$E_4 : w = 0, \quad C_4 : t + 1 + w^3t^2 = 0.$$

The total transform of the original curve is

$$w^{12}t^8(t + 1 + w^3t^2) = 0,$$

and we compute δ_P using (3.9.3):

$$\delta_P = \frac{1}{2}3(3-1) + \frac{1}{2}1(1-1) + \frac{1}{2}1(1-1) + \frac{1}{2}1(1-1) = 3.$$

(d) The substitution $x = uy$ gives

$$y^3(u^3 + y^2 + y^3) = 0,$$

so the first blow-up gives

$$E_1 : y = 0, \quad C_1 : u^3 + y^2 + y^3 = 0.$$

Next, the substitution $y = uv$ gives

$$u^2(u + v^2 + uv^3) = 0,$$

so the second blow-up gives

$$E_2 : u = 0, \quad C_2 : u + v^2 + uv^3 = 0.$$

The substitution $u = vw$ then gives

$$v(w + v + v^3w) = 0,$$

so the third blow-up gives

$$E_3 : v = 0, \quad C_3 : w + v + v^3w = 0.$$

Finally, the substitution $v = wt$ gives

$$w(1 + t + w^3t^3) = 0,$$

so the fourth blow-up gives

$$E_4 : w = 0, \quad C_4 : 1 + t + w^3t^3 = 0.$$

The total transform of the original curve is

$$w^{15}t^9(1 + t + w^3t^3) = 0,$$

and we compute δ_P using (3.9.3):

$$\delta_P = \frac{1}{2}3(3-1) + \frac{1}{2}2(2-1) + \frac{1}{2}1(1-1) + \frac{1}{2}1(1-1) = 4.$$

(e) The substitution $x = uy$ gives

$$y^3(u^3 + uy + y^2) = 0,$$

so the first blow-up gives

$$E_1 : y = 0, \quad C_1 : u^3 + uy + y^2 = 0.$$

Next, the substitution $y = uv$ gives

$$u^2(u + v + v^2) = 0,$$

so the second blow-up gives

$$E_2 : u = 0, \quad C_2 : u + v + v^2 = 0.$$

The substitution $u = vw$ then gives

$$v(w + 1 + v) = 0,$$

so the third blow-up gives

$$E_3 : v = 0, \quad C_3 : w + 1 + v = 0.$$

The total transform of the original curve is

$$v^9w^5(vw)^2(v^2w)^3(w + 1 + v) = 0,$$

and we compute δ_P using (3.9.3):

$$\delta_P = \frac{1}{2}3(3-1) + \frac{1}{2}2(2-1) + \frac{1}{2}1(1-1) = 4.$$

The singularities in (a), (b), and (d) are equivalent, while the singularities in (c) and (e) are not equivalent to any of the others. □

Chapter 5, Section 4

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1. The linear system of conics in $\mathbb{P}^2$ with two assigned base points P_1 and P_2 (4.1) determines a morphism ψ of X' (which is $\mathbb{P}^2$ with P_1 and P_2 blown up) to a nonsingular quadric surface Y in $\mathbb{P}^3$, and furthermore X' via ψ is isomorphic to Y with one point blown up.

Proof. We have the commutative diagram

$$\begin{array}{ccccc} & & X' & & \\ & \swarrow^{\pi} & & \searrow^{\psi} & \\ \mathbb{P}^2 & & \overset{\varphi}{\dashrightarrow} & & Y \hookrightarrow \mathbb{P}^3 \end{array},$$

where φ is defined by the linear system $\mathfrak{d}$ of conics on $\mathbb{P}^2$ with assigned base-points P_1 and P_2, Y is the image of φ, and X' is the blow-up of $\mathbb{P}^2$ at P_1 and P_2. We may take the two points to be $P_1 = (1:0:0)$ and $P_2 = (0:1:0)$. Then the vector space $V \subseteq \mathrm{H}^0(\mathbb{P}^2, \mathcal{O}_{\mathbb{P}^2}(2))$ corresponding to $\mathfrak{d}$ is spanned by

$$y_0 = x_1x_2, \quad y_1 = x_0x_2, \quad y_2 = x_0x_1, \quad y_3 = x_2^2,$$

where y_i are the homogenous coordinates of $\mathbb{P}^3$. In particular, φ is defined as

$$\varphi(x_0 : x_1 : x_2) = (y_0 : y_1 : y_2 : y_3),$$

and the points in the image of ψ satisfies the equation

$$y_0y_1 - y_2y_3 = 0,$$

which corresponds to a nonsingular quadric surface $Y \subset \mathbb{P}^3$. It is a birational map since if we restrict to the affine patch defined by $x_2 \neq 0$, then we can identify φ with a quadratic transformation (4.2.3).

For each $i = 1, 2$, let E_i be the strict transform of P_i, let M be the line through P_1 and P_2, which is defined by $x_2 = 0$ in $\mathbb{P}^2$, and let $\tilde{M}$ be the strict transform of M in X'. By (3.6), we have

$$\tilde{M} \sim \pi^*M - E_1 - E_2,$$

so by (3.2), $\tilde{M}$ has self intersection

$$\tilde{M}^2 = (M)^2 + (E_1)^2 + (E_2)^2 = -1.$$

Let λ be a local coordinate on M such that M consists of the points $(\lambda : 1 : 0)$. The image of M under φ for $\lambda \neq 0$ is computed to be

$$\varphi(\lambda : 1 : 0) = (0 : 0 : \lambda : 0),$$

so denote by Q the point $\varphi(M - \{P_1, P_2\}) = (0 : 0 : 1 : 0)$. We claim X' can be identified with the blowing-up of Y at Q. Indeed, φ induces an isomorphism

$$\mathbb{P}^2 - M \cong Y - \{Q\} \cup \psi(E_1) \cup \psi(E_2),$$

and π induces an isomorphism

$$X' - E_1 \cup E_2 \cong \mathbb{P}^2 - \{P_1, P_2\},$$

so ψ induces an isomorphism

$$X' - \tilde{M} \cup E_1 \cup E_2 \cong Y - \{Q\} \cup \psi(E_1) \cup \psi(E_2),$$

and since $\psi(E_i)$ are lines in $\mathbb{P}^3$ which intersect at Q, we conclude that ψ induces an isomorphism

$$X' - \tilde{M} \cong Y - \{Q\}.$$

Hence by (5.7), X' can be identified with Y blown-up at Q. (See [Manin, §18] for more details.) □

2. Let φ be the quadratic transformation of (4.2.3), centered at P_1, P_2, P_3. If C is an irreducible curve of degree d in $\mathbb{P}^2$, with points of multiplicity r_1, r_2, r_3 at P_1, P_2, P_3, then the strict transform C' of C by φ has degree $d' = 2d - r_1 - r_2 - r_3$, and has points of multiplicity $d - r_2 - r_3$ at Q_1, $d - r_1 - r_3$ at Q_2, and $d - r_1 - r_2$ at Q_3. The curve may have arbitrary singularities.

Proof. We have a commutative diagram

$$\begin{array}{ccc} & X & \\ {\scriptstyle\pi}\swarrow & & \searrow{\scriptstyle\psi} \\ \mathbb{P}^2 & \overset{\varphi}{\dashrightarrow} & \mathbb{P}^2 \end{array} .$$

Let H be a hyperplane section of $\mathbb{P}^2$, so that φ is given by the linear system $\mathfrak{d} = |2H - P_1 - P_2 - P_3|$, and let E_1, E_2, E_3 be the exceptional divisors of X via the blowing-up of $\mathbb{P}^2$ at P_1, P_2, P_3, respectively. By (3.6), the strict transform $\tilde{C}$ of C by π is linearly equivalent to

$$\tilde{C} = \pi^* C - r_1 E_1 - r_2 E_2 - r_3 E_3.$$

Since ψ is given by the linear system $\mathfrak{d}' = |D|$ with $D = 2\pi^* H - E_1 - E_2 - E_3$, φ is a birational map such that $C' = \varphi(C) = \psi(\tilde{C})$, and $C \sim dH$ in $\mathbb{P}^2$, the degree of C' in $\mathbb{P}^2$ can be computed using (Ex. 1.2) and (3.2)

$$\begin{aligned} \tilde{C}.D &= (d\pi^* H - r_1 E_1 - r_2 E_2 - r_3 E_3).(2\pi^* H - E_1 - E_2 - E_3) \\ &= 2d(H^2) + r_1 (E_1)^2 + r_2 (E_2)^2 + r_3 (E_3)^2 \\ &= 2d - r_1 - r_2 - r_3. \end{aligned}$$

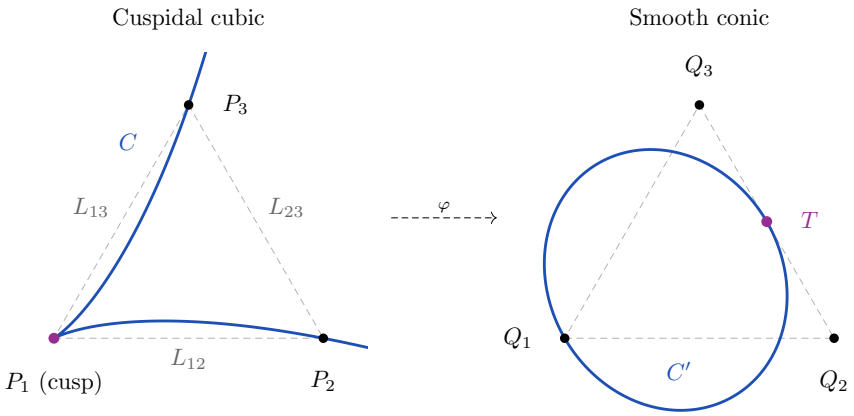
For each $i \neq j$, let L_{ij} be the line which goes through P_i and P_j. By (3.6), for each $k = 1, 2, 3$, the multiplicity of C' at Q_k is the intersection of $\tilde{C}$ with the strict transform $\tilde{L}_{ij}$ of L_{ij} with $i, j \neq k$ via π (4.2.3), which by (3.6) is linearly equivalent to

$$\tilde{L}_{ij} = \pi^* L_{ij} - E_i - E_j.$$

Again, we have $L_{ij} \sim H$ in $\mathbb{P}^2$, so

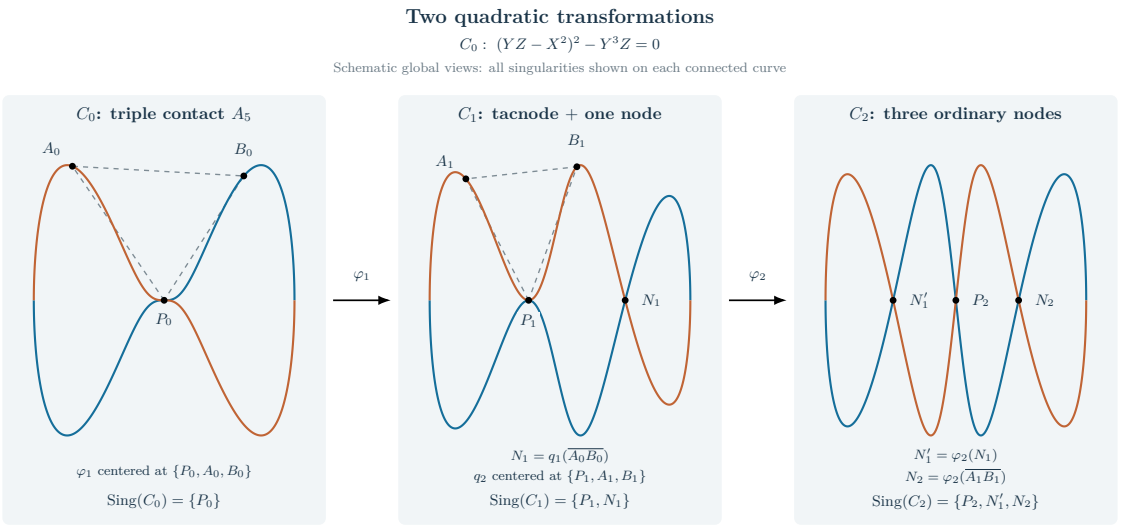
$$\begin{aligned} \tilde{C}.\tilde{L}_{ij} &= (d\pi^* H - r_1 E_1 - r_2 E_2 - r_3 E_3).(\pi^* H - E_i - E_j) \\ &= d(H)^2 + r_i (E_i)^2 + r_j (E_j)^2 \\ &= d - r_i - r_j. \end{aligned}$$

Hence, C' has points of multiplicity $d - r_i - r_j$ at Q_k. □



3. Let C be an irreducible curve in $\mathbb{P}^2$. Then there exists a finite sequence of quadratic transformations, centered at suitable triples of points, so that the strict transform C' of C has only *ordinary* singularities, i.e., multiple points with all distinct tangent directions.

Proof. If C is a conic in $\mathbb{P}^2$, then it is nonsingular, so assume C has degree ≥ 3, and let P_1 be a nonordinary singular point of C. Our goal is to find points $P_2, P_3 \in \mathbb{P}^2$ such that the quadratic transformation centered at P_i improves the singularity at P_1 and only introduces ordinary singularities on the strict transform of C. To that end, for each $i \neq j$, let L_{ij} be the line through P_i and P_j. The restriction of the quadratic transformation $\varphi : \mathbb{P}^2 \to \mathbb{P}^2$ centered at P_i to $\mathbb{P}^2 - \bigcup L_{ij}$ is an isomorphism onto its image, and each L_{ij} is sent to a point Q_k with $k \neq i, j$ such that if M is a curve meeting L_{ij} transversally away from the centers, then the strict transform of M has ordinary singularities at Q_k. Therefore, we require that away from P_1, the lines L_{ij} meets C transversally, and L_{12}, L_{13} are distinct from the tangent directions of C at P. There are only finitely many tangent directions of C at P_1, and since C is not strange (IV, 3.8) (IV, 3.9), we can find many lines L_{12} and L_{13} through P_1 with tangent vectors distinct from that of C at P_1 which meets C transversally. Lastly, pick L_{23} which avoids P_1, not parallel to L_{12}, L_{13}, meets C transversally, and the points P_2, P_3 of intersection of L_{23} with L_{12}, L_{13}, respectively, are not on C, which again exists because C is not strange. The strict transform of C under the quadratic transformation centered at P_i improves the singularity at P_1 (3.7) and only introduces ordinary singularities at Q_k. By (3.9), we can repeat this process for each nonordinary singularity until the strict transform only has ordinary singularities introduced by the quadratic transformations. □



4. (a) Use (4.5) to prove the following lemma on cubics: If C is an irreducible plane cubic curve, if L is a line meeting C in points P, Q, R, and L' is a line meeting C in points P', Q', R', let P'' be the third intersection of the line PP' with C, and define Q'', R'' similarly. Then P'', Q'', R'' are collinear.

(b) Let P_0 be an inflection point of C, and *define* the group operation on the set of regular points of C by the geometric recipe "let the line PQ meet C at R, and let P_0R meet C at T, then $P + Q = T$" as in (II, 6.10.2) and (II, 6.11.4). Use (a) to show that this operation is associative.

Proof.

(a) Let L_P be the line PP', and similarly for L_Q, L_R, and let M be the line going through P'' and Q''. The divisors C and $L_P + L_Q + L_R$ belong to the linear system of conics in $\mathbb{P}^2$ with assigned base points $P, P', P'', Q, Q'', Q''', R, R'$, and they intersect at R'', so by (4.5) R'' is the unique point in $\mathbb{P}^2$ such that every cubic curve in $\mathbb{P}^2$ going through $P, P', P'', Q, Q'', Q''', R, R'$ also goes through R''. In particular, $L + L' + M$ belongs to this linear system, so it must go through R''. Therefore, M must go through R''.

(b) Let P_1, P_2, P_3 be regular points of C, let L_{12} be the line P_1P_2, let R_{12} be the third point L_{12} meets C, let L'_{12} be the line P_0R_{12}, let $Q_{12} = P_1 + P_2$, and similarly for L_{23}, L'_{23}, and R_{23}, Q_{23}. Let $M_{12,3}$ be the line $Q_{12}P_3$, let $S_{12,3}$ be the third point $M_{12,3}$ meets C, let $M'_{12,3}$ be the line $P_0S_{12,3}$, let $T_{12,3} = Q_{12} + P_3$, and similarly for $M_{1,23}, M'_{1,23}$, and $S_{1,23}, T_{1,23}$. We want to show $T_{12,3} = T_{1,23}$. Applying (a) to

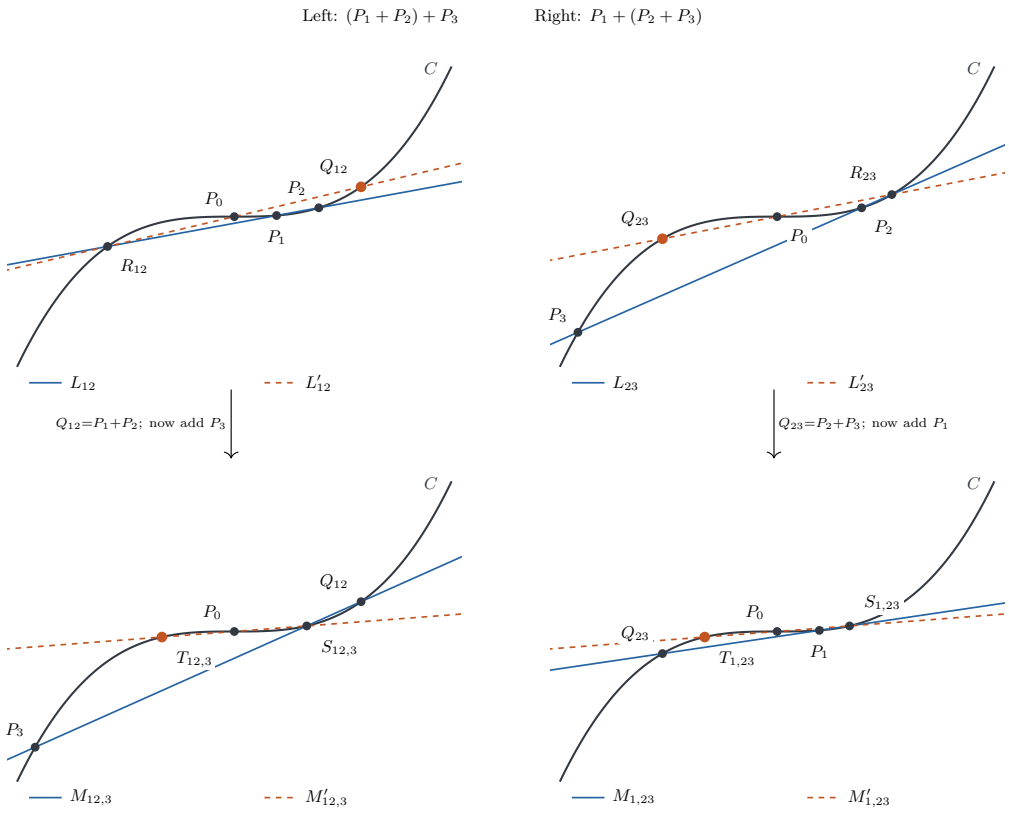
$$L = L_{12}, \quad L' = L'_{23},$$

and

$$\begin{aligned} PP' &= M_{1,23}, & P'' &= S_{1,23}, \\ QQ' &= L_{23}, & Q'' &= P_3, \\ RR' &= L'_{12}, & R'' &= Q_{12}, \end{aligned}$$

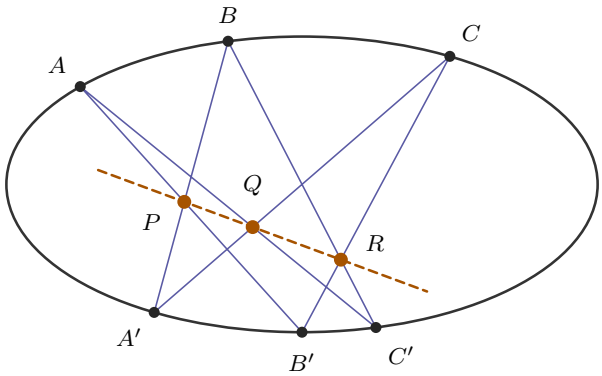
shows $S_{1,23}$, P_3, and Q_{12} are collinear. Therefore, $S_{1,23} = S_{12,3}$, and so $T_{1,23} = T_{12,3}$.

□



$T_{12,3} = (P_1+P_2)+P_3 = P_1+(P_2+P_3) = T_{1,23}$.

5. Prove Pascal's theorem: if A, B, C, A', B', C' are any six points on a conic, then the points $P = AB'.A'B$, $Q = AC'.A'C$, and $R = BC'.B'C$ are collinear.



Proof. Denote the conic by C, and let L be the line PQ. By (4.5), any cubic curve through the points $A, B, C, A', B', C', P, Q$ passes through a uniquely determined point, which we claim to be R. Indeed, the divisors $AB' + A'C + BC'$ and $AC' + A'B + B'C$ intersect at R, so $C + L$ passes through R. Hence, L passes through R. □

6. Generalize (4.5) as follows: given 13 points $P_1, \ldots, P_{13}$ in the plane, there are three additional determined points P_{14}, P_{15}, P_{16}, such that all quartic curves through $P_1, \ldots, P_{13}$ also pass through P_{14}, P_{15}, P_{16}. What hypotheses are necessary on $P_1, \ldots, P_{13}$ for this to be true?

7. If D is any divisor of degree d on the cubic surface (4.7.3), show that

$$p_a(D) \leq \begin{cases} \frac{1}{6}(d-1)(d-2) & \text{if } d \equiv 1, 2 \mod 3 \\ \frac{1}{6}(d-1)(d-2) + \frac{2}{3} & \text{if } d \equiv 0 \mod 3. \end{cases}$$

Show furthermore that for every $d > 0$, this maximum is achieved by some irreducible nonsingular curve.

Proof. Let $D = a\ell - \sum b_i e_i$ be a divisor of degree d. By (4.8), we have

$$d = 3a - \sum b_i,$$
$$p_a(D) = \frac{1}{2}(a-1)(a-2) - \frac{1}{2}\sum b_i(b_i - 1),$$

so to find an upper bound on $p_a(D)$, we want to maximize a and minimize b_i. If $d \equiv 0 \mod 3$, then we can take $a = d/3$ and $b_i = 0$ for all i. □

8. Show that a divisor class D on the cubic surface contains an irreducible curve $\iff$ it contains an irreducible nonsingular curve $\iff$ it is either (a) one of the 27 lines, or (b) a conic (meaning a curve of degree 2) with $D^2 = 0$, or (c) $D.L \geq 0$ for every line L, and $D^2 > 0$.

9. If C is an irreducible non-singular curve of degree d on the cubic surface, and if the genus $g > 0$, then

$$g \geq \begin{cases} \frac{1}{2}(d-6) & \text{if } d \text{ is even, } d \geq 8, \\ \frac{1}{2}(d-5) & \text{if } d \text{ is odd, } d \geq 13, \end{cases}$$

and this minimum value of $g > 0$ is achieved for each d in the range given.

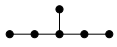
10. A curious consequence of the implications (iv) $\implies$ (iii) of (4.11) is the following numerical fact: Given integers $a, b_1, \ldots, b_6$ such that $b_i > 0$ for each i, $a - b_i - b_j > 0$ for each i, j, and $2a - \sum_{i \neq j} b_i > 0$ for each j, we must necessarily have $a^2 - \sum b_i^2 > 0$. Prove this directly (for $a, b_1, \ldots, b_6 \in \mathbb{R}$) using methods of freshman calculus.

11. *The Weyl Groups.* Given any diagram consisting of points and line segments joining some of them, we define an abstract group, given by the generators and relations, as follows: each point represents a generator x_i. The relations are $x_i^2 = 1$ for each i; $(x_i x_j)^2 = 1$ if i and j are not joined by a line segment, and $(x_i x_j)^3 = 1$ if i and j are joined by a line segment.

(a) The *Weyl group* $\mathbf{A}_n$ is defined using the diagram

of $n-1$ points, each joined to the next. Show that it is isomorphic to the symmetric group Σ_n as follows: map the generators of $\mathbf{A}_n$ to the elements $(12), (23), \ldots, (n-1, n)$ of Σ_n, to get a surjective homomorphism $\mathbf{A}_n \to \Sigma_n$. Then estimate the number of elements of $\mathbf{A}_n$ to show in fact it is an isomorphism.

(b) The *Weyl group* $\mathbf{E}_6$ is defined using the diagram



Call the generators $x_1, \ldots, x_5$ and y. Show that one obtains a surjective homomorphism $\mathbf{E}_6 \to G$, the group of automorphisms of the configuration of 27 lines (4.10.1), by sending $x_1, \ldots, x_5$ to the permutations $(12), (23), \ldots, (56)$ of the E_i, respectively, and y to the element associated with the quadratic transformation based at P_1, P_2, P_3.

(c) Estimate the number of elements in $\mathbf{E}_6$, and thus conclude that $\mathbf{E}_6 \cong G$.

12. Use (4.11) to show that if D is any ample divisor on the cubic surface X, then $\mathrm{H}^1(X, \mathcal{O}_X(-D)) = 0$. This is Kodaira's vanishing theorem for the cubic surface (III, 7.15).

Proof. By (4.11), D is ample if and only if D is very ample, so by Bertini's theorem, we can assume D is irreducible and nonsingular as a closed subvariety of X. In particular, $\mathcal{O}_X(-D)$ has no global sections, and we have a short exact sequence

$$0 \longrightarrow \mathcal{O}_X(-D) \longrightarrow \mathcal{O}_X \longrightarrow \mathcal{O}_D \longrightarrow 0 \ ,$$

where $\mathcal{O}_D$ is the structure sheaf of D as a variety. Since X is a hypersurface of $\mathbb{P}^3$, we have

$$\mathrm{H}^1(X, \mathcal{O}_X) = \mathrm{H}^1(\mathbb{P}^3, \mathcal{O}_{\mathbb{P}^3}) = 0,$$

so the long exact sequence in cohomology gives

$$0 \longrightarrow \mathrm{H}^0(X, \mathcal{O}_X) \longrightarrow \mathrm{H}^0(D, \mathcal{O}_D) \longrightarrow \mathrm{H}^1(X, \mathcal{O}_X(-D)) \longrightarrow 0 \ ,$$

and since D is irreducible, we have

$$h^0(X, \mathcal{O}_X) = h^0(D, \mathcal{O}_D) = 1.$$

Hence by exactness, we have $\mathrm{H}^1(X, \mathcal{O}_X(-D)) = 0$. □

13. Let X be the Del Pezzo surface of degree 4 in $\mathbb{P}^4$ obtained by blowing up 5 points of $\mathbb{P}^2$ (4.7).

(a) Show that X contains 16 lines.

(b) Show that X is a complete intersection of two quadric hypersurfaces in $\mathbb{P}^4$ (the converse follows from (4.7.1)).

14. Using the method of (4.13.1), verify that there are nonsingular curves in $\mathbb{P}^3$ with $d = 8$, $g = 6, 7$; $d = 9$, $g = 7, 8, 9$; $d = 10$, $g = 8, 9, 10, 11$. Combining with (IV, §6), this completes the determination of all possible g for curves of degree $d \geq 10$ in $\mathbb{P}^3$.

15. Let $P_1, \ldots, P_4$ be a fintie set of (ordinary) points of $\mathbb{P}^2$, no 3 collinear. We define an *admissible transformation* to be a quadratic transformation (4.2.3) centered at some three of the P_i (call them P_1, P_2, P_3). This gives a new $\mathbb{P}^2$, and a new set of r points, namely Q_1, Q_2, Q_3, and the images of $P_4, \ldots, P_r$. We say that $P_1, \ldots, P_r$ are in *general position* if no three are collinear, and furthermore after any finite sequence of admissible transformations, the new set of r points also have no three collinear.

(a) A set of 6 points is in general position if and only if no three are collinear and not all six lie on a conic.

(b) If $P_1, \ldots, P_r$ are in general position, then the r points obtained by any finite sequence of admissible transformations are also in general position.

(c) Assume the ground field k is uncountable. Then given $P_1, \ldots, P_r$ in general position, there is a dense $V \subseteq \mathbb{P}^2$ such that for any $P_{r+1} \in V$, $P_1, \ldots, P_{r+1}$ will be in general position.

(d) Now take $P_1, \ldots, P_r \in \mathbb{P}^2$ in general position, and let X be the surface obtained by blowing up $P_1, \ldots, P_r$. If $r = 7$, show that X has exactly 56 irreducible nonsingular curves C with $g = 0$, $C^2 = -1$, and that these are the only irreducible curves with negative self-intersection. Ditto for $r = 8$, the number being 240.

(e) For $r = 9$, show that the surface X defined in (d) has infinitely many irreducible nonsingular curves C with $g = 0$ and $C^2 = -1$.

16. For the *Fermat cubic surface.* $x_0^3 + x_1^3 + x_2^3 + x_3^3 = 0$, find the equations of the 27 lines explicitly, and verify their incidence relations. What is the group of automorphisms of this surface?

Chapter 5, Section 5

James Lee

October 11, 2026

1. Let f be a rational function on the surface X. Show that it is possible to "resolve the singularities of f" in the following sense: there is a birational moprhism $g : X' \to X$ so that f induces a morphism of X' to $\mathbb{P}^1$.

2. Let $Y \cong \mathbb{P}^1$ be a curve in a surface X, with $Y^2 < 0$. Show that Y is contractible (5.7.2) to a point on a projective variety X_0 (in general singular).

3. If $\pi : \tilde{X} \to X$ is a monoidal transformation with center P, show that $\mathrm{H}^1(\tilde{X}, \Omega_{\tilde{X}}) \cong \mathrm{H}^1(X, \Omega_X) \oplus k$. This gives another proof of (5.8).

4. Let $f : X \to X'$ be a birational morphism of nonsingular surfaces.

(a) If $Y \subseteq X$ is an irreducible curve such that $f(Y)$ is a point, then $Y \cong \mathbb{P}^1$ and $Y^2 < 0$.

(b) (Mumford [6].) Let $P' \in X'$ be a fundamental point of f^{-1}, and let $Y_1, \ldots, Y_r$ be the irreducible components of $f^{-1}(P')$. Show that the matrix $||Y_i.Y_j||$ is negative definite.

5. Let C be a curve, and let $\pi : X \to C$ and $\pi' : X' \to C$ be two geometrically ruled surfaces over C. Show that there is a finite sequence of elementary transformations (5.7.1) which transform X into X'.

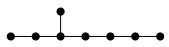
6. Let X be a surface with function field K. Show that every valuation ring R of K/k is one of the three kinds described in (II, Ex 4.12).

7. Let Y be an irreducible curve on a surface X, and suppose there is a morphism $f : X \to X_0$ to a projective variety X_0 of dimension 2, such that $f(Y)$ is a point P and $f^{-1}(P) = Y$. Then show that $Y^2 < 0$.

8. *A surface singularity.* Let k be an algebraically closed field, and let X be the surface in $\mathbb{A}^3_k$ defined by the equation $x^2 + y^3 + z^5 = 0$. It has an isolated singularity at the origin $P = (0, 0, 0)$.

(a) Show that the affine ring $A = k[x, y, z]/(x^2 + y^3 + z^5)$ of X is a unique factorization domain, as follows. Let $t = z^{-1}$, $u = t^3x$, and $v = t^2y$. Show that z is irreducible in A, $t \in k[u, v]$, and $A[z^{-1}] = k[u, v, t^{-1}]$. Conclude that A is a UFD.

(b) Show that the singularity at P can be resolved by eight successive blowing-up. If $\tilde{X}$ is the resulting nonsingular surface, then the inverse image of P is a union of eight projective lines, which intersect each other according to the Dynkin diagram $\mathbb{E}_8$:



Here each circle denotes a line, and two circles are joined by a line segment whenever the corresponding lines intersect.

Chapter 5, Section 6

James Lee

October 11, 2026

1. Let X be a surface in $\mathbb{P}^n$, $n \geq 3$, defined as the complete intersection of hypersurfaces of degree $d_1, \ldots, d_{n-2}$, with each $d_i \geq 2$. Show that for all but finitely many choices of $(n, d_1, \ldots, d_{n-2})$, the surface X is of general type. List the exceptional cases, and where they fit into the classification picture.

2. Prove the following theorem of Chern and Griffiths. Let X be a nonsingular surface of degree d in $\mathbb{P}^{n+1}_{\mathbb{C}}$, which is not contained in any hyperplane. If $d < 2n$, then $p_g(X) = 0$. If $d = 2n$, then either $p_g(X) = 0$, or $p_g(X) = 1$ and X is a K3 surface.

Appendix A

Intersection Theory

Appendix A

James Lee

October 9, 2026

1. Show that the definition of rational equivalence in §1 is equivalent to the equivalence relation generated by the following relation: two cycles Y, Z of codimension r on X are equivalent if there exists a cycle W of codimension r on $X \times \mathbb{A}^1$ which intersects $X \times \{0\}$ and $X \times \{1\}$ properly, and such that $Y = W.(X \times \{0\})$, $Z = W.(X \times \{1\})$.

2. Prove the following result about Weil divisors, which generalizes (IV, Ex. 2.6), and which is needed to show that f_* is well-defined modulo ratonal equivalence (A3): Let $f : X \to X'$ be a proper, generically finite map of normal varieties, and let D_1 and D_2 be linearly equivalent Weil divisors on X. Then f_*D_1 and f_*D_2 are linearly equivalent Weil divisors on X'.

3. Show directly that any subvariety of degree d in $\mathbb{P}^n$ is rationally equivalent to d times a linear space of the same dimension, by using a projection argument similar to (III, 9.8.3).

4. Let $\pi : X \to C$ be a ruled surface (V, §2) over a nonsingular curve C. Show that the group $A^2(X)$ of zero-cycles modulo rational equivalence is isomorphic to $\operatorname{Pic} C$.

5. Let X be a surface, let $P \in X$ be a point, and let $\pi : \tilde{X} \to X$ be the monoidal transformation with center P (V, §3). Show that $A(\tilde{X}) \cong \pi^* A(X) \oplus \mathbb{Z}$, where $\mathbb{Z}$ is generated by the exceptional curve $E \in A^1(X)$, and the intersection theory is determined by $E^2 = -\pi^*(P)$.

6. Let X be a nonsingular projective variety of dimension n, and let $\Delta \subset X \times X$ be the diagonal. Show that $c_n(\mathcal{T}_X) = \Delta^2$ in $A^n(X)$, under the natural isomorphism of X with Δ.

7. Let X be a nonsingular projective 3-fold, with Chern classes c_1, c_2, c_3. Show that

$$1 - p_a = \frac{1}{24} c_1 c_2,$$

and for any divisor D,

$$\chi(\mathcal{O}_X(D)) = \frac{1}{12} D.(D - K).(2D - K) + \frac{1}{12} D.c_2 + 1 - p_a.$$

8. Let $\mathcal{E}$ be a locally free sheaf of rank 2 on $\mathbb{P}^3$, with Chern classes c_1, c_2. Since $A(\mathbb{P}^3) \cong \mathbb{Z}[h]/(h^4)$, we can think of c_1 and c_2 as integers. Show that $c_1 c_2 \equiv 0 \mod 2$.

9. *Surfaces in* $\mathbb{P}^4$

(a) Verify the formula of (4.1.3) for the rational cubic scroll in $\mathbb{P}^4$ (V, 2.19.1).

(b) If X is a K3 surface in $\mathbb{P}^4$, show that its degree must be 4 or 6.

(c) If X is an Abelian surface in $\mathbb{P}^4$, show that its degree must be 10.

(d) Determine which of the rational ruled surfaces X_e, $e \geq 0$ (V, §2), admit an embedding in $\mathbb{P}^4$.

10. Use the fact that the tangent sheaf of an Abelian variety is free, to show that it is impossible to embed an Abelian 3-fold in $\mathbb{P}^5$.

Appendix B

Transcendental Methods

Appendix B

James Lee

October 9, 2026

1. Show that the unit disc in $\mathbb{C}$ is not isomorphic to X_h for any scheme X.

2. Let $z_1, z_2, \ldots$ be an infinite sequence of complex numbers with $|z_n| \to \infty$ as $n \to \infty$. Let $\mathfrak{J} \subset \mathcal{O}_{\mathbb{C}}$ be the sheaf of ideals of holomorphic functions vanishing at all of the z_n. Show that there is no coherent algebraic sheaf of ideals $\mathcal{I} \subset \mathcal{O}_X$, where $X = \mathbb{A}^1_{\mathbb{C}}$ is the affine line, such that $\mathfrak{J} = \mathcal{I}_h$ as an ideal in $\mathcal{O}_{\mathbb{C}}$. Show on the other hand that there is a coherent sheaf $\mathcal{F}$ on X such that $\mathcal{F}_h = \mathfrak{J}$ as coherent sheaves.

3. (Serre [12].) On $\mathbb{C}^2 - (0,0)$, we define an invertible analytic sheaf $\mathfrak{L}$ as follows: $\mathfrak{L} \cong \mathcal{O}$ when $z \neq 0$, $\mathfrak{L} \cong \mathcal{O}$ when $w \neq 0$, and when both $z, w \neq 0$, the two copies of $\mathcal{O}$ are glued by multiplication by $e^{-1/zw}$ in the local ring at the point (z, w). Show that there is no invertible algebraic sheaf $\mathcal{L}$ on $\mathbb{A}^2 - (0,0)$ with $\mathcal{L}_h \cong \mathfrak{L}$.

4. Show directly that if X is a scheme which is reduced and proper over $\mathbb{C}$, then $\mathrm{H}^0(X, \mathcal{O}_X) \cong \mathrm{H}^0(X_h, \mathcal{O}_{X_h})$. Conversely, show that if X is not proper over $\mathbb{C}$, then there is a coherent sheaf $\mathcal{F}$ on X with $\mathrm{H}^0(X, \mathcal{F}) \neq \mathrm{H}^0(X_h, \mathcal{F}_h)$.

5. If X, X' are nonsingular affine algebraic curves, with $X_h \cong X'_h$, show that $X \cong X'$.

6. Show that if X and Y are projective schemes over $\mathbb{C}$, and $\mathfrak{f} : X_h \to Y_h$ is a morphism of analytic spaces, then there exists a (unique) morphism $f : X \to Y$ with $f_h = \mathfrak{f}$.

Appendix C

The Weil Conjectures

Appendix C

James Lee

October 9, 2026

1. Let X be a disjoint union of locally closed subschemes X_i. Then show that

$$Z(X,t) = \prod Z(X_i,t).$$

2. Let $X = \mathbb{P}^n_k$, where $k = \mathbb{F}_q$, and show from the definition of the zeta function that

$$Z(\mathbb{P}^n_k, t) = \frac{1}{(1-t)(1-qt)\cdots(1-q^n t)}.$$

Verify the Weil conjectures for $\mathbb{P}^n_k$.

3. Let X be a scheme of finite type over $\mathbb{F}_q$, and let $\mathbb{A}^1$ be teh affine lines. Show that

$$Z(X \times \mathbb{A}^1, t) = Z(X, qt).$$

4. The *Riemann zeta function* is defined as

$$\zeta(s) = \prod \frac{1}{1-p^{-s}},$$

for $s \in C$, the product being taken over all prime integers p. If we regard this function as being associated with the scheme $\operatorname{Spec}\mathbb{Z}$, it is natural to define, for any scheme X of finite type over $\operatorname{Spec}\mathbb{Z}$,

$$\zeta_X(s) = \prod (1 - N(x)^{-s})^{-1}$$

where the product is taken over all closed points $x \in X$, and $N(x)$ denotes the number of elements in the residue field $k(x)$. Show that if X is of finite type over $\mathbb{F}_q$, then this function is connected to $Z(X,t)$ by the formula

$$\zeta_X(s) = Z(X, q^{-s}).$$

5. Let X be a curve of genus g over k. Assuming the statements (1.1) to (1.4) of the Weil conjectures, show that $N_1, N_2, \ldots, N_g$ determine N_r for all $r \geq 1$.

6. Use (IV, Ex. 4.16) to prove the Weil conjectures for elliptic curves. First note that for any r,

$$N_r = q^r - (f^r + \hat{f}^r) + 1,$$

where $f = F'$. Then calculate $Z(t)$ formally and conclude that

$$Z(t) = \frac{(1-ft)(1-\hat{f}t)}{(1-t)(1-qt)}$$

and hence

$$Z(t) = \frac{1 - at + qt^2}{(1-t)(1-qt)},$$

where $f + \hat{f} = a_X$. This proves rationality immediately. Verify the functional equation. Finally, if we write

$$1 - at + qt^2 = (1-\alpha t)(1-\beta t),$$

show that $|a| \leq 2\sqrt{q}$ if and only if $|\alpha| = |\beta| = \sqrt{q}$. Thus the analogue of the Riemann hypothesis is just (IV, Ex. 4.16d).

7. Use (V, Ex. 1.10) to prove the analogue of the Riemann hypothesis (1.3) for any curve C of genus g defined over $\mathbb{F}_q$. Write $N_r = 1 - a_r + q^r$. Then according to (V, Ex. 1.10),

$$|a_r| \le 2g\sqrt{q^r}.$$

On the other hand, by (4.2) the zeta function of C can be written

$$Z(t) = \frac{P_1(t)}{(1-t)(1-qt)}$$

where

$$P_1(t) = \prod_{i=1}^{2g}(1 - \alpha_i t)$$

is a polynomial of degree $2g = \dim \mathrm{H}^1(C, \mathbb{Q}_\ell)$.

(a) Using the definition of the zeta function and taking logs, show that

$$a_r = \sum_{i=1}^{2g}(\alpha_i)^r$$

for each r.

(b) Next show that $|a_r| \le 2g\sqrt{q^r}$ for all $r \iff |\alpha_i| \le \sqrt{q}$ for all i.

(c) Finally, use the functional equation (4.4) to show that $|\alpha_i| \le \sqrt{q}$ for all i implies that $|\alpha_i| = \sqrt{q}$ for all i.